

MatrixMultiplicationBias

March 26, 2026

1 Triton Kernel Profiling on AMD GPUs

1.1 Get the System Hardware Info

```
[1]: !amd-smi
```

```
+-----+
| AMD-SMI 26.0.2+39589fda      amdgpu version: 6.16.13  ROCm version: 7.0.3  |
| Platform: Linux Guest                                             |
+-----+
| BDF                        GPU-Name | Mem-Uti   Temp   UEC      Power-Usage |
| GPU  HIP-ID  OAM-ID  Partition-Mode | GFX-Uti   Fan      Mem-Usage |
+=====+
| 0000:ff:00.0 AMD Instinct MI300X VF | N/A       N/A    0        N/A/750 W |
| 0      0      7      SPX/NPS1 | N/A       N/A      285/196288 MB |
+-----+

+-----+
| Processes:                                                         |
| GPU      PID  Process Name          GTT_MEM  VRAM_MEM  MEM_USAGE  CU % |
+=====+
| No running processes found                                         |
+-----+
```

1.2 Check profiling environment

```
[2]: !rocprow-compute --version
```

```
-----
rocprow-compute version: 3.2.3 (release)
Git revision:      1d722c14
-----
```

1.3 Triton MatMulBias Kernel Program

```
[3]: %%writefile triton/matmulbias.py
import argparse
import torch

import triton
```

```

import triton.language as tl
import triton.profiler as proton

def _matmul_launch_metadata(grid, kernel, args):
    ret = {}
    M, N, K, WS = args["M"], args["N"], args["K"], args.get("WARP_SPECIALIZE",
↪False)
    BM, BN, BK = args["BLOCK_SIZE_M"], args["BLOCK_SIZE_N"],
↪args["BLOCK_SIZE_K"]
    ws_str = "_ws" if WS else ""
    ret["name"] = f"{kernel.name}{ws_str} [M={M}, N={N}, K={K}] [BM={BM},
↪BN={BN} BK={BK}]"
    if "output_ptr" in args:
        bytes_per_elem = args["output_ptr"].element_size()
    else:
        bytes_per_elem = 2
    ret[f"flops{bytes_per_elem * 8}"] = 2. * M * N * K
    ret["bytes"] = bytes_per_elem * (M * K + N * K + M * N)
    return ret

def _bias_launch_metadata(grid, kernel, args):
    ret = {}
    M, N, WS = args["M"], args["N"], args.get("WARP_SPECIALIZE", False)
    BM, BN = args["BLOCK_SIZE_M"], args["BLOCK_SIZE_N"]
    ws_str = "_ws" if WS else ""
    ret["name"] = f"{kernel.name}{ws_str} [M={M}, N={N}] [BM={BM}, BN={BN}]"
    if "output_ptr" in args:
        bytes_per_elem = args["output_ptr"].element_size()
    else:
        bytes_per_elem = 2
    ret[f"flops{bytes_per_elem * 8}"] = 2. * M * N
    ret["bytes"] = bytes_per_elem * (M + N + M * N)
    return ret

def matmul_autotune_config(pre_hook=None):
    return [
        triton.Config({'BLOCK_SIZE_M': BM, 'BLOCK_SIZE_N': BN, "BLOCK_SIZE_K":
↪BK, "GROUP_SIZE_M": 8}, num_stages=s,
        num_warps=w, pre_hook=pre_hook)
        for BM in [128]
        for BN in [128, 256]
        for BK in [64, 128]
        for s in ([3, 4, 5])
        for w in [4, 8]
    ]

```

```

def bias_autotune_config(pre_hook=None):
    return [
        triton.Config({'BLOCK_SIZE_M': BM, 'BLOCK_SIZE_N': BN, "GROUP_SIZE_M": 8}, num_stages=s,
        ↪8}, num_stages=s,
            num_warps=w, pre_hook=pre_hook)
        for BM in [128]
        for BN in [128, 256]
        for s in ([3, 4, 5])
        for w in [4, 8]
    ]

# MatMul kernel
@triton.autotune(
    configs=matmul_autotune_config(),
    key=['M', 'N', 'K'],
)
@triton.jit(launch_metadata=_matmul_launch_metadata)
def matmul_kernel(
    a_ptr, b_ptr, c_ptr,
    M, N, K,
    stride_am, stride_ak,
    stride_bk, stride_bn,
    stride_cm, stride_cn,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, BLOCK_SIZE_K: 8,
    ↪8 tl.constexpr,
    GROUP_SIZE_M: tl.constexpr
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)
    pid_n = (pid % num_pid_in_group) // group_size_m

    tl.assume(pid_m >= 0)
    tl.assume(pid_n >= 0)
    tl.assume(stride_am > 0)
    tl.assume(stride_ak > 0)
    tl.assume(stride_bn > 0)
    tl.assume(stride_bk > 0)
    tl.assume(stride_cm > 0)
    tl.assume(stride_cn > 0)

    offs_am = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M

```

```

    offs_bn = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
    offs_k = tl.arange(0, BLOCK_SIZE_K)
    a_ptrs = a_ptr + (offs_am[:, None] * stride_am + offs_k[None, :] *
↳stride_ak)
    b_ptrs = b_ptr + (offs_k[:, None] * stride_bk + offs_bn[None, :] *
↳stride_bn)

    accumulator = tl.zeros((BLOCK_SIZE_M, BLOCK_SIZE_N), dtype=tl.float32)
    for k in range(0, tl.cdiv(K, BLOCK_SIZE_K)):
        a = tl.load(a_ptrs, mask=offs_k[None, :] < K - k * BLOCK_SIZE_K,
↳other=0.0)
        b = tl.load(b_ptrs, mask=offs_k[:, None] < K - k * BLOCK_SIZE_K,
↳other=0.0)
        accumulator = tl.dot(a, b, accumulator)
        a_ptrs += BLOCK_SIZE_K * stride_ak
        b_ptrs += BLOCK_SIZE_K * stride_bk
    c = accumulator.to(tl.float16)

    offs_cm = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
    offs_cn = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
    c_ptrs = c_ptr + stride_cm * offs_cm[:, None] + stride_cn * offs_cn[None, :]
    c_mask = (offs_cm[:, None] < M) & (offs_cn[None, :] < N)
    tl.store(c_ptrs, c, mask=c_mask)

# Bias kernel
@triton.autotune(
    configs=bias_autotune_config(),
    key=['M', 'K'],
)
@triton.jit(launch_metadata=_bias_launch_metadata)
def bias_kernel(
    a_ptr,
    bias_ptr,
    output_ptr,
    M, N,
    stride_m, stride_n,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, GROUP_SIZE_M:
↳tl.constexpr,
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)

```

```

pid_n = (pid % num_pid_in_group) // group_size_m

offs_m = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M
offs_n = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
a_ptrs = a_ptr + (offs_m[:, None] * stride_m + offs_n[None, :] * stride_n)

a = tl.load(a_ptrs)

bias = tl.load(bias_ptr + offs_n, mask=offs_n < N, other=0.0)

output = a + bias[None, :]

offs_m = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
offs_n = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
output_ptrs = output_ptr + stride_m * offs_m[:, None] + stride_n *
↪offs_n[None, :]
output_mask = (offs_m[:, None] < M) & (offs_n[None, :] < N)
tl.store(output_ptrs, output, mask=output_mask)

# MatMulBias Fusion kernel
@triton.autotune(
    configs=matmul_autotune_config(),
    key=['M', 'N', 'K'],
)
@triton.jit(launch_metadata=_matmul_launch_metadata)
def matmulbias_kernel(
    a_ptr, b_ptr, c_ptr,
    bias_ptr,
    M, N, K,
    stride_am, stride_ak,
    stride_bk, stride_bn,
    stride_cm, stride_cn,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, BLOCK_SIZE_K:
↪tl.constexpr,
    GROUP_SIZE_M: tl.constexpr
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)
    pid_n = (pid % num_pid_in_group) // group_size_m

    tl.assert(pid_m >= 0)

```

```

tl.assume(pid_n >= 0)
tl.assume(stride_am > 0)
tl.assume(stride_ak > 0)
tl.assume(stride_bn > 0)
tl.assume(stride_bk > 0)
tl.assume(stride_cm > 0)
tl.assume(stride_cn > 0)

offs_am = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M
offs_bn = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
offs_k = tl.arange(0, BLOCK_SIZE_K)
a_ptrs = a_ptr + (offs_am[:, None] * stride_am + offs_k[None, :] *
↪stride_ak)
b_ptrs = b_ptr + (offs_k[:, None] * stride_bk + offs_bn[None, :] *
↪stride_bn)

accumulator = tl.zeros((BLOCK_SIZE_M, BLOCK_SIZE_N), dtype=tl.float32)
for k in range(0, tl.cdiv(K, BLOCK_SIZE_K)):
    a = tl.load(a_ptrs, mask=offs_k[None, :] < K - k * BLOCK_SIZE_K,
↪other=0.0)
    b = tl.load(b_ptrs, mask=offs_k[:, None] < K - k * BLOCK_SIZE_K,
↪other=0.0)
    accumulator = tl.dot(a, b, accumulator)
    a_ptrs += BLOCK_SIZE_K * stride_ak
    b_ptrs += BLOCK_SIZE_K * stride_bk
    c = accumulator.to(tl.float16)

bias = tl.load(bias_ptr + offs_bn, mask=offs_bn < N, other=0.0)
output = c + bias[None, :]

offs_cm = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
offs_cn = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
c_ptrs = c_ptr + stride_cm * offs_cm[:, None] + stride_cn * offs_cn[None, :]
c_mask = (offs_cm[:, None] < M) & (offs_cn[None, :] < N)
tl.store(c_ptrs, output, mask=c_mask)

# MatMulBias kernel wrapper function
def nonfusion_matmulbias(a: torch.Tensor, b: torch.Tensor, bias: torch.Tensor):
    assert a.shape[1] == b.shape[0], "Incompatible dimensions"
    assert a.is_contiguous(), "Matrix A must be contiguous"
    M, K = a.shape
    K, N = b.shape
    assert bias.shape[0] == N, "BIAS has incompatible dimensions"
    c = torch.empty((M, N), device=a.device, dtype=torch.float16)
    matmul_grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.
↪cdiv(N, META['BLOCK_SIZE_N']), )

```

```

matmul_kernel[matmul_grid](
    a, b, c,
    M, N, K,
    a.stride(0), a.stride(1),
    b.stride(0), b.stride(1),
    c.stride(0), c.stride(1),
)

o = torch.empty((M, N), device=c.device, dtype=torch.float16)
bias_grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.
↪cdiv(N, META['BLOCK_SIZE_N']), )
bias_kernel[bias_grid](
    c, bias, o,
    M, N,
    c.stride(0), c.stride(1),
)

return o

# MatMulBias kernel wrapper function
def fusion_matmulbias(a: torch.Tensor, b: torch.Tensor, bias: torch.Tensor):
    assert a.shape[1] == b.shape[0], "Incompatible dimensions"
    assert a.is_contiguous(), "Matrix A must be contiguous"
    M, K = a.shape
    K, N = b.shape
    assert bias.shape[0] == N, "BIAS has incompatible dimensions"
    o = torch.empty((M, N), device=a.device, dtype=torch.float16)
    grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.cdiv(N, ↪
↪META['BLOCK_SIZE_N']), )
    matmulbias_kernel[grid](
        a, b, o,
        bias,
        M, N, K,
        a.stride(0), a.stride(1),
        b.stride(0), b.stride(1),
        o.stride(0), o.stride(1),
    )

    return o

def get_rtol():
    target = triton.runtime.driver.active.get_current_target()
    if target.backend == "hip":
        if target.arch == "gfx90a":
            return 1e-2
        elif target.arch == "gfx942":

```

```

        return 1e-2
    else:
        return 0

def show_profile(profile_name):
    import triton.profiler.viewer as proton_viewer
    metric_names = ["time/ms", "flop16/s"]
    file_name = f"profiles/{profile_name}.hatchet"
    tree, metrics = proton_viewer.parse(metric_names, file_name)

    print(f"Proton profile results for {profile_name}")
    proton_viewer.print_tree(tree, metrics)

if __name__ == "__main__":
    parser = argparse.ArgumentParser()
    parser.add_argument("--profile", action="store_true", help="run the proton_
↳ profiler")
    parser.add_argument("--trace", action="store_true", help="enable python_
↳ tracing for the proton profiler")
    parser.add_argument("--verify", action="store_true", help="run a torch_
↳ version of the kernel and compare the results")
    parser.add_argument("--M", default=1024, type=int, help="M size of the_
↳ input matrices, default: %(default)d")
    parser.add_argument("--N", default=1024, type=int, help="N size of the_
↳ input matrices, default: %(default)d")
    parser.add_argument("kernel", choices=["fusion", "nonfusion"], help="select_
↳ which version of the kernel to run")
    args = parser.parse_args()

    # Test matrices
    torch.manual_seed(0)
    M = args.M
    N = args.N
    a = torch.randn((M, N), device='cuda', dtype=torch.float16)
    b = torch.randn((N, M), device='cuda', dtype=torch.float16)
    bias = torch.randn(N, device='cuda', dtype=torch.float16)

    if args.profile:
        if args.trace:
            context = "python"
        else:
            context = "shadow"

        print(f"Profiling the {args.kernel} matmulbias kernel (input size M:
↳ {args.M} N:{args.N}) ...")

```



```

        proton.start(f"profiles/{args.kernel}_matmulbias_{args.M}x{args.N}",
↪context=context, hook="triton")
    else:
        print(f"Running the {args.kernel} matmulbias kernel (input size M:{args.
↪M} N:{args.N}) ...")

    if args.kernel == "fusion":
        triton_output = fusion_matmulbias(a, b, bias)
    elif args.kernel == "nonfusion":
        triton_output = nonfusion_matmulbias(a, b, bias)

    if args.profile:
        proton.finalize()
    print(f"triton_output: {triton_output}")

    if args.profile:
        show_profile(f"{args.kernel}_matmulbias_{args.M}x{args.N}")

    if args.verify:
        print(f"Running the torch matmulbias kernel ...")
        torch_output = torch.addmm(bias, a, b)
        print(f"torch_output: {torch_output}")
        print("Verifying triton results with torch ...")
        if torch.allclose(triton_output, torch_output, atol=1e-2,
↪rtol=get_rtol()):
            print("Verified")
        else:
            print("Differ")

```

Overwriting triton/matmulbias.py

1.3.1 Help

```
[4]: !python triton/matmulbias.py --help
```

```
usage: matmulbias.py [-h] [--profile] [--trace] [--verify] [--M M] [--N N]
                    {fusion,nonfusion}
```

positional arguments:

{fusion,nonfusion} select which version of the kernel to run

options:

-h, --help	show this help message and exit
--profile	run the proton profiler
--trace	enable python tracing for the proton profiler
--verify	run a torch version of the kernel and compare the results
--M M	M size of the input matrices, default: 1024

--N N N size of the input matrices, default: 1024

1.4 Non-Fusion MatMulBias Kernel

1.4.1 Verify the Non-Fusion MatMulBias Kernel

This also ensures the kernel has been built and cached. The results from the triton kernel are compared against torch.

Small Input Default size is M: 1024 and N: 1024

```
[5]: !python triton/matmulbias.py --verify nonfusion
```

Running the nonfusion matmulbias kernel (input size M:1024 N:1024) ...

```
triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500,  8.8203,
 6.4609],
 [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
 [13.8125, -6.4609,  0.5146, ..., 48.0000, 25.1562, -27.5156],
 ...,
 [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
 [ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
 [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1250]],
 device='cuda:0', dtype=torch.float16)
```

Running the torch matmulbias kernel ...

```
torch_output: tensor([[ 2.7480, 43.6562, 30.9531, ..., 26.2500,  8.8203,
 6.4609],
 [ 7.1602, -21.3906, 17.7969, ..., -16.8125, 29.6406, -48.3125],
 [13.8125, -6.4609,  0.5142, ..., 48.0312, 25.1406, -27.5156],
 ...,
 [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
 [ 9.6094, -52.2500, 43.0000, ..., -45.8125, -49.3125, 32.7188],
 [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1562]],
 device='cuda:0', dtype=torch.float16)
```

Verifying triton results with torch ...

Verified

Large Input

```
[6]: !python triton/matmulbias.py --M 32768 --N 32768 --verify nonfusion
```

Running the nonfusion matmulbias kernel (input size M:32768 N:32768) ...

```
triton_output: tensor([[ -60.8750,  56.6562,  57.0938, ..., 274.0000,
295.2500,
 14.8828],
 [205.3750, -353.7500, -102.1875, ..., -24.4219,  57.0625,
-30.7969],
 [ 56.2188, 258.7500,  46.6562, ..., -366.0000, 395.2500,
285.2500],
 ...,
 [ -5.0977,  99.9375, -73.5625, ..., -70.5000, 204.2500,
```

```

        -156.8750],
    [ 86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
      100.8750],
    [-27.6406, 93.5625, 77.9375, ..., -213.1250, -323.2500,
      -16.8438]], device='cuda:0', dtype=torch.float16)
Running the torch matmulbias kernel ...
torch_output: tensor([[ -60.8750,  56.6562,  57.1250, ..., 274.0000,
 295.2500,
    14.8828],
 [ 205.3750, -353.7500, -102.1875, ..., -24.4219,  57.0312,
   -30.7969],
 [  56.2188,  258.7500,  46.6562, ..., -366.0000,  395.2500,
   285.2500],
 ...,
 [  -5.0977,  99.9375, -73.5625, ..., -70.5000,  204.2500,
   -156.8750],
 [ 86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
   100.8750],
 [-27.6406, 93.5625, 77.9375, ..., -213.0000, -323.2500,
   -16.8438]], device='cuda:0', dtype=torch.float16)
Verifying triton results with torch ...
Verified

```

1.4.2 Profile the Non-Fusion MatMulBias Kernel

Using the Triton Proton CLI Generates a hatchet file, `nonfusion_matmulbias_x.hatchet`, under the profiles directory.

Help

```
[7]: !proton --help
```

```

usage:
  proton [options] script.py [script_args] [script_options]
  proton [options] pytest [pytest_args] [script_options]
  python -m triton.profiler.proton [options] script.py [script_args]
[script_options]

```

The proton command utility for profiling scripts and pytest tests.

positional arguments:

```
target_args          Subcommand and its arguments
```

options:

```

-h, --help          show this help message and exit
-n NAME, --name NAME Name of the profiling session
-b {cupti,roctracer,instrumentation}, --backend
{cupti,roctracer,instrumentation}
                    Profiling backend

```

```

-c {shadow,python}, --context {shadow,python}
    Profiling context
-m MODE, --mode MODE Profiling mode
-d {tree,trace}, --data {tree,trace}
    Profiling data
-k {None,triton}, --hook {None,triton}
    Profiling hook

```

Small Input

```
[8]: !proton -d tree -n profiles/nonfusion_matmulbias_1024x1024 -k triton triton/
    ↪matmulbias.py nonfusion
```

Running the nonfusion matmulbias kernel (input size M:1024 N:1024) ...

```

triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500,  8.8203,
 6.4609],
 [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
 [13.8125, -6.4609,  0.5146, ..., 48.0000, 25.1562, -27.5156],
 ...,
 [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
 [ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
 [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1250]],
 device='cuda:0', dtype=torch.float16)

```

Large Input

```
[9]: !proton -d tree -n profiles/nonfusion_matmulbias_32768x32768 -k triton triton/
    ↪matmulbias.py --M 32768 --N 32768 nonfusion
```

Running the nonfusion matmulbias kernel (input size M:32768 N:32768) ...

```

triton_output: tensor([[ -60.8750,  56.6562,  57.0938, ..., 274.0000,
295.2500,
 14.8828],
 [205.3750, -353.7500, -102.1875, ..., -24.4219,  57.0625,
 -30.7969],
 [ 56.2188, 258.7500,  46.6562, ..., -366.0000, 395.2500,
285.2500],
 ...,
 [ -5.0977,  99.9375, -73.5625, ..., -70.5000, 204.2500,
 -156.8750],
 [ 86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
100.8750],
 [-27.6406,  93.5625,  77.9375, ..., -213.1250, -323.2500,
 -16.8438]], device='cuda:0', dtype=torch.float16)

```

Using the Triton Proton API Generates a hatchet file, `nonfusion_matmulbias_x.hatchet`, under the profiles directory and outputs the profile.

Small Input

```
[10]: !python triton/matmulbias.py --profile nonfusion
```

```
Profiling the nonfusion matmulbias kernel (input size M:1024 N:1024) ...
triton_output: tensor([[ 2.7480,  43.6250,  30.9531, ...,  26.2500,   8.8203,
  6.4609],
      [ 7.1562, -21.3906,  17.7969, ..., -16.8125,  29.6562, -48.3125],
      [ 13.8125, -6.4609,   0.5146, ...,  48.0000,  25.1562, -27.5156],
      ...,
      [-54.4375, -1.4375, -0.4214, ...,   5.4297,  17.2812, -39.0625],
      [ 9.6094, -52.2500,  43.0000, ..., -45.8438, -49.3125,  32.6875],
      [-12.8516,  9.5391,  53.3750, ..., -9.3438,   7.6797,  54.1250]],
      device='cuda:0', dtype=torch.float16)
Proton profile results for nonfusion_matmulbias_1024x1024
1019.654 3635813321220.598 ROOT
  890.020 nan _ZN2at6native29vectorized_elementwise_kern
elILi4ENS0_11FillFunctorIiEEST5arrayIPcLm1EEEEviT0_T1_
  44.832 587901839487.468 bias_kernel [M=1024, N=1024]
[BM=128, BN=128]
  46.623 555476064829.953 bias_kernel [M=1024, N=1024]
[BM=128, BN=256]
  38.180 95732061307624.141 matmul_kernel [M=1024,
N=1024, K=1024] [BM=128, BN=128 BK=64]

Legend (Metric: time/ms (inc) Min: 38.18 Max: 1019.65)
  921.51 - 1019.65
  725.21 - 921.51
  528.92 - 725.21
  332.62 - 528.92
  136.33 - 332.62
  38.18 - 136.33

name User code      Only in left graph
Only in right graph
```

Large Input

```
[11]: !python triton/matmulbias.py --M 32768 --N 32768 --profile nonfusion
```

```
Profiling the nonfusion matmulbias kernel (input size M:32768 N:32768) ...
triton_output: tensor([[ -60.8750,   56.6562,   57.0938, ...,  274.0000,
 295.2500,
      14.8828],
      [ 205.3750, -353.7500, -102.1875, ..., -24.4219,   57.0625,
     -30.7969],
      [  56.2188,  258.7500,   46.6562, ..., -366.0000,  395.2500,
     285.2500],
      ...,
      [  -5.0977,   99.9375,  -73.5625, ..., -70.5000,  204.2500,
```


will be removed from rocprof-compute in ROCm 7.1. Please migrate to amd-smi for compute partition parsing. For migration help, see <https://github.com/ROCm/amdsmi>

INFO Rocprofiler-Compute version: 3.2.3

INFO Profiler choice: rocprofiler-sdk

INFO Path:

/workspace/user/workloads/nonfusion_matmulbias_1024x1024/MI300X_A1

INFO Target: MI300X_A1

INFO Command: python triton/matmulbias.py nonfusion

INFO Kernel Selection: None

INFO Dispatch Selection: None

INFO Hardware Blocks: []

INFO Report Sections: []

INFO

INFO ~~~~~

INFO Collecting Performance Counters

INFO ~~~~~

INFO

INFO [Run 1/12][Approximate profiling time left: pending first measurement...]

INFO [profiling] Current input file: /workspace/user/workloads/nonfusion_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_IFETCH_LEVEL.txt

INFO |-> [rocprofiler-sdk] W20260316 20:19:26.962586

140634747031104 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178960 sec

INFO |-> [rocprofiler-sdk] W20260316 20:19:26.962692

140634747031104 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py nonfusion' :: 0.000000 sec

INFO |-> [rocprofiler-sdk] W20260316 20:19:28.224911

140634747031104 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO |-> [rocprofiler-sdk] Running the nonfusion matmulbias kernel (input size M:1024 N:1024) ...

INFO |-> [rocprofiler-sdk] triton_output: tensor([[2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],

INFO |-> [rocprofiler-sdk] [7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],

INFO |-> [rocprofiler-sdk] [13.8125, -6.4609, 0.5146, ..., 48.0000, 25.1562, -27.5156],

INFO |-> [rocprofiler-sdk] ...,

INFO |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ..., 5.4297, 17.2812, -39.0625],

INFO |-> [rocprofiler-sdk] [9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],

```

INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260316 20:19:32.039907
140634747031104 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      5.077216 sec
INFO      |-> [rocprofiler-sdk] E20260316 20:19:32.053541
140634747031104 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/250_kernel_
trace.csv
INFO      |-> [rocprofiler-sdk] E20260316 20:19:32.074180
140634747031104 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/250_counter
_collection.csv
INFO      |-> [rocprofiler-sdk] E20260316 20:19:34.356269
140634747031104 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/250_agent_i
nfo.csv
INFO      |-> [rocprofiler-sdk] W20260316 20:19:34.359280
140634747031104 simple_timer.cpp:55] [rocprofv3] output generation ::
2.313767 sec
INFO      |-> [rocprofiler-sdk] W20260316 20:19:34.360381
140634747031104 simple_timer.cpp:55] [rocprofv3] tool finalization ::
2.314938 sec
INFO      |-> [rocprofiler-sdk]
INFO [Run 2/12][Approximate profiling time left: 2 minutes and 33
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260316 20:19:42.373238
140305936600640 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.179849 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:19:42.373345
140305936600640 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:19:43.632716
140305936600640 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:19:46.603820
140305936600640 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.230475 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:19:46.617406
140305936600640 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/272_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:19:46.638357
140305936600640 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/272_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:19:49.000715
140305936600640 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/272_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:19:49.003719
140305936600640 simple_timer.cpp:55] [rocprofv3] output generation ::
2.394512 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:19:49.004784
140305936600640 simple_timer.cpp:55] [rocprofv3] tool finalization ::
2.395647 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 3/12] [Approximate profiling time left: 2 minutes and 16
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:19:57.034349
139818232688192 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.153715 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:19:57.034457
139818232688192 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:19:58.294791
139818232688192 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:01.051793
139818232688192 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.017336 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:20:01.065197
139818232688192 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/293_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:01.087148
139818232688192 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/293_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:02.465019
139818232688192 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/293_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:20:02.468165
139818232688192 simple_timer.cpp:55] [rocprofv3] output generation ::
1.410796 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:02.469232
139818232688192 simple_timer.cpp:55] [rocprofv3] tool finalization ::
1.411931 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 4/12] [Approximate profiling time left: 1 minute and 49
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:07.535093
139986148781632 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.129498 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:07.535198
139986148781632 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:08.798842
139986148781632 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:11.570462
139986148781632 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.035264 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:20:11.583935
139986148781632 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/314_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:11.606753
139986148781632 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/314_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:11.744579
139986148781632 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/314_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:20:11.747720
139986148781632 simple_timer.cpp:55] [rocprofv3] output generation ::
0.171558 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:11.748719
139986148781632 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.172626 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 5/12] [Approximate profiling time left: 1 minute and 21
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:13.820021
140493456154176 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178293 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:13.820130
140493456154176 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:15.109730
140493456154176 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:18.090712
140493456154176 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.270583 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:20:18.104180
140493456154176 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/335_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:18.125472
140493456154176 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/335_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:20.455784
140493456154176 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/335_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:20:20.458609
140493456154176 simple_timer.cpp:55] [rocprofv3] output generation ::
2.362198 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:20.459707
140493456154176 simple_timer.cpp:55] [rocprofv3] tool finalization ::
2.363362 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 6/12] [Approximate profiling time left: 1 minute and 14
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_0.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:28.327263
140372240825920 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.130644 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:28.327373
140372240825920 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:29.601622
140372240825920 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:32.377991
140372240825920 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.050618 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:20:32.391314
140372240825920 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/356_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:32.414705
140372240825920 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/356_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:32.537427
140372240825920 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/356_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:20:32.540401
140372240825920 simple_timer.cpp:55] [rocprofv3] output generation ::
0.156884 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:32.541351
140372240825920 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.157899 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 7/12] [Approximate profiling time left: 56 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_1.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:34.169346
1399039333296192 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128400 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:34.169444
1399039333296192 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:35.426312
1399039333296192 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:38.183270
139903933296192 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.013826 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:20:38.196697
139903933296192 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/377_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:38.219466
139903933296192 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/377_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:38.342006
139903933296192 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/377_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:20:38.344975
139903933296192 simple_timer.cpp:55] [rocprofv3] output generation ::
0.156118 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:38.345969
139903933296192 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.157182 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 8/12] [Approximate profiling time left: 41 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_2.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:39.988836
140193707937344 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128548 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:39.988942
140193707937344 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:41.249412
140193707937344 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:44.006830
140193707937344 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.017888 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:20:44.020296
140193707937344 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/397_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:44.042997
140193707937344 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/397_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:44.165312
140193707937344 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/397_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:20:44.168146
140193707937344 simple_timer.cpp:55] [rocprofv3] output generation ::
0.155616 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:44.169140
140193707937344 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.156677 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 9/12] [Approximate profiling time left: 29 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_3.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:45.804358
139938370170432 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128461 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:45.804456
139938370170432 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:47.060440
139938370170432 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:49.844073
139938370170432 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.039617 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:20:49.857427
139938370170432 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/418_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:49.883984
139938370170432 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/418_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:50.028690
139938370170432 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/418_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:20:50.031559
139938370170432 simple_timer.cpp:55] [rocprofv3] output generation ::
0.181907 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:50.032545
139938370170432 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.182960 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 10/12][Approximate profiling time left: 18 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_4.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:51.770888
139771344559680 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128482 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:51.770991
139771344559680 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:53.024729
139771344559680 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:55.869108
139771344559680 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.098117 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:20:55.882511
139771344559680 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/438_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:55.919063
139771344559680 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/438_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:20:56.122705
139771344559680 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/438_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:20:56.125647
139771344559680 simple_timer.cpp:55] [rocprofv3] output generation ::
0.250896 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:56.126600
139771344559680 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.251918 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 11/12][Approximate profiling time left: 9 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_5.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:20:58.082415
139954994138688 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128767 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:58.082515
139954994138688 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:20:59.338176
139954994138688 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:21:02.197090
139954994138688 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.114575 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:21:02.210538
139954994138688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/459_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:21:02.246839
139954994138688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/459_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:21:02.450553
139954994138688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/459_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:21:02.453508
139954994138688 simple_timer.cpp:55] [rocprofv3] output generation ::
0.250718 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:21:02.454492
139954994138688 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.251771 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 12/12][Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_6.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:21:04.414898
140324462005824 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128245 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:21:04.414996
140324462005824 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:21:05.667761
140324462005824 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:21:08.277258
140324462005824 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      3.862262 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:21:08.291046
140324462005824 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/480_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:21:08.337729
140324462005824 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/480_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:21:08.480087
140324462005824 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/480_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:21:08.483090
140324462005824 simple_timer.cpp:55] [rocprofv3] output generation ::
0.199903 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:21:08.484091
140324462005824 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.200976 sec

INFO      |-> [rocprofiler-sdk]
WARNING Detected differing Workgroup_Size values while joining
pmc_perf.csv
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/nonfusion_matmulbias_1024x1024/MI300X_A1
INFO [roofline] No roofline data found. Generating...
Empirical Roofline Calculation
Copyright © 2025 Advanced Micro Devices, Inc. All rights reserved.
Total detected GPU devices: 1
GPU Device 0 (gfx942) with 304 CUs: Profiling..
 99% [||||| ]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.1 ms, mean:4201.4 GB/sec, stdev=13.0
GB/sec
 99% [||||| ]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,

```

traffic:2611340115968 bytes, duration:410.7 ms, mean:6332.1 GB/sec, stdev=16.8 GB/sec
 99% [|||||]
 L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
 traffic:1632087572480 bytes, duration:63.7 ms, mean:25643.6 GB/sec, stdev=70.7 GB/sec
 99% [|||||]
 L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
 traffic:127506841600 bytes, duration:3.8 ms, mean:33474.5 GB/sec, stdev=299.4 GB/sec
 99% [|||||]
 LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
 traffic:79691776000 bytes, duration:1.3 ms, mean:62666.5 GB/sec, stdev=663.3 GB/sec
 99% [|||||]
 Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:1081258016768, duration:13.0 ms, mean:82920.9 GFLOPS,
 stdev=171.0 GFLOPS
 99% [|||||]
 Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39621.3 GFLOPS,
 stdev=57.3 GFLOPS
 99% [|||||]
 Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:530428461056, duration:91.770 ms, mean:5770.6 GFLOPS,
 stdev=8.6 GFLOPS
 99% [|||||]
 Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:265214230528, duration:2.022 ms, mean:131312.5 GFLOPS,
 stdev=1552.3 GFLOPS
 99% [|||||]
 Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:122406567936, duration:1.1 ms, mean:106719.2 GFLOPS,
 stdev=1606.3 GFLOPS
 99% [|||||]
 Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82574.7 GOPS,
 stdev=169.9 GOPS
 99% [|||||]
 Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, IOP:265214230528, duration:3.7 ms, mean:71018.0 GOPS,
 stdev=389.5 GOPS
 99% [|||||]
 Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, IOP:122406567936, duration:6.2 ms, mean:19777.7 GOPS,
 stdev=58.8 GOPS
 100% [|||||]
 Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,

```
INFO [roofline] Empirical Roofline PDFs saved!
```

```
[13]: !ROCPROF=rocprofiler-sdk rocprof-compute profile -n
      ↪ nonfusion_matmulbias_32768x32768 -- python triton/matmulbias.py --M 32768
      ↪ --N 32768 nonfusion
```

----- /-|----- -|-----
 |' _/ _ \ / _| ' \ | ' _/ _ \ | |----- / _/ _ \ | ' _ \ _ \ | ' \ | | | | _/ _
 \
 | | | () | (_ |) | | | () | _|-----| (_ () | | | | | |) | | _ | | |
 _/
 | _ \ _/ \ _| . _/ | _ \ _/ | _ \ _ \ _/ | _ | | | _ _/
 \ _ , _ | \ _ \ _|
 | _| | _|


```

WARNING DEPRECATION WARNING: rocm-smi is deprecated in ROCm 7.0 and
will be removed from rocprof-compute in ROCm 7.1. Please migrate to amd-smi for
compute partition parsing. For migration help, see
https://github.com/ROCm/amdsmi
INFO Rocprofiler-Compute version: 3.2.3
INFO Profiler choice: rocprofiler-sdk
INFO Path:
/workspace/user/workloads/nonfusion_matmulbias_32768x32768/MI300X_A1
INFO Target: MI300X_A1
INFO Command: python triton/matmulbias.py --M 32768 --N 32768
nonfusion
INFO Kernel Selection: None
INFO Dispatch Selection: None
INFO Hardware Blocks: []
INFO Report Sections: []
INFO
INFO ~~~~~
INFO Collecting Performance Counters
INFO ~~~~~
INFO
INFO [Run 1/12][Approximate profiling time left: pending first
measurement...]
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_IFETCH_LEVEL.txt
INFO      |-> [rocprofiler-sdk] W20260316 20:22:56.918563
140029976456768 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.152039 sec
INFO      |-> [rocprofiler-sdk] W20260316 20:22:56.918662
140029976456768 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec
INFO      |-> [rocprofiler-sdk] W20260316 20:22:58.173384
140029976456768 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)
INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],

```

```

INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000, 204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [ 86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260316 20:23:05.030611
140029976456768 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.111949 sec
INFO      |-> [rocprofiler-sdk] E20260316 20:23:05.044090
140029976456768 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/591_kerne
l_trace.csv
INFO      |-> [rocprofiler-sdk] E20260316 20:23:05.060634
140029976456768 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/591_count
er_collection.csv
INFO      |-> [rocprofiler-sdk] E20260316 20:23:05.650284
140029976456768 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/591_agent
_info.csv
INFO      |-> [rocprofiler-sdk] W20260316 20:23:05.653159
140029976456768 simple_timer.cpp:55] [rocprofv3] output generation ::
0.616861 sec
INFO      |-> [rocprofiler-sdk] W20260316 20:23:05.654227
140029976456768 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.617998 sec
INFO      |-> [rocprofiler-sdk]
INFO [Run 2/12] [Approximate profiling time left: 1 minute and 59
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:23:09.075212
140465698315840 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178714 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:09.075313
140465698315840 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:10.333595
140465698315840 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:23:17.215607
140465698315840 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.140295 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:23:17.228949
140465698315840 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/612_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:23:17.245413
140465698315840 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/612_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:23:18.315022
140465698315840 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/612_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:23:18.317870
140465698315840 simple_timer.cpp:55] [rocprofv3] output generation ::
1.096697 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:18.318956
140465698315840 simple_timer.cpp:55] [rocprofv3] tool finalization ::
1.097850 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 3/12] [Approximate profiling time left: 1 minute and 55
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:23:22.775264
140449047461440 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178454 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:22.775369
140449047461440 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:24.031468
140449047461440 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:23:30.924802
140449047461440 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.149433 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:23:30.938147
140449047461440 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/635_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:23:30.954431
140449047461440 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/635_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:23:32.076349
140449047461440 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/635_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:23:32.079203
140449047461440 simple_timer.cpp:55] [rocprofv3] output generation ::
1.148905 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:32.080288
140449047461440 simple_timer.cpp:55] [rocprofv3] tool finalization ::
1.150057 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 4/12] [Approximate profiling time left: 1 minute and 46
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:23:36.616467
140369419800128 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128457 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:36.616570
140369419800128 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000001 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:37.871694
140369419800128 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:23:44.730546
140369419800128 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.113976 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:23:44.743980
140369419800128 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/659_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:23:44.760675
140369419800128 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/659_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:23:44.819776
140369419800128 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/659_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:23:44.822616
140369419800128 simple_timer.cpp:55] [rocprofv3] output generation ::
0.086442 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:44.823615
140369419800128 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.087510 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 5/12] [Approximate profiling time left: 1 minute and 26
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260316 20:23:46.612507
140104020983360 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.179553 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:46.612610
140104020983360 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:47.870313
140104020983360 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:23:54.741312
140104020983360 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.128702 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:23:54.754627
140104020983360 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/680_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:23:54.770965
140104020983360 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/680_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:23:55.857357
140104020983360 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/680_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:23:55.860264
140104020983360 simple_timer.cpp:55] [rocprofv3] output generation ::
1.113388 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:23:55.861327
140104020983360 simple_timer.cpp:55] [rocprofv3] tool finalization ::
1.114522 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 6/12] [Approximate profiling time left: 1 minute and 16
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_0.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:00.176776
140402541813312 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.127762 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:00.176879
140402541813312 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:01.431625
140402541813312 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:08.305464
140402541813312 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.128585 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:24:08.318856
140402541813312 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/701_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:08.335572
140402541813312 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/701_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:08.388695
140402541813312 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/701_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:24:08.391566
140402541813312 simple_timer.cpp:55] [rocprofv3] output generation ::
0.080488 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:08.392562
140402541813312 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.081553 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 7/12] [Approximate profiling time left: 1 minute]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_1.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:09.819451
139967990627904 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128155 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:09.819554
139967990627904 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:11.079212
139967990627904 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:17.964484
139967990627904 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.144930 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:24:17.978102
139967990627904 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/722_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:17.994792
139967990627904 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/722_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:18.047291
139967990627904 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/722_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:24:18.050143
139967990627904 simple_timer.cpp:55] [rocprofv3] output generation ::
0.080116 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:18.051121
139967990627904 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.081168 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 8/12] [Approximate profiling time left: 47 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_2.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:19.475445
139819405766208 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128618 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:19.475549
139819405766208 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:20.731253
139819405766208 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:27.608266
139819405766208 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.132718 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:24:27.621639
139819405766208 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/743_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:27.638339
139819405766208 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/743_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:27.690990
139819405766208 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/743_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:24:27.693944
139819405766208 simple_timer.cpp:55] [rocprofv3] output generation ::
0.080094 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:27.694919
139819405766208 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.081135 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 9/12] [Approximate profiling time left: 34 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_3.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:29.113343
139913092558400 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128329 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:29.113446
139913092558400 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:30.368314
139913092558400 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:37.205433
139913092558400 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.091988 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:24:37.218732
139913092558400 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/764_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:37.235985
139913092558400 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/764_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:37.286718
139913092558400 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/764_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:24:37.289630
139913092558400 simple_timer.cpp:55] [rocprofv3] output generation ::
0.078654 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:37.290622
139913092558400 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.079715 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 10/12][Approximate profiling time left: 22 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_4.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:38.714299
139712097443392 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128070 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:38.714399
139712097443392 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:39.969193
139712097443392 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:46.819289
139712097443392 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.104891 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:24:46.832757
139712097443392 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/785_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:46.850700
139712097443392 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/785_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:46.898884
139712097443392 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/785_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:24:46.901734
139712097443392 simple_timer.cpp:55] [rocprofv3] output generation ::
0.076957 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:46.902730
139712097443392 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.078019 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 11/12][Approximate profiling time left: 11 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_5.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:48.321763
140351425588800 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128023 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:48.321867
140351425588800 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:49.578417
140351425588800 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:56.445452
140351425588800 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.123585 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:24:56.458682
140351425588800 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/806_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:56.476499
140351425588800 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/806_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:24:56.526074
140351425588800 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/806_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:24:56.528925
140351425588800 simple_timer.cpp:55] [rocprofv3] output generation ::
0.077985 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:56.529937
140351425588800 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.079065 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 12/12][Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_6.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:24:57.961049
140342103244352 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128229 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:57.961147
140342103244352 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:24:59.214252
140342103244352 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:25:06.083145
140342103244352 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.121999 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:25:06.096549
140342103244352 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/827_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:25:06.114557
140342103244352 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/827_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:25:06.142563
140342103244352 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/827_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:25:06.145465
140342103244352 simple_timer.cpp:55] [rocprofv3] output generation ::
0.056720 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:25:06.146441
140342103244352 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.057766 sec

INFO      |-> [rocprofiler-sdk]
WARNING Detected differing Workgroup_Size values while joining
pmc_perf.csv
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/nonfusion_matmulbias_32768x32768/MI300X_A1
INFO [roofline] No roofline data found. Generating...
Empirical Roofline Calculation
Copyright © 2025 Advanced Micro Devices, Inc. All rights reserved.
Total detected GPU devices: 1
GPU Device 0 (gfx942) with 304 CUs: Profiling..
 99% [||||| ]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.1 ms, mean:4215.1 GB/sec, stdev=18.7
GB/sec
 99% [||||| ]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,

```


traffic:2611340115968 bytes, duration:410.8 ms, mean:6344.3 GB/sec, stdev=13.8 GB/sec

99% [|||||]

L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, traffic:1632087572480 bytes, duration:63.4 ms, mean:25701.0 GB/sec, stdev=61.1 GB/sec

99% [|||||]

L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, traffic:127506841600 bytes, duration:3.8 ms, mean:33616.5 GB/sec, stdev=155.9 GB/sec

99% [|||||]

LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, traffic:79691776000 bytes, duration:1.3 ms, mean:59671.3 GB/sec, stdev=324.7 GB/sec

99% [|||||]

Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, FLOP:1081258016768, duration:13.0 ms, mean:82964.1 GFLOPS, stdev=188.6 GFLOPS

99% [|||||]

Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39610.1 GFLOPS, stdev=58.2 GFLOPS

99% [|||||]

Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, FLOP:530428461056, duration:91.877 ms, mean:5767.6 GFLOPS, stdev=8.9 GFLOPS

99% [|||||]

Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, FLOP:265214230528, duration:2.025 ms, mean:131314.8 GFLOPS, stdev=1532.4 GFLOPS

99% [|||||]

Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, FLOP:122406567936, duration:1.2 ms, mean:106786.9 GFLOPS, stdev=1462.6 GFLOPS

99% [|||||]

Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82581.1 GOPS, stdev=175.5 GOPS

99% [|||||]

Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, IOP:265214230528, duration:3.7 ms, mean:71537.9 GOPS, stdev=790.8 GOPS

99% [|||||]

Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, IOP:122406567936, duration:6.2 ms, mean:19762.6 GOPS, stdev=55.2 GOPS

100% [|||||]

Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,

```

experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
100% [|||||]
Peak MFMA FLOPs (F6), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
99% [|||||]
Peak MFMA FLOPs (F8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:10200547328000, duration:5.3 ms, mean:1926918.5 GFLOPS,
stdev=16024.3 GFLOPS
99% [|||||]
Peak MFMA FLOPs (F16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:5100273664000, duration:5.4 ms, mean:946216.5 GFLOPS,
stdev=2997.9 GFLOPS
99% [|||||]
Peak MFMA FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:2550136832000, duration:4.8 ms, mean:525058.1 GFLOPS,
stdev=3495.1 GFLOPS
99% [|||||]
Peak MFMA FLOPs (F32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1275068416000, duration:9.1 ms, mean:140385.8 GFLOPS,
stdev=612.9 GFLOPS
99% [|||||]
Peak MFMA FLOPs (F64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:637534208000, duration:4.4 ms, mean:145679.1 GFLOPS,
stdev=683.2 GFLOPS
99% [|||||]
Peak MFMA IOPs (I8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:5100273664000, duration:5.0 ms, mean:1006894.3 GOPS,
stdev=5190.8 GOPS

```

INFO [roofline] Empirical Roofline PDFs saved!

1.4.3 Analyze the Non-Fusion MatMulBias Kernel Profiles

Using the Triton Proton Viewer CLI

Help

```
[14]: !proton-viewer --help
```

```

usage: proton-viewer [-h] [-l] [-m METRICS] [-i INCLUDE] [-e EXCLUDE]
                    [-t THRESHOLD] [-d DEPTH]
                    [-f {full,file_function_line,function_line,file_function}]
                    [--print-sorted] [--diff-profile DIFF_PROFILE]

```

Performance data viewer for proton profiles.

options:

```

-h, --help          show this help message and exit
-l, --list          List available metrics. Metric names are case

```

insensitive and ignore units.

Derived metrics can be created when source metrics are available.

- time/s, time/ms, time/us, time/ns: time
- avg_time/s, avg_time/ms, avg_time/us, avg_time/ns:
time / count
- flop[<8/16/32/64>]/s, gflop[<8/16/32/64>]/s,
tflop[<8/16/32/64>]/s: flops / time
- byte/s, gbyte/s, tbyte/s: bytes / time
- util: max(sum(flops<width>) / peak_flops<width>_time,
sum(bytes) / peak_bandwidth_time)
- <metric>/%: frame(metric) / sum(metric). Only
available for inclusive metrics (e.g. time)

-m METRICS, --metrics METRICS

At maximum two metrics can be specified, separated by comma.

There are two modes:

1) Choose the output metric to display. It's case insensitive and ignore units.

2) Derive a new metric from existing metrics.

-i INCLUDE, --include INCLUDE

Find frames that match the given regular expression and return all nodes in the paths that pass through the matching frames.

For example, the following command will display all paths that contain frames that contains "test":

```
...  
proton-viewer -i ".*test.*" path/to/file.json  
...
```

-e EXCLUDE, --exclude EXCLUDE

Exclude frames that match the given regular expression and their children.

For example, the following command will exclude all paths starting from frames that contains "test":

```
...  
proton-viewer -e ".*test.*" path/to/file.json  
...
```

-t THRESHOLD, --threshold THRESHOLD

Exclude frames(kernels) whose metrics are below the given threshold. This filter only applies on the first metric.

-d DEPTH, --depth DEPTH

The depth of the tree to display

-f {full,file_function_line,function_line,file_function}, --format {full,file_function_line,function_line,file_function}

Formatting the frame name.

- full: include the path, file name, function name and line number.

- file_function_line: include the file name, function name and line number.

- function_line: include the function name and line number.
- file_function: include the file name and function name.
- print-sorted Sort output by metric value instead of chronologically
- diff-profile DIFF_PROFILE, -diff DIFF_PROFILE Compare two profiles. When used as 'proton-viewer -m time -diff file1.log file2.log', computes the difference: file2['time'] - file1['time']

Small Input

```
[15]: !proton-viewer -m time/ms,flops16 profiles/nonfusion_matmulbias_1024x1024.
      ↪hatchet
```

```
1019.654 3707271905280.000 ROOT
 890.020 nan _ZN2at6native29vectorized_elementwise_kern
elILi4ENS0_11FillFunctorIiEEST5arrayIPcLm1EEEEviT0_T1_
 44.832 26357006336.000 bias_kernel [M=1024, N=1024]
[BM=128, BN=128]
 46.623 25897730048.000 bias_kernel [M=1024, N=1024]
[BM=128, BN=256]
 38.180 3655017168896.000 matmul_kernel [M=1024, N=1024,
K=1024] [BM=128, BN=128 BK=64]
```

Legend (Metric: time/ms (inc) Min: 38.18 Max: 1019.65)

```
921.51 - 1019.65
725.21 - 921.51
528.92 - 725.21
332.62 - 528.92
136.33 - 332.62
38.18 - 136.33
```

name User code Only in left graph
Only in right graph

Large Input

```
[16]: !proton-viewer -m time/ms,flops16 profiles/nonfusion_matmulbias_32768x32768.
      ↪hatchet
```

```
5961.749 1199021725057024.000 ROOT
 51.216 nan _ZN2at6native29vectorized_elementwise_kernel
ILi4ENS0_11FillFunctorIiEEST5arrayIPcLm1EEEEviT0_T1_
 809.386 1389421920256.000 bias_kernel [M=32768,
N=32768] [BM=128, BN=128]
 802.369 1363652116480.000 bias_kernel [M=32768,
N=32768] [BM=128, BN=256]
4298.778 1196268651020288.000 matmul_kernel [M=32768,
```



```

42424.34      42332.67  85.56  *
      kernel<4, at::native::FillFunctor<int>,

      std::array<char*, 1ul> >(int, at::nat...

1  bias_kernel                                1605.00  5873062.69
3659.23      3541.33  8.37  *

2  matmul_kernel                                153.00  3891466.15
25434.42      24495.33  5.55

3  __amd_rocclr_copyBuffer                    89.00  266733.59
2997.01      2973.25  0.38

4  void at::native::(anonymous namespace)::    3.00  19053.25
6351.08      7383.42  0.03
      distribution_elementwise_grid_stride_ker

      nel<float, 4, at::native::templates::...

5  void at::native::(anonymous namespace)::    7.00  16420.00
2345.71      2251.83  0.02
      CatArrayBatchedCopy_contig<at::native::(

      anonymous namespace)::OpaqueType<2u>,...

6  void at::native::vectorized_elementwise_    3.00  10537.33
3512.44      3511.33  0.02
      kernel<4, at::native::compare_scalar_ker

      nel<double>(at::TensorIteratorBase&, ...

7  void at::native::vectorized_elementwise_    2.00  6267.50
3133.75      3133.75  0.01
      kernel<8, at::native::AUnaryFunctor<c10:

      :Half, c10::Half, bool, at::native::(...

```

8	void at::native::vectorized_elementwise_	2.00	5332.08
2666.04	2666.04 0.01		
	kernel<8, at::native::BinaryFunctor<c10:		
	:Half, c10::Half, bool, at::native::(...		
9	void at::native::vectorized_elementwise_	2.00	5011.00
2505.50	2505.50 0.01		
	kernel<8, at::native::AbsFunctor<c10::Ha		
	lf>, std::array<char*, 2ul> >(int, at...		

0.2 Dispatch List

	Dispatch_ID	Kernel_Name	
	GPU_ID		
0	0	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*):...	0
1	1	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*):...	0
2	2	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*):...	0

```

        nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
        , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
        eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*):...

```

```

3          3  matmul_kernel
0

```

```

4          4  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
        l>)

```

```

5          5  matmul_kernel
0

```

```

6          6  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
        l>)

```

```

7          7  matmul_kernel
0

```

```

8          8  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
        l>)

```

```

9          9  matmul_kernel
0

```

1. System Info

	Info
workload_name	nonfusion_matmulbias_1024x1024
command	python triton/matmulbias.py nonfusion
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Mon Mar 16 20:19:23 2026 (UTC)
version	3
hostname	b031b949a562
cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1
linux_distro	CentOS Stream 10 (Coughlan)
linux_kernel_version	5.15.0-170-generic
amd_gpu_kernel_version	nan
cpu_memory	230893520
gpu_memory	nan
rocm_version	7.0.3
vbios	113-M3000108-103
compute_partition	SPX
memory_partition	nan
gpu_series	MI300
gpu_model	MI300X_A1
gpu_arch	gfx942
gpu_chip_id	29877

gpu_l1	32
gpu_l2	4096
cu_per_gpu	304
simd_per_cu	4
se_per_gpu	32
wave_size	64
workgroup_max_size	1024
max_waves_per_cu	32
max_sclk	2100
max_mclk	1300
cur_sclk	2100
cur_mclk	1300
total_l2_chan	128
lds_banks_per_cu	32
sqc_per_gpu	160
pipes_per_gpu	4
num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
Peak	Pct of Peak		

2.1.0 81715.20	VALU FLOPs 0.12	99.37	Gflop/s
2.1.1 81715.20	VALU IOPs 0.78	640.69	Giop/s
2.1.2 2614886.40	MFMA FLOPs (F8) 0.00	0.00	Gflop/s
2.1.3 1307443.20	MFMA FLOPs (BF16) 0.00	0.00	Gflop/s
2.1.4 1307443.20	MFMA FLOPs (F16) 0.00	0.00	Gflop/s
2.1.5 163430.40	MFMA FLOPs (F32) 0.00	0.00	Gflop/s
2.1.6 163430.40	MFMA FLOPs (F64) 0.00	0.00	Gflop/s
2.1.7	MFMA FLOPs (F6F4)		Gflop/s
2.1.8 2614886.40	MFMA IOPs (Int8) 0.00	0.00	Giop/s
2.1.9 304.00	Active CUs 40.46	123.00	Cus
2.1.10 100.00	SALU Utilization 1.66	1.66	Pct
2.1.11 100.00	VALU Utilization 1.57	1.57	Pct

2.1.12	MFMA Utilization	0.00	Pct
100.00	0.00		
2.1.13	VMEM Utilization	0.42	Pct
100.00	0.42		
2.1.14	Branch Utilization	0.30	Pct
100.00	0.30		
2.1.15	VALU Active Threads	63.74	Threads
64.00	99.59		
2.1.16	IPC	0.25	Instr/cycle
5.00	5.00		
2.1.17	Wavefront Occupancy	6886.88	Wavefronts
9728.00	70.79		
2.1.18	Theoretical LDS Bandwidth	9.08	Gb/s
81715.20	0.01		
2.1.19	LDS Bank Conflicts/Access	1.00	Conflicts/access
32.00	3.12		
2.1.20	vL1D Cache Hit Rate	15.54	Pct
100.00	15.54		
2.1.21	vL1D Cache BW	7246.88	Gb/s
81715.20	8.87		
2.1.22	L2 Cache Hit Rate	42.14	Pct
100.00	42.14		
2.1.23	L2 Cache BW	6893.90	Gb/s
34406.40	20.04		

2.1.24 5324.80	L2-Fabric Read BW 5.89	313.37	Gb/s
2.1.25 5324.80	L2-Fabric Write BW 61.48	3273.86	Gb/s
2.1.26	L2-Fabric Read Latency	473.80	Cycles
2.1.27	L2-Fabric Write Latency	380.90	Cycles
2.1.28 100.00	sL1D Cache Hit Rate 99.74	99.74	Pct
2.1.29 21504.00	sL1D Cache BW 0.22	46.35	Gb/s
2.1.30 100.00	L1I Hit Rate 99.82	99.82	Pct
2.1.31 21504.00	L1I BW 2.25	483.00	Gb/s
2.1.32	L1I Fetch Latency		Cycles

3. Memory Chart

3.1 Memory Chart

(Normalization: per_kernel)

Instr Buff Instr Dispatch Exec
LDS L2 Cache

SALU: 443825

-----> Active
CUs Req : 152 Util : 0 %
 xGMI/PCIE

Wave 0 Instr Buf 12
3: 304 <-----

SMEM:
30706 Lat :
N/A Hit : 42
%

>
~
|
RVGPRseq: 36.0

Wave N-1 Instr Buf VALU:
371691 Vector L1 Cache
Request |
|

>
|
v
SGPRs : 36

MFMA: 0
Rd : 78648 Hit : 16 % Rd :
8780 Rd : 11532

```

-----
> <----- <-----
-
Fabric

Wt : 7930507 Lat : N/A Wr : 1982627 Wr :
1982627 Rd :
9054

VMEM: 125156 LDS Alloc: -
-----> -----
> <----- Latency (cycles) Rd:
9055

----->
0 Atomic: 0 Coales: 25 % Atomic: 0 Atomic:
0 Wr : 1982627 <----- HBM
<
-----> <-----
> -----> Rd : 474 Wr:
1982627

LDS: 152
Stall : 36
cycles Atomic: 0 Wr :
381 ----->
-----> Scratch Alloc:

-----> Atomic: 0
0

Wave Occupancy GWS: 0
Scalar L1D
Cache Latency (cycles)

56 -----
---> Rd :
75

Wave Life
<-----
^

|

7577 BRANCH:
61413 Wavefronts: Rd : 30706 Hit : 93.0 % Wr
: 0 Rd : N/A |
|
-----> 0 <

```

```

-----
> |
|
Lat : N/A Atomic: 0 Wr :
N/A |
v
^
<-----
>
| Workgroups:
|
| Instr L1 Cache
| GMI
|
|
| Fetch :
223823 Hit : 99 % Req
: 2669
-----
-----
<-----

Lat : 1042.0 cycles

```

4. Roofline

4.1 Roofline

140918.1
140918GFLOP/s

36422.6

9414.0

2433.2
@ @

628.9 62667 GB/s

33474 GB/s

162.6

42.0 4201 GB/s

0.01	0.18	3.16	56.23	1000.00
------	------	------	-------	---------

Performance (GFLOP/sec) Arithmetic Intensity (FLOP)s/Byte)

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct
5.1.1	CPF Stall	1.59	0.06	7.82	Pct
5.1.2	CPF-L2 Utilization	24.27	3.08	46.77	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	2.28	0.06	9.46	Pct

5.2 Packet Processor

Metric_ID	Metric	Avg	Min	Max
Unit				
5.2.0	CPC SYNC FIFO Full Rate			
Pct				
5.2.1	CPC CANE Stall Rate			
Pct				
5.2.2	CPC ADC Utilization			
Pct				
5.2.3	CPC Utilization	100.00	100.00	100.00
Pct				
5.2.4	CPC Stall Rate	23.35	7.62	48.20

Pct

5.2.5	CPC Packet Decoding Utilization	53.39	24.60	114.57
-------	---------------------------------	-------	-------	--------

Pct

5.2.6	CPC-Workgroup Manager Utilization	32.42	1.31	70.12
-------	-----------------------------------	-------	------	-------

Pct

5.2.7	CPC-L2 Utilization	0.68	0.17	1.30
-------	--------------------	------	------	------

Pct

5.2.8	CPC-UTCL1 Stall	0.42	0.11	4.82
-------	-----------------	------	------	------

Pct

5.2.9	CPC-UTCL2 Utilization	0.41	0.10	4.60
-------	-----------------------	------	------	------

Pct

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID	Metric	Avg	Min	Max	Unit
6.1.0	Schedule-Pipe Wave Occupancy				Wave
6.1.1	Accelerator Utilization	100.22	64.52	127.39	Pct
6.1.2	Scheduler-Pipe Utilization	0.00	0.00	0.00	Pct
6.1.3	Scheduler-Pipe Wave Utilization				Pct

6.1.4	Workgroup Manager Utilization	52.82	24.07	95.10	Pct
6.1.5	Shader Engine Utilization	47.91	15.91	93.94	Pct
6.1.6	SIMD Utilization	38.07	2.27	296.12	Pct
6.1.7 Workgroups	Dispatched Workgroups	0.00	0.00	0.00	
6.1.8 Wavefronts	Dispatched Wavefronts	0.00	0.00	0.00	
6.1.9 Cycles/wave	VGPR Writes				
6.1.10 Cycles/wave	SGPR Writes				

6.2 Workgroup Manager - Resource Allocation

Metric_ID Unit	Metric	Avg	Min	Max
6.2.0 Pct	Not-scheduled Rate (Workgroup Manager)	0.00	0.00	0.00
6.2.1 Pct	Not-scheduled Rate (Scheduler-Pipe)	0.00	0.00	0.00
6.2.2 Pct	Scheduler-Pipe FIFO Full Rate			

6.2.3 Pct	Scheduler-Pipe Stall Rate	0.00	0.00	0.00
6.2.4 Pct	Scratch Stall Rate	0.00	0.00	0.00
6.2.5 Pct	Insufficient SIMD Waveslots	0.00	0.00	0.00
6.2.6 Pct	Insufficient SIMD VGPRs	0.00	0.00	0.00
6.2.7 Pct	Insufficient SIMD SGPRs	0.00	0.00	0.00
6.2.8 Pct	Insufficient CU LDS	0.00	0.00	0.00
6.2.9 Pct	Insufficient CU Barriers	0.00	0.00	0.00
6.2.10 Pct	Reached CU Workgroup Limit	0.00	0.00	0.00
6.2.11 Pct	Reached CU Wavefront Limit	0.00	0.00	0.00

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID	Metric	Avg	Min	Max	Unit
-----------	--------	-----	-----	-----	------

7.1.0	Grid Size	1975012.54	8192.00	4194304.00	Work items
7.1.1	Workgroup Size	297.71	256.00	512.00	Work items
7.1.2	Total Wavefronts	0.00	0.00	0.00	
7.1.3	Saved Wavefronts	0.00	0.00	0.00	
7.1.4	Restored Wavefronts	0.00	0.00	0.00	
7.1.5	VGPRs	36.43	12.00	88.00	
7.1.6	AGPRs	1.87	0.00	4.00	
7.1.7	SGPRs	36.26	32.00	48.00	
7.1.8	LDS Allocation	0.00	0.00	0.00	Bytes
7.1.9	Scratch Allocation	0.00	0.00	0.00	Bytes/workitem

7.2 Wavefront Runtime Stats

Metric_ID	Metric	Avg	Min
Max Unit			
7.2.0	Kernel Time (Nanosec)	21822.35	3020.33

57747.50 Ns

7.2.1	Kernel Time (Cycles)	61078.18	19930.12
137149.12	Cycle		

7.2.2	Instructions per wavefront	228.54	37.00
620.00	Instr/wavefront		

7.2.3	Wave Cycles	323687610.64	703940.00
948861620.00	Cycles per kernel		

7.2.4	Dependency Wait Cycles	71221142.75	280892.00
203915840.00	Cycles per kernel		

7.2.5	Issue Wait Cycles	248500866.68	92608.00
738183632.00	Cycles per kernel		

7.2.6	Active Cycles	4262698.89	310784.00
8650752.00	Cycles per kernel		

7.2.7	Wavefront Occupancy	6886.88	7.05
34010.20	Wavefronts		

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.1.0	VALU	371691.02	58240.00	720896.00	Instr per kernel
10.1.1	VMEM	125155.73	4224.00	262144.00	Instr per kernel
10.1.2	LDS	152.20	0.00	1152.00	Instr per kernel
10.1.3	MFMA	0.00	0.00	0.00	Instr per kernel
10.1.4	SALU	443825.46	12160.00	917504.00	Instr per kernel

10.1.5	SMEM	30706.44	0.00	65536.00	Instr per kernel
10.1.6	Branch	61412.87	0.00	131072.00	Instr per kernel

10.2 VALU Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.2.0	INT32	202062.24	31488.00	393216.00	Instr per kernel
10.2.1	INT64	30706.44	0.00	65536.00	Instr per kernel
10.2.2	F16-ADD	4353.70	0.00	8192.00	Instr per kernel
10.2.3	F16-MUL	0.00	0.00	0.00	Instr per kernel
10.2.4	F16-FMA	0.00	0.00	0.00	Instr per kernel
10.2.5	F16-Trans	0.00	0.00	0.00	Instr per kernel
10.2.6	F32-ADD	0.00	0.00	0.00	Instr per kernel
10.2.7	F32-MUL	612.54	0.00	2048.00	Instr per kernel
10.2.8	F32-FMA	0.00	0.00	0.00	Instr per kernel

10.2.9	F32-Trans	612.54	0.00	2048.00	Instr per kernel
10.2.10	F64-ADD	0.00	0.00	0.00	Instr per kernel
10.2.11	F64-MUL	0.00	0.00	0.00	Instr per kernel
10.2.12	F64-FMA	0.00	0.00	0.00	Instr per kernel
10.2.13	F64-Trans	0.00	0.00	0.00	Instr per kernel
10.2.14	Conversion	1225.07	0.00	4096.00	Instr per kernel

10.3 VMEM Instr Mix

Metric_ID	Metric	Avg	Min	Max
Unit				
10.3.0	Global/Generic Instr	122825.75	0.00	262144.00
Instr per kernel				
10.3.1	Global/Generic Read	0.00	0.00	0.00
Instr per kernel				
10.3.2	Global/Generic Write	122825.75	0.00	262144.00
Instr per kernel				
10.3.3	Global/Generic Atomic	0.00	0.00	0.00
Instr per kernel				
10.3.4	Spill/Stack Instr	2329.98	0.00	4608.00

Instr per kernel

10.3.5 Spill/Stack Coalesceable Instr
Instr per kernel

10.3.6 Spill/Stack Read 1241.56 0.00 2560.00
Instr per kernel

10.3.7 Spill/Stack Write 1088.42 0.00 2048.00
Instr per kernel

10.3.8 Spill/Stack Atomic 0.00 0.00 0.00
Instr per kernel

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.4.0	MFMA-I8	0.00	0.00	0.00	Instr per kernel
10.4.1	MFMA-F8	0.00	0.00	0.00	Instr per kernel
10.4.2	MFMA-F16	0.00	0.00	0.00	Instr per kernel
10.4.3	MFMA-BF16	0.00	0.00	0.00	Instr per kernel
10.4.4	MFMA-F32	0.00	0.00	0.00	Instr per kernel
10.4.5	MFMA-F64	0.00	0.00	0.00	Instr per kernel
10.4.6	MFMA-F6F4				Instr per kernel

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak	Pct of Peak
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11.1.0	VALU FLOPs	99.37	Gflop	81715.20	0.12
11.1.1	VALU IOPs	640.69	Giop	81715.20	0.78
11.1.2	MFMA FLOPs (F8)	0.00	Gflop	2614886.40	0.00
11.1.3	MFMA FLOPs (BF16)	0.00	Gflop	1307443.20	0.00
11.1.4	MFMA FLOPs (F16)	0.00	Gflop	1307443.20	0.00
11.1.5	MFMA FLOPs (F32)	0.00	Gflop	163430.40	0.00
11.1.6	MFMA FLOPs (F64)	0.00	Gflop	163430.40	0.00
11.1.7	MFMA FLOPs (F6F4)		Gflop		
11.1.8	MFMA IOPs (INT8)	0.00	Giop	2614886.40	0.00

11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
11.2.0	IPC	0.25	0.03	0.53	Instr/cycle
11.2.1	IPC (Issued)	0.98	0.95	1.00	Instr/cycle

11.2.2	SALU Utilization	1.66	0.15	3.24	Pct
11.2.3	VALU Utilization	1.57	0.71	2.38	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	0.42	0.05	0.86	Pct
11.2.6	Branch Utilization	0.30	0.00	0.65	Pct
11.2.7	VALU Active Threads	63.74	63.19	64.01	Threads
11.2.8	MFMA Utilization	0.00	0.00	0.00	Pct
11.2.9	MFMA Instr Cycles				Cycles/instr
11.2.10	VMEM Latency				Cycles
11.2.11	SMEM Latency				Cycles

11.3 Arithmetic Operations

Metric_ID	Metric	Avg	Min	Max	Unit
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11.3.0 kernel	FLOPs (Total)	357041.00	0.00	786432.00	Ops per
11.3.1 kernel	IOPs (Total)	14897195.40	2015232.00	29360128.00	Ops per
11.3.2 kernel	F8 OPs	0.00	0.00	0.00	Ops per
11.3.3 kernel	F16 OPs	278636.50	0.00	524288.00	Ops per
11.3.4 kernel	BF16 OPs	0.00	0.00	0.00	Ops per
11.3.5 kernel	F32 OPs	78404.49	0.00	262144.00	Ops per
11.3.6 kernel	F64 OPs	0.00	0.00	0.00	Ops per
11.3.7 kernel	F6F4 OPs				Ops per
11.3.8 kernel	INT8 OPs	0.00	0.00	0.00	Ops per

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	0.01	Pct of peak

12.1.1	Access Rate	0.00	Pct of peak		
12.1.2	Theoretical Bandwidth (% of Peak)	0.01	Pct of peak		
12.1.3	Bank Conflict Rate	3.12	Pct of peak		
12.2 LDS Stats					
Metric_ID	Metric	Avg	Min	Max	Unit
12.2.0	LDS Instrs	152.20	0.00	1152.00	
Instr per kernel					
12.2.1	LDS LOAD				
Instr per kernel					
12.2.2	LDS STORE				
Instr per kernel					
12.2.3	LDS ATOMIC				
Instr per kernel					
12.2.4	LDS LOAD Bandwidth				Gbps
12.2.5	LDS STORE Bandwidth				Gbps
12.2.6	LDS ATOMIC Bandwidth				Gbps
12.2.7	Theoretical Bandwidth	38963.54	0.00	294912.00	
Bytes per kernel					
12.2.8	LDS Latency				
Cycles					

12.2.9	Bank Conflicts/Access	1.00	1.00	1.00
	Conflicts/access			
12.2.10	Index Accesses	608.81	0.00	4608.00
	Cycles per kernel			
12.2.11	Atomic Return Cycles	0.00	0.00	0.00
	Cycles per kernel			
12.2.12	Bank Conflict	304.40	0.00	2304.00
	Cycles per kernel			
12.2.13	Addr Conflict	0.00	0.00	0.00
	Cycles per kernel			
12.2.14	Unaligned Stall	0.00	0.00	0.00
	Cycles per kernel			
12.2.15	Mem Violations	0.00	0.00	0.00
	Accesses per kernel			
12.2.16	LDS Command FIFO Full Rate			
	Cycles per kernel			
12.2.17	LDS Data FIFO Full Rate			
	Cycles per kernel			

13. Instruction Cache

13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	2.25	Pct of peak
13.1.1	Cache Hit Rate	98.67	Pct of peak

13.1.2 L1I-L2 Bandwidth 0.04 Pct of peak

13.2 Instruction Cache Accesses

Metric_ID	Metric	Avg	Min	Max
Unit				

13.2.0	Req	223823.43	14080.00	458752.00
Req per kernel				

13.2.1	Hits	222205.08	13754.00	455775.00
Hits per kernel				

13.2.2	Misses - Non Duplicated	101.14	32.00	160.00
Misses per kernel				

13.2.3	Misses - Duplicated	1602.12	192.00	4126.00
Misses per kernel				

13.2.4	Cache Hit Rate	98.67	97.35	99.33
Pct				

13.2.5	Instruction Fetch Latency			
Cycles				

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
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13.3.0	L1I-L2 Bandwidth	170831.00	65536.00	245760.00	Bytes per kernel
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14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.22	Pct of peak
14.1.1	Cache Hit Rate	92.72	Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
14.2.0 kernel	Req	30706.44	0.00	65536.00	Req per
14.2.1 kernel	Hits	28472.07	0.00	61141.00	Req per
14.2.2 kernel	Misses - Non Duplicated	74.97	0.00	160.00	Req per
14.2.3 kernel	Misses- Duplicated	2159.40	0.00	4736.00	Req per
14.2.4	Cache Hit Rate	92.72	92.53	93.29	Pct
14.2.5 kernel	Read Req (Total)	30706.44	0.00	65536.00	Req per
14.2.6 kernel	Atomic Req	0.00	0.00	0.00	Req per
14.2.7 kernel	Read Req (1 DWord)	0.00	0.00	0.00	Req per

14.2.8 kernel	Read Req (2 DWord)	0.00	0.00	0.00	Req per
14.2.9 kernel	Read Req (4 DWord)	30706.44	0.00	65536.00	Req per
14.2.10 kernel	Read Req (8 DWord)	0.00	0.00	0.00	Req per
14.2.11 kernel	Read Req (16 DWord)	0.00	0.00	0.00	Req per

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	4797.88	0.00	10240.00	Bytes per kernel
14.3.1	Read Req	74.97	0.00	160.00	Req per kernel
14.3.2	Write Req	0.00	0.00	0.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel
14.3.4	Stall Cycles	0.00	0.00	0.00	Cycles per kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric_ID	Metric	Avg	Min
Max Unit			
15.1.0 89.78 Pct	Address Processing Unit Busy	35.81	0.91
15.1.1	Address Stall	0.15	0.00

1.23 Pct

15.1.2	Data Stall	28.84	0.01
78.74	Pct		

15.1.3	Data-Processor → Address Stall	0.00	0.00
0.00	Pct		

15.1.4 Sequencer → TA Address Stall
Cycles per kernel

15.1.5 Sequencer → TA Command Stall
Cycles per kernel

15.1.6 Sequencer → TA Data Stall
Cycles per kernel

15.1.7	Total Instructions	125155.73	4224.00
262144.00	Instructions per kernel		

15.1.8	Global/Generic Instructions	122825.75	0.00
262144.00	Instructions per kernel		

15.1.9	Global/Generic Read Instructions	0.00	0.00
0.00	Instructions per kernel		

15.1.10 Global/Generic Read Instructions for LDS
Instructions per kernel

15.1.11	Global/Generic Write Instructions	122825.75	0.00
262144.00	Instructions per kernel		

15.1.12	Global/Generic Atomic Instructions	0.00	0.00
0.00	Instructions per kernel		

15.1.13	Spill/Stack Instructions	2329.98	0.00
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4608.00	Instructions per kernel		
15.1.14	Spill/Stack Read Instructions	1241.56	0.00
2560.00	Instructions per kernel		
15.1.15	Spill/Stack Read Instructions for LDS		
	Instructions per kernel		
15.1.16	Spill/Stack Write Instructions	1088.42	0.00
2048.00	Instructions per kernel		
15.1.17	Spill/Stack Atomic Instructions	0.00	0.00
0.00	Instructions per kernel		
15.1.18	Spill/Stack Total Cycles	37076.77	0.00
73728.00	Cycles per kernel		
15.1.19	Spill/Stack Coalesced Read	67.65	0.00
512.00	Cycles per kernel		
15.1.20	Spill/Stack Coalesced Write	0.00	0.00
0.00	Cycles per kernel		

15.2 Data-Return Path

Metric_ID	Metric	Avg	Min
Max	Unit		
15.2.0	Data-Return Busy	36.49	1.25
91.67	Pct		
15.2.1	Cache RAM → Data-Return Stall	35.80	0.75
78.70	Pct		
15.2.2	Workgroup manager → Data-Return Stall	0.00	0.00
0.00	Pct		

15.2.3	Coalescable Instructions	16.91	0.00
128.00	Instructions per kernel		
15.2.4	Read Instructions	1241.56	0.00
2560.00	Instructions per kernel		
15.2.5	Write Instructions	123914.17	2048.00
262144.00	Instructions per kernel		
15.2.6	Atomic Instructions	0.00	0.00
0.00	Instructions per kernel		
15.2.7	Write Ack Instructions		
	Instructions per kernel		

INFO Not showing table with empty column(s): 16.6 L1D Addr
Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
16.1.0	Hit rate	15.54	Pct of peak
16.1.1	Bandwidth	8.87	Pct of peak
16.1.2	Utilization	47.35	Pct of peak
16.1.3	Coalescing	25.08	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3
Max					
16.2.0	Stalled on L2 Data	1.42	2.59	6.53	93.05

94.61

16.2.1	Stalled on L2 Req	0.00	0.00	0.00	76.78
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96.86

16.2.2	Stalled on Address
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16.2.3	Stalled on Data
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16.2.4	Stalled on Latency FIFO
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16.2.5	Stalled on Request FIFO
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16.2.6	Stalled on Read Return
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16.2.7	Tag RAM Stall (Read)	0.00	0.00	0.00	0.00
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0.00

16.2.8	Tag RAM Stall (Write)	0.00	0.00	0.00	0.00
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0.00

16.2.9	Tag RAM Stall (Atomic)	0.00	0.00	0.00	0.00
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0.00

16.3 L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max
Unit				

16.3.0	Total Req	8009154.97	264192.00	16777216.00
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Req per kernel

16.3.1	Read Req	78647.95	0.00	163840.00
Req per kernel				
16.3.2	Write Req	7930507.02	131072.00	16777216.00
Req per kernel				
16.3.3	Atomic Req	0.00	0.00	0.00
Req per kernel				
16.3.4	Cache BW	256292958.94	8454144.00	536870912.00
Bytes per kernel				
16.3.5	Cache Hit Rate	15.54	0.00	33.16
Pct				
16.3.6	Cache Accesses	2002288.74	66048.00	4194304.00
Req per kernel				
16.3.7	Cache Hits	10882.38	0.00	24448.00
Req per kernel				
16.3.8	Invalidations	169.02	32.00	304.00
Req per kernel				
16.3.9	L1-L2 BW	128011902.14	4210688.00	268435456.00
Bytes per kernel				
16.3.10	Tag RAM 0 Req			
Req per kernel				
16.3.11	Tag RAM 1 Req			
Req per kernel				
16.3.12	Tag RAM 2 Req			
Req per kernel				

16.3.13 Tag RAM 3 Req
Req per kernel

16.3.14 L1-L2 Read 8779.61 0.00 16632.00
Req per kernel

16.3.15 L1-L2 Write 1982626.75 32768.00 4194304.00
Req per kernel

16.3.16 L1-L2 Atomic 0.00 0.00 0.00
Req per kernel

16.3.17 L1 Access Latency
Cycles

16.3.18 L1-L2 Read Latency
Cycles

16.3.19 L1-L2 Write Latency
Cycles

16.4 L1D - L2 Transactions

Metric_ID	Metric	Xfer	Coherency	Avg	Min
Max Unit					

16.4.0	NC - Read	Read	NC	0.00	0.00
0.00	Req per kernel				

16.4.1	UC - Read	Read	UC	0.00	0.00
0.00	Req per kernel				

16.4.2	CC - Read	Read	CC	0.00	0.00
0.00	Req per kernel				

16.4.3	RW - Read	Read	RW	8778.45	0.00
16624.00	Req per kernel				
16.4.4	RW - Write	Write	RW	1982626.75	32768.00
4194304.00	Req per kernel				
16.4.5	NC - Write	Write	NC	0.00	0.00
0.00	Req per kernel				
16.4.6	UC - Write	Write	UC	0.00	0.00
0.00	Req per kernel				
16.4.7	CC - Write	Write	CC	0.00	0.00
0.00	Req per kernel				
16.4.8	NC - Atomic	Atomic	NC	0.00	0.00
0.00	Req per kernel				
16.4.9	UC - Atomic	Atomic	UC	0.00	0.00
0.00	Req per kernel				
16.4.10	CC - Atomic	Atomic	CC	0.00	0.00
0.00	Req per kernel				
16.4.11	RW - Atomic	Atomic	RW	0.00	0.00
0.00	Req per kernel				

16.5 L1D Addr Translation

Metric_ID	Metric	Avg	Min
Max	Units		
16.5.0	Req	2002288.74	66048.00
4194304.00	Req per kernel		

16.5.1	Inflight Req		
Req per kernel			
16.5.2	Hit Ratio	94.29	91.69
100.00 Pct			
16.5.3	Hits	1839448.67	64512.00
3896856.00 Req per kernel			
16.5.4	Translation Misses	7742.66	0.00
16384.00 Req per kernel			
16.5.5	Misses under Translation Miss		
Req per kernel			
16.5.6	Permission Misses	0.00	0.00
0.00 Req per kernel			

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache

17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
17.1.0	Utilization	52.68	Pct
17.1.1	Bandwidth	20.04	Pct
17.1.2	Hit Rate	42.14	Pct
17.1.3	L2-Fabric Read BW	313.37	Gb/s
17.1.4	L2-Fabric Write and Atomic BW	3273.86	Gb/s
17.1.5	HBM Bandwidth	5324.80	Gb/s

17.2 L2 - Fabric Transactions

Metric_ID	Metric	Avg	Min
Max Unit			
17.2.0	Read BW	1141084.80	12800.00
2150912.00	Bytes per kernel		
17.2.1	HBM Read Traffic	100.35	86.21
138.85 Pct			
17.2.2	Remote Read Traffic	0.69	0.00
13.79 Pct			
17.2.3	Uncached Read Traffic	4.95	0.00
97.24 Pct			
17.2.4	Write and Atomic BW	126888112.32	2097152.00
268435456.00	Bytes per kernel		
17.2.5	HBM Write and Atomic Traffic	100.00	100.00
100.00 Pct			
17.2.6	Remote Write and Atomic Traffic	0.00	0.00
0.00 Pct			
17.2.7	Atomic Traffic	0.00	0.00
0.00 Pct			
17.2.8	Uncached Write and Atomic Traffic	0.00	0.00
0.00 Pct			
17.2.9	Read Latency	473.80	239.04
990.98 Cycles			
17.2.10	Write and Atomic Latency	380.90	320.68
555.88 Cycles			

17.2.11 Atomic Latency
Cycles

17.2.12 Read Stall
Pct

17.2.13 Write Stall
Pct

17.3 L2 Cache Accesses

Metric_ID	Metric	Avg	Min	Max
Unit				
17.3.0	Bandwidth	255251889.25	6438912.00	537405696.00
	Bytes per kernel			
17.3.1	Read Bandwidth			
	Bytes per kernel			
17.3.2	Write Bandwidth			
	Bytes per kernel			
17.3.3	Atomic Bandwidth			
	Bytes per kernel			
17.3.4	Req	1994155.38	50304.00	4198482.00
	Req per kernel			
17.3.5	Read Req	11532.03	4000.00	18808.00
	Req per kernel			
17.3.6	Write Req	1982626.75	32768.00	4194304.00
	Req per kernel			
17.3.7	Atomic Req	0.00	0.00	0.00

Req per kernel

17.3.8	Streaming Req	0.00	0.00	0.00
Req per kernel				

17.3.9	Bypasss Req
Req per kernel	

17.3.10	Probe Req	0.00	0.00	5.00
Req per kernel				

17.3.11	Input Buffer Req
Req per kernel	

17.3.12	Cache Hit	42.14	34.60	50.05
Pct				

17.3.13	Hits	993970.18	17472.00	2101048.00
Hits per kernel				

17.3.14	Misses	1000236.02	33024.00	2097450.00
Misses per kernel				

17.3.15	Writeback	991397.83	16496.00	2097208.00
Cachelines per kernel				

17.3.16	Writeback (Internal)	921193.11	0.00	1966087.00
Cachelines per kernel				

17.3.17	Writeback (vL1D Req)	70120.26	16384.00	131072.00
Cachelines per kernel				

17.3.18	Evict (Internal)	859832.52	0.00	1835120.00
Cachelines per kernel				

17.3.19	Evict (vL1D Req)	0.00	0.00	0.00
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Cachelines per kernel

17.3.20	NC Req	0.00	0.00	0.00
Req per kernel				
17.3.21	UC Req	0.15	0.00	195.00
Req per kernel				
17.3.22	CC Req	0.00	0.00	0.00
Req per kernel				
17.3.23	RW Req	1994144.19	50304.00	4198304.00
Req per kernel				

17.5 L2 - Fabric Interface Stalls

Metric_ID	Metric	Min	Max	Unit	Type	
Transaction	Avg					
17.5.0	Read - PCIe Stall Pct				PCIe Stall	Read
17.5.1	Read - Infinity Fabric Stall Pct				Infinity Fabric Stall	Read
17.5.2	Read - HBM Stall Pct				HBM Stall	Read
17.5.3	Write - PCIe Stall Pct				PCIe Stall	Write
17.5.4	Write - Infinity Fabric Stall Pct				Infinity Fabric Stall	Write
17.5.5	Write - HBM Stall Pct				HBM Stall	Write

17.5.6	Write - Credit Starvation	Credit Starvation	Write
0.00	0.00 0.00 Pct		

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID	Metric	Avg	Min
Max Unit			
17.6.0	Read (32B)	0.00	0.00
0.00 Req per kernel			
17.6.1	Read (64B)	278.62	200.00
584.00 Req per kernel			
17.6.2	Read (128B)	8775.42	0.00
16512.00 Req per kernel			
17.6.3	Read (Uncached)	18.67	0.00
388.00 Req per kernel			
17.6.4	HBM Read	9055.18	200.00
17120.00 Req per kernel			
17.6.5	Remote Read	2.88	0.00
32.00 Req per kernel			
17.6.6	Read Bandwidth - PCIe		
Bytes per kernel			
17.6.7	Read Bandwidth - Infinity Fabric		
Bytes per kernel			
17.6.8	Read Bandwidth - HBM		
Bytes per kernel			

17.6.9	Write and Atomic (32B)	0.00	0.00
0.00	Req per kernel		
17.6.10	Write and Atomic (Uncached)	0.00	0.00
0.00	Req per kernel		
17.6.11	Write and Atomic (64B)	1982626.75	32768.00
4194304.00	Req per kernel		
17.6.12	HBM Write and Atomic	1982626.75	32768.00
4194304.00	Req per kernel		
17.6.13	Remote Write and Atomic	0.00	0.00
0.00	Req per kernel		
17.6.14	Write Bandwidth - PCIe		
	Bytes per kernel		
17.6.15	Write Bandwidth - Infinity Fabric		
	Bytes per kernel		
17.6.16	Write Bandwidth - HBM		
	Bytes per kernel		
17.6.17	Atomic	0.00	0.00
0.00	Req per kernel		
17.6.18	Atomic - HBM		
	Req per kernel		
17.6.19	Atomic Bandwidth - PCIe		
	Bytes per kernel		
17.6.20	Atomic Bandwidth - Infinity Fabric		
	Bytes per kernel		

17.6.21 Atomic Bandwidth - HBM
Bytes per kernel

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read
Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write
and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	42.05	7.52	34.33	50.05	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.2.0	0.00	32.90	35.32	37.20	50.12	50.14
18.2.1	1.00	32.90	35.32	37.20	50.12	50.14
18.2.2	2.00	32.82	35.32	37.20	50.12	50.14
18.2.3	3.00	32.90	35.32	37.20	50.12	50.14
18.2.4	4.00	32.90	35.56	37.32	50.12	50.14
18.2.5	5.00	32.90	35.50	37.20	50.00	50.03
18.2.6	6.00	32.90	35.96	37.50	50.00	50.03
18.2.7	7.00	32.90	35.96	37.50	50.00	50.03
18.2.8	8.00	32.99	33.33	36.32	50.00	50.03
18.2.9	9.00	33.07	33.33	36.32	50.00	50.03
18.2.10	10.00	32.99	33.33	36.32	50.00	50.03
18.2.11	11.00	33.07	33.33	36.32	50.12	50.14
18.2.12	12.00	33.07	33.33	36.32	50.00	50.03

18.2.13	13.00	33.07	33.33	36.32	50.00	50.03
18.2.14	14.00	33.07	33.33	36.32	50.00	50.03
18.2.15	15.00	33.07	33.33	36.32	50.00	50.03
18.2.16	16.00	32.99	33.33	36.86	50.00	50.03
18.2.17	17.00	33.07	33.33	36.95	50.00	50.03
18.2.18	18.00	32.99	33.33	36.86	50.00	50.03
18.2.19	19.00	33.07	33.33	36.95	50.12	50.14
18.2.20	20.00	33.07	33.33	36.90	50.00	50.03
18.2.21	21.00	33.07	33.33	36.95	50.00	50.03
18.2.22	22.00	33.07	33.33	36.95	50.00	50.03
18.2.23	23.00	33.07	33.33	36.95	50.00	50.03
18.2.24	24.00	32.90	35.32	37.20	50.12	50.14
18.2.25	25.00	32.90	35.32	37.20	50.12	50.14
18.2.26	26.00	32.82	35.32	37.20	50.12	50.14
18.2.27	27.00	32.90	35.32	37.20	50.12	50.14
18.2.28	28.00	32.90	35.32	37.32	50.12	50.14
18.2.29	29.00	32.90	35.32	37.20	50.00	50.03
18.2.30	30.00	32.90	35.96	37.50	50.00	50.03
18.2.31	31.00	32.90	35.96	37.50	50.00	50.03
18.2.32	32.00	32.90	35.32	37.38	50.12	50.14
18.2.33	33.00	32.90	35.32	37.38	50.12	50.14
18.2.34	34.00	32.82	35.32	37.29	50.12	50.14
18.2.35	35.00	32.90	35.32	37.38	50.12	50.14
18.2.36	36.00	32.90	35.32	37.38	50.12	50.14

18.2.37	37.00	32.90	35.32	37.38	50.00	50.03
18.2.38	38.00	32.90	35.96	37.50	50.00	50.03
18.2.39	39.00	32.90	35.96	37.50	50.00	50.03
18.2.40	40.00	32.99	33.33	36.95	50.00	50.03
18.2.41	41.00	33.07	33.33	36.95	50.00	50.03
18.2.42	42.00	32.99	33.33	36.86	50.00	50.03
18.2.43	43.00	33.07	33.33	36.95	50.12	50.14
18.2.44	44.00	33.07	33.33	36.95	50.00	50.03
18.2.45	45.00	33.07	33.33	36.95	50.00	50.03
18.2.46	46.00	33.07	33.33	36.95	50.00	50.03
18.2.47	47.00	33.07	33.33	36.95	50.00	50.03
18.2.48	48.00	32.99	33.33	36.32	50.00	50.03
18.2.49	49.00	33.07	33.33	36.32	50.00	50.03
18.2.50	50.00	32.99	33.33	36.32	50.00	50.03
18.2.51	51.00	33.07	33.33	36.32	50.12	50.14
18.2.52	52.00	33.07	33.33	36.32	50.00	50.03
18.2.53	53.00	33.07	33.33	36.32	50.00	50.03
18.2.54	54.00	33.07	33.33	36.32	50.00	50.03
18.2.55	55.00	33.07	33.33	36.32	50.00	50.03
18.2.56	56.00	32.90	35.32	37.20	50.12	50.14
18.2.57	57.00	32.90	35.32	37.20	50.12	50.14
18.2.58	58.00	32.82	35.32	37.20	50.12	50.14
18.2.59	59.00	32.90	35.32	37.20	50.12	50.14
18.2.60	60.00	32.90	35.32	37.32	50.12	50.14

18.2.61	61.00	32.90	35.32	37.20	50.00	50.03
18.2.62	62.00	32.90	35.96	37.50	50.00	50.03
18.2.63	63.00	32.90	35.96	37.50	50.00	50.03
18.2.64	64.00	32.99	33.33	36.90	50.00	50.03
18.2.65	65.00	33.07	33.33	36.95	50.00	50.03
18.2.66	66.00	32.99	33.33	36.86	50.00	50.03
18.2.67	67.00	33.07	33.33	36.95	50.12	50.14
18.2.68	68.00	33.07	33.33	36.95	50.00	50.03
18.2.69	69.00	33.07	33.33	36.95	50.00	50.03
18.2.70	70.00	33.07	33.33	36.95	50.00	50.03
18.2.71	71.00	33.07	33.33	36.95	50.00	50.03
18.2.72	72.00	32.90	35.32	37.20	50.12	50.14
18.2.73	73.00	32.90	35.32	37.20	50.12	50.14
18.2.74	74.00	32.82	35.32	37.20	50.12	50.14
18.2.75	75.00	32.90	35.32	37.20	50.12	50.14
18.2.76	76.00	32.90	35.32	37.32	50.12	50.14
18.2.77	77.00	32.90	35.32	37.20	50.00	50.03
18.2.78	78.00	32.90	35.96	37.50	50.00	50.03
18.2.79	79.00	32.90	35.96	37.50	50.00	50.03
18.2.80	80.00	32.90	35.32	37.20	50.12	50.14
18.2.81	81.00	32.90	35.32	37.20	50.12	50.14
18.2.82	82.00	32.82	35.32	37.20	50.12	50.14
18.2.83	83.00	32.90	35.32	37.20	50.12	50.14
18.2.84	84.00	32.90	35.32	37.32	50.12	50.14

18.2.85	85.00	32.90	35.32	37.20	50.00	50.03
18.2.86	86.00	32.90	35.96	37.50	50.00	50.03
18.2.87	87.00	32.90	35.96	37.50	50.00	50.03
18.2.88	88.00	32.99	33.33	36.32	50.00	50.03
18.2.89	89.00	33.07	33.33	36.32	50.00	50.03
18.2.90	90.00	32.99	33.33	36.32	50.00	50.03
18.2.91	91.00	33.07	33.33	36.32	50.12	50.14
18.2.92	92.00	33.07	33.33	36.32	50.00	50.03
18.2.93	93.00	33.07	33.33	36.32	50.00	50.03
18.2.94	94.00	33.07	33.33	36.32	50.00	50.03
18.2.95	95.00	33.07	33.33	36.32	50.00	50.03
18.2.96	96.00	32.99	33.33	36.86	50.00	50.03
18.2.97	97.00	33.07	33.33	36.95	50.00	50.03
18.2.98	98.00	32.99	33.33	36.86	50.00	50.03
18.2.99	99.00	33.07	33.33	36.95	50.12	50.14
18.2.100	100.00	33.07	33.33	36.95	50.00	50.03
18.2.101	101.00	33.07	33.33	36.95	50.00	50.03
18.2.102	102.00	33.07	33.33	36.95	50.00	50.03
18.2.103	103.00	33.07	33.33	36.95	50.00	50.03
18.2.104	104.00	32.90	35.32	37.38	50.12	50.14
18.2.105	105.00	32.90	35.32	37.38	50.12	50.14
18.2.106	106.00	32.82	35.32	37.29	50.12	50.14
18.2.107	107.00	32.90	35.32	37.29	50.12	50.14
18.2.108	108.00	32.90	35.46	37.38	50.12	50.14

18.2.109	109.00	32.90	35.32	37.38	50.00	50.03
18.2.110	110.00	32.90	35.96	37.50	50.00	50.03
18.2.111	111.00	32.90	35.96	37.50	50.00	50.03
18.2.112	112.00	32.90	35.32	37.20	50.12	50.14
18.2.113	113.00	32.90	35.32	37.20	50.12	50.14
18.2.114	114.00	32.82	35.32	37.20	50.12	50.14
18.2.115	115.00	32.90	35.32	37.20	50.12	50.14
18.2.116	116.00	32.90	35.32	37.32	50.12	50.14
18.2.117	117.00	32.90	35.32	37.20	50.00	50.03
18.2.118	118.00	32.90	35.96	37.50	50.00	50.03
18.2.119	119.00	32.90	35.96	37.50	50.00	50.03
18.2.120	120.00	32.99	33.33	36.81	50.00	50.03
18.2.121	121.00	33.07	33.33	36.86	50.00	50.03
18.2.122	122.00	32.99	33.33	36.86	50.00	50.03
18.2.123	123.00	33.07	33.33	36.90	50.12	50.14
18.2.124	124.00	33.07	33.33	36.86	50.00	50.03
18.2.125	125.00	33.07	33.33	36.86	50.00	50.03
18.2.126	126.00	33.07	33.33	36.86	50.00	50.03
18.2.127	127.00	33.07	33.33	36.86	50.00	50.03

18.3 L2 Requests (per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.3.0	0.00	386.00	402.00	414.00	32848.00	32870.00
18.3.1	1.00	386.00	402.00	414.00	32848.00	32869.00
18.3.2	2.00	386.00	402.00	414.00	32848.00	32869.00

18.3.3	3.00	386.00	402.00	414.00	32848.00	32869.00
18.3.4	4.00	386.00	402.00	414.00	32848.00	32869.00
18.3.5	5.00	386.00	402.00	414.00	32768.00	32789.00
18.3.6	6.00	386.00	402.00	418.00	32768.00	32789.00
18.3.7	7.00	386.00	402.00	418.00	32768.00	32790.00
18.3.8	8.00	384.00	400.00	412.00	32768.00	32789.00
18.3.9	9.00	384.00	400.00	412.00	32768.00	32789.00
18.3.10	10.00	384.00	400.00	412.00	32768.00	32789.00
18.3.11	11.00	384.00	400.00	416.00	32848.00	32869.00
18.3.12	12.00	384.00	400.00	412.00	32768.00	32789.00
18.3.13	13.00	384.00	400.00	412.00	32768.00	32789.00
18.3.14	14.00	384.00	400.00	412.00	32768.00	32790.00
18.3.15	15.00	384.00	400.00	412.00	32768.00	32789.00
18.3.16	16.00	384.00	400.00	412.00	32768.00	32789.00
18.3.17	17.00	384.00	400.00	412.00	32768.00	32789.00
18.3.18	18.00	384.00	400.00	412.00	32768.00	32789.00
18.3.19	19.00	384.00	400.00	416.00	32848.00	32869.00
18.3.20	20.00	384.00	400.00	412.00	32768.00	32789.00
18.3.21	21.00	384.00	400.00	412.00	32768.00	32789.00
18.3.22	22.00	384.00	400.00	412.00	32768.00	32790.00
18.3.23	23.00	384.00	400.00	412.00	32768.00	32789.00
18.3.24	24.00	386.00	402.00	414.00	32848.00	32870.00
18.3.25	25.00	386.00	402.00	414.00	32848.00	32869.00
18.3.26	26.00	386.00	402.00	414.00	32848.00	32869.00

18.3.27	27.00	386.00	402.00	414.00	32848.00	32869.00
18.3.28	28.00	386.00	402.00	414.00	32848.00	32869.00
18.3.29	29.00	386.00	402.00	414.00	32768.00	32789.00
18.3.30	30.00	386.00	402.00	418.00	32768.00	32789.00
18.3.31	31.00	386.00	402.00	418.00	32768.00	32790.00
18.3.32	32.00	386.00	402.00	414.00	32848.00	32870.00
18.3.33	33.00	386.00	402.00	414.00	32848.00	32869.00
18.3.34	34.00	386.00	402.00	414.00	32848.00	32869.00
18.3.35	35.00	386.00	402.00	414.00	32848.00	32869.00
18.3.36	36.00	386.00	402.00	414.00	32848.00	32869.00
18.3.37	37.00	386.00	402.00	414.00	32768.00	32789.00
18.3.38	38.00	386.00	402.00	418.00	32768.00	32789.00
18.3.39	39.00	386.00	402.00	418.00	32768.00	32790.00
18.3.40	40.00	384.00	400.00	412.00	32768.00	32789.00
18.3.41	41.00	384.00	400.00	412.00	32768.00	32789.00
18.3.42	42.00	384.00	400.00	412.00	32768.00	32789.00
18.3.43	43.00	384.00	400.00	416.00	32848.00	32869.00
18.3.44	44.00	384.00	400.00	412.00	32768.00	32789.00
18.3.45	45.00	384.00	400.00	412.00	32768.00	32789.00
18.3.46	46.00	384.00	400.00	412.00	32768.00	32790.00
18.3.47	47.00	384.00	400.00	412.00	32768.00	32789.00
18.3.48	48.00	384.00	400.00	412.00	32768.00	32789.00
18.3.49	49.00	384.00	400.00	412.00	32768.00	32789.00
18.3.50	50.00	384.00	400.00	412.00	32768.00	32789.00

18.3.51	51.00	384.00	400.00	416.00	32848.00	32869.00
18.3.52	52.00	384.00	400.00	412.00	32768.00	32789.00
18.3.53	53.00	384.00	400.00	412.00	32768.00	32789.00
18.3.54	54.00	384.00	400.00	412.00	32768.00	32790.00
18.3.55	55.00	384.00	400.00	412.00	32768.00	32789.00
18.3.56	56.00	386.00	402.00	414.00	32848.00	32870.00
18.3.57	57.00	386.00	402.00	414.00	32848.00	32869.00
18.3.58	58.00	386.00	402.00	414.00	32848.00	32869.00
18.3.59	59.00	386.00	402.00	414.00	32848.00	32869.00
18.3.60	60.00	386.00	402.00	414.00	32848.00	32869.00
18.3.61	61.00	386.00	402.00	414.00	32768.00	32789.00
18.3.62	62.00	386.00	402.00	418.00	32768.00	32789.00
18.3.63	63.00	386.00	402.00	418.00	32768.00	32790.00
18.3.64	64.00	384.00	400.00	412.00	32768.00	32789.00
18.3.65	65.00	384.00	400.00	412.00	32768.00	32789.00
18.3.66	66.00	384.00	400.00	412.00	32768.00	32789.00
18.3.67	67.00	384.00	400.00	416.00	32848.00	32869.00
18.3.68	68.00	384.00	400.00	412.00	32768.00	32789.00
18.3.69	69.00	384.00	400.00	412.00	32768.00	32789.00
18.3.70	70.00	384.00	400.00	412.00	32768.00	32790.00
18.3.71	71.00	384.00	400.00	412.00	32768.00	32789.00
18.3.72	72.00	386.00	402.00	414.00	32848.00	32870.00
18.3.73	73.00	386.00	402.00	414.00	32848.00	32869.00
18.3.74	74.00	386.00	402.00	414.00	32848.00	32869.00

18.3.75	75.00	386.00	402.00	414.00	32848.00	32869.00
18.3.76	76.00	386.00	402.00	414.00	32848.00	32869.00
18.3.77	77.00	386.00	402.00	414.00	32768.00	32789.00
18.3.78	78.00	386.00	402.00	418.00	32768.00	32789.00
18.3.79	79.00	386.00	402.00	418.00	32768.00	32790.00
18.3.80	80.00	386.00	402.00	414.00	32848.00	32870.00
18.3.81	81.00	386.00	402.00	414.00	32848.00	32869.00
18.3.82	82.00	386.00	402.00	414.00	32848.00	32869.00
18.3.83	83.00	386.00	402.00	414.00	32848.00	32869.00
18.3.84	84.00	386.00	402.00	414.00	32848.00	32869.00
18.3.85	85.00	386.00	402.00	414.00	32768.00	32789.00
18.3.86	86.00	386.00	402.00	418.00	32768.00	32789.00
18.3.87	87.00	386.00	402.00	418.00	32768.00	32790.00
18.3.88	88.00	384.00	400.00	412.00	32768.00	32789.00
18.3.89	89.00	384.00	400.00	412.00	32768.00	32789.00
18.3.90	90.00	384.00	400.00	412.00	32768.00	32789.00
18.3.91	91.00	384.00	400.00	416.00	32848.00	32869.00
18.3.92	92.00	384.00	400.00	412.00	32768.00	32789.00
18.3.93	93.00	384.00	400.00	412.00	32768.00	32789.00
18.3.94	94.00	384.00	400.00	412.00	32768.00	32790.00
18.3.95	95.00	384.00	400.00	412.00	32768.00	32789.00
18.3.96	96.00	384.00	400.00	412.00	32768.00	32789.00
18.3.97	97.00	384.00	400.00	412.00	32768.00	32789.00
18.3.98	98.00	384.00	400.00	412.00	32768.00	32789.00

18.3.99	99.00	384.00	400.00	416.00	32848.00	32869.00
18.3.100	100.00	384.00	400.00	412.00	32768.00	32789.00
18.3.101	101.00	384.00	400.00	412.00	32768.00	32789.00
18.3.102	102.00	384.00	400.00	412.00	32768.00	32790.00
18.3.103	103.00	384.00	400.00	412.00	32768.00	32789.00
18.3.104	104.00	386.00	402.00	414.00	32848.00	32870.00
18.3.105	105.00	386.00	402.00	414.00	32848.00	32869.00
18.3.106	106.00	386.00	402.00	414.00	32848.00	32869.00
18.3.107	107.00	386.00	402.00	414.00	32848.00	32869.00
18.3.108	108.00	386.00	402.00	414.00	32848.00	32869.00
18.3.109	109.00	386.00	402.00	414.00	32768.00	32789.00
18.3.110	110.00	386.00	402.00	418.00	32768.00	32789.00
18.3.111	111.00	386.00	402.00	418.00	32768.00	32790.00
18.3.112	112.00	386.00	402.00	414.00	32848.00	32870.00
18.3.113	113.00	386.00	402.00	414.00	32848.00	32869.00
18.3.114	114.00	386.00	402.00	414.00	32848.00	32869.00
18.3.115	115.00	386.00	402.00	414.00	32848.00	32869.00
18.3.116	116.00	386.00	402.00	414.00	32848.00	32869.00
18.3.117	117.00	386.00	402.00	414.00	32768.00	32789.00
18.3.118	118.00	386.00	402.00	418.00	32768.00	32789.00
18.3.119	119.00	386.00	402.00	418.00	32768.00	32790.00
18.3.120	120.00	384.00	400.00	412.00	32768.00	32789.00
18.3.121	121.00	384.00	400.00	412.00	32768.00	32789.00
18.3.122	122.00	384.00	400.00	412.00	32768.00	32789.00

18.3.123	123.00	384.00	400.00	416.00	32848.00	32869.00
18.3.124	124.00	384.00	400.00	412.00	32768.00	32789.00
18.3.125	125.00	384.00	400.00	412.00	32768.00	32789.00
18.3.126	126.00	384.00	400.00	412.00	32768.00	32790.00
18.3.127	127.00	384.00	400.00	412.00	32768.00	32789.00

18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	116.81	15489.27	0.00
18.4.1	1.00	116.80	15489.27	0.00
18.4.2	2.00	118.50	15489.27	0.00
18.4.3	3.00	118.06	15489.27	0.00
18.4.4	4.00	117.36	15489.27	0.00
18.4.5	5.00	80.07	15489.27	0.00
18.4.6	6.00	79.34	15489.27	0.00
18.4.7	7.00	79.36	15489.27	0.00
18.4.8	8.00	72.22	15489.27	0.00
18.4.9	9.00	71.90	15489.27	0.00
18.4.10	10.00	72.16	15489.27	0.00
18.4.11	11.00	109.39	15489.27	0.00
18.4.12	12.00	71.83	15489.27	0.00
18.4.13	13.00	71.79	15489.27	0.00
18.4.14	14.00	71.90	15489.27	0.00
18.4.15	15.00	71.82	15489.27	0.00
18.4.16	16.00	72.33	15489.27	0.00

18.4.17	17.00	72.01	15489.27	0.00
18.4.18	18.00	72.27	15489.27	0.00
18.4.19	19.00	109.50	15489.27	0.00
18.4.20	20.00	71.93	15489.27	0.00
18.4.21	21.00	71.90	15489.27	0.00
18.4.22	22.00	72.01	15489.27	0.00
18.4.23	23.00	71.93	15489.27	0.00
18.4.24	24.00	116.74	15489.27	0.00
18.4.25	25.00	116.73	15489.27	0.00
18.4.26	26.00	118.42	15489.27	0.00
18.4.27	27.00	117.97	15489.27	0.00
18.4.28	28.00	117.27	15489.27	0.00
18.4.29	29.00	79.98	15489.27	0.00
18.4.30	30.00	79.27	15489.27	0.00
18.4.31	31.00	79.29	15489.27	0.00
18.4.32	32.00	116.86	15489.27	0.00
18.4.33	33.00	116.85	15489.27	0.00
18.4.34	34.00	118.54	15489.27	0.00
18.4.35	35.00	118.10	15489.27	0.00
18.4.36	36.00	117.41	15489.27	0.00
18.4.37	37.00	80.08	15489.27	0.00
18.4.38	38.00	79.40	15489.27	0.00
18.4.39	39.00	79.37	15489.27	0.00
18.4.40	40.00	72.29	15489.27	0.00

18.4.41	41.00	71.97	15489.27	0.00
18.4.42	42.00	72.24	15489.27	0.00
18.4.43	43.00	109.47	15489.27	0.00
18.4.44	44.00	71.90	15489.27	0.00
18.4.45	45.00	71.86	15489.27	0.00
18.4.46	46.00	71.98	15489.27	0.00
18.4.47	47.00	71.89	15489.27	0.00
18.4.48	48.00	72.31	15489.27	0.00
18.4.49	49.00	71.99	15489.27	0.00
18.4.50	50.00	72.25	15489.27	0.00
18.4.51	51.00	109.48	15489.27	0.00
18.4.52	52.00	71.91	15489.27	0.00
18.4.53	53.00	71.88	15489.27	0.00
18.4.54	54.00	71.99	15489.27	0.00
18.4.55	55.00	71.91	15489.27	0.00
18.4.56	56.00	116.84	15489.27	0.00
18.4.57	57.00	116.83	15489.27	0.00
18.4.58	58.00	118.53	15489.27	0.00
18.4.59	59.00	118.09	15489.27	0.00
18.4.60	60.00	117.41	15489.27	0.00
18.4.61	61.00	80.07	15489.27	0.00
18.4.62	62.00	79.41	15489.27	0.00
18.4.63	63.00	79.36	15489.27	0.00
18.4.64	64.00	72.33	15489.27	0.00

18.4.65	65.00	72.01	15489.27	0.00
18.4.66	66.00	72.28	15489.27	0.00
18.4.67	67.00	109.51	15489.27	0.00
18.4.68	68.00	71.94	15489.27	0.00
18.4.69	69.00	71.90	15489.27	0.00
18.4.70	70.00	72.02	15489.27	0.00
18.4.71	71.00	71.94	15489.27	0.00
18.4.72	72.00	116.87	15489.27	0.00
18.4.73	73.00	116.85	15489.27	0.00
18.4.74	74.00	118.55	15489.27	0.00
18.4.75	75.00	118.10	15489.27	0.00
18.4.76	76.00	117.42	15489.27	0.00
18.4.77	77.00	80.08	15489.27	0.00
18.4.78	78.00	79.41	15489.27	0.00
18.4.79	79.00	79.37	15489.27	0.00
18.4.80	80.00	116.87	15489.27	0.00
18.4.81	81.00	116.86	15489.27	0.00
18.4.82	82.00	118.55	15489.27	0.00
18.4.83	83.00	118.11	15489.27	0.00
18.4.84	84.00	117.42	15489.27	0.00
18.4.85	85.00	80.10	15489.27	0.00
18.4.86	86.00	79.42	15489.27	0.00
18.4.87	87.00	79.38	15489.27	0.00
18.4.88	88.00	72.28	15489.27	0.00

18.4.89	89.00	71.96	15489.27	0.00
18.4.90	90.00	72.22	15489.27	0.00
18.4.91	91.00	109.45	15489.27	0.00
18.4.92	92.00	71.88	15489.27	0.00
18.4.93	93.00	71.85	15489.27	0.00
18.4.94	94.00	71.96	15489.27	0.00
18.4.95	95.00	71.88	15489.27	0.00
18.4.96	96.00	72.28	15489.27	0.00
18.4.97	97.00	71.96	15489.27	0.00
18.4.98	98.00	72.22	15489.27	0.00
18.4.99	99.00	109.45	15489.27	0.00
18.4.100	100.00	71.88	15489.27	0.00
18.4.101	101.00	71.84	15489.27	0.00
18.4.102	102.00	71.96	15489.27	0.00
18.4.103	103.00	71.88	15489.27	0.00
18.4.104	104.00	116.88	15489.27	0.00
18.4.105	105.00	116.88	15489.27	0.00
18.4.106	106.00	118.57	15489.27	0.00
18.4.107	107.00	118.13	15489.27	0.00
18.4.108	108.00	117.45	15489.27	0.00
18.4.109	109.00	80.13	15489.27	0.00
18.4.110	110.00	79.45	15489.27	0.00
18.4.111	111.00	79.44	15489.27	0.00
18.4.112	112.00	116.77	15489.27	0.00

18.4.113	113.00	116.77	15489.27	0.00
18.4.114	114.00	118.46	15489.27	0.00
18.4.115	115.00	118.02	15489.27	0.00
18.4.116	116.00	117.32	15489.27	0.00
18.4.117	117.00	80.02	15489.27	0.00
18.4.118	118.00	79.30	15489.27	0.00
18.4.119	119.00	79.32	15489.27	0.00
18.4.120	120.00	72.33	15489.27	0.00
18.4.121	121.00	72.01	15489.27	0.00
18.4.122	122.00	72.28	15489.27	0.00
18.4.123	123.00	109.50	15489.27	0.00
18.4.124	124.00	71.94	15489.27	0.00
18.4.125	125.00	71.90	15489.27	0.00
18.4.126	126.00	72.02	15489.27	0.00
18.4.127	127.00	71.93	15489.27	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
L2-Fabric Atomic			
18.5.0 0.00	0.00	72.89	15489.27
18.5.1 0.00	1.00	72.55	15489.27
18.5.2 0.00	2.00	72.55	15489.27

18.5.3 0.00	3.00	72.54	15489.27
18.5.4 0.00	4.00	72.56	15489.27
18.5.5 0.00	5.00	70.66	15489.27
18.5.6 0.00	6.00	70.96	15489.27
18.5.7 0.00	7.00	71.03	15489.27
18.5.8 0.00	8.00	69.24	15489.27
18.5.9 0.00	9.00	69.24	15489.27
18.5.10 0.00	10.00	69.45	15489.27
18.5.11 0.00	11.00	71.31	15489.27
18.5.12 0.00	12.00	69.24	15489.27
18.5.13 0.00	13.00	69.23	15489.27
18.5.14 0.00	14.00	69.25	15489.27

18.5.15 0.00	15.00	69.23	15489.27
18.5.16 0.00	16.00	69.24	15489.27
18.5.17 0.00	17.00	69.24	15489.27
18.5.18 0.00	18.00	69.45	15489.27
18.5.19 0.00	19.00	71.31	15489.27
18.5.20 0.00	20.00	69.24	15489.27
18.5.21 0.00	21.00	69.23	15489.27
18.5.22 0.00	22.00	69.25	15489.27
18.5.23 0.00	23.00	69.23	15489.27
18.5.24 0.00	24.00	72.89	15489.27
18.5.25 0.00	25.00	72.55	15489.27
18.5.26 0.00	26.00	72.55	15489.27

18.5.27 0.00	27.00	72.54	15489.27
18.5.28 0.00	28.00	72.56	15489.27
18.5.29 0.00	29.00	70.66	15489.27
18.5.30 0.00	30.00	70.96	15489.27
18.5.31 0.00	31.00	71.03	15489.27
18.5.32 0.00	32.00	72.89	15489.27
18.5.33 0.00	33.00	72.55	15489.27
18.5.34 0.00	34.00	72.55	15489.27
18.5.35 0.00	35.00	72.54	15489.27
18.5.36 0.00	36.00	72.56	15489.27
18.5.37 0.00	37.00	70.66	15489.27
18.5.38 0.00	38.00	70.96	15489.27

18.5.39 0.00	39.00	71.03	15489.27
18.5.40 0.00	40.00	69.24	15489.27
18.5.41 0.00	41.00	69.24	15489.27
18.5.42 0.00	42.00	69.45	15489.27
18.5.43 0.00	43.00	71.31	15489.27
18.5.44 0.00	44.00	69.24	15489.27
18.5.45 0.00	45.00	69.23	15489.27
18.5.46 0.00	46.00	69.25	15489.27
18.5.47 0.00	47.00	69.23	15489.27
18.5.48 0.00	48.00	69.24	15489.27
18.5.49 0.00	49.00	69.24	15489.27
18.5.50 0.00	50.00	69.45	15489.27

18.5.51 0.00	51.00	71.31	15489.27
18.5.52 0.00	52.00	69.24	15489.27
18.5.53 0.00	53.00	69.23	15489.27
18.5.54 0.00	54.00	69.25	15489.27
18.5.55 0.00	55.00	69.23	15489.27
18.5.56 0.00	56.00	72.89	15489.27
18.5.57 0.00	57.00	72.55	15489.27
18.5.58 0.00	58.00	72.55	15489.27
18.5.59 0.00	59.00	72.54	15489.27
18.5.60 0.00	60.00	72.56	15489.27
18.5.61 0.00	61.00	70.66	15489.27
18.5.62 0.00	62.00	70.96	15489.27

18.5.63 0.00	63.00	71.03	15489.27
18.5.64 0.00	64.00	69.24	15489.27
18.5.65 0.00	65.00	69.24	15489.27
18.5.66 0.00	66.00	69.45	15489.27
18.5.67 0.00	67.00	71.31	15489.27
18.5.68 0.00	68.00	69.24	15489.27
18.5.69 0.00	69.00	69.23	15489.27
18.5.70 0.00	70.00	69.25	15489.27
18.5.71 0.00	71.00	69.23	15489.27
18.5.72 0.00	72.00	72.89	15489.27
18.5.73 0.00	73.00	72.55	15489.27
18.5.74 0.00	74.00	72.55	15489.27

18.5.75 0.00	75.00	72.54	15489.27
18.5.76 0.00	76.00	72.56	15489.27
18.5.77 0.00	77.00	70.66	15489.27
18.5.78 0.00	78.00	70.96	15489.27
18.5.79 0.00	79.00	71.03	15489.27
18.5.80 0.00	80.00	72.89	15489.27
18.5.81 0.00	81.00	72.55	15489.27
18.5.82 0.00	82.00	72.55	15489.27
18.5.83 0.00	83.00	72.54	15489.27
18.5.84 0.00	84.00	72.57	15489.27
18.5.85 0.00	85.00	70.66	15489.27
18.5.86 0.00	86.00	70.96	15489.27

18.5.87 0.00	87.00	71.03	15489.27
18.5.88 0.00	88.00	69.24	15489.27
18.5.89 0.00	89.00	69.24	15489.27
18.5.90 0.00	90.00	69.45	15489.27
18.5.91 0.00	91.00	71.31	15489.27
18.5.92 0.00	92.00	69.24	15489.27
18.5.93 0.00	93.00	69.23	15489.27
18.5.94 0.00	94.00	69.25	15489.27
18.5.95 0.00	95.00	69.23	15489.27
18.5.96 0.00	96.00	69.24	15489.27
18.5.97 0.00	97.00	69.24	15489.27
18.5.98 0.00	98.00	69.45	15489.27

18.5.99 0.00	99.00	71.31	15489.27
18.5.100 0.00	100.00	69.24	15489.27
18.5.101 0.00	101.00	69.23	15489.27
18.5.102 0.00	102.00	69.25	15489.27
18.5.103 0.00	103.00	69.23	15489.27
18.5.104 0.00	104.00	72.89	15489.27
18.5.105 0.00	105.00	72.55	15489.27
18.5.106 0.00	106.00	72.55	15489.27
18.5.107 0.00	107.00	72.54	15489.27
18.5.108 0.00	108.00	72.56	15489.27
18.5.109 0.00	109.00	70.66	15489.27
18.5.110 0.00	110.00	70.96	15489.27

18.5.111 0.00	111.00	71.03	15489.27
18.5.112 0.00	112.00	72.89	15489.27
18.5.113 0.00	113.00	72.55	15489.27
18.5.114 0.00	114.00	72.55	15489.27
18.5.115 0.00	115.00	72.54	15489.27
18.5.116 0.00	116.00	72.56	15489.27
18.5.117 0.00	117.00	70.66	15489.27
18.5.118 0.00	118.00	70.96	15489.27
18.5.119 0.00	119.00	71.03	15489.27
18.5.120 0.00	120.00	69.24	15489.27
18.5.121 0.00	121.00	69.24	15489.27
18.5.122 0.00	122.00	69.45	15489.27

18.5.123 0.00	123.00	71.31	15489.27
18.5.124 0.00	124.00	69.24	15489.27
18.5.125 0.00	125.00	69.23	15489.27
18.5.126 0.00	126.00	69.25	15489.27
18.5.127 0.00	127.00	69.23	15489.27

18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	0.00	0.00	458.18	482.25	1047.57
18.6.1	1.00	0.00	0.00	474.13	495.71	974.73
18.6.2	2.00	179.17	270.50	483.42	503.18	1119.61
18.6.3	3.00	0.00	0.00	479.92	497.95	1045.91
18.6.4	4.00	0.00	0.00	481.57	497.50	1166.55
18.6.5	5.00	0.00	483.97	493.83	530.58	1185.63
18.6.6	6.00	0.00	514.18	523.32	564.92	1271.72
18.6.7	7.00	0.00	491.75	504.20	559.57	1196.97
18.6.8	8.00	451.67	473.97	489.69	572.08	2312.00
18.6.9	9.00	473.21	495.27	506.46	564.51	1854.00
18.6.10	10.00	466.61	489.91	504.90	597.02	2109.00

18.6.11	11.00	182.00	275.50	491.59	519.55	1098.55
18.6.12	12.00	0.00	498.11	512.34	561.33	1256.80
18.6.13	13.00	474.52	497.09	518.31	609.55	1969.00
18.6.14	14.00	0.00	516.20	529.88	579.76	1151.21
18.6.15	15.00	0.00	501.01	515.02	559.44	1205.12
18.6.16	16.00	456.08	478.32	496.52	589.38	2292.00
18.6.17	17.00	477.58	497.46	512.44	566.59	2111.00
18.6.18	18.00	469.67	494.17	507.97	612.33	2112.00
18.6.19	19.00	183.00	276.75	494.84	517.36	1213.74
18.6.20	20.00	0.00	499.17	517.07	575.96	1216.50
18.6.21	21.00	476.97	498.77	522.58	651.15	1935.00
18.6.22	22.00	0.00	518.96	533.23	592.55	1322.00
18.6.23	23.00	0.00	502.91	517.92	572.62	1263.90
18.6.24	24.00	0.00	0.00	457.17	478.52	1098.24
18.6.25	25.00	0.00	0.00	473.19	490.64	1052.31
18.6.26	26.00	182.17	277.00	484.46	504.35	1173.18
18.6.27	27.00	0.00	0.00	483.38	504.99	1125.34
18.6.28	28.00	0.00	0.00	482.39	498.76	1214.20
18.6.29	29.00	0.00	486.42	494.94	528.33	1251.16
18.6.30	30.00	0.00	514.92	525.44	569.66	1217.26
18.6.31	31.00	0.00	494.09	505.24	549.66	1186.45
18.6.32	32.00	0.00	0.00	468.40	486.10	1165.25
18.6.33	33.00	0.00	0.00	487.54	504.73	1125.76
18.6.34	34.00	241.17	373.00	489.24	513.32	1214.55

18.6.35	35.00	0.00	0.00	493.35	513.01	1286.58
18.6.36	36.00	0.00	0.00	493.06	513.07	1381.56
18.6.37	37.00	0.00	496.48	507.39	546.60	1317.36
18.6.38	38.00	0.00	512.54	523.84	574.22	1474.02
18.6.39	39.00	0.00	493.59	503.53	532.38	1307.46
18.6.40	40.00	449.53	475.94	493.18	590.87	2687.00
18.6.41	41.00	469.09	491.74	504.21	560.20	2430.00
18.6.42	42.00	480.31	499.70	511.46	628.88	2587.00
18.6.43	43.00	253.67	432.25	497.24	523.26	1307.71
18.6.44	44.00	0.00	502.05	517.89	573.10	1322.00
18.6.45	45.00	479.07	502.00	524.30	606.80	2814.00
18.6.46	46.00	0.00	534.19	549.64	606.74	1240.65
18.6.47	47.00	0.00	521.41	535.92	567.55	1319.66
18.6.48	48.00	451.69	472.48	488.52	552.33	2864.00
18.6.49	49.00	466.43	489.30	502.70	551.89	2435.00
18.6.50	50.00	477.42	499.28	514.35	613.40	2589.00
18.6.51	51.00	254.83	429.25	496.07	518.02	1289.54
18.6.52	52.00	0.00	498.33	516.73	573.69	1283.56
18.6.53	53.00	475.47	500.34	521.37	613.84	2694.00
18.6.54	54.00	0.00	530.82	548.88	599.54	1298.45
18.6.55	55.00	0.00	513.29	528.23	576.81	1275.44
18.6.56	56.00	0.00	0.00	462.62	481.72	1092.79
18.6.57	57.00	0.00	0.00	485.06	504.91	1002.36
18.6.58	58.00	239.17	365.00	485.29	505.08	1235.87

18.6.59	59.00	0.00	0.00	490.86	511.55	1181.38
18.6.60	60.00	0.00	0.00	490.44	510.44	1140.81
18.6.61	61.00	0.00	493.77	504.37	546.18	1145.88
18.6.62	62.00	0.00	511.83	523.19	569.07	1179.52
18.6.63	63.00	0.00	490.02	499.43	535.16	1222.41
18.6.64	64.00	456.97	482.51	508.05	582.51	2170.00
18.6.65	65.00	475.04	497.16	510.54	566.97	1910.00
18.6.66	66.00	483.37	505.47	519.65	611.15	2049.00
18.6.67	67.00	160.83	298.25	499.79	523.84	1224.95
18.6.68	68.00	0.00	504.22	520.19	574.08	1257.67
18.6.69	69.00	478.03	504.71	527.19	629.73	2093.00
18.6.70	70.00	0.00	535.00	546.76	607.63	1186.34
18.6.71	71.00	0.00	515.30	531.14	592.02	1309.11
18.6.72	72.00	0.00	0.00	473.84	497.84	1149.70
18.6.73	73.00	0.00	0.00	496.92	516.07	1075.47
18.6.74	74.00	158.00	247.50	496.30	518.10	1221.74
18.6.75	75.00	0.00	0.00	502.16	524.91	1221.06
18.6.76	76.00	0.00	0.00	501.57	517.64	1207.96
18.6.77	77.00	0.00	503.42	513.09	548.50	1256.80
18.6.78	78.00	0.00	524.85	535.13	572.50	1386.35
18.6.79	79.00	0.00	502.57	511.77	543.64	1281.85
18.6.80	80.00	0.00	0.00	473.28	493.23	1097.27
18.6.81	81.00	0.00	0.00	494.90	517.36	1018.45
18.6.82	82.00	150.83	246.44	492.69	515.01	1163.23

18.6.83	83.00	0.00	0.00	500.10	522.42	1177.75
18.6.84	84.00	0.00	0.00	499.00	520.09	1374.95
18.6.85	85.00	0.00	500.15	509.91	554.77	1267.80
18.6.86	86.00	0.00	519.40	533.10	593.44	1355.98
18.6.87	87.00	0.00	496.98	506.68	550.49	1287.54
18.6.88	88.00	452.32	479.44	504.12	635.88	2227.00
18.6.89	89.00	473.77	497.20	512.79	556.25	2046.00
18.6.90	90.00	477.39	504.16	518.22	616.02	1852.00
18.6.91	91.00	168.83	304.00	496.91	527.75	1260.50
18.6.92	92.00	0.00	498.77	515.04	564.23	1319.59
18.6.93	93.00	477.08	499.43	524.23	629.24	2246.00
18.6.94	94.00	0.00	530.53	546.38	612.26	1316.51
18.6.95	95.00	0.00	513.62	531.03	585.63	1275.67
18.6.96	96.00	241.00	476.17	492.77	545.65	1847.00
18.6.97	97.00	262.50	499.21	509.98	551.11	1523.00
18.6.98	98.00	262.50	491.68	508.59	554.02	1620.00
18.6.99	99.00	88.83	156.75	491.85	522.44	1330.55
18.6.100	100.00	0.00	499.78	518.12	566.50	1247.16
18.6.101	101.00	257.00	504.69	528.38	593.45	1686.00
18.6.102	102.00	0.00	520.83	537.14	595.93	1191.56
18.6.103	103.00	0.00	501.29	515.83	564.66	1258.77
18.6.104	104.00	0.00	0.00	467.10	490.18	1070.19
18.6.105	105.00	0.00	0.00	485.80	502.25	1010.76
18.6.106	106.00	85.00	139.15	494.12	514.27	1148.42

18.6.107	107.00	0.00	0.00	492.88	514.32	1178.52
18.6.108	108.00	0.00	0.00	493.46	508.78	1229.87
18.6.109	109.00	0.00	493.63	500.04	537.92	1171.56
18.6.110	110.00	0.00	526.92	539.54	590.18	1296.40
18.6.111	111.00	0.00	505.47	515.98	556.21	1184.71
18.6.112	112.00	0.00	0.00	468.70	489.75	1106.64
18.6.113	113.00	0.00	0.00	484.12	499.96	1046.48
18.6.114	114.00	87.50	133.50	494.57	512.75	1173.39
18.6.115	115.00	0.00	0.00	492.06	511.39	1132.48
18.6.116	116.00	0.00	0.00	491.48	505.94	1226.31
18.6.117	117.00	0.00	489.99	497.01	544.57	1227.17
18.6.118	118.00	0.00	526.01	537.39	579.17	1327.02
18.6.119	119.00	0.00	502.19	514.47	553.27	1227.02
18.6.120	120.00	280.50	476.27	493.11	539.20	1624.00
18.6.121	121.00	269.00	496.40	507.02	541.00	1449.00
18.6.122	122.00	259.00	489.56	505.84	560.37	1341.00
18.6.123	123.00	89.00	150.00	489.73	520.98	1302.76
18.6.124	124.00	0.00	501.55	515.29	563.48	1278.85
18.6.125	125.00	257.00	497.50	518.84	573.02	1594.00
18.6.126	126.00	0.00	516.27	529.16	588.64	1227.97
18.6.127	127.00	0.00	501.91	517.45	569.74	1238.60

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	299.27	332.86	343.92	353.93	486.91

18.7.1	1.00	305.48	327.34	335.34	362.71	469.61
18.7.2	2.00	309.44	344.06	353.58	368.89	494.49
18.7.3	3.00	310.57	341.95	351.55	367.58	489.56
18.7.4	4.00	305.26	354.34	371.39	380.99	538.01
18.7.5	5.00	307.25	350.69	363.19	373.40	516.38
18.7.6	6.00	321.36	373.46	387.42	396.92	545.49
18.7.7	7.00	317.26	364.70	374.07	386.37	519.45
18.7.8	8.00	298.62	347.00	374.97	462.40	654.13
18.7.9	9.00	308.33	368.80	398.73	464.81	652.98
18.7.10	10.00	306.45	356.48	382.25	458.25	657.17
18.7.11	11.00	313.93	371.77	398.14	450.91	660.17
18.7.12	12.00	307.31	358.92	387.43	416.81	609.06
18.7.13	13.00	309.06	362.57	387.76	409.09	594.97
18.7.14	14.00	316.90	364.21	389.16	429.58	613.36
18.7.15	15.00	310.63	368.72	396.51	452.27	649.35
18.7.16	16.00	300.69	351.36	374.82	459.95	650.21
18.7.17	17.00	310.61	373.19	399.34	465.12	653.79
18.7.18	18.00	307.95	360.20	382.59	456.68	659.65
18.7.19	19.00	315.57	374.86	398.66	448.59	656.51
18.7.20	20.00	308.41	363.29	387.66	416.97	616.33
18.7.21	21.00	311.08	366.27	388.50	409.35	593.10
18.7.22	22.00	318.37	367.78	389.46	429.68	617.49
18.7.23	23.00	311.61	372.28	396.91	451.70	659.20
18.7.24	24.00	302.96	335.86	346.29	357.73	490.22

18.7.25	25.00	308.36	328.52	337.11	370.14	467.48
18.7.26	26.00	315.57	346.13	355.78	374.70	501.47
18.7.27	27.00	314.50	343.28	352.61	372.58	496.37
18.7.28	28.00	311.65	361.43	373.55	382.35	533.34
18.7.29	29.00	312.70	354.69	365.62	375.66	511.84
18.7.30	30.00	325.31	378.74	389.30	399.24	543.86
18.7.31	31.00	321.30	366.60	376.00	392.51	518.05
18.7.32	32.00	301.76	350.54	380.55	460.02	659.19
18.7.33	33.00	311.84	370.57	403.14	469.09	658.26
18.7.34	34.00	308.34	357.09	385.33	456.13	651.80
18.7.35	35.00	315.89	373.06	404.05	451.67	648.21
18.7.36	36.00	309.88	362.71	393.42	421.32	641.23
18.7.37	37.00	311.89	365.27	393.06	414.94	626.19
18.7.38	38.00	318.43	375.64	403.48	424.71	603.45
18.7.39	39.00	311.73	361.38	388.30	445.37	649.72
18.7.40	40.00	301.91	338.72	348.48	358.38	502.71
18.7.41	41.00	308.44	332.24	340.76	368.62	478.14
18.7.42	42.00	316.48	347.48	356.52	372.59	505.04
18.7.43	43.00	315.15	345.36	354.56	371.75	500.29
18.7.44	44.00	311.12	363.60	376.03	385.61	544.66
18.7.45	45.00	313.27	357.28	368.37	377.97	525.82
18.7.46	46.00	325.76	380.59	392.08	401.56	564.07
18.7.47	47.00	320.23	363.20	372.65	388.76	526.71
18.7.48	48.00	302.25	334.48	345.27	354.75	488.39

18.7.49	49.00	308.14	329.24	337.61	363.99	474.59
18.7.50	50.00	313.76	343.47	352.94	367.84	494.89
18.7.51	51.00	315.21	342.55	351.63	367.06	493.66
18.7.52	52.00	309.33	357.38	372.68	382.19	537.66
18.7.53	53.00	311.05	353.13	364.95	374.99	517.77
18.7.54	54.00	324.23	375.00	388.54	398.30	547.42
18.7.55	55.00	318.48	359.63	369.25	383.18	515.59
18.7.56	56.00	302.18	350.26	377.83	454.36	651.25
18.7.57	57.00	310.98	370.42	400.12	464.84	653.40
18.7.58	58.00	309.00	357.29	383.20	451.02	638.65
18.7.59	59.00	316.18	373.28	401.63	447.25	636.38
18.7.60	60.00	310.32	363.09	390.76	418.25	633.17
18.7.61	61.00	312.62	365.89	391.07	411.79	616.78
18.7.62	62.00	318.74	375.44	401.00	421.55	603.74
18.7.63	63.00	312.18	360.78	386.56	442.71	646.31
18.7.64	64.00	303.68	344.29	353.81	363.92	507.80
18.7.65	65.00	309.83	336.84	345.34	375.56	486.12
18.7.66	66.00	316.40	352.47	361.47	379.60	514.11
18.7.67	67.00	317.24	351.31	360.11	377.99	507.21
18.7.68	68.00	310.50	366.96	382.10	391.93	557.11
18.7.69	69.00	312.52	361.73	374.16	383.89	537.44
18.7.70	70.00	326.60	383.78	397.29	406.86	569.24
18.7.71	71.00	322.41	368.35	376.68	393.51	528.61
18.7.72	72.00	302.82	361.72	386.37	466.25	654.46

18.7.73	73.00	311.87	383.03	409.90	469.25	659.59
18.7.74	74.00	310.42	368.44	391.18	467.17	672.75
18.7.75	75.00	317.06	385.67	410.56	460.83	673.13
18.7.76	76.00	310.91	373.95	398.27	426.17	617.38
18.7.77	77.00	313.52	376.95	398.10	416.69	613.31
18.7.78	78.00	320.68	388.12	408.92	426.70	614.79
18.7.79	79.00	314.05	371.25	393.79	451.27	670.66
18.7.80	80.00	301.78	359.87	383.57	459.46	653.75
18.7.81	81.00	311.87	381.68	407.48	464.38	652.59
18.7.82	82.00	308.45	365.94	387.98	461.26	666.15
18.7.83	83.00	316.78	383.67	407.24	454.18	665.53
18.7.84	84.00	309.70	372.64	395.56	421.55	609.85
18.7.85	85.00	312.90	374.95	395.29	412.72	599.22
18.7.86	86.00	318.53	385.77	405.70	422.49	611.86
18.7.87	87.00	312.11	368.71	390.33	446.29	643.88
18.7.88	88.00	303.57	342.41	352.23	363.95	494.83
18.7.89	89.00	308.91	334.08	343.10	378.08	474.37
18.7.90	90.00	317.87	350.13	359.47	380.66	496.93
18.7.91	91.00	316.58	348.20	357.71	378.97	492.24
18.7.92	92.00	311.72	368.82	379.91	389.09	555.37
18.7.93	93.00	313.40	361.47	372.20	382.02	538.67
18.7.94	94.00	326.00	384.92	395.14	404.42	568.26
18.7.95	95.00	322.22	365.25	374.26	396.16	525.48
18.7.96	96.00	302.77	352.79	381.27	455.07	657.05

18.7.97	97.00	313.30	373.79	405.13	466.75	656.88
18.7.98	98.00	309.61	361.45	388.12	449.72	639.22
18.7.99	99.00	318.42	375.25	403.03	442.50	604.26
18.7.100	100.00	310.29	363.88	391.90	413.30	609.58
18.7.101	101.00	312.99	367.38	391.68	406.60	593.43
18.7.102	102.00	319.25	368.31	395.40	434.40	623.76
18.7.103	103.00	313.96	373.00	403.04	454.17	659.73
18.7.104	104.00	304.16	336.36	345.34	358.48	483.48
18.7.105	105.00	308.79	329.61	338.93	372.02	462.76
18.7.106	106.00	316.37	346.68	355.20	376.48	494.68
18.7.107	107.00	316.05	343.92	352.41	374.99	488.22
18.7.108	108.00	311.45	358.79	372.59	382.44	542.49
18.7.109	109.00	312.96	353.68	363.90	375.46	522.22
18.7.110	110.00	325.77	378.91	390.77	399.90	560.60
18.7.111	111.00	322.03	363.18	371.93	394.39	521.55
18.7.112	112.00	302.77	332.62	342.42	354.85	479.21
18.7.113	113.00	308.97	326.99	335.09	362.40	458.46
18.7.114	114.00	314.11	343.10	351.68	369.04	491.06
18.7.115	115.00	315.08	340.92	349.47	366.26	483.32
18.7.116	116.00	308.90	354.23	369.65	379.89	535.63
18.7.117	117.00	310.45	349.43	360.88	371.93	516.38
18.7.118	118.00	324.02	374.47	388.04	397.00	555.55
18.7.119	119.00	320.34	359.91	368.78	386.17	520.08
18.7.120	120.00	302.88	347.89	380.90	451.76	655.22

18.7.121	121.00	312.47	365.85	403.61	464.27	655.37
18.7.122	122.00	310.16	355.89	386.66	446.04	639.09
18.7.123	123.00	316.69	368.20	402.13	438.99	596.41
18.7.124	124.00	309.67	357.61	390.17	410.35	605.51
18.7.125	125.00	312.45	361.23	389.98	404.37	591.38
18.7.126	126.00	319.30	363.10	394.41	431.80	619.27
18.7.127	127.00	313.87	365.57	401.95	453.20	657.13

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00
18.8.2	2.00	0.00	0.00	0.00	0.00	0.00
18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00
18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00
18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00

18.8.15	15.00	0.00	0.00	0.00	0.00	0.00
18.8.16	16.00	0.00	0.00	0.00	0.00	0.00
18.8.17	17.00	0.00	0.00	0.00	0.00	0.00
18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00
18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00
18.8.26	26.00	0.00	0.00	0.00	0.00	0.00
18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00
18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00
18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00

18.8.39	39.00	0.00	0.00	0.00	0.00	0.00
18.8.40	40.00	0.00	0.00	0.00	0.00	0.00
18.8.41	41.00	0.00	0.00	0.00	0.00	0.00
18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00
18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00
18.8.50	50.00	0.00	0.00	0.00	0.00	0.00
18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00
18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00
18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00

18.8.63	63.00	0.00	0.00	0.00	0.00	0.00
18.8.64	64.00	0.00	0.00	0.00	0.00	0.00
18.8.65	65.00	0.00	0.00	0.00	0.00	0.00
18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00
18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00
18.8.74	74.00	0.00	0.00	0.00	0.00	0.00
18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00
18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00
18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00

18.8.87	87.00	0.00	0.00	0.00	0.00	0.00
18.8.88	88.00	0.00	0.00	0.00	0.00	0.00
18.8.89	89.00	0.00	0.00	0.00	0.00	0.00
18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00
18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00
18.8.98	98.00	0.00	0.00	0.00	0.00	0.00
18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00
18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00
18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00

18.8.111	111.00	0.00	0.00	0.00	0.00	0.00
18.8.112	112.00	0.00	0.00	0.00	0.00	0.00
18.8.113	113.00	0.00	0.00	0.00	0.00	0.00
18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00
18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00
18.8.122	122.00	0.00	0.00	0.00	0.00	0.00
18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00
18.8.127	127.00	0.00	0.00	0.00	0.00	0.00

18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.12.0	0.00	0.00	0.00	130.00	130.00	130.00
18.12.1	1.00	0.00	0.00	130.00	130.00	130.00
18.12.2	2.00	0.00	0.00	130.00	130.00	130.00
18.12.3	3.00	0.00	0.00	130.00	130.00	130.00
18.12.4	4.00	0.00	0.00	130.00	130.00	130.00

18.12.5	5.00	0.00	0.00	130.00	130.00	130.00
18.12.6	6.00	0.00	0.00	130.00	130.00	130.00
18.12.7	7.00	0.00	0.00	130.00	130.00	130.00
18.12.8	8.00	0.00	0.00	128.00	128.00	128.00
18.12.9	9.00	0.00	0.00	128.00	128.00	128.00
18.12.10	10.00	0.00	0.00	128.00	128.00	128.00
18.12.11	11.00	0.00	0.00	128.00	128.00	128.00
18.12.12	12.00	0.00	0.00	128.00	128.00	128.00
18.12.13	13.00	0.00	0.00	128.00	128.00	128.00
18.12.14	14.00	0.00	0.00	128.00	128.00	128.00
18.12.15	15.00	0.00	0.00	128.00	128.00	128.00
18.12.16	16.00	0.00	0.00	128.00	128.00	128.00
18.12.17	17.00	0.00	0.00	128.00	128.00	128.00
18.12.18	18.00	0.00	0.00	128.00	128.00	128.00
18.12.19	19.00	0.00	0.00	128.00	128.00	128.00
18.12.20	20.00	0.00	0.00	128.00	128.00	128.00
18.12.21	21.00	0.00	0.00	128.00	128.00	128.00
18.12.22	22.00	0.00	0.00	128.00	128.00	128.00
18.12.23	23.00	0.00	0.00	128.00	128.00	128.00
18.12.24	24.00	0.00	0.00	130.00	130.00	130.00
18.12.25	25.00	0.00	0.00	130.00	130.00	130.00
18.12.26	26.00	0.00	0.00	130.00	130.00	130.00
18.12.27	27.00	0.00	0.00	130.00	130.00	130.00
18.12.28	28.00	0.00	0.00	130.00	130.00	130.00

18.12.29	29.00	0.00	0.00	130.00	130.00	130.00
18.12.30	30.00	0.00	0.00	130.00	130.00	130.00
18.12.31	31.00	0.00	0.00	130.00	130.00	130.00
18.12.32	32.00	0.00	0.00	130.00	130.00	130.00
18.12.33	33.00	0.00	0.00	130.00	130.00	130.00
18.12.34	34.00	0.00	0.00	130.00	130.00	130.00
18.12.35	35.00	0.00	0.00	130.00	130.00	130.00
18.12.36	36.00	0.00	0.00	130.00	130.00	130.00
18.12.37	37.00	0.00	0.00	130.00	130.00	130.00
18.12.38	38.00	0.00	0.00	130.00	130.00	130.00
18.12.39	39.00	0.00	0.00	130.00	130.00	130.00
18.12.40	40.00	0.00	0.00	128.00	128.00	128.00
18.12.41	41.00	0.00	0.00	128.00	128.00	128.00
18.12.42	42.00	0.00	0.00	128.00	128.00	128.00
18.12.43	43.00	0.00	0.00	128.00	128.00	128.00
18.12.44	44.00	0.00	0.00	128.00	128.00	128.00
18.12.45	45.00	0.00	0.00	128.00	128.00	128.00
18.12.46	46.00	0.00	0.00	128.00	128.00	128.00
18.12.47	47.00	0.00	0.00	128.00	128.00	128.00
18.12.48	48.00	0.00	0.00	128.00	128.00	128.00
18.12.49	49.00	0.00	0.00	128.00	128.00	128.00
18.12.50	50.00	0.00	0.00	128.00	128.00	128.00
18.12.51	51.00	0.00	0.00	128.00	128.00	128.00
18.12.52	52.00	0.00	0.00	128.00	128.00	128.00

18.12.53	53.00	0.00	0.00	128.00	128.00	128.00
18.12.54	54.00	0.00	0.00	128.00	128.00	128.00
18.12.55	55.00	0.00	0.00	128.00	128.00	128.00
18.12.56	56.00	0.00	0.00	130.00	130.00	130.00
18.12.57	57.00	0.00	0.00	130.00	130.00	130.00
18.12.58	58.00	0.00	0.00	130.00	130.00	130.00
18.12.59	59.00	0.00	0.00	130.00	130.00	130.00
18.12.60	60.00	0.00	0.00	130.00	130.00	130.00
18.12.61	61.00	0.00	0.00	130.00	130.00	130.00
18.12.62	62.00	0.00	0.00	130.00	130.00	130.00
18.12.63	63.00	0.00	0.00	130.00	130.00	130.00
18.12.64	64.00	0.00	0.00	128.00	128.00	128.00
18.12.65	65.00	0.00	0.00	128.00	128.00	128.00
18.12.66	66.00	0.00	0.00	128.00	128.00	128.00
18.12.67	67.00	0.00	0.00	128.00	128.00	128.00
18.12.68	68.00	0.00	0.00	128.00	128.00	128.00
18.12.69	69.00	0.00	0.00	128.00	128.00	128.00
18.12.70	70.00	0.00	0.00	128.00	128.00	128.00
18.12.71	71.00	0.00	0.00	128.00	128.00	128.00
18.12.72	72.00	0.00	0.00	130.00	130.00	130.00
18.12.73	73.00	0.00	0.00	130.00	130.00	130.00
18.12.74	74.00	0.00	0.00	130.00	130.00	130.00
18.12.75	75.00	0.00	0.00	130.00	130.00	130.00
18.12.76	76.00	0.00	0.00	130.00	130.00	130.00

18.12.77	77.00	0.00	0.00	130.00	130.00	130.00
18.12.78	78.00	0.00	0.00	130.00	130.00	130.00
18.12.79	79.00	0.00	0.00	130.00	130.00	130.00
18.12.80	80.00	0.00	0.00	130.00	130.00	130.00
18.12.81	81.00	0.00	0.00	130.00	130.00	130.00
18.12.82	82.00	0.00	0.00	130.00	130.00	130.00
18.12.83	83.00	0.00	0.00	130.00	130.00	130.00
18.12.84	84.00	0.00	0.00	130.00	130.00	130.00
18.12.85	85.00	0.00	0.00	130.00	130.00	130.00
18.12.86	86.00	0.00	0.00	130.00	130.00	130.00
18.12.87	87.00	0.00	0.00	130.00	130.00	130.00
18.12.88	88.00	0.00	0.00	128.00	128.00	128.00
18.12.89	89.00	0.00	0.00	128.00	128.00	128.00
18.12.90	90.00	0.00	0.00	128.00	128.00	128.00
18.12.91	91.00	0.00	0.00	128.00	128.00	128.00
18.12.92	92.00	0.00	0.00	128.00	128.00	128.00
18.12.93	93.00	0.00	0.00	128.00	128.00	128.00
18.12.94	94.00	0.00	0.00	128.00	128.00	128.00
18.12.95	95.00	0.00	0.00	128.00	128.00	128.00
18.12.96	96.00	0.00	0.00	128.00	128.00	128.00
18.12.97	97.00	0.00	0.00	128.00	128.00	128.00
18.12.98	98.00	0.00	0.00	128.00	128.00	128.00
18.12.99	99.00	0.00	0.00	128.00	128.00	128.00
18.12.100	100.00	0.00	0.00	128.00	128.00	128.00

18.12.101	101.00	0.00	0.00	128.00	128.00	128.00
18.12.102	102.00	0.00	0.00	128.00	128.00	128.00
18.12.103	103.00	0.00	0.00	128.00	128.00	128.00
18.12.104	104.00	0.00	0.00	130.00	130.00	130.00
18.12.105	105.00	0.00	0.00	130.00	130.00	130.00
18.12.106	106.00	0.00	0.00	130.00	130.00	130.00
18.12.107	107.00	0.00	0.00	130.00	130.00	130.00
18.12.108	108.00	0.00	0.00	130.00	130.00	130.00
18.12.109	109.00	0.00	0.00	130.00	130.00	130.00
18.12.110	110.00	0.00	0.00	130.00	130.00	130.00
18.12.111	111.00	0.00	0.00	130.00	130.00	130.00
18.12.112	112.00	0.00	0.00	130.00	130.00	130.00
18.12.113	113.00	0.00	0.00	130.00	130.00	130.00
18.12.114	114.00	0.00	0.00	130.00	130.00	130.00
18.12.115	115.00	0.00	0.00	130.00	130.00	130.00
18.12.116	116.00	0.00	0.00	130.00	130.00	130.00
18.12.117	117.00	0.00	0.00	130.00	130.00	130.00
18.12.118	118.00	0.00	0.00	130.00	130.00	130.00
18.12.119	119.00	0.00	0.00	130.00	130.00	130.00
18.12.120	120.00	0.00	0.00	128.00	128.00	128.00
18.12.121	121.00	0.00	0.00	128.00	128.00	128.00
18.12.122	122.00	0.00	0.00	128.00	128.00	128.00
18.12.123	123.00	0.00	0.00	128.00	128.00	128.00
18.12.124	124.00	0.00	0.00	128.00	128.00	128.00

Large Input

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 |' _ / _ \ / _ | ' \ | ' _ / _ \ | | _ _ _ / _ / _ \ | ' _ \ _ \ | ' \ | | | _ / _
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 | _ \ _ / \ _ | _ / | _ \ _ / | _ \ _ \ _ / | _ | _ | _ _ /
 \ _ , _ | \ _ \ _ |
 | _ |
 | _ |

```
INFO [analysis] deriving rocprofiler-compute metrics...
```

```
WARNING PC sampling: can not detect pc sampling method without /workspa
ce/user/workloads/nonfusion_matmulbias_32768x32768/MI300X_A1/ps_file_pc_sampling
_host_trap.csv
```

0.1 Top Kernels

162

42920.61	42820.42	0.53		
kernel<4, at::native::FillFunctor<int>, std::array<char*, 1ul> >(int, at::nat...				
3	void at::native::(anonymous namespace)::	3.00	3341656.33	
1113885.44	1563214.92	0.07		
distribution_elementwise_grid_stride_ker nel<float, 4, at::native::templates::...				
4	__amd_rocclr_copyBuffer	89.00	266208.18	
2991.10	2963.08	0.01		
5	void at::native::(anonymous namespace)::	7.00	16630.83	
2375.83	2278.42	0.00		
CatArrayBatchedCopy_contig<at::native::(anonymous namespace)::OpaqueType<2u>,...				
6	void at::native::vectorized_elementwise_	3.00	10483.75	
3494.58	3497.83	0.00		
kernel<4, at::native::compare_scalar_ker nel<double>(at::TensorIteratorBase&, ...				
7	void at::native::vectorized_elementwise_	2.00	6137.17	
3068.58	3068.58	0.00		
kernel<8, at::native::UnaryFunctor<c10: :Half, c10::Half, bool, at::native::(...				
8	void at::native::vectorized_elementwise_	2.00	5308.83	
2654.42	2654.42	0.00		
kernel<8, at::native::BinaryFunctor<c10: :Half, c10::Half, bool, at::native::(...				

```

9 void at::native::vectorized_elementwise_      2.00      5121.75
2560.88      2560.88      0.00
kernel<8, at::native::AbsFunctor<c10::Ha

lf>, std::array<char*, 2ul> >(int, at...
```

0.2 Dispatch List

Dispatch_ID	Kernel_Name	GPU_ID
0	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
1	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
2	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0

```

3          3  matmul_kernel
0

4          4  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)

5          5  matmul_kernel
0

6          6  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)

7          7  matmul_kernel
0

8          8  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)

9          9  matmul_kernel
0

```

1. System Info

Info

workload_name	nonfusion_matmulbias_32768x32768
command nonfusion	python triton/matmulbias.py --M 32768 --N 32768
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Mon Mar 16 20:22:53 2026 (UTC)
version	3
hostname	b031b949a562
cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1
linux_distro	CentOS Stream 10 (Coughlan)
linux_kernel_version	5.15.0-170-generic
amd_gpu_kernel_version	nan
cpu_memory	230893520

gpu_memory	nan
rocm_version	7.0.3
vbios	113-M3000108-103
compute_partition	SPX
memory_partition	nan
gpu_series	MI300
gpu_model	MI300X_A1
gpu_arch	gfx942
gpu_chip_id	29877
gpu_l1	32
gpu_l2	4096
cu_per_gpu	304

<code>simd_per_cu</code>	4
<code>se_per_gpu</code>	32
<code>wave_size</code>	64
<code>workgroup_max_size</code>	1024
<code>max_waves_per_cu</code>	32
<code>max_sclk</code>	2100
<code>max_mclk</code>	1300
<code>cur_sclk</code>	2100
<code>cur_mclk</code>	1300
<code>total_l2_chan</code>	128
<code>lds_banks_per_cu</code>	32
<code>sqc_per_gpu</code>	160

pipes_per_gpu	4
num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
Peak	Pct of Peak		
2.1.0	VALU FLOPs	586.47	Gflop/s
81715.20	0.72		
2.1.1	VALU IOPs	2294.85	Giop/s
81715.20	2.81		
2.1.2	MFMA FLOPs (F8)	0.00	Gflop/s
2614886.40	0.00		
2.1.3	MFMA FLOPs (BF16)	0.00	Gflop/s
1307443.20	0.00		
2.1.4	MFMA FLOPs (F16)	6300.95	Gflop/s
1307443.20	0.48		
2.1.5	MFMA FLOPs (F32)	0.00	Gflop/s
163430.40	0.00		

2.1.6 163430.40	MFMA FLOPs (F64) 0.00	0.00	Gflop/s
2.1.7	MFMA FLOPs (F6F4)		Gflop/s
2.1.8 2614886.40	MFMA IOPs (Int8) 0.00	0.00	Giop/s
2.1.9 304.00	Active CUs 100.00	304.00	Cus
2.1.10 100.00	SALU Utilization 4.80	4.80	Pct
2.1.11 100.00	VALU Utilization 12.77	12.77	Pct
2.1.12 100.00	MFMA Utilization 0.86	0.86	Pct
2.1.13 100.00	VMEM Utilization 0.79	0.79	Pct
2.1.14 100.00	Branch Utilization 0.41	0.41	Pct
2.1.15 64.00	VALU Active Threads 99.25	63.52	Threads
2.1.16 5.00	IPC 3.98	0.20	Instr/cycle
2.1.17 9728.00	Wavefront Occupancy 42.24	4109.59	Wavefronts

2.1.18 81715.20	Theoretical LDS Bandwidth 0.50	407.26	Gb/s
2.1.19 32.00	LDS Bank Conflicts/Access 2.87	0.92	Conflicts/access
2.1.20 100.00	vL1D Cache Hit Rate 29.61	29.61	Pct
2.1.21 81715.20	vL1D Cache BW 9.79	7997.58	Gb/s
2.1.22 100.00	L2 Cache Hit Rate 33.66	33.66	Pct
2.1.23 34406.40	L2 Cache BW 16.33	5618.88	Gb/s
2.1.24 5324.80	L2-Fabric Read BW 35.65	1898.19	Gb/s
2.1.25 5324.80	L2-Fabric Write BW 34.41	1832.51	Gb/s
2.1.26	L2-Fabric Read Latency	2378.71	Cycles
2.1.27	L2-Fabric Write Latency	461.10	Cycles
2.1.28 100.00	sL1D Cache Hit Rate		Pct
2.1.29 21504.00	sL1D Cache BW 0.00	0.00	Gb/s

2.1.30	L1I Hit Rate	100.00	Pct
100.00	100.00		
2.1.31	L1I BW	1041.16	Gb/s
21504.00	4.84		
2.1.32	L1I Fetch Latency		Cycles

3. Memory Chart

3.1 Memory Chart

(Normalization: per_kernel)

Instr Buff	Instr Dispatch	Exec
LDS	L2 Cache	

SALU: 139722921

		-----> Active
CUs	Req : 11602341Util : 1 %	
	xGMI/PCIE	

Wave	0 Instr Buf	30
4: 304	<-----	

SMEM: 0

```

Lat      :
N/A      Hit      :      34
%
-----
>
^
|
RVGPRseq:  58.0

|  |
Wave N-1 Instr Buf      VALU:
457495231      Vector L1 Cache
Request      |
|
-----
>
|
v
SGPRs :      32

MFMA:
94212186      Rd      : 16555350Hit      :      30 %      Rd
: 205022215 Rd      : 205024415

-----
>      <-----      <-----
-
Fabric

Wt      : 13716185Lat      :      N/A      Wr      : 35762530 Wr      :
35762530      Rd      :
115125125

VMEM: 28035371      LDS Alloc:      -
----->
>      <----- Latency (cycles)      Rd:
115103313

----->
0      Atomic:      0      Coales:      25 %      Atomic:      0      Atomic:
0      Wr      : 33554467      <-----      HBM
<
----->      <-----
>      ----->      Rd      :      2379      Wr:
33554466

LDS:
116023412      Stall :      40
cycles      Atomic:      0      Wr      :
461      ----->

```

```

-----> Scratch Alloc:

-----> Atomic:      0
                                0

Wave Occupancy      GWS: 0
Scalar L1D
Cache      Latency (cycles)

14      -----
--->      Rd      :
0

Wave Life
      <-----
      ^

|
78790
BRANCH: 34031365 Wavefronts: Rd      :      0 Hit      :
N/A      Wr      :      0 Rd      :
N/A      |
|
      ----->      0 <
-----
>      |
|

Lat      :      N/A Atomic:      0 Wr      :
N/A      |
v
      ^
      <-----

>

|      Workgroups:

|
0      Instr L1 Cache
      GMI
|
|
|
|      Fetch :

```

151667893 Hit : 100 % Req
: 8623

-----<-----

Lat : 0.0 cycles

4. Roofline
4.1 Roofline

286672.8

140386GFLOP/s

131315 GFLOP/s
65860.3

15130.8

3476.1

798.6

59671 GB/s

33616 GB/s

183.5 25701 GB/s

42.2 4215 GB/s

0.01

0.18

3.16

56.23

1000.00

Performance (GFLOP/sec) Arithmetic Intensity (FLOP)s/Byte)

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct
5.1.1	CPF Stall	0.03	0.00	0.11	Pct
5.1.2	CPF-L2 Utilization	0.45	0.00	0.94	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	0.01	0.00	0.10	Pct

5.2 Packet Processor

Metric_ID Unit	Metric	Avg	Min	Max
5.2.0 Pct	CPC SYNC FIFO Full Rate			
5.2.1 Pct	CPC CANE Stall Rate			
5.2.2 Pct	CPC ADC Utilization			
5.2.3 Pct	CPC Utilization	100.00	100.00	100.00
5.2.4 Pct	CPC Stall Rate	0.84	0.09	1.38
5.2.5 Pct	CPC Packet Decoding Utilization	3.18	0.44	5.29
5.2.6 Pct	CPC-Workgroup Manager Utilization	96.78	78.24	119.65
5.2.7 Pct	CPC-L2 Utilization	0.01	0.00	0.02
5.2.8 Pct	CPC-UTCL1 Stall	0.01	0.00	0.07
5.2.9 Pct	CPC-UTCL2 Utilization	0.01	0.00	0.06

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID	Metric	Avg	Min	Max	Unit
6.1.0	Schedule-Pipe Wave Occupancy				Wave
6.1.1	Accelerator Utilization	100.03	81.82	121.94	Pct
6.1.2	Scheduler-Pipe Utilization	0.00	0.00	0.00	Pct
6.1.3	Scheduler-Pipe Wave Utilization				Pct
6.1.4	Workgroup Manager Utilization	98.08	95.76	100.00	Pct
6.1.5	Shader Engine Utilization	98.56	77.22	117.40	Pct
6.1.6	SIMD Utilization	98.46	77.69	117.06	Pct
6.1.7 Workgroups	Dispatched Workgroups	0.00	0.00	0.00	
6.1.8 Wavefronts	Dispatched Wavefronts	0.00	0.00	0.00	
6.1.9 Cycles/wave	VGPR Writes				

6.1.10 SGPR Writes
Cycles/wave

6.2 Workgroup Manager - Resource Allocation

Metric_ID Unit	Metric	Avg	Min	Max
6.2.0 Pct	Not-scheduled Rate (Workgroup Manager)	0.00	0.00	0.00
6.2.1 Pct	Not-scheduled Rate (Scheduler-Pipe)	0.00	0.00	0.00
6.2.2 Pct	Scheduler-Pipe FIFO Full Rate			
6.2.3 Pct	Scheduler-Pipe Stall Rate	0.00	0.00	0.00
6.2.4 Pct	Scratch Stall Rate	0.00	0.00	0.00
6.2.5 Pct	Insufficient SIMD Waveslots	0.00	0.00	0.00
6.2.6 Pct	Insufficient SIMD VGPRs	0.00	0.00	0.00
6.2.7 Pct	Insufficient SIMD SGPRs	0.00	0.00	0.00
6.2.8 Pct	Insufficient CU LDS	0.00	0.00	0.00
6.2.9	Insufficient CU Barriers	0.00	0.00	0.00

Pct

6.2.10	Reached CU Workgroup Limit	0.00	0.00	0.00
--------	----------------------------	------	------	------

Pct

6.2.11	Reached CU Wavefront Limit	0.00	0.00	0.00
--------	----------------------------	------	------	------

Pct

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID Unit	Metric	Avg	Min	Max
7.1.0 Work items	Grid Size	18974490.10	8388608.00	33554432.00
7.1.1 Work items	Workgroup Size	367.65	256.00	512.00
7.1.2 Wavefronts	Total Wavefronts	0.00	0.00	0.00
7.1.3 Wavefronts	Saved Wavefronts	0.00	0.00	0.00
7.1.4 Wavefronts	Restored Wavefronts	0.00	0.00	0.00
7.1.5 Registers	VGPRs	58.18	12.00	96.00
7.1.6 Registers	AGPRs	4.57	0.00	192.00

7.1.7	SGPRs	32.35	32.00	48.00
Registers				

7.1.8	LDS Allocation	0.00	0.00	0.00
Bytes				

7.1.9	Scratch Allocation	0.00	0.00	0.00
Bytes/workitem				

7.2 Wavefront Runtime Stats

Metric_ID	Metric	Avg	Min
Max Unit			

7.2.0	Kernel Time (Nanosec)	6506612.29	1087659.08
258000510.50	Ns		

7.2.1	Kernel Time (Cycles)	8231188.95	1641651.12
389934410.00	Cycle		

7.2.2	Instructions per wavefront	2431.60	262.00
109366.00	Instr/wavefront		

7.2.3	Wave Cycles	22767438703.05	7546294336.00
620512045572.00	Cycles per kernel		

7.2.4	Dependency Wait Cycles	9986085830.21	2478911260.00
192927753384.00	Cycles per kernel		

7.2.5	Issue Wait Cycles	9814747146.61	2317802012.00
283385006508.00	Cycles per kernel		

7.2.6	Active Cycles	3541859475.67	341311984.00
161027429048.00	Cycles per kernel		

7.2.7	Wavefront Occupancy	4109.59	266.82
9999.04	Wavefronts		

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.1.0 per kernel	VALU	363283045.08	63569920.00	15155593216.00	Instr
10.1.1 per kernel	VMEM	28035370.61	4325376.00	1077936128.00	Instr
10.1.2 per kernel	LDS	116023412.27	0.00	6979321856.00	Instr
10.1.3 per kernel	MFMA	94212185.85	0.00	4294967296.00	Instr
10.1.4 per kernel	SALU	139722921.13	13238272.00	6218579968.00	Instr
10.1.5 per kernel	SMEM	0.00	0.00	0.00	Instr
10.1.6 per kernel	Branch	34031364.95	1048576.00	1611661312.00	Instr

10.2 VALU Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
-----------	--------	-----	-----	-----	------

10.2.0 per kernel	INT32	153907279.28	30146560.00	7536115712.00	Instr
10.2.1 per kernel	INT64	31845929.62	8388608.00	1078460416.00	Instr
10.2.2 per kernel	F16-ADD	8204599.82	0.00	8388608.00	Instr
10.2.3 per kernel	F16-MUL	0.00	0.00	0.00	Instr
10.2.4 per kernel	F16-FMA	0.00	0.00	0.00	Instr
10.2.5 per kernel	F16-Trans	0.00	0.00	0.00	Instr
10.2.6 per kernel	F32-ADD	0.00	0.00	0.00	Instr
10.2.7 per kernel	F32-MUL	1185905.63	524288.00	2097152.00	Instr
10.2.8 per kernel	F32-FMA	0.00	0.00	0.00	Instr
10.2.9 per kernel	F32-Trans	1185905.63	524288.00	2097152.00	Instr
10.2.10 per kernel	F64-ADD	0.00	0.00	0.00	Instr
10.2.11	F64-MUL	0.00	0.00	0.00	Instr

per kernel

10.2.12	F64-FMA	0.00	0.00	0.00	Instr
---------	---------	------	------	------	-------

per kernel

10.2.13	F64-Trans	0.00	0.00	0.00	Instr
---------	-----------	------	------	------	-------

per kernel

10.2.14	Conversion	2739827.61	1048576.00	20971520.00	Instr
---------	------------	------------	------------	-------------	-------

per kernel

10.3 VMEM Instr Mix

Metric_ID	Metric	Avg	Min
Max	Unit		

10.3.0	Global/Generic Instr	27747350.46	4194304.00
1077936128.00	Instr per kernel		

10.3.1	Global/Generic Read	25604196.42	2097152.00
1073741824.00	Instr per kernel		

10.3.2	Global/Generic Write	2143154.04	2097152.00
4194304.00	Instr per kernel		

10.3.3	Global/Generic Atomic	0.00	0.00
0.00	Instr per kernel		

10.3.4	Spill/Stack Instr	288020.15	0.00
524288.00	Instr per kernel		

10.3.5	Spill/Stack Coalesceable Instr		
	Instr per kernel		

10.3.6	Spill/Stack Read	288020.15	0.00
524288.00	Instr per kernel		

10.3.7	Spill/Stack Write	0.00	0.00
0.00	Instr per kernel		

10.3.8	Spill/Stack Atomic	0.00	0.00
0.00	Instr per kernel		

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.4.0 kernel	MFMA-I8	0.00	0.00	0.00	Instr per
10.4.1 kernel	MFMA-F8	0.00	0.00	0.00	Instr per
10.4.2 kernel	MFMA-F16	94212185.85	0.00	4294967296.00	Instr per
10.4.3 kernel	MFMA-BF16	0.00	0.00	0.00	Instr per
10.4.4 kernel	MFMA-F32	0.00	0.00	0.00	Instr per
10.4.5 kernel	MFMA-F64	0.00	0.00	0.00	Instr per
10.4.6 kernel	MFMA-F6F4				Instr per

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID Peak	Metric	Avg	Unit	Peak	Pct of
11.1.0 0.72	VALU FLOPs	586.47	Gflop	81715.20	
11.1.1 2.81	VALU IOPs	2294.85	Giop	81715.20	
11.1.2 0.00	MFMA FLOPs (F8)	0.00	Gflop	2614886.40	
11.1.3 0.00	MFMA FLOPs (BF16)	0.00	Gflop	1307443.20	
11.1.4 0.48	MFMA FLOPs (F16)	6300.95	Gflop	1307443.20	
11.1.5 0.00	MFMA FLOPs (F32)	0.00	Gflop	163430.40	
11.1.6 0.00	MFMA FLOPs (F64)	0.00	Gflop	163430.40	
11.1.7	MFMA FLOPs (F6F4)		Gflop		
11.1.8 0.00	MFMA IOPs (INT8)	0.00	Giop	2614886.40	

11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
-----------	--------	-----	-----	-----	------

11.2.0	IPC	0.20	0.15	0.48	Instr/cycle
11.2.1	IPC (Issued)	0.97	0.85	0.98	Instr/cycle
11.2.2	SALU Utilization	4.80	2.15	9.03	Pct
11.2.3	VALU Utilization	12.77	9.75	24.51	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	0.79	0.62	1.36	Pct
11.2.6	Branch Utilization	0.41	0.21	2.71	Pct
11.2.7	VALU Active Threads	63.52	63.23	64.00	Threads
11.2.8	MFMA Utilization	0.86	0.00	43.28	Pct
11.2.9	MFMA Instr Cycles	32.00	32.00	32.00	Cycles/instr
11.2.10	VMEM Latency				Cycles
11.2.11	SMEM Latency				Cycles

11.3 Arithmetic Operations

Metric_ID	Metric	Avg	Min
Max	Unit		
11.3.0	FLOPs (Total)	1544249343238.94	603979776.00
70369012613120.00	Ops per kernel		
11.3.1	IOPs (Total)	11888205369.48	2466250752.00
551332872192.00	Ops per kernel		
11.3.2	F8 OPs	0.00	0.00
0.00	Ops per kernel		
11.3.3	F16 OPs	1544097547318.17	536870912.00
70368744177664.00	Ops per kernel		
11.3.4	BF16 OPs	0.00	0.00
0.00	Ops per kernel		
11.3.5	F32 OPs	151795920.76	67108864.00
268435456.00	Ops per kernel		
11.3.6	F64 OPs	0.00	0.00
0.00	Ops per kernel		
11.3.7	F6F4 OPs		
	Ops per kernel		
11.3.8	INT8 OPs	0.00	0.00
0.00	Ops per kernel		

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	0.97	Pct of peak
12.1.1	Access Rate	0.62	Pct of peak
12.1.2	Theoretical Bandwidth (% of Peak)	0.50	Pct of peak
12.1.3	Bank Conflict Rate	2.87	Pct of peak

12.2 LDS Stats

Metric_ID	Metric	Avg	Min
Max	Unit		
12.2.0	LDS Instrs	116023412.27	0.00
6979321856.00	Instr per kernel		

12.2.1 LDS LOAD
Instr per kernel

12.2.2 LDS STORE
Instr per kernel

12.2.3 LDS ATOMIC
Instr per kernel

12.2.4 LDS LOAD Bandwidth
Gbps

12.2.5 LDS STORE Bandwidth
Gbps

12.2.6 LDS ATOMIC Bandwidth
Gbps

12.2.7	Theoretical Bandwidth	83779961402.80	0.00
4398046511104.00	Bytes per kernel		
12.2.8	LDS Latency		
	Cycles		
12.2.9	Bank Conflicts/Access	0.92	0.00
1.00	Conflicts/access		
12.2.10	Index Accesses	655118489.27	0.00
34359738368.00	Cycles per kernel		
12.2.11	Atomic Return Cycles	0.00	0.00
0.00	Cycles per kernel		
12.2.12	Bank Conflict	587540.81	0.00
2359296.00	Cycles per kernel		
12.2.13	Addr Conflict	0.00	0.00
0.00	Cycles per kernel		
12.2.14	Unaligned Stall	0.00	0.00
0.00	Cycles per kernel		
12.2.15	Mem Violations	0.00	0.00
0.00	Accesses per kernel		
12.2.16	LDS Command FIFO Full Rate		
	Cycles per kernel		
12.2.17	LDS Data FIFO Full Rate		
	Cycles per kernel		

13. Instruction Cache

13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	4.84	Pct of peak
13.1.1	Cache Hit Rate	99.98	Pct of peak
13.1.2	L1I-L2 Bandwidth	0.00	Pct of peak

13.2 Instruction Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
13.2.0	Req	151667893.02	15597568.00	7006584832.00	Req per kernel
13.2.1	Hits	151664100.27	15592068.00	7006581851.00	Hits per kernel
13.2.2	Misses - Non Duplicated	283.78	160.00	1368.00	Misses per kernel
13.2.3	Misses - Duplicated	3641.74	2247.00	8314.00	Misses per kernel
13.2.4	Cache Hit Rate	99.98	99.96	100.00	Pct
13.2.5	Instruction Fetch Latency				Cycles

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
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13.3.0	L1I-L2 Bandwidth	551902.97	327680.00	3276800.00	Bytes
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per kernel

14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.00	Pct of peak
14.1.1	Cache Hit Rate		Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
14.2.0	Req	0.00	0.00	0.00	Req per kernel
14.2.1	Hits	0.00	0.00	0.00	Req per kernel
14.2.2	Misses - Non Duplicated	0.00	0.00	0.00	Req per kernel
14.2.3	Misses- Duplicated	0.00	0.00	0.00	Req per kernel
14.2.4	Cache Hit Rate				Pct
14.2.5	Read Req (Total)	0.00	0.00	0.00	Req per kernel

14.2.6	Atomic Req	0.00	0.00	0.00	Req per kernel
14.2.7	Read Req (1 DWord)	0.00	0.00	0.00	Req per kernel
14.2.8	Read Req (2 DWord)	0.00	0.00	0.00	Req per kernel
14.2.9	Read Req (4 DWord)	0.00	0.00	0.00	Req per kernel
14.2.10	Read Req (8 DWord)	0.00	0.00	0.00	Req per kernel
14.2.11	Read Req (16 DWord)	0.00	0.00	0.00	Req per kernel

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	0.00	0.00	0.00	Bytes per kernel
14.3.1	Read Req	0.00	0.00	0.00	Req per kernel
14.3.2	Write Req	0.00	0.00	0.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel
14.3.4	Stall Cycles	0.02	0.00	12.00	Cycles per kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric_ID	Metric	Avg
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Min	Max	Unit	
15.1.0	Address Processing Unit Busy		91.36
35.47	116.75 Pct		
15.1.1	Address Stall		2.52
1.28	14.83 Pct		
15.1.2	Data Stall		57.83
0.65	72.24 Pct		
15.1.3	Data-Processor → Address Stall		0.00
0.00	0.00 Pct		
15.1.4	Sequencer → TA Address Stall		
	Cycles per kernel		
15.1.5	Sequencer → TA Command Stall		
	Cycles per kernel		
15.1.6	Sequencer → TA Data Stall		
	Cycles per kernel		
15.1.7	Total Instructions		28035370.61
4325376.00	1077936128.00 Instructions per kernel		
15.1.8	Global/Generic Instructions		27747350.46
4194304.00	1077936128.00 Instructions per kernel		
15.1.9	Global/Generic Read Instructions		25604196.42
2097152.00	1073741824.00 Instructions per kernel		
15.1.10	Global/Generic Read Instructions for LDS		
	Instructions per kernel		
15.1.11	Global/Generic Write Instructions		2143154.04

2097152.00	4194304.00	Instructions per kernel	
15.1.12	Global/Generic Atomic Instructions		0.00
0.00	0.00	Instructions per kernel	
15.1.13	Spill/Stack Instructions		288020.15
0.00	524288.00	Instructions per kernel	
15.1.14	Spill/Stack Read Instructions		288020.15
0.00	524288.00	Instructions per kernel	
15.1.15	Spill/Stack Read Instructions for LDS		
	Instructions per kernel		
15.1.16	Spill/Stack Write Instructions		0.00
0.00	0.00	Instructions per kernel	
15.1.17	Spill/Stack Atomic Instructions		0.00
0.00	0.00	Instructions per kernel	
15.1.18	Spill/Stack Total Cycles		4216628.52
0.00	8388608.00	Cycles per kernel	
15.1.19	Spill/Stack Coalesced Read		130564.62
0.00	524288.00	Cycles per kernel	
15.1.20	Spill/Stack Coalesced Write		0.00
0.00	0.00	Cycles per kernel	

15.2 Data-Return Path

Metric_ID	Metric		Avg	Min
	Max	Unit		
15.2.0	Data-Return Busy		96.49	69.18
	98.03	Pct		

15.2.1	Cache RAM → Data-Return Stall	90.56	45.43
108.21	Pct		
15.2.2	Workgroup manager → Data-Return Stall	0.00	0.00
0.00	Pct		
15.2.3	Coalescable Instructions	32641.16	0.00
131072.00	Instructions per kernel		
15.2.4	Read Instructions	25892216.57	2228224.00
1073741824.00	Instructions per kernel		
15.2.5	Write Instructions	2143154.04	2097152.00
4194304.00	Instructions per kernel		
15.2.6	Atomic Instructions	0.00	0.00
0.00	Instructions per kernel		
15.2.7	Write Ack Instructions		
	Instructions per kernel		

INFO Not showing table with empty column(s): 16.6 L1D Addr
Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
16.1.0	Hit rate	29.61	Pct of peak
16.1.1	Bandwidth	9.79	Pct of peak
16.1.2	Utilization	99.30	Pct of peak
16.1.3	Coalescing	25.14	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3
Max					
16.2.0 112.60	Stalled on L2 Data	38.72	88.71	92.29	96.10
16.2.1 94.75	Stalled on L2 Req	0.00	8.85	10.66	79.34
16.2.2	Stalled on Address				
16.2.3	Stalled on Data				
16.2.4	Stalled on Latency FIFO				
16.2.5	Stalled on Request FIFO				
16.2.6	Stalled on Read Return				
16.2.7 0.00	Tag RAM Stall (Read)	0.00	0.00	0.00	0.00
16.2.8 0.26	Tag RAM Stall (Write)	0.00	0.00	0.00	0.00
16.2.9 0.00	Tag RAM Stall (Atomic)	0.00	0.00	0.00	0.00

16.3 L1D Cache Accesses

Metric_ID	Metric	Avg	Min
Max	Unit		
16.3.0	Total Req	1792696943.65	270532608.00
68987912192.00	Req per kernel		
16.3.1	Read Req	1655535084.84	136314880.00
68719476736.00	Req per kernel		
16.3.2	Write Req	137161858.81	134217728.00
268435456.00	Req per kernel		
16.3.3	Atomic Req	0.00	0.00
0.00	Req per kernel		
16.3.4	Cache BW	57931575311.86	8657043456.00
2233382993920.00	Bytes per kernel		
16.3.5	Cache Hit Rate	29.61	25.39
50.00	Pct		
16.3.6	Cache Accesses	452590432.12	67633152.00
17448304640.00	Req per kernel		
16.3.7	Cache Hits	211805687.27	17170432.00
8724152320.00	Req per kernel		
16.3.8	Invalidations	304.44	304.00
646.00	Req per kernel		
16.3.9	L1-L2 BW	28531645414.57	4311744384.00
1108101624704.00	Bytes per kernel		
16.3.10	Tag RAM 0 Req		
	Req per kernel		

16.3.11 Tag RAM 1 Req
Req per kernel

16.3.12 Tag RAM 2 Req
Req per kernel

16.3.13 Tag RAM 3 Req
Req per kernel

16.3.14	L1-L2 Read	205022214.75	16908287.00
8589935079.00	Req per kernel		

16.3.15	L1-L2 Write	35762530.11	33554432.00
134217728.00	Req per kernel		

16.3.16	L1-L2 Atomic	0.00	0.00
0.00	Req per kernel		

16.3.17 L1 Access Latency
Cycles

16.3.18 L1-L2 Read Latency
Cycles

16.3.19 L1-L2 Write Latency
Cycles

16.4 L1D - L2 Transactions

Metric_ID	Metric	Xfer	Coherency	Avg	Min
	Max Unit				
16.4.0	NC - Read	Read	NC	0.00	0.00
	0.00 Req per kernel				

16.4.1	UC - Read	Read	UC	0.00	0.00
	0.00	Req per kernel			
16.4.2	CC - Read	Read	CC	0.00	0.00
	0.00	Req per kernel			
16.4.3	RW - Read	Read	RW	205022749.10	16908286.00
8589935042.00		Req per kernel			
16.4.4	RW - Write	Write	RW	35762530.11	33554432.00
134217728.00		Req per kernel			
16.4.5	NC - Write	Write	NC	0.00	0.00
	0.00	Req per kernel			
16.4.6	UC - Write	Write	UC	0.00	0.00
	0.00	Req per kernel			
16.4.7	CC - Write	Write	CC	0.00	0.00
	0.00	Req per kernel			
16.4.8	NC - Atomic	Atomic	NC	0.00	0.00
	0.00	Req per kernel			
16.4.9	UC - Atomic	Atomic	UC	0.00	0.00
	0.00	Req per kernel			
16.4.10	CC - Atomic	Atomic	CC	0.00	0.00
	0.00	Req per kernel			
16.4.11	RW - Atomic	Atomic	RW	0.00	0.00
	0.00	Req per kernel			

16.5 L1D Addr Translation

Metric_ID	Metric	Avg	Min
Max Units			
16.5.0	Req	452590432.12	67633152.00
17448304640.00	Req per kernel		
16.5.1	Inflight Req		
Req per kernel			
16.5.2	Hit Ratio	99.82	99.61
99.90 Pct			
16.5.3	Hits	450989776.78	67474181.00
17380894720.00	Req per kernel		
16.5.4	Translation Misses	1555022.00	77816.00
67411260.00	Req per kernel		
16.5.5	Misses under Translation Miss		
Req per kernel			
16.5.6	Permission Misses	0.00	0.00
0.00	Req per kernel		

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache
17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
17.1.0	Utilization	98.35	Pct
17.1.1	Bandwidth	16.33	Pct
17.1.2	Hit Rate	33.66	Pct
17.1.3	L2-Fabric Read BW	1898.19	Gb/s

17.1.4	L2-Fabric Write and Atomic BW	1832.51	Gb/s
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17.1.5	HBM Bandwidth	5324.80	Gb/s
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17.2 L2 - Fabric Transactions

Metric_ID	Metric	Avg
Min	Max Unit	
17.2.0	Read BW	14727379238.98
2164157312.00	589268213184.00 Bytes per kernel	
17.2.1	HBM Read Traffic	100.00
97.53	102.39 Pct	
17.2.2	Remote Read Traffic	0.00
0.00	2.47 Pct	
17.2.3	Uncached Read Traffic	0.00
0.00	0.00 Pct	
17.2.4	Write and Atomic BW	2147484813.30
2147483648.00	2147615872.00 Bytes per kernel	
17.2.5	HBM Write and Atomic Traffic	100.00
100.00	100.00 Pct	
17.2.6	Remote Write and Atomic Traffic	0.00
0.00	0.00 Pct	
17.2.7	Atomic Traffic	0.00
0.00	0.00 Pct	
17.2.8	Uncached Write and Atomic Traffic	0.00
0.00	0.00 Pct	

17.2.9	Read Latency	2378.71
474.12	2963.00 Cycles	

17.2.10	Write and Atomic Latency	461.10
155.94	577.78 Cycles	

17.2.11	Atomic Latency
	Cycles

17.2.12	Read Stall
	Pct

17.2.13	Write Stall
	Pct

17.3 L2 Cache Accesses

Metric_ID	Metric	Avg	Min
Max	Unit		

17.3.0	Bandwidth	30821620435.08	6460784384.00
1116698420096.00	Bytes per kernel		

17.3.1	Read Bandwidth
	Bytes per kernel

17.3.2	Write Bandwidth
	Bytes per kernel

17.3.3	Atomic Bandwidth
	Bytes per kernel

17.3.4	Req	240793909.65	50474878.00
8724206407.00	Req per kernel		

17.3.5	Read Req	205024414.92	16920447.00
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8589988862.00	Req per kernel		
17.3.6	Write Req	35762530.11	33554432.00
134217728.00	Req per kernel		
17.3.7	Atomic Req	0.00	0.00
0.00	Req per kernel		
17.3.8	Streaming Req	0.00	0.00
0.00	Req per kernel		
17.3.9	Bypasss Req		
	Req per kernel		
17.3.10	Probe Req	0.00	0.00
0.00	Req per kernel		
17.3.11	Input Buffer Req		
	Req per kernel		
17.3.12	Cache Hit	33.66	33.26
49.10	Pct		
17.3.13	Hits	108645102.05	16789590.00
4283451715.00	Hits per kernel		
17.3.14	Misses	131976004.68	33684024.00
4654291636.00	Misses per kernel		
17.3.15	Writeback	16777361.12	16777328.00
16782145.00	Cachelines per kernel		
17.3.16	Writeback (Internal)	16662095.79	16649117.00
16753138.00	Cachelines per kernel		
17.3.17	Writeback (vL1D Req)	114300.22	28317.00

128450.00 Cachelines per kernel

17.3.18 Evict (Internal) 131630420.87 33422169.00
4748682138.00 Cachelines per kernel

17.3.19 Evict (vL1D Req) 0.00 0.00
0.00 Cachelines per kernel

17.3.20 NC Req 0.00 0.00
0.00 Req per kernel

17.3.21 UC Req 11.19 0.00
2600.00 Req per kernel

17.3.22 CC Req 0.00 0.00
0.00 Req per kernel

17.3.23 RW Req 240786779.82 50474879.00
8724203761.00 Req per kernel

17.5 L2 - Fabric Interface Stalls

Metric_ID	Metric	Min	Max	Unit	Type	
Transaction	Avg					
17.5.0	Read - PCIe Stall Pct				PCIe Stall	Read
17.5.1	Read - Infinity Fabric Stall Pct				Infinity Fabric Stall	Read
17.5.2	Read - HBM Stall Pct				HBM Stall	Read
17.5.3	Write - PCIe Stall Pct				PCIe Stall	Write

17.5.4	Write - Infinity Fabric Stall Pct	Infinity Fabric Stall	Write
17.5.5	Write - HBM Stall Pct	HBM Stall	Write
17.5.6	Write - Credit Starvation Pct	Credit Starvation	Write
0.00	0.00	0.00	

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID	Metric	Avg
Min	Max Unit	
17.6.0	Read (32B)	0.00
0.00	0.00 Req per kernel	
17.6.1	Read (64B)	134948.81
-16185562.00	108672105.00 Req per kernel	
17.6.2	Read (128B)	114990175.90
16906897.00	4578900715.00 Req per kernel	
17.6.3	Read (Uncached)	22.37
0.00	5200.00 Req per kernel	
17.6.4	HBM Read	115103312.74
16907481.00	4664039916.00 Req per kernel	
17.6.5	Remote Read	194678.62
0.00	114869189.00 Req per kernel	
17.6.6	Read Bandwidth - PCIe Bytes per kernel	

17.6.7	Read Bandwidth - Infinity Fabric Bytes per kernel	
17.6.8	Read Bandwidth - HBM Bytes per kernel	
17.6.9 -18.00	Write and Atomic (32B) 4102.00 Req per kernel	32.78
17.6.10 0.00	Write and Atomic (Uncached) 0.00 Req per kernel	0.00
17.6.11 33554432.00	Write and Atomic (64B) 33554456.00 Req per kernel	33554433.82
17.6.12 33554432.00	HBM Write and Atomic 33558308.00 Req per kernel	33554465.59
17.6.13 0.00	Remote Write and Atomic 1117.00 Req per kernel	4.39
17.6.14	Write Bandwidth - PCIe Bytes per kernel	
17.6.15	Write Bandwidth - Infinity Fabric Bytes per kernel	
17.6.16	Write Bandwidth - HBM Bytes per kernel	
17.6.17 0.00	Atomic 0.00 Req per kernel	0.00
17.6.18	Atomic - HBM Req per kernel	

17.6.19 Atomic Bandwidth - PCIe
Bytes per kernel

17.6.20 Atomic Bandwidth - Infinity Fabric
Bytes per kernel

17.6.21 Atomic Bandwidth - HBM
Bytes per kernel

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	33.67	2.21	33.26	49.10	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.2.0	0.00	33.26	33.28	33.38	33.41	49.10
18.2.1	1.00	33.26	33.29	33.38	33.41	49.10
18.2.2	2.00	33.26	33.28	33.38	33.41	49.10
18.2.3	3.00	33.27	33.28	33.38	33.41	49.10
18.2.4	4.00	33.26	33.27	33.38	33.41	49.10
18.2.5	5.00	33.26	33.27	33.38	33.41	49.10
18.2.6	6.00	33.26	33.28	33.38	33.41	49.10
18.2.7	7.00	33.26	33.27	33.38	33.41	49.10
18.2.8	8.00	33.26	33.28	33.37	33.41	49.10

18.2.9	9.00	33.26	33.29	33.37	33.41	49.10
18.2.10	10.00	33.26	33.28	33.37	33.40	49.10
18.2.11	11.00	33.26	33.28	33.37	33.41	49.10
18.2.12	12.00	33.26	33.27	33.38	33.40	49.10
18.2.13	13.00	33.26	33.27	33.38	33.40	49.10
18.2.14	14.00	33.26	33.27	33.38	33.41	49.10
18.2.15	15.00	33.26	33.28	33.38	33.40	49.10
18.2.16	16.00	33.26	33.28	33.37	33.41	49.10
18.2.17	17.00	33.26	33.28	33.37	33.41	49.10
18.2.18	18.00	33.26	33.28	33.37	33.41	49.10
18.2.19	19.00	33.26	33.28	33.37	33.41	49.10
18.2.20	20.00	33.26	33.27	33.38	33.40	49.10
18.2.21	21.00	33.26	33.27	33.38	33.41	49.10
18.2.22	22.00	33.26	33.27	33.38	33.41	49.10
18.2.23	23.00	33.26	33.28	33.38	33.40	49.10
18.2.24	24.00	33.26	33.28	33.38	33.41	49.10
18.2.25	25.00	33.26	33.28	33.38	33.41	49.10
18.2.26	26.00	33.26	33.28	33.38	33.41	49.10
18.2.27	27.00	33.27	33.28	33.38	33.41	49.10
18.2.28	28.00	33.26	33.27	33.38	33.41	49.10
18.2.29	29.00	33.26	33.27	33.38	33.41	49.10
18.2.30	30.00	33.26	33.28	33.38	33.41	49.10
18.2.31	31.00	33.26	33.27	33.38	33.41	49.10
18.2.32	32.00	33.26	33.29	33.38	33.41	49.10

18.2.33	33.00	33.26	33.29	33.38	33.41	49.10
18.2.34	34.00	33.26	33.29	33.38	33.41	49.10
18.2.35	35.00	33.27	33.28	33.38	33.41	49.10
18.2.36	36.00	33.26	33.27	33.38	33.41	49.10
18.2.37	37.00	33.26	33.27	33.38	33.41	49.10
18.2.38	38.00	33.26	33.28	33.38	33.41	49.10
18.2.39	39.00	33.26	33.27	33.38	33.41	49.10
18.2.40	40.00	33.26	33.29	33.37	33.41	49.10
18.2.41	41.00	33.26	33.28	33.37	33.41	49.10
18.2.42	42.00	33.26	33.28	33.37	33.41	49.10
18.2.43	43.00	33.26	33.29	33.37	33.41	49.10
18.2.44	44.00	33.26	33.27	33.38	33.41	49.10
18.2.45	45.00	33.26	33.27	33.38	33.41	49.10
18.2.46	46.00	33.26	33.27	33.38	33.41	49.10
18.2.47	47.00	33.26	33.28	33.38	33.41	49.10
18.2.48	48.00	33.26	33.28	33.37	33.40	49.10
18.2.49	49.00	33.26	33.28	33.37	33.41	49.10
18.2.50	50.00	33.26	33.28	33.37	33.41	49.10
18.2.51	51.00	33.26	33.28	33.37	33.41	49.10
18.2.52	52.00	33.26	33.28	33.38	33.41	49.10
18.2.53	53.00	33.26	33.27	33.38	33.41	49.10
18.2.54	54.00	33.26	33.27	33.38	33.41	49.10
18.2.55	55.00	33.26	33.28	33.38	33.41	49.10
18.2.56	56.00	33.26	33.28	33.38	33.41	49.10

18.2.57	57.00	33.26	33.28	33.38	33.41	49.10
18.2.58	58.00	33.26	33.29	33.38	33.41	49.10
18.2.59	59.00	33.27	33.28	33.38	33.41	49.10
18.2.60	60.00	33.26	33.27	33.38	33.41	49.10
18.2.61	61.00	33.26	33.27	33.38	33.41	49.10
18.2.62	62.00	33.26	33.28	33.38	33.41	49.10
18.2.63	63.00	33.26	33.27	33.38	33.41	49.10
18.2.64	64.00	33.26	33.29	33.37	33.41	49.10
18.2.65	65.00	33.26	33.29	33.37	33.41	49.10
18.2.66	66.00	33.26	33.29	33.37	33.40	49.10
18.2.67	67.00	33.26	33.28	33.37	33.41	49.10
18.2.68	68.00	33.26	33.27	33.38	33.41	49.10
18.2.69	69.00	33.26	33.27	33.38	33.41	49.10
18.2.70	70.00	33.26	33.27	33.38	33.40	49.10
18.2.71	71.00	33.26	33.28	33.38	33.40	49.10
18.2.72	72.00	33.26	33.29	33.38	33.41	49.10
18.2.73	73.00	33.26	33.28	33.38	33.41	49.10
18.2.74	74.00	33.26	33.28	33.38	33.41	49.10
18.2.75	75.00	33.27	33.28	33.38	33.41	49.10
18.2.76	76.00	33.26	33.27	33.38	33.41	49.10
18.2.77	77.00	33.26	33.27	33.38	33.41	49.10
18.2.78	78.00	33.26	33.28	33.38	33.41	49.10
18.2.79	79.00	33.26	33.27	33.38	33.41	49.10
18.2.80	80.00	33.26	33.28	33.38	33.41	49.10

18.2.81	81.00	33.26	33.28	33.38	33.41	49.10
18.2.82	82.00	33.26	33.29	33.38	33.41	49.10
18.2.83	83.00	33.27	33.29	33.38	33.41	49.10
18.2.84	84.00	33.26	33.27	33.38	33.41	49.10
18.2.85	85.00	33.26	33.27	33.38	33.41	49.10
18.2.86	86.00	33.26	33.28	33.38	33.41	49.10
18.2.87	87.00	33.26	33.27	33.38	33.41	49.10
18.2.88	88.00	33.26	33.28	33.37	33.41	49.10
18.2.89	89.00	33.26	33.28	33.37	33.41	49.10
18.2.90	90.00	33.26	33.29	33.37	33.41	49.10
18.2.91	91.00	33.26	33.29	33.37	33.41	49.10
18.2.92	92.00	33.26	33.28	33.38	33.41	49.10
18.2.93	93.00	33.26	33.27	33.38	33.41	49.10
18.2.94	94.00	33.26	33.27	33.38	33.41	49.10
18.2.95	95.00	33.26	33.28	33.38	33.40	49.10
18.2.96	96.00	33.26	33.28	33.37	33.41	49.10
18.2.97	97.00	33.26	33.28	33.37	33.40	49.10
18.2.98	98.00	33.26	33.28	33.37	33.41	49.10
18.2.99	99.00	33.26	33.28	33.37	33.41	49.10
18.2.100	100.00	33.26	33.27	33.38	33.41	49.10
18.2.101	101.00	33.26	33.27	33.38	33.41	49.10
18.2.102	102.00	33.26	33.27	33.38	33.41	49.10
18.2.103	103.00	33.26	33.28	33.38	33.41	49.10
18.2.104	104.00	33.26	33.28	33.38	33.41	49.10

18.2.105	105.00	33.26	33.29	33.38	33.41	49.10
18.2.106	106.00	33.26	33.28	33.38	33.41	49.10
18.2.107	107.00	33.27	33.28	33.38	33.41	49.10
18.2.108	108.00	33.26	33.27	33.38	33.41	49.10
18.2.109	109.00	33.26	33.27	33.38	33.41	49.10
18.2.110	110.00	33.26	33.28	33.38	33.41	49.10
18.2.111	111.00	33.26	33.27	33.38	33.41	49.10
18.2.112	112.00	33.26	33.28	33.38	33.41	49.10
18.2.113	113.00	33.26	33.28	33.38	33.41	49.10
18.2.114	114.00	33.26	33.28	33.38	33.41	49.10
18.2.115	115.00	33.27	33.28	33.38	33.41	49.10
18.2.116	116.00	33.26	33.27	33.38	33.41	49.10
18.2.117	117.00	33.26	33.27	33.38	33.41	49.10
18.2.118	118.00	33.26	33.28	33.38	33.41	49.10
18.2.119	119.00	33.26	33.27	33.38	33.41	49.10
18.2.120	120.00	33.26	33.28	33.37	33.41	49.10
18.2.121	121.00	33.26	33.28	33.37	33.41	49.10
18.2.122	122.00	33.26	33.28	33.37	33.41	49.10
18.2.123	123.00	33.26	33.28	33.37	33.41	49.10
18.2.124	124.00	33.26	33.27	33.38	33.41	49.10
18.2.125	125.00	33.26	33.27	33.38	33.41	49.10
18.2.126	126.00	33.26	33.27	33.38	33.41	49.10
18.2.127	127.00	33.26	33.28	33.38	33.41	49.10

18.3 L2 Requests (per normUnit)

Metric_ID Max	Channel	Min	Q1	Median	Q3
18.3.0 68157780.00	0.00	394320.00	394464.00	395106.00	395252.00
18.3.1 68157768.00	1.00	394320.00	394464.00	395106.00	395258.00
18.3.2 68157774.00	2.00	394400.00	394462.00	395116.00	395254.00
18.3.3 68157958.00	3.00	394320.00	394456.00	395116.00	395256.00
18.3.4 68157767.00	4.00	394320.00	394451.00	395104.00	395252.00
18.3.5 68157768.00	5.00	394320.00	394460.00	395104.00	395258.00
18.3.6 68157767.00	6.00	394320.00	394448.00	395124.00	395256.00
18.3.7 68157771.00	7.00	394320.00	394445.00	395124.00	395254.00
18.3.8 68157934.00	8.00	394320.00	394360.00	395036.00	395178.00
18.3.9 68157937.00	9.00	394320.00	394410.00	395036.00	395180.00
18.3.10 68157933.00	10.00	394320.00	394372.00	395046.00	395186.00

18.3.11 68157933.00	11.00	394320.00	394382.00	395054.00	395182.00
18.3.12 68157932.00	12.00	394320.00	394393.00	395040.00	395184.00
18.3.13 68157938.00	13.00	394320.00	394405.00	395080.00	395186.00
18.3.14 68157933.00	14.00	394319.00	394394.50	395042.00	395183.00
18.3.15 68157936.00	15.00	394320.00	394428.00	395042.00	395182.00
18.3.16 68157927.00	16.00	394320.00	394391.50	395026.00	395180.00
18.3.17 68157929.00	17.00	394320.00	394438.00	395026.00	395184.00
18.3.18 68157927.00	18.00	394320.00	394386.00	395044.00	395186.00
18.3.19 68157928.00	19.00	394320.00	394386.00	395046.00	395186.00
18.3.20 68157932.00	20.00	394320.00	394413.00	395034.00	395188.00
18.3.21 68157931.00	21.00	394320.00	394397.00	395074.00	395186.00
18.3.22 68157929.00	22.00	394320.00	394407.00	395038.00	395188.00

18.3.23 68157930.00	23.00	394320.00	394442.00	395038.00	395188.00
18.3.24 68157772.00	24.00	394320.00	394461.00	395098.00	395257.50
18.3.25 68157767.00	25.00	394320.00	394468.00	395098.00	395264.00
18.3.26 68157768.00	26.00	394400.00	394477.00	395120.00	395257.00
18.3.27 68157909.00	27.00	394320.00	394465.50	395120.00	395258.50
18.3.28 68157768.00	28.00	394320.00	394471.00	395110.00	395256.00
18.3.29 68157768.00	29.00	394320.00	394470.00	395110.00	395262.00
18.3.30 68157767.00	30.00	394320.00	394463.00	395121.00	395262.00
18.3.31 68157769.00	31.00	394320.00	394454.00	395120.00	395258.00
18.3.32 68157774.00	32.00	394320.00	394466.00	395108.00	395262.00
18.3.33 68157768.00	33.00	394320.00	394450.00	395108.00	395258.00
18.3.34 68157768.00	34.00	394400.00	394464.00	395114.00	395256.00

18.3.35 68157953.00	35.00	394320.00	394465.00	395114.00	395256.00
18.3.36 68157767.00	36.00	394319.00	394463.00	395104.00	395252.00
18.3.37 68157768.00	37.00	394320.00	394469.00	395104.00	395252.00
18.3.38 68157768.00	38.00	394320.00	394467.00	395104.00	395257.00
18.3.39 68157772.00	39.00	394320.00	394462.50	395104.00	395255.00
18.3.40 68157933.00	40.00	394320.00	394390.50	395034.00	395183.00
18.3.41 68157937.00	41.00	394320.00	394431.00	395034.00	395181.00
18.3.42 68157930.00	42.00	394320.00	394382.00	395048.00	395186.00
18.3.43 68157931.00	43.00	394319.00	394387.00	395052.00	395185.00
18.3.44 68157935.00	44.00	394320.00	394397.00	395034.00	395190.00
18.3.45 68157935.00	45.00	394320.00	394391.00	395074.00	395188.00
18.3.46 68157930.00	46.00	394320.00	394386.00	395050.00	395190.00

18.3.47 68157935.00	47.00	394320.00	394438.00	395050.00	395186.00
18.3.48 68157928.00	48.00	394320.00	394393.00	395042.00	395182.00
18.3.49 68157936.00	49.00	394320.00	394423.00	395042.00	395183.00
18.3.50 68157929.00	50.00	394320.00	394378.00	395052.00	395188.00
18.3.51 68157941.00	51.00	394320.00	394378.00	395054.00	395186.00
18.3.52 68157929.00	52.00	394320.00	394390.00	395036.00	395184.00
18.3.53 68157938.00	53.00	394320.00	394395.00	395076.00	395186.00
18.3.54 68157928.00	54.00	394320.00	394397.00	395044.00	395186.00
18.3.55 68157943.00	55.00	394320.00	394437.00	395044.00	395190.00
18.3.56 68157771.00	56.00	394320.00	394469.50	395110.00	395262.00
18.3.57 68157772.00	57.00	394320.00	394463.00	395110.00	395261.00
18.3.58 68157767.00	58.00	394400.00	394470.00	395116.00	395258.00

18.3.59 68157921.00	59.00	394320.00	394465.50	395117.00	395258.00
18.3.60 68157765.00	60.00	394320.00	394467.00	395104.00	395254.00
18.3.61 68157775.00	61.00	394320.00	394465.00	395104.00	395256.00
18.3.62 68157767.00	62.00	394320.00	394457.50	395106.00	395249.00
18.3.63 68157780.00	63.00	394320.00	394463.00	395106.00	395256.50
18.3.64 68157932.00	64.00	394320.00	394396.00	395032.00	395180.50
18.3.65 68157935.00	65.00	394320.00	394430.00	395032.00	395184.00
18.3.66 68157931.00	66.00	394319.00	394382.00	395046.00	395182.00
18.3.67 68157934.00	67.00	394320.00	394393.00	395034.00	395182.00
18.3.68 68157934.00	68.00	394320.00	394387.00	395040.00	395186.50
18.3.69 68157932.00	69.00	394319.00	394401.00	395080.00	395182.00
18.3.70 68157932.00	70.00	394320.00	394396.50	395038.00	395184.00

18.3.71 68157933.00	71.00	394320.00	394437.00	395038.00	395184.00
18.3.72 68157774.00	72.00	394320.00	394470.00	395104.00	395254.00
18.3.73 68157770.00	73.00	394320.00	394476.00	395104.00	395257.00
18.3.74 68157770.00	74.00	394400.00	394474.00	395128.00	395257.00
18.3.75 68157941.00	75.00	394320.00	394470.00	395128.00	395256.00
18.3.76 68157768.00	76.00	394320.00	394465.00	395114.00	395256.00
18.3.77 68157768.00	77.00	394320.00	394462.00	395114.00	395254.00
18.3.78 68157766.00	78.00	394320.00	394471.00	395122.00	395256.00
18.3.79 68157773.00	79.00	394320.00	394461.00	395122.00	395260.00
18.3.80 68157778.00	80.00	394320.00	394473.00	395114.00	395257.00
18.3.81 68157776.00	81.00	394319.00	394461.00	395114.00	395260.00
18.3.82 68157773.00	82.00	394400.00	394455.00	395116.00	395262.00

18.3.83 68157916.00	83.00	394320.00	394466.00	395116.00	395260.00
18.3.84 68157775.00	84.00	394320.00	394461.50	395122.00	395255.00
18.3.85 68157773.00	85.00	394320.00	394469.00	395122.00	395259.00
18.3.86 68157773.00	86.00	394320.00	394468.00	395122.00	395257.00
18.3.87 68157778.00	87.00	394320.00	394469.00	395122.00	395261.00
18.3.88 68157940.00	88.00	394320.00	394384.00	395036.00	395183.00
18.3.89 68157940.00	89.00	394320.00	394439.00	395036.00	395184.00
18.3.90 68157937.00	90.00	394320.00	394395.50	395048.00	395189.00
18.3.91 68157937.00	91.00	394319.00	394393.00	395044.00	395184.00
18.3.92 68157937.00	92.00	394320.00	394397.00	395044.00	395186.00
18.3.93 68157943.00	93.00	394320.00	394404.00	395084.00	395188.00
18.3.94 68157935.00	94.00	394320.00	394395.00	395060.00	395187.50

18.3.95 68157938.00	95.00	394320.00	394437.00	395060.00	395188.00
18.3.96 68157935.00	96.00	394320.00	394381.00	395030.00	395182.00
18.3.97 68157936.00	97.00	394320.00	394433.00	395030.00	395182.00
18.3.98 68157931.00	98.00	394319.00	394381.00	395056.00	395186.00
18.3.99 68157934.00	99.00	394320.00	394376.00	395058.00	395184.00
18.3.100 68157934.00	100.00	394320.00	394394.00	395038.00	395182.00
18.3.101 68157940.00	101.00	394320.00	394393.00	395078.00	395184.00
18.3.102 68157932.00	102.00	394320.00	394401.50	395064.00	395186.00
18.3.103 68157932.00	103.00	394319.00	394425.00	395064.00	395190.00
18.3.104 68157775.00	104.00	394320.00	394462.00	395106.00	395258.50
18.3.105 68157770.00	105.00	394320.00	394437.00	395106.00	395259.00
18.3.106 68157768.00	106.00	394399.00	394462.00	395114.00	395254.00

18.3.107 68157953.00	107.00	394320.00	394461.00	395114.00	395257.00
18.3.108 68157771.00	108.00	394320.00	394449.00	395104.00	395257.00
18.3.109 68157771.00	109.00	394320.00	394450.00	395104.00	395258.00
18.3.110 68157767.00	110.00	394320.00	394451.00	395118.00	395252.00
18.3.111 68157772.00	111.00	394320.00	394455.00	395118.00	395259.50
18.3.112 68157781.00	112.00	394320.00	394470.00	395112.00	395258.00
18.3.113 68157773.00	113.00	394320.00	394467.00	395112.00	395259.00
18.3.114 68157777.00	114.00	394399.00	394481.00	395112.00	395255.00
18.3.115 68157885.00	115.00	394320.00	394465.00	395112.00	395256.00
18.3.116 68157775.00	116.00	394320.00	394473.00	395102.00	395255.00
18.3.117 68157773.00	117.00	394320.00	394460.00	395102.00	395252.00
18.3.118 68157776.00	118.00	394320.00	394477.00	395116.00	395254.00

18.3.119 68157778.00	119.00	394320.00	394474.00	395116.00	395256.00
18.3.120 68157937.00	120.00	394319.00	394387.00	395038.00	395186.00
18.3.121 68157939.00	121.00	394320.00	394422.00	395038.00	395184.00
18.3.122 68157938.00	122.00	394320.00	394391.00	395056.00	395186.00
18.3.123 68157939.00	123.00	394320.00	394399.00	395048.00	395186.00
18.3.124 68157938.00	124.00	394320.00	394406.00	395050.00	395184.00
18.3.125 68157937.00	125.00	394320.00	394394.00	395090.00	395182.00
18.3.126 68157938.00	126.00	394320.00	394407.00	395058.00	395188.00
18.3.127 68157939.00	127.00	394320.00	394438.00	395058.00	395188.00

18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	1601834.42	279394.77	0.00
18.4.1	1.00	1601833.90	279394.77	0.00
18.4.2	2.00	1601838.12	279394.77	0.00

18.4.3	3.00	1601843.77	279394.77	0.00
18.4.4	4.00	1601807.81	279394.77	0.00
18.4.5	5.00	1601808.96	279394.77	0.00
18.4.6	6.00	1601816.31	279394.77	0.00
18.4.7	7.00	1601809.74	279394.77	0.00
18.4.8	8.00	1601801.31	279394.77	0.00
18.4.9	9.00	1601800.66	279394.77	0.00
18.4.10	10.00	1601801.18	279394.77	0.00
18.4.11	11.00	1601798.31	279394.77	0.00
18.4.12	12.00	1601809.06	279394.77	0.00
18.4.13	13.00	1601809.35	279394.77	0.00
18.4.14	14.00	1601811.61	279394.77	0.00
18.4.15	15.00	1601814.92	279394.77	0.00
18.4.16	16.00	1601801.41	279394.77	0.00
18.4.17	17.00	1601800.62	279394.77	0.00
18.4.18	18.00	1601803.27	279394.77	0.00
18.4.19	19.00	1601800.11	279394.77	0.00
18.4.20	20.00	1601810.06	279394.77	0.00
18.4.21	21.00	1601810.22	279394.77	0.00
18.4.22	22.00	1601812.89	279394.77	0.00
18.4.23	23.00	1601815.15	279394.77	0.00
18.4.24	24.00	1601835.85	279394.77	0.00
18.4.25	25.00	1601836.45	279394.77	0.00
18.4.26	26.00	1601838.89	279394.77	0.00

18.4.27	27.00	1601844.58	279394.77	0.00
18.4.28	28.00	1601809.53	279394.77	0.00
18.4.29	29.00	1601809.61	279394.77	0.00
18.4.30	30.00	1601818.46	279394.77	0.00
18.4.31	31.00	1601811.27	279394.77	0.00
18.4.32	32.00	1601834.36	279394.77	0.00
18.4.33	33.00	1601834.20	279394.77	0.00
18.4.34	34.00	1601837.36	279394.77	0.00
18.4.35	35.00	1601842.26	279394.77	0.00
18.4.36	36.00	1601809.43	279394.77	0.00
18.4.37	37.00	1601809.28	279394.77	0.00
18.4.38	38.00	1601816.56	279394.77	0.00
18.4.39	39.00	1601809.34	279394.77	0.00
18.4.40	40.00	1601802.26	279394.77	0.00
18.4.41	41.00	1601801.41	279394.77	0.00
18.4.42	42.00	1601803.48	279394.77	0.00
18.4.43	43.00	1601800.29	279394.77	0.00
18.4.44	44.00	1601811.57	279394.77	0.00
18.4.45	45.00	1601811.64	279394.77	0.00
18.4.46	46.00	1601813.98	279394.77	0.00
18.4.47	47.00	1601817.64	279394.77	0.00
18.4.48	48.00	1601802.66	279394.77	0.00
18.4.49	49.00	1601801.95	279394.77	0.00
18.4.50	50.00	1601802.62	279394.77	0.00

18.4.51	51.00	1601800.19	279394.77	0.00
18.4.52	52.00	1601812.64	279394.77	0.00
18.4.53	53.00	1601813.07	279394.77	0.00
18.4.54	54.00	1601814.03	279394.77	0.00
18.4.55	55.00	1601818.55	279394.77	0.00
18.4.56	56.00	1601834.76	279394.77	0.00
18.4.57	57.00	1601834.23	279394.77	0.00
18.4.58	58.00	1601836.66	279394.77	0.00
18.4.59	59.00	1601842.39	279394.77	0.00
18.4.60	60.00	1601809.66	279394.77	0.00
18.4.61	61.00	1601809.47	279394.77	0.00
18.4.62	62.00	1601817.44	279394.77	0.00
18.4.63	63.00	1601810.41	279394.77	0.00
18.4.64	64.00	1601801.79	279394.77	0.00
18.4.65	65.00	1601801.20	279394.77	0.00
18.4.66	66.00	1601803.27	279394.77	0.00
18.4.67	67.00	1601799.96	279394.77	0.00
18.4.68	68.00	1601810.12	279394.77	0.00
18.4.69	69.00	1601811.06	279394.77	0.00
18.4.70	70.00	1601813.23	279394.77	0.00
18.4.71	71.00	1601815.61	279394.77	0.00
18.4.72	72.00	1601836.49	279394.77	0.00
18.4.73	73.00	1601836.56	279394.77	0.00
18.4.74	74.00	1601838.51	279394.77	0.00

18.4.75	75.00	1601844.83	279394.77	0.00
18.4.76	76.00	1601809.35	279394.77	0.00
18.4.77	77.00	1601810.01	279394.77	0.00
18.4.78	78.00	1601818.80	279394.77	0.00
18.4.79	79.00	1601811.35	279394.77	0.00
18.4.80	80.00	1601837.56	279394.77	0.00
18.4.81	81.00	1601837.58	279394.77	0.00
18.4.82	82.00	1601839.82	279394.77	0.00
18.4.83	83.00	1601846.05	279394.77	0.00
18.4.84	84.00	1601810.38	279394.77	0.00
18.4.85	85.00	1601811.14	279394.77	0.00
18.4.86	86.00	1601818.69	279394.77	0.00
18.4.87	87.00	1601811.71	279394.77	0.00
18.4.88	88.00	1601803.85	279394.77	0.00
18.4.89	89.00	1601802.60	279394.77	0.00
18.4.90	90.00	1601802.52	279394.77	0.00
18.4.91	91.00	1601801.09	279394.77	0.00
18.4.92	92.00	1601811.54	279394.77	0.00
18.4.93	93.00	1601811.71	279394.77	0.00
18.4.94	94.00	1601813.03	279394.77	0.00
18.4.95	95.00	1601816.20	279394.77	0.00
18.4.96	96.00	1601801.38	279394.77	0.00
18.4.97	97.00	1601801.00	279394.77	0.00
18.4.98	98.00	1601803.45	279394.77	0.00

18.4.99	99.00	1601800.54	279394.77	0.00
18.4.100	100.00	1601811.13	279394.77	0.00
18.4.101	101.00	1601810.92	279394.77	0.00
18.4.102	102.00	1601812.43	279394.77	0.00
18.4.103	103.00	1601816.80	279394.77	0.00
18.4.104	104.00	1601833.28	279394.77	0.00
18.4.105	105.00	1601834.20	279394.77	0.00
18.4.106	106.00	1601835.62	279394.77	0.00
18.4.107	107.00	1601842.63	279394.77	0.00
18.4.108	108.00	1601807.27	279394.77	0.00
18.4.109	109.00	1601808.71	279394.77	0.00
18.4.110	110.00	1601815.83	279394.77	0.00
18.4.111	111.00	1601808.02	279394.77	0.00
18.4.112	112.00	1601832.50	279394.77	0.00
18.4.113	113.00	1601832.78	279394.77	0.00
18.4.114	114.00	1601834.88	279394.77	0.00
18.4.115	115.00	1601842.04	279394.77	0.00
18.4.116	116.00	1601807.62	279394.77	0.00
18.4.117	117.00	1601807.99	279394.77	0.00
18.4.118	118.00	1601815.70	279394.77	0.00
18.4.119	119.00	1601808.38	279394.77	0.00
18.4.120	120.00	1601800.13	279394.77	0.00
18.4.121	121.00	1601799.86	279394.77	0.00
18.4.122	122.00	1601803.37	279394.77	0.00

18.4.123	123.00	1601800.31	279394.77	0.00
18.4.124	124.00	1601810.69	279394.77	0.00
18.4.125	125.00	1601810.59	279394.77	0.00
18.4.126	126.00	1601813.05	279394.77	0.00
18.4.127	127.00	1601816.59	279394.77	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID L2-Fabric Atomic	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
18.5.0 0.00	0.00	898631.21	262144.05
18.5.1 0.00	1.00	898629.90	262144.06
18.5.2 0.00	2.00	898633.38	262144.05
18.5.3 0.00	3.00	898630.86	262144.06
18.5.4 0.00	4.00	898630.36	262144.06
18.5.5 0.00	5.00	898629.39	262144.07
18.5.6 0.00	6.00	898632.18	262144.06
18.5.7 0.00	7.00	898629.69	262144.07

18.5.8 0.00	8.00	898625.74	262144.07
18.5.9 0.00	9.00	898624.94	262144.10
18.5.10 0.00	10.00	898627.84	262144.08
18.5.11 0.00	11.00	898625.17	262144.11
18.5.12 0.00	12.00	898625.78	262144.07
18.5.13 0.00	13.00	898624.72	262144.07
18.5.14 0.00	14.00	898627.39	262144.06
18.5.15 0.00	15.00	898624.93	262144.07
18.5.16 0.00	16.00	900960.89	262144.35
18.5.17 0.00	17.00	900962.06	262144.32
18.5.18 0.00	18.00	900962.62	262144.34
18.5.19 0.00	19.00	900961.15	262144.35

18.5.20 0.00	20.00	900961.31	262144.34
18.5.21 0.00	21.00	900962.33	262144.30
18.5.22 0.00	22.00	900962.88	262144.32
18.5.23 0.00	23.00	900961.16	262144.31
18.5.24 0.00	24.00	900967.26	262144.32
18.5.25 0.00	25.00	900968.64	262144.29
18.5.26 0.00	26.00	900968.64	262144.32
18.5.27 0.00	27.00	900967.68	262144.30
18.5.28 0.00	28.00	900965.93	262144.33
18.5.29 0.00	29.00	900967.53	262144.30
18.5.30 0.00	30.00	900967.28	262144.31
18.5.31 0.00	31.00	900966.06	262144.30

18.5.32 0.00	32.00	896594.10	262144.06
18.5.33 0.00	33.00	896593.46	262144.10
18.5.34 0.00	34.00	896595.35	262144.09
18.5.35 0.00	35.00	896595.06	262144.09
18.5.36 0.00	36.00	896592.36	262144.05
18.5.37 0.00	37.00	896591.82	262144.08
18.5.38 0.00	38.00	896593.56	262144.07
18.5.39 0.00	39.00	896593.27	262144.08
18.5.40 0.00	40.00	896591.96	262144.05
18.5.41 0.00	41.00	896592.36	262144.07
18.5.42 0.00	42.00	896593.37	262144.06
18.5.43 0.00	43.00	896592.69	262144.07

18.5.44 0.00	44.00	896592.86	262144.05
18.5.45 0.00	45.00	896593.20	262144.07
18.5.46 0.00	46.00	896593.90	262144.06
18.5.47 0.00	47.00	896593.27	262144.07
18.5.48 0.00	48.00	898865.51	262144.14
18.5.49 0.00	49.00	898868.21	262144.19
18.5.50 0.00	50.00	898866.53	262144.13
18.5.51 0.00	51.00	898867.63	262144.18
18.5.52 0.00	52.00	898865.63	262144.14
18.5.53 0.00	53.00	898867.98	262144.18
18.5.54 0.00	54.00	898866.00	262144.14
18.5.55 0.00	55.00	898867.23	262144.18

18.5.56 0.00	56.00	898865.82	262144.15
18.5.57 0.00	57.00	898867.65	262144.20
18.5.58 0.00	58.00	898867.21	262144.17
18.5.59 0.00	59.00	898867.98	262144.21
18.5.60 0.00	60.00	898864.18	262144.14
18.5.61 0.00	61.00	898866.24	262144.18
18.5.62 0.00	62.00	898866.08	262144.15
18.5.63 0.00	63.00	898866.57	262144.18
18.5.64 0.00	64.00	913737.31	262144.73
18.5.65 0.00	65.00	913740.16	262144.63
18.5.66 0.00	66.00	913740.22	262144.71
18.5.67 0.00	67.00	913739.64	262144.67

18.5.68 0.00	68.00	913737.30	262144.74
18.5.69 0.00	69.00	913739.72	262144.64
18.5.70 0.00	70.00	913740.56	262144.71
18.5.71 0.00	71.00	913740.30	262144.67
18.5.72 0.00	72.00	913733.37	262144.75
18.5.73 0.00	73.00	913734.50	262144.69
18.5.74 0.00	74.00	913736.13	262144.71
18.5.75 0.00	75.00	913735.24	262144.71
18.5.76 0.00	76.00	913732.39	262144.75
18.5.77 0.00	77.00	913733.67	262144.67
18.5.78 0.00	78.00	913735.29	262144.69
18.5.79 0.00	79.00	913734.13	262144.67

18.5.80 0.00	80.00	898830.32	262144.22
18.5.81 0.00	81.00	898827.50	262144.25
18.5.82 0.00	82.00	898830.99	262144.26
18.5.83 0.00	83.00	898829.33	262144.20
18.5.84 0.00	84.00	898828.07	262144.21
18.5.85 0.00	85.00	898825.77	262144.20
18.5.86 0.00	86.00	898828.58	262144.24
18.5.87 0.00	87.00	898826.71	262144.19
18.5.88 0.00	88.00	898825.17	262144.22
18.5.89 0.00	89.00	898823.62	262144.22
18.5.90 0.00	90.00	898825.86	262144.23
18.5.91 0.00	91.00	898823.98	262144.18

18.5.92 0.00	92.00	898825.87	262144.22
18.5.93 0.00	93.00	898824.08	262144.21
18.5.94 0.00	94.00	898826.94	262144.23
18.5.95 0.00	95.00	898825.12	262144.18
18.5.96 0.00	96.00	900902.37	262144.24
18.5.97 0.00	97.00	900904.24	262144.25
18.5.98 0.00	98.00	900905.34	262144.24
18.5.99 0.00	99.00	900904.34	262144.28
18.5.100 0.00	100.00	900902.60	262144.22
18.5.101 0.00	101.00	900904.07	262144.23
18.5.102 0.00	102.00	900906.31	262144.25
18.5.103 0.00	103.00	900905.45	262144.26

18.5.104 0.00	104.00	900905.25	262144.23
18.5.105 0.00	105.00	900906.57	262144.22
18.5.106 0.00	106.00	900907.60	262144.22
18.5.107 0.00	107.00	900907.28	262144.23
18.5.108 0.00	108.00	900903.96	262144.21
18.5.109 0.00	109.00	900905.56	262144.21
18.5.110 0.00	110.00	900905.93	262144.22
18.5.111 0.00	111.00	900905.56	262144.23
18.5.112 0.00	112.00	897441.42	262144.22
18.5.113 0.00	113.00	897439.86	262144.36
18.5.114 0.00	114.00	897442.68	262144.21
18.5.115 0.00	115.00	897442.27	262144.34

18.5.116 0.00	116.00	897440.63	262144.22
18.5.117 0.00	117.00	897439.27	262144.37
18.5.118 0.00	118.00	897442.48	262144.23
18.5.119 0.00	119.00	897441.78	262144.35
18.5.120 0.00	120.00	897433.33	262144.26
18.5.121 0.00	121.00	897432.31	262144.40
18.5.122 0.00	122.00	897436.27	262144.26
18.5.123 0.00	123.00	897434.68	262144.38
18.5.124 0.00	124.00	897433.34	262144.24
18.5.125 0.00	125.00	897432.24	262144.37
18.5.126 0.00	126.00	897435.86	262144.25
18.5.127 0.00	127.00	897434.54	262144.35

18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	343.33	1923.61	2080.39	2287.21	2847.04
18.6.1	1.00	352.97	1945.70	2119.81	2325.04	2925.24
18.6.2	2.00	356.19	1965.12	2152.23	2362.48	2969.62
18.6.3	3.00	356.77	1962.12	2146.66	2351.05	2981.00
18.6.4	4.00	363.51	1946.70	2110.61	2270.86	2816.01
18.6.5	5.00	357.40	1948.58	2115.11	2280.60	2874.63
18.6.6	6.00	384.64	2016.88	2218.85	2396.51	3019.56
18.6.7	7.00	369.31	2012.49	2209.02	2392.91	2935.74
18.6.8	8.00	342.88	2382.73	2479.46	2591.79	2929.55
18.6.9	9.00	357.65	2470.57	2593.95	2705.64	3238.03
18.6.10	10.00	350.84	2433.68	2569.49	2778.14	3446.91
18.6.11	11.00	356.60	2487.95	2637.33	2828.78	3304.01
18.6.12	12.00	363.10	2122.78	2292.98	2457.35	2931.85
18.6.13	13.00	356.56	2110.74	2295.52	2470.00	3012.87
18.6.14	14.00	378.93	2207.98	2405.33	2652.52	3152.73
18.6.15	15.00	363.46	2209.96	2423.48	2665.06	3121.60
18.6.16	16.00	449.25	2385.25	2488.98	2598.74	2985.06
18.6.17	17.00	457.94	2475.59	2591.47	2719.98	3303.67
18.6.18	18.00	452.40	2439.78	2578.42	2782.42	3383.48
18.6.19	19.00	457.60	2491.80	2636.85	2827.92	3275.93
18.6.20	20.00	467.83	2120.75	2295.63	2468.33	2971.63
18.6.21	21.00	457.24	2114.24	2298.27	2479.54	2971.02

18.6.22	22.00	479.31	2206.53	2410.89	2654.09	3070.25
18.6.23	23.00	463.10	2208.89	2423.19	2668.31	3020.23
18.6.24	24.00	444.74	1917.35	2085.54	2270.04	2816.45
18.6.25	25.00	453.27	1943.56	2127.04	2316.55	2892.33
18.6.26	26.00	457.68	1967.49	2165.06	2370.80	2972.11
18.6.27	27.00	457.69	1962.07	2154.21	2350.67	2956.05
18.6.28	28.00	467.99	1944.26	2118.72	2280.90	2811.80
18.6.29	29.00	457.94	1942.06	2120.52	2290.94	2844.77
18.6.30	30.00	485.82	2020.63	2224.05	2407.85	2945.16
18.6.31	31.00	470.13	2010.20	2219.50	2403.74	2914.96
18.6.32	32.00	332.61	2392.69	2494.60	2611.16	3100.06
18.6.33	33.00	345.71	2483.59	2605.54	2722.74	3273.77
18.6.34	34.00	339.65	2453.95	2593.73	2778.44	3455.62
18.6.35	35.00	345.11	2512.21	2660.22	2832.72	3305.14
18.6.36	36.00	351.46	2118.48	2299.61	2478.86	2980.21
18.6.37	37.00	345.22	2116.12	2305.68	2491.78	3034.28
18.6.38	38.00	366.79	2199.02	2395.28	2671.14	3110.49
18.6.39	39.00	350.65	2138.06	2324.48	2545.80	2935.73
18.6.40	40.00	328.43	1924.37	2087.20	2287.17	2840.03
18.6.41	41.00	340.13	1953.93	2126.27	2332.18	2978.89
18.6.42	42.00	342.58	1968.26	2170.55	2361.99	2891.70
18.6.43	43.00	343.83	1960.08	2161.94	2348.32	2874.77
18.6.44	44.00	351.79	1956.47	2117.93	2290.61	2779.67
18.6.45	45.00	346.28	1958.22	2127.45	2303.18	2798.57

18.6.46	46.00	372.43	2019.74	2223.90	2402.60	2950.45
18.6.47	47.00	357.36	2008.03	2214.95	2383.95	2933.56
18.6.48	48.00	460.26	1906.84	2073.64	2273.31	2791.98
18.6.49	49.00	469.46	1942.08	2108.13	2314.29	2942.86
18.6.50	50.00	471.42	1949.68	2143.88	2348.24	2884.94
18.6.51	51.00	472.80	1940.69	2136.54	2326.77	2883.20
18.6.52	52.00	480.51	1940.40	2105.28	2269.31	2759.44
18.6.53	53.00	474.53	1947.91	2106.99	2285.65	2782.95
18.6.54	54.00	500.84	2007.06	2199.77	2377.61	2914.78
18.6.55	55.00	486.02	1992.72	2195.17	2360.96	2871.49
18.6.56	56.00	459.27	2376.90	2475.09	2594.35	3028.55
18.6.57	57.00	474.32	2464.64	2581.21	2698.54	3181.97
18.6.58	58.00	468.59	2434.51	2573.87	2744.49	3354.06
18.6.59	59.00	474.07	2497.21	2634.44	2819.73	3225.51
18.6.60	60.00	480.44	2099.89	2278.36	2466.86	2923.32
18.6.61	61.00	474.01	2098.95	2279.46	2465.46	3005.57
18.6.62	62.00	495.33	2181.91	2378.48	2645.53	3082.53
18.6.63	63.00	479.24	2116.69	2314.83	2542.39	2934.15
18.6.64	64.00	531.47	1925.05	2088.02	2291.11	2787.05
18.6.65	65.00	541.52	1957.43	2126.08	2332.46	2878.59
18.6.66	66.00	542.89	1964.51	2157.36	2368.44	2889.25
18.6.67	67.00	544.58	1954.26	2150.27	2354.64	2865.95
18.6.68	68.00	548.78	1951.32	2107.76	2285.76	2783.16
18.6.69	69.00	543.23	1953.98	2116.85	2298.20	2787.54

18.6.70	70.00	569.37	2011.30	2218.12	2401.69	2937.48
18.6.71	71.00	554.38	1998.91	2201.52	2388.71	2954.06
18.6.72	72.00	528.07	2388.04	2491.87	2597.45	3069.38
18.6.73	73.00	543.17	2484.85	2598.14	2717.34	3424.44
18.6.74	74.00	537.19	2446.78	2586.59	2755.13	3331.77
18.6.75	75.00	542.57	2501.13	2646.38	2828.40	3340.95
18.6.76	76.00	548.35	2114.97	2296.16	2478.23	2892.69
18.6.77	77.00	542.08	2108.31	2297.45	2483.00	2942.72
18.6.78	78.00	563.77	2197.36	2413.55	2661.95	3159.58
18.6.79	79.00	547.66	2130.72	2319.23	2541.65	2906.76
18.6.80	80.00	469.10	2390.11	2494.09	2593.98	3019.76
18.6.81	81.00	481.83	2477.44	2599.48	2710.10	3353.14
18.6.82	82.00	475.93	2447.10	2578.48	2754.03	3309.87
18.6.83	83.00	481.20	2494.86	2647.88	2824.53	3303.64
18.6.84	84.00	487.37	2107.74	2288.11	2473.34	2923.59
18.6.85	85.00	480.91	2103.21	2295.02	2481.00	2995.28
18.6.86	86.00	502.55	2181.83	2399.86	2663.13	3144.82
18.6.87	87.00	486.41	2120.27	2315.54	2543.66	2937.07
18.6.88	88.00	467.86	1916.35	2081.94	2292.29	2784.53
18.6.89	89.00	480.20	1947.43	2121.50	2344.19	2890.39
18.6.90	90.00	481.61	1960.03	2142.03	2362.28	2843.78
18.6.91	91.00	483.05	1948.70	2134.68	2340.91	2864.38
18.6.92	92.00	487.77	1946.52	2103.77	2294.50	2739.85
18.6.93	93.00	481.98	1946.32	2112.69	2304.66	2819.40

18.6.94	94.00	508.04	2005.17	2198.00	2400.56	2898.97
18.6.95	95.00	492.91	1993.45	2189.76	2382.91	2880.60
18.6.96	96.00	412.23	2408.26	2507.93	2635.88	3179.12
18.6.97	97.00	424.61	2503.20	2626.80	2745.92	3415.13
18.6.98	98.00	418.19	2449.06	2593.19	2812.74	3388.61
18.6.99	99.00	423.59	2499.02	2648.89	2866.71	3359.20
18.6.100	100.00	431.27	2127.98	2321.01	2489.13	2976.99
18.6.101	101.00	424.20	2128.57	2318.93	2506.74	3027.49
18.6.102	102.00	444.86	2216.92	2420.87	2685.26	3114.76
18.6.103	103.00	429.00	2207.30	2422.01	2689.53	3116.11
18.6.104	104.00	410.60	1929.98	2094.22	2292.10	2859.41
18.6.105	105.00	422.55	1963.48	2145.30	2347.59	2911.53
18.6.106	106.00	424.43	1976.25	2184.12	2376.60	2988.15
18.6.107	107.00	425.57	1976.03	2177.16	2364.08	2954.80
18.6.108	108.00	431.60	1952.53	2128.56	2311.40	2893.12
18.6.109	109.00	424.78	1951.12	2135.07	2316.06	2923.64
18.6.110	110.00	451.20	2028.04	2238.78	2430.39	2986.95
18.6.111	111.00	435.47	2011.58	2227.08	2416.64	2972.42
18.6.112	112.00	452.37	1914.31	2088.61	2276.58	2825.58
18.6.113	113.00	462.39	1948.45	2131.00	2330.06	2894.54
18.6.114	114.00	464.71	1961.50	2153.65	2368.14	2947.43
18.6.115	115.00	465.96	1960.18	2151.71	2346.97	2872.68
18.6.116	116.00	471.30	1936.12	2115.20	2288.83	2813.82
18.6.117	117.00	464.74	1931.12	2111.68	2299.60	2861.20

18.6.118	118.00	491.88	2013.75	2220.39	2415.22	2976.76
18.6.119	119.00	476.52	1998.83	2210.26	2394.90	2973.02
18.6.120	120.00	451.73	2386.82	2489.55	2611.49	3131.63
18.6.121	121.00	465.91	2475.37	2602.70	2732.51	3413.69
18.6.122	122.00	458.85	2432.51	2571.54	2788.37	3322.59
18.6.123	123.00	464.31	2483.08	2629.51	2849.05	3361.15
18.6.124	124.00	471.37	2115.30	2302.17	2470.40	2954.63
18.6.125	125.00	464.07	2103.93	2299.05	2484.62	2986.40
18.6.126	126.00	486.51	2198.41	2401.68	2656.93	3105.82
18.6.127	127.00	470.87	2194.23	2402.39	2677.19	3082.34

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	151.75	348.11	386.66	428.74	543.42
18.7.1	1.00	156.58	360.16	399.42	441.49	563.31
18.7.2	2.00	157.54	359.71	412.71	452.10	564.89
18.7.3	3.00	159.85	359.64	407.98	449.00	567.83
18.7.4	4.00	161.22	367.98	404.30	437.25	549.69
18.7.5	5.00	159.02	368.24	408.75	439.22	559.30
18.7.6	6.00	171.18	395.21	444.14	477.27	571.97
18.7.7	7.00	164.46	388.24	441.29	473.63	571.59
18.7.8	8.00	152.05	487.71	511.12	541.66	627.17
18.7.9	9.00	158.50	512.07	539.75	573.56	721.07
18.7.10	10.00	154.47	500.09	532.39	587.07	752.61
18.7.11	11.00	158.41	515.58	551.24	600.72	737.89

18.7.12	12.00	159.80	423.00	455.79	491.04	594.34
18.7.13	13.00	157.33	419.63	455.32	494.63	578.48
18.7.14	14.00	166.84	449.65	490.01	557.09	666.13
18.7.15	15.00	159.26	447.53	488.71	555.28	661.78
18.7.16	16.00	147.05	488.11	511.67	541.20	634.31
18.7.17	17.00	153.62	511.80	541.01	573.48	709.75
18.7.18	18.00	149.65	501.83	532.74	586.67	742.93
18.7.19	19.00	154.86	516.48	551.28	600.70	748.10
18.7.20	20.00	156.23	424.88	455.60	493.87	590.92
18.7.21	21.00	154.09	421.75	456.33	496.32	581.51
18.7.22	22.00	162.94	450.76	489.75	554.71	671.02
18.7.23	23.00	155.62	448.21	490.94	554.79	665.72
18.7.24	24.00	146.87	348.68	384.05	428.98	544.90
18.7.25	25.00	151.89	360.79	399.16	440.35	573.30
18.7.26	26.00	152.61	360.53	411.74	451.87	565.08
18.7.27	27.00	154.02	358.81	407.62	447.65	570.32
18.7.28	28.00	155.57	367.76	404.92	435.02	547.38
18.7.29	29.00	153.63	367.44	405.13	439.92	559.23
18.7.30	30.00	165.16	393.51	444.15	476.47	573.43
18.7.31	31.00	158.49	387.88	437.04	474.29	576.68
18.7.32	32.00	152.10	488.05	515.68	542.95	645.39
18.7.33	33.00	158.50	512.53	543.01	575.23	688.37
18.7.34	34.00	154.54	501.57	537.33	595.87	757.04
18.7.35	35.00	159.69	519.38	553.72	608.70	750.81

18.7.36	36.00	161.25	425.29	460.67	496.62	600.63
18.7.37	37.00	158.95	422.08	462.15	500.24	606.19
18.7.38	38.00	168.08	446.89	488.63	559.20	675.91
18.7.39	39.00	160.34	425.92	465.58	529.36	623.49
18.7.40	40.00	150.64	347.09	385.68	429.83	538.41
18.7.41	41.00	155.71	358.67	399.23	441.97	552.31
18.7.42	42.00	156.39	355.38	405.15	450.44	563.62
18.7.43	43.00	157.90	354.79	402.93	446.29	543.10
18.7.44	44.00	160.39	368.41	405.77	438.76	537.81
18.7.45	45.00	158.60	369.19	408.78	442.72	548.26
18.7.46	46.00	170.33	388.08	438.65	475.27	590.38
18.7.47	47.00	163.67	382.21	431.01	470.58	574.46
18.7.48	48.00	145.80	344.31	384.61	424.62	532.18
18.7.49	49.00	150.99	355.37	396.52	440.53	547.40
18.7.50	50.00	151.51	352.34	406.31	445.78	554.35
18.7.51	51.00	154.32	352.15	400.52	445.33	543.27
18.7.52	52.00	156.57	365.88	402.16	435.31	543.27
18.7.53	53.00	154.75	367.19	405.90	440.17	542.36
18.7.54	54.00	166.48	386.72	436.57	471.70	571.14
18.7.55	55.00	160.15	381.22	429.82	466.89	567.00
18.7.56	56.00	147.40	482.97	509.68	537.23	638.26
18.7.57	57.00	153.86	508.81	538.81	568.48	674.29
18.7.58	58.00	150.21	495.77	532.41	591.07	728.66
18.7.59	59.00	153.82	512.86	549.01	604.28	746.63

18.7.60	60.00	155.39	420.68	456.20	492.03	585.75
18.7.61	61.00	152.96	418.95	458.78	495.84	599.83
18.7.62	62.00	162.30	442.61	485.67	555.59	667.38
18.7.63	63.00	154.59	422.51	460.36	523.57	615.79
18.7.64	64.00	149.25	349.67	391.27	432.97	547.32
18.7.65	65.00	154.25	361.33	405.64	445.89	552.90
18.7.66	66.00	155.03	357.90	410.61	451.62	558.05
18.7.67	67.00	157.68	359.47	408.97	451.10	556.28
18.7.68	68.00	159.28	367.46	404.63	441.65	540.66
18.7.69	69.00	157.06	368.71	407.90	444.12	552.16
18.7.70	70.00	168.91	389.05	442.46	474.40	572.57
18.7.71	71.00	162.05	383.11	435.39	469.95	568.46
18.7.72	72.00	150.20	488.51	514.22	542.28	634.94
18.7.73	73.00	156.55	512.58	542.82	574.49	672.87
18.7.74	74.00	152.28	496.99	533.93	587.62	743.10
18.7.75	75.00	156.24	517.88	551.77	608.00	767.33
18.7.76	76.00	157.81	423.79	458.64	496.15	585.73
18.7.77	77.00	155.27	423.91	461.40	497.21	591.87
18.7.78	78.00	164.99	444.24	486.96	559.59	685.28
18.7.79	79.00	157.31	425.40	464.23	525.67	640.50
18.7.80	80.00	146.85	486.41	514.14	541.08	629.21
18.7.81	81.00	153.27	511.20	540.87	572.28	678.55
18.7.82	82.00	149.51	499.07	533.28	586.22	722.31
18.7.83	83.00	154.32	520.52	553.22	606.08	765.16

18.7.84	84.00	155.97	424.62	459.46	495.07	589.17
18.7.85	85.00	153.80	423.62	458.97	497.15	584.56
18.7.86	86.00	162.83	444.51	485.75	558.91	674.00
18.7.87	87.00	155.08	425.90	464.87	524.72	636.31
18.7.88	88.00	146.31	348.82	390.07	431.70	539.37
18.7.89	89.00	151.51	360.82	402.16	444.74	552.03
18.7.90	90.00	151.91	358.63	406.20	451.08	551.64
18.7.91	91.00	153.70	358.08	405.21	447.78	540.23
18.7.92	92.00	155.23	365.10	402.63	439.72	534.07
18.7.93	93.00	153.31	367.61	405.64	443.02	553.42
18.7.94	94.00	165.06	387.36	439.53	472.72	584.71
18.7.95	95.00	158.58	382.27	430.86	468.45	559.78
18.7.96	96.00	150.94	489.63	516.48	543.76	656.27
18.7.97	97.00	157.57	515.53	545.16	576.30	686.34
18.7.98	98.00	153.23	498.82	532.05	585.06	742.36
18.7.99	99.00	158.53	515.94	550.20	603.69	741.98
18.7.100	100.00	159.84	424.16	457.33	497.13	607.58
18.7.101	101.00	157.62	421.12	457.86	498.18	590.23
18.7.102	102.00	166.69	449.85	490.58	556.08	674.85
18.7.103	103.00	159.38	447.88	490.10	555.36	667.30
18.7.104	104.00	150.65	349.31	387.05	431.99	566.14
18.7.105	105.00	155.75	361.47	402.79	445.03	565.02
18.7.106	106.00	156.26	363.63	414.96	454.26	578.04
18.7.107	107.00	157.81	362.80	410.73	451.29	577.49

18.7.108	108.00	159.29	366.91	405.66	438.33	556.21
18.7.109	109.00	157.12	366.45	405.92	440.07	554.91
18.7.110	110.00	169.16	395.47	445.68	478.40	587.61
18.7.111	111.00	162.22	387.15	438.78	475.00	587.19
18.7.112	112.00	147.41	346.42	383.28	426.40	549.85
18.7.113	113.00	152.33	358.30	399.18	439.40	556.53
18.7.114	114.00	153.14	359.68	410.65	451.42	576.78
18.7.115	115.00	155.63	361.00	409.49	448.30	566.71
18.7.116	116.00	157.35	364.81	403.25	436.30	547.93
18.7.117	117.00	154.98	364.49	403.31	437.58	553.28
18.7.118	118.00	166.93	392.38	441.33	475.88	580.83
18.7.119	119.00	159.96	385.99	434.41	471.78	571.59
18.7.120	120.00	147.87	484.04	512.98	539.52	642.27
18.7.121	121.00	154.23	511.63	540.26	571.81	687.00
18.7.122	122.00	150.38	496.37	529.27	581.86	726.55
18.7.123	123.00	154.25	511.79	547.23	599.42	733.39
18.7.124	124.00	155.71	418.59	452.12	489.70	593.63
18.7.125	125.00	153.34	418.65	453.23	494.53	582.36
18.7.126	126.00	162.65	447.85	484.86	551.25	668.43
18.7.127	127.00	155.10	443.41	486.45	551.13	673.71

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00

18.8.2	2.00	0.00	0.00	0.00	0.00	0.00
18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00
18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00
18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00
18.8.15	15.00	0.00	0.00	0.00	0.00	0.00
18.8.16	16.00	0.00	0.00	0.00	0.00	0.00
18.8.17	17.00	0.00	0.00	0.00	0.00	0.00
18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00
18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00

18.8.26	26.00	0.00	0.00	0.00	0.00	0.00
18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00
18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00
18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00
18.8.39	39.00	0.00	0.00	0.00	0.00	0.00
18.8.40	40.00	0.00	0.00	0.00	0.00	0.00
18.8.41	41.00	0.00	0.00	0.00	0.00	0.00
18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00
18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00

18.8.50	50.00	0.00	0.00	0.00	0.00	0.00
18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00
18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00
18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00
18.8.63	63.00	0.00	0.00	0.00	0.00	0.00
18.8.64	64.00	0.00	0.00	0.00	0.00	0.00
18.8.65	65.00	0.00	0.00	0.00	0.00	0.00
18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00
18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00

18.8.74	74.00	0.00	0.00	0.00	0.00	0.00
18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00
18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00
18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00
18.8.87	87.00	0.00	0.00	0.00	0.00	0.00
18.8.88	88.00	0.00	0.00	0.00	0.00	0.00
18.8.89	89.00	0.00	0.00	0.00	0.00	0.00
18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00
18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00

18.8.98	98.00	0.00	0.00	0.00	0.00	0.00
18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00
18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00
18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00
18.8.111	111.00	0.00	0.00	0.00	0.00	0.00
18.8.112	112.00	0.00	0.00	0.00	0.00	0.00
18.8.113	113.00	0.00	0.00	0.00	0.00	0.00
18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00
18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00

18.8.122	122.00	0.00	0.00	0.00	0.00	0.00
18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00
18.8.127	127.00	0.00	0.00	0.00	0.00	0.00

18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID Max	Channel	Min	Q1	Median	Q3
18.12.0 35901832.00	0.00	132057.00	132096.00	132096.00	132098.00
18.12.1 35901905.00	1.00	132056.00	132096.00	132096.00	132098.00
18.12.2 35901846.00	2.00	132063.00	132096.00	132096.00	132098.00
18.12.3 35901398.00	3.00	132056.00	132096.00	132096.00	132098.00
18.12.4 35901792.00	4.00	132055.00	132096.00	132096.00	132098.00
18.12.5 35901860.00	5.00	132057.00	132096.00	132096.00	132098.00
18.12.6 35901790.00	6.00	132056.00	132096.00	132096.00	132098.00

18.12.7 35901391.00	7.00	132044.00	132096.00	132096.00	132098.00
18.12.8 35902729.00	8.00	132052.00	132096.00	132096.00	132098.00
18.12.9 35902620.00	9.00	132059.00	132096.00	132096.00	132098.00
18.12.10 35902611.00	10.00	132063.00	132096.00	132096.00	132098.00
18.12.11 35902095.00	11.00	132050.00	132096.00	132096.00	132098.00
18.12.12 35902704.00	12.00	132046.00	132096.00	132096.00	132098.00
18.12.13 35902671.00	13.00	132052.00	132096.00	132096.00	132098.00
18.12.14 35902648.00	14.00	132054.00	132096.00	132096.00	132098.00
18.12.15 35902103.00	15.00	132056.00	132096.00	132096.00	132098.00
18.12.16 36543229.00	16.00	132050.00	132096.00	132096.00	132098.00
18.12.17 36543061.00	17.00	132046.00	132096.00	132096.00	132098.00
18.12.18 36543587.00	18.00	132065.00	132096.00	132096.00	132098.00

18.12.19 36542927.00	19.00	132068.00	132096.00	132096.00	132098.00
18.12.20 36543264.00	20.00	132043.00	132096.00	132096.00	132098.00
18.12.21 36543028.00	21.00	132055.00	132096.00	132096.00	132098.00
18.12.22 36543578.00	22.00	132067.00	132096.00	132096.00	132098.00
18.12.23 36542884.00	23.00	132060.00	132096.00	132096.00	132098.00
18.12.24 36541573.00	24.00	132053.00	132096.00	132096.00	132098.00
18.12.25 36541550.00	25.00	132041.00	132096.00	132096.00	132098.00
18.12.26 36541956.00	26.00	132068.00	132096.00	132096.00	132098.00
18.12.27 36541289.00	27.00	132065.00	132096.00	132096.00	132098.00
18.12.28 36541588.00	28.00	132052.00	132096.00	132096.00	132098.00
18.12.29 36541524.00	29.00	132050.00	132096.00	132096.00	132098.00
18.12.30 36541933.00	30.00	132060.00	132096.00	132096.00	132098.00

18.12.31 36541326.00	31.00	132068.00	132096.00	132096.00	132098.00
18.12.32 36283992.00	32.00	132052.00	132096.00	132096.00	132098.00
18.12.33 36283234.00	33.00	132047.00	132096.00	132096.00	132098.00
18.12.34 36283327.00	34.00	132064.00	132096.00	132096.00	132098.00
18.12.35 36283327.00	35.00	132057.00	132096.00	132096.00	132098.00
18.12.36 36284004.00	36.00	132051.00	132096.00	132096.00	132098.00
18.12.37 36283247.00	37.00	132047.00	132096.00	132096.00	132098.00
18.12.38 36283320.00	38.00	132057.00	132096.00	132096.00	132098.00
18.12.39 36283261.00	39.00	132063.00	132096.00	132096.00	132098.00
18.12.40 36284102.00	40.00	132059.00	132096.00	132096.00	132098.00
18.12.41 36283476.00	41.00	132053.00	132096.00	132096.00	132098.00
18.12.42 36283459.00	42.00	132062.00	132096.00	132096.00	132098.00

18.12.43 36283515.00	43.00	132062.00	132096.00	132096.00	132098.00
18.12.44 36284173.00	44.00	132054.00	132096.00	132096.00	132098.00
18.12.45 36283459.00	45.00	132061.00	132096.00	132096.00	132098.00
18.12.46 36283524.00	46.00	132062.00	132096.00	132096.00	132098.00
18.12.47 36283534.00	47.00	132062.00	132096.00	132096.00	132098.00
18.12.48 35765688.00	48.00	132054.00	132096.00	132096.00	132098.00
18.12.49 35766392.00	49.00	132055.00	132096.00	132096.00	132098.00
18.12.50 35765571.00	50.00	132062.00	132096.00	132096.00	132098.00
18.12.51 35766189.00	51.00	132060.00	132096.00	132096.00	132098.00
18.12.52 35765642.00	52.00	132054.00	132096.00	132096.00	132098.00
18.12.53 35766394.00	53.00	132047.00	132096.00	132096.00	132098.00
18.12.54 35765551.00	54.00	132061.00	132096.00	132096.00	132098.00

18.12.55 35766158.00	55.00	132049.00	132096.00	132096.00	132098.00
18.12.56 35766773.00	56.00	132053.00	132096.00	132096.00	132098.00
18.12.57 35767521.00	57.00	132052.00	132096.00	132096.00	132098.00
18.12.58 35766758.00	58.00	132053.00	132096.00	132096.00	132098.00
18.12.59 35767526.00	59.00	132057.00	132096.00	132096.00	132098.00
18.12.60 35766783.00	60.00	132054.00	132096.00	132096.00	132098.00
18.12.61 35767496.00	61.00	132052.00	132096.00	132096.00	132098.00
18.12.62 35766763.00	62.00	132061.00	132096.00	132096.00	132098.00
18.12.63 35767512.00	63.00	132056.00	132096.00	132096.00	132098.00
18.12.64 37470348.00	64.00	132063.00	132096.00	132096.00	132098.00
18.12.65 37470645.00	65.00	132068.00	132096.00	132096.00	132098.00
18.12.66 37471196.00	66.00	132064.00	132096.00	132096.00	132098.00

18.12.67 37470018.00	67.00	132068.00	132096.00	132096.00	132098.00
18.12.68 37470301.00	68.00	132061.00	132096.00	132096.00	132098.00
18.12.69 37470714.00	69.00	132052.00	132096.00	132096.00	132098.00
18.12.70 37471202.00	70.00	132063.00	132096.00	132096.00	132098.00
18.12.71 37469985.00	71.00	132066.00	132096.00	132096.00	132098.00
18.12.72 37469519.00	72.00	132059.00	132096.00	132096.00	132098.00
18.12.73 37470095.00	73.00	132058.00	132096.00	132096.00	132098.00
18.12.74 37470323.00	74.00	132064.00	132096.00	132096.00	132098.00
18.12.75 37469350.00	75.00	132062.00	132096.00	132096.00	132098.00
18.12.76 37469541.00	76.00	132064.00	132096.00	132096.00	132098.00
18.12.77 37470081.00	77.00	132063.00	132096.00	132096.00	132098.00
18.12.78 37470306.00	78.00	132065.00	132096.00	132096.00	132098.00

18.12.79 37469329.00	79.00	132071.00	132096.00	132096.00	132098.00
18.12.80 35935009.00	80.00	132054.00	132096.00	132096.00	132098.00
18.12.81 35935423.00	81.00	132052.00	132096.00	132096.00	132098.00
18.12.82 35934987.00	82.00	132059.00	132096.00	132096.00	132098.00
18.12.83 35934880.00	83.00	132057.00	132096.00	132096.00	132098.00
18.12.84 35934995.00	84.00	132049.00	132096.00	132096.00	132098.00
18.12.85 35935406.00	85.00	132040.00	132096.00	132096.00	132098.00
18.12.86 35934988.00	86.00	132047.00	132096.00	132096.00	132098.00
18.12.87 35934855.00	87.00	132053.00	132096.00	132096.00	132098.00
18.12.88 35934710.00	88.00	132049.00	132096.00	132096.00	132098.00
18.12.89 35935044.00	89.00	132046.00	132096.00	132096.00	132098.50
18.12.90 35934467.00	90.00	132052.00	132096.00	132096.00	132098.00

18.12.91 35934628.00	91.00	132054.00	132096.00	132096.00	132098.00
18.12.92 35934750.00	92.00	132048.00	132096.00	132096.00	132098.00
18.12.93 35935077.00	93.00	132044.00	132096.00	132096.00	132098.00
18.12.94 35934463.00	94.00	132051.00	132096.00	132096.00	132098.00
18.12.95 35934605.00	95.00	132053.00	132096.00	132096.00	132098.00
18.12.96 36674683.00	96.00	132043.00	132096.00	132096.00	132098.00
18.12.97 36673978.00	97.00	132036.00	132096.00	132096.00	132098.00
18.12.98 36674419.00	98.00	132067.00	132096.00	132096.00	132098.00
18.12.99 36674229.00	99.00	132066.00	132096.00	132096.00	132098.00
18.12.100 36674679.00	100.00	132040.00	132096.00	132096.00	132098.00
18.12.101 36673968.00	101.00	132039.00	132096.00	132096.00	132098.00
18.12.102 36674409.00	102.00	132064.00	132096.00	132096.00	132098.00

18.12.103 36674192.00	103.00	132060.00	132096.00	132096.00	132098.00
18.12.104 36675074.00	104.00	132039.00	132096.00	132096.00	132098.00
18.12.105 36674572.00	105.00	132032.00	132096.00	132096.00	132098.00
18.12.106 36674951.00	106.00	132065.00	132096.00	132096.00	132098.00
18.12.107 36674828.00	107.00	132059.00	132096.00	132096.00	132098.00
18.12.108 36675065.00	108.00	132033.00	132096.00	132096.00	132098.00
18.12.109 36674575.00	109.00	132026.00	132096.00	132096.00	132098.00
18.12.110 36674954.00	110.00	132061.00	132096.00	132096.00	132098.00
18.12.111 36674807.00	111.00	132068.00	132096.00	132096.00	132098.00
18.12.112 35923122.00	112.00	132050.00	132096.00	132096.00	132098.00
18.12.113 35923978.00	113.00	132053.00	132096.00	132096.00	132098.00
18.12.114 35923258.00	114.00	132058.00	132096.00	132096.00	132098.00

18.12.115 35923851.00	115.00	132051.00	132096.00	132096.00	132098.00
18.12.116 35923087.00	116.00	132063.00	132096.00	132096.00	132098.00
18.12.117 35924002.00	117.00	132062.00	132096.00	132096.00	132098.00
18.12.118 35923294.00	118.00	132056.00	132096.00	132096.00	132098.00
18.12.119 35923839.00	119.00	132057.00	132096.00	132096.00	132098.00
18.12.120 35925358.00	120.00	132051.00	132096.00	132096.00	132098.00
18.12.121 35926206.00	121.00	132060.00	132096.00	132096.00	132098.00
18.12.122 35925471.00	122.00	132061.00	132096.00	132096.00	132098.00
18.12.123 35925907.00	123.00	132053.00	132096.00	132096.00	132098.00
18.12.124 35925385.00	124.00	132050.00	132096.00	132096.00	132098.00
18.12.125 35926183.00	125.00	132063.00	132096.00	132096.00	132098.00
18.12.126 35925463.00	126.00	132058.00	132096.00	132096.00	132098.00

```

18.12.127      127.00   132053.00   132096.00   132096.00   132098.00
35925909.00

```

1.5 Fusion MatMulBias Kernel

1.5.1 Verify the Fusion MatMulBias Kernel

This also ensures the kernel has been built and cached. The results from the triton kernel are compared against torch.

Small Input

```
[19]: !python triton/matmulbias.py --verify fusion
```

```

Running the fusion matmulbias kernel (input size M:1024 N:1024) ...
triton_output: tensor([[ 2.7480,  43.6250,  30.9531, ...,  26.2500,   8.8203,
  6.4609],
        [  7.1562, -21.3906,  17.7969, ..., -16.8125,  29.6562, -48.3125],
        [ 13.8125,  -6.4609,   0.5146, ...,  48.0000,  25.1562, -27.5156],
        ...,
        [-54.4375,  -1.4375,  -0.4214, ...,   5.4297,  17.2812, -39.0625],
        [  9.6094, -52.2500,  43.0000, ..., -45.8438, -49.3125,  32.6875],
        [-12.8516,   9.5391,  53.3750, ...,  -9.3438,   7.6797,  54.1250]],
        device='cuda:0', dtype=torch.float16)
Running the torch matmulbias kernel ...
torch_output: tensor([[ 2.7480,  43.6562,  30.9531, ...,  26.2500,   8.8203,
  6.4609],
        [  7.1602, -21.3906,  17.7969, ..., -16.8125,  29.6406, -48.3125],
        [ 13.8125,  -6.4609,   0.5142, ...,  48.0312,  25.1406, -27.5156],
        ...,
        [-54.4375,  -1.4375,  -0.4214, ...,   5.4297,  17.2812, -39.0625],
        [  9.6094, -52.2500,  43.0000, ..., -45.8125, -49.3125,  32.7188],
        [-12.8516,   9.5391,  53.3750, ...,  -9.3438,   7.6797,  54.1562]],
        device='cuda:0', dtype=torch.float16)
Verifying triton results with torch ...
Verified

```

Large Input

```
[20]: !python triton/matmulbias.py --M 32768 --N 32768 --verify fusion
```

```

Running the fusion matmulbias kernel (input size M:32768 N:32768) ...
triton_output: tensor([[ -60.8750,   56.6562,   57.0938, ...,  274.0000,
 295.2500,
        14.8828],
        [ 205.3750, -353.7500, -102.1875, ...,  -24.4219,   57.0625,
        -30.7969],
        [  56.2188,  258.7500,   46.6562, ..., -366.0000,  395.2500,

```

```

285.2500],
...,
[ -5.0977,  99.9375, -73.5625, ..., -70.5000,  204.2500,
-156.8750],
[ 86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
100.8750],
[ -27.6406,  93.5625,  77.9375, ..., -213.1250, -323.2500,
-16.8438]], device='cuda:0', dtype=torch.float16)
Running the torch matmulbias kernel ...
torch_output: tensor([[ -60.8750,  56.6562,  57.1250, ..., 274.0000,
295.2500,
14.8828],
[ 205.3750, -353.7500, -102.1875, ..., -24.4219,  57.0312,
-30.7969],
[ 56.2188,  258.7500,  46.6562, ..., -366.0000,  395.2500,
285.2500],
...,
[ -5.0977,  99.9375, -73.5625, ..., -70.5000,  204.2500,
-156.8750],
[ 86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
100.8750],
[ -27.6406,  93.5625,  77.9375, ..., -213.0000, -323.2500,
-16.8438]], device='cuda:0', dtype=torch.float16)
Verifying triton results with torch ...
Verified

```

1.5.2 Profile the Fusion MatMulBias Kernel

Using the Triton Proton CLI Generates a hatchet file, `fusion_matmulbias_x.hatchet`, under the profiles directory.

Small Input

```
[21]: !proton -d tree -n profiles/fusion_matmulbias_1024x1024 -k triton triton/
      ↪matmulbias.py fusion
```

```

Running the fusion matmulbias kernel (input size M:1024 N:1024) ...
triton_output: tensor([[ 2.7480,  43.6250,  30.9531, ..., 26.2500,  8.8203,
6.4609],
[ 7.1562, -21.3906,  17.7969, ..., -16.8125,  29.6562, -48.3125],
[ 13.8125, -6.4609,  0.5146, ..., 48.0000,  25.1562, -27.5156],
...,
[-54.4375, -1.4375, -0.4214, ...,  5.4297,  17.2812, -39.0625],
[ 9.6094, -52.2500,  43.0000, ..., -45.8438, -49.3125,  32.6875],
[-12.8516,  9.5391,  53.3750, ..., -9.3438,  7.6797,  54.1250]],
device='cuda:0', dtype=torch.float16)

```

Large Input

```
[22]: !proton -d tree -n profiles/fusion_matmulbias_32768x32768 -k triton triton/
      ↪matmulbias.py --M 32768 --N 32768 fusion
```

Running the fusion matmulbias kernel (input size M:32768 N:32768) ...

```
triton_output: tensor([[ -60.8750,  56.6562,  57.0938, ..., 274.0000,
295.2500,
    14.8828],
 [ 205.3750, -353.7500, -102.1875, ..., -24.4219,  57.0625,
   -30.7969],
 [  56.2188,  258.7500,  46.6562, ..., -366.0000,  395.2500,
   285.2500],
 ...,
 [  -5.0977,  99.9375, -73.5625, ..., -70.5000,  204.2500,
  -156.8750],
 [  86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
   100.8750],
 [ -27.6406,  93.5625,  77.9375, ..., -213.1250, -323.2500,
  -16.8438]], device='cuda:0', dtype=torch.float16)
```

Using the Triton Proton API Generates a hatchet file, fusion_matmulbias_x.hatchet, under the profiles directory and outputs the profile.

Small Input

```
[23]: !python triton/matmulbias.py --profile fusion
```

Profiling the fusion matmulbias kernel (input size M:1024 N:1024) ...

```
triton_output: tensor([[ 2.7480,  43.6250,  30.9531, ..., 26.2500,  8.8203,
6.4609],
 [  7.1562, -21.3906,  17.7969, ..., -16.8125,  29.6562, -48.3125],
 [ 13.8125,  -6.4609,   0.5146, ...,  48.0000,  25.1562, -27.5156],
 ...,
 [-54.4375,  -1.4375,  -0.4214, ...,   5.4297,  17.2812, -39.0625],
 [  9.6094, -52.2500,  43.0000, ..., -45.8438, -49.3125,  32.6875],
 [-12.8516,   9.5391,  53.3750, ...,  -9.3438,   7.6797,  54.1250]],
 device='cuda:0', dtype=torch.float16)
```

Proton profile results for fusion_matmulbias_1024x1024

```
96.044 37765020802695.430 ROOT
 56.706 nan _ZN2at6native29vectorized_elementwise_kernel
ILi4ENS0_11FillFunctorIiEES5arrayIPcLm1EEEEviT0_T1_
 39.338 92202873718740.719 matmulbias_kernel [M=1024,
N=1024, K=1024] [BM=128, BN=128 BK=64]
```

Legend (Metric: time/ms (inc) Min: 39.34 Max: 96.04)

```
90.37 - 96.04
79.03 - 90.37
67.69 - 79.03
56.35 - 67.69
45.01 - 56.35
```

39.34 - 45.01

name User code Only in left graph
Only in right graph

Large Input

```
[24]: !python triton/matmulbias.py --M 32768 --N 32768 --profile fusion
```

Profiling the fusion matmulbias kernel (input size M:32768 N:32768) ...
triton_output: tensor([[-60.8750, 56.6562, 57.0938, ..., 274.0000,
295.2500,

14.8828],
[205.3750, -353.7500, -102.1875, ..., -24.4219, 57.0625,
-30.7969],
[56.2188, 258.7500, 46.6562, ..., -366.0000, 395.2500,
285.2500],
...,
[-5.0977, 99.9375, -73.5625, ..., -70.5000, 204.2500,
-156.8750],
[86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
100.8750],
[-27.6406, 93.5625, 77.9375, ..., -213.1250, -323.2500,
-16.8438]], device='cuda:0', dtype=torch.float16)

Proton profile results for fusion_matmulbias_32768x32768

4268.561 280251057697337.656 ROOT
0.557 nan _ZN2at6native29vectorized_elementwise_kernelI
Li4ENS0_11FillFunctorIiEESt5arrayIPcLm1EEEEviT0_T1_
4268.003 280287657419565.031 matmulbias_kernel
[M=32768, N=32768, K=32768] [BM=128, BN=128 BK=64]

Legend (Metric: time/ms (inc) Min: 0.56 Max: 4268.56)

3841.76 - 4268.56
2988.16 - 3841.76
2134.56 - 2988.16
1280.96 - 2134.56
427.36 - 1280.96
0.56 - 427.36

name User code Only in left graph
Only in right graph

Using the ROCm Systems Compute profiler Generates files found under the work-
loads/fusion_matmulbias_x/ directory.

Small Input


```

INFO      |-> [rocprofiler-sdk] W20260316 20:36:56.518509
139668751539776 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178110 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:36:56.518628
139668751539776 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:36:57.777962
139668751539776 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:36:59.045226
139668751539776 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.526598 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:36:59.058242
139668751539776 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1201_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:36:59.072112
139668751539776 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1201_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:36:59.390775
139668751539776 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1201_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:36:59.393602
139668751539776 simple_timer.cpp:55] [rocprofv3] output generation ::
0.343173 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:36:59.394674
139668751539776 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.344311 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 2/12][Approximate profiling time left: 53 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:01.912483
140285895417408 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178001 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:01.912590
140285895417408 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:03.171562
140285895417408 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:04.440965
140285895417408 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.528375 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:37:04.454028
140285895417408 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1221_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:04.467919
140285895417408 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1221_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:04.784145
140285895417408 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1221_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:37:04.786981
140285895417408 simple_timer.cpp:55] [rocprofv3] output generation ::
0.340731 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:04.788068
140285895417408 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.341886 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 3/12][Approximate profiling time left: 48 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:06.917662
140258689437248 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128381 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:06.917768
140258689437248 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:08.171576
140258689437248 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:09.421514
140258689437248 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.503746 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:37:09.434653
140258689437248 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1242_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:09.448572
140258689437248 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1242_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:09.470074
140258689437248 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1242_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:37:09.473050
140258689437248 simple_timer.cpp:55] [rocprofv3] output generation ::
0.046545 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:09.473929
140258689437248 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.047490 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 4/12][Approximate profiling time left: 38 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:11.010796
139910121811520 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.164180 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:11.010905
139910121811520 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:12.269283
139910121811520 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:13.531488
139910121811520 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.520584 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:37:13.544460
139910121811520 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1262_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:13.558359
139910121811520 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1262_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:13.811716
139910121811520 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1262_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:37:13.814488
139910121811520 simple_timer.cpp:55] [rocprofv3] output generation ::
0.277811 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:13.815600
139910121811520 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.278991 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 5/12] [Approximate profiling time left: 34 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:16.037366
140096055578176 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.166941 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:16.037470
140096055578176 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:17.293851
140096055578176 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:18.555931
140096055578176 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.518461 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:37:18.568940
140096055578176 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1282_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:18.582753
140096055578176 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1282_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:18.821081
140096055578176 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1282_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:37:18.823962
140096055578176 simple_timer.cpp:55] [rocprofv3] output generation ::
0.262817 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:18.825034
140096055578176 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.263959 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 6/12][Approximate profiling time left: 29 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_0.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:20.716705
139982254255680 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128347 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:20.716804
139982254255680 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:21.969012
139982254255680 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:23.218883
139982254255680 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.502080 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:37:23.232016
139982254255680 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1303_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:23.245984
139982254255680 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1303_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:23.266085
139982254255680 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1303_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:37:23.268967
139982254255680 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044858 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:23.269921
139982254255680 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045881 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 7/12][Approximate profiling time left: 23 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_1.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:24.574933
140343767739968 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128635 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:24.575031
140343767739968 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:25.832375
140343767739968 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:27.078909
140343767739968 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.503878 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:37:27.091949
140343767739968 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1323_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:27.105880
140343767739968 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1323_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:27.125848
140343767739968 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1323_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:37:27.128774
140343767739968 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044661 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:27.129831
140343767739968 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045790 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 8/12][Approximate profiling time left: 18 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_2.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:28.430536
140533304831552 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128125 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:28.430651
140533304831552 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:29.687606
140533304831552 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:30.935619
140533304831552 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.504968 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:37:30.948802
140533304831552 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1348_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:30.962693
140533304831552 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1348_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:30.982722
140533304831552 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1348_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:37:30.985680
140533304831552 simple_timer.cpp:55] [rocprofv3] output generation ::
0.045105 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:30.986603
140533304831552 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.046099 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 9/12][Approximate profiling time left: 13 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_3.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:32.285071
139970835639872 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128063 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:32.285169
139970835639872 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:33.539832
139970835639872 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:34.773081
139970835639872 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.487912 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:37:34.786120
139970835639872 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1368_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:34.800365
139970835639872 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1368_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:34.821256
139970835639872 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1368_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:37:34.824252
139970835639872 simple_timer.cpp:55] [rocprofv3] output generation ::
0.045926 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:34.825212
139970835639872 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.046956 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 10/12][Approximate profiling time left: 8 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_4.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:36.128784
140422938676800 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128255 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:36.128886
140422938676800 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:37.380891
140422938676800 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:38.593745
140422938676800 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.464859 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:37:38.606773
140422938676800 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1388_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:38.621710
140422938676800 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1388_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:38.646467
140422938676800 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1388_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:37:38.649349
140422938676800 simple_timer.cpp:55] [rocprofv3] output generation ::
0.050371 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:38.650289
140422938676800 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.051380 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 11/12][Approximate profiling time left: 4 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_5.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:39.970345
139976304090688 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128385 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:39.970444
139976304090688 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:41.221167
139976304090688 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:42.440115
139976304090688 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.469671 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:37:42.453137
139976304090688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1409_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:42.468121
139976304090688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1409_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:42.493398
139976304090688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1409_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:37:42.496280
139976304090688 simple_timer.cpp:55] [rocprofv3] output generation ::
0.050951 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:42.497243
139976304090688 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.051983 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 12/12][Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_6.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:43.818818
140536525852224 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128443 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:43.818921
140536525852224 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:45.074438
140536525852224 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:37:46.269658
140536525852224 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.450737 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:37:46.282706
140536525852224 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1429_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:46.298337
140536525852224 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1429_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:37:46.315913
140536525852224 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/b031b949a562/1429_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:37:46.318798
140536525852224 simple_timer.cpp:55] [rocprofv3] output generation ::
0.043897 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:37:46.319743
140536525852224 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.044914 sec

INFO      |-> [rocprofiler-sdk]
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/fusion_matmulbias_1024x1024/MI300X_A1
INFO [roofline] No roofline data found. Generating...
Empirical Roofline Calculation
Copyright © 2025 Advanced Micro Devices, Inc. All rights reserved.
Total detected GPU devices: 1
GPU Device 0 (gfx942) with 304 CUs: Profiling...
99% [|||||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.0 ms, mean:4210.6 GB/sec, stdev=8.7 GB/sec
99% [|||||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:2611340115968 bytes, duration:413.7 ms, mean:6318.6 GB/sec, stdev=16.5
GB/sec
99% [|||||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]

```


L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:1632087572480 bytes, duration:63.6 ms, mean:25697.1 GB/sec, stdev=58.3
GB/sec
99% [|||||]
L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:127506841600 bytes, duration:3.8 ms, mean:33556.2 GB/sec, stdev=150.5
GB/sec
99% [|||||]
LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:79691776000 bytes, duration:1.3 ms, mean:63528.9 GB/sec, stdev=153.0
GB/sec
99% [|||||]
Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1081258016768, duration:13.1 ms, mean:82917.3 GFLOPS,
stdev=182.2 GFLOPS
99% [|||||]
Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39616.4 GFLOPS,
stdev=77.6 GFLOPS
99% [|||||]
Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:92.013 ms, mean:5770.5 GFLOPS,
stdev=9.2 GFLOPS
99% [|||||]
Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:265214230528, duration:2.014 ms, mean:131344.4 GFLOPS,
stdev=1485.7 GFLOPS
99% [|||||]
Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:122406567936, duration:1.1 ms, mean:106934.4 GFLOPS,
stdev=1858.8 GFLOPS
99% [|||||]
Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82596.3 GOPS,
stdev=198.9 GOPS
99% [|||||]
Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:265214230528, duration:3.7 ms, mean:72197.1 GOPS,
stdev=485.9 GOPS
99% [|||||]
Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:122406567936, duration:6.2 ms, mean:19773.2 GOPS,
stdev=65.8 GOPS
100% [|||||]
Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
100% [|||||]
Peak MFMA FLOPs (F6), GPU ID: 0, workgroupSize:256, workgroups:38912,

```
INFO [roofline] Empirical Roofline PDFs saved!
```

```
[26]: !ROCPROF=rocprofiler-sdk rocprof-compute profile -n
      ↪ fusion_matmulbias_32768x32768 -- python triton/matmulbias.py --M 32768 --N
      ↪ 32768 fusion
```

```

- - - - -   - - - - -   - - - - -   / _ |           - - - - -   - - - - -   - - -   -   | _ |
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| | | ( ) | ( _ | | ) | | | ( ) | _ | _ _ _ | ( _ | ( ) | | | | | | | | | | | | | | | |
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| _ | \ _ _ / \ _ _ | . _ / | _ | \ _ _ / | _ |           \ _ _ \ _ _ / | _ | | _ | | _ | . _ /
\ _ _ , _ | \ _ _ \ _ _ |
           | |
                                   | |

```

```

https://github.com/ROCm/amdsmi
INFO Rocprofiler-Compute version: 3.2.3
INFO Profiler choice: rocprofiler-sdk
INFO Path:
/workspace/user/workloads/fusion_matmulbias_32768x32768/MI300X_A1
INFO Target: MI300X_A1
INFO Command: python triton/matmulbias.py --M 32768 --N 32768 fusion
INFO Kernel Selection: None
INFO Dispatch Selection: None
INFO Hardware Blocks: []
INFO Report Sections: []
INFO
INFO ~~~~~
INFO Collecting Performance Counters
INFO ~~~~~
INFO
INFO [Run 1/12][Approximate profiling time left: pending first
measurement...]
INFO [profiling] Current input file: /workspace/user/workloads/fusion
n_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_IFETCH_LEVEL.txt
INFO      |-> [rocprofiler-sdk] W20260316 20:39:36.950833
140655005294144 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178356 sec
INFO      |-> [rocprofiler-sdk] W20260316 20:39:36.950941
140655005294144 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec
INFO      |-> [rocprofiler-sdk] W20260316 20:39:38.206977
140655005294144 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)
INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000, 395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [  -5.0977,  99.9375, -73.5625,
..., -70.5000, 204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],

```

```

INFO      |-> [rocprofiler-sdk] [ 86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406, 93.5625, 77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260316 20:39:43.311391
140655005294144 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' :: 6.360451 sec
INFO      |-> [rocprofiler-sdk] E20260316 20:39:43.324763
140655005294144 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1539_kernel_
trace.csv
INFO      |-> [rocprofiler-sdk] E20260316 20:39:43.337836
140655005294144 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1539_counter
_collection.csv
INFO      |-> [rocprofiler-sdk] E20260316 20:39:43.451533
140655005294144 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1539_agent_i
nfo.csv
INFO      |-> [rocprofiler-sdk] W20260316 20:39:43.454288
140655005294144 simple_timer.cpp:55] [rocprofv3] output generation ::
0.137282 sec
INFO      |-> [rocprofiler-sdk] W20260316 20:39:43.455348
140655005294144 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.138406 sec
INFO      |-> [rocprofiler-sdk]
INFO [Run 2/12] [Approximate profiling time left: 1 minute and 24
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:39:45.243750
140650447232576 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.152661 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:39:45.243851
140650447232576 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:39:46.502624
140650447232576 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:39:51.618457
140650447232576 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.374607 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:39:51.631948
140650447232576 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1560_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:39:51.645021
140650447232576 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1560_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:39:51.706606
140650447232576 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1560_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:39:51.709396
140650447232576 simple_timer.cpp:55] [rocprofv3] output generation ::
0.085266 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:39:51.710438
140650447232576 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.086373 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 3/12] [Approximate profiling time left: 1 minute and 14
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:39:53.147040
140155014909504 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128325 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:39:53.147143
140155014909504 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:39:54.402700
140155014909504 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,   57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,   57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,   46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [  -5.0977,   99.9375,  -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125,  -26.3594, -128.8750,
..., -60.7500,  -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,   93.5625,   77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:39:59.478662
140155014909504 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.331519 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:39:59.492506
140155014909504 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1581_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:39:59.505567
140155014909504 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1581_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:39:59.516248
140155014909504 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1581_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:39:59.519171
140155014909504 simple_timer.cpp:55] [rocprofv3] output generation ::
0.034802 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:39:59.520162
140155014909504 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.035858 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 4/12] [Approximate profiling time left: 1 minute and 4
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:01.110939
140513623846464 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178082 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:01.111043
140513623846464 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:02.367307
140513623846464 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:07.470042
140513623846464 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.358999 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:40:07.483506
140513623846464 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1602_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:07.496590
140513623846464 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1602_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:07.611778
140513623846464 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1602_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:40:07.614576
140513623846464 simple_timer.cpp:55] [rocprofv3] output generation ::
0.138854 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:07.615661
140513623846464 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.140003 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 5/12][Approximate profiling time left: 57 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:09.564354
139866448924224 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178488 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:09.564455
139866448924224 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:10.821353
139866448924224 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:15.962374
139866448924224 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.397919 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:40:15.975733
139866448924224 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1623_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:15.988906
139866448924224 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1623_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:16.105463
139866448924224 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1623_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:40:16.108264
139866448924224 simple_timer.cpp:55] [rocprofv3] output generation ::
0.140289 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:16.109305
139866448924224 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.141388 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 6/12][Approximate profiling time left: 49 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_0.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:17.713872
140498169278016 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128276 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:17.713981
140498169278016 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:18.970319
140498169278016 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,   57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,   57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,   46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [  -5.0977,   99.9375,  -73.5625,
...,  -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125,  -26.3594, -128.8750,
...,  -60.7500,  -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,   93.5625,   77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:24.084125
140498169278016 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.370145 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:40:24.097598
140498169278016 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1644_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:24.110540
140498169278016 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1644_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:24.120692
140498169278016 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1644_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:40:24.123586
140498169278016 simple_timer.cpp:55] [rocprofv3] output generation ::
0.033792 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:24.124555
140498169278016 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.034826 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 7/12][Approximate profiling time left: 40 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_1.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:25.409001
140590752046656 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.127777 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:25.409104
140590752046656 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:26.667018
140590752046656 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,   57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,   57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,   46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [  -5.0977,   99.9375,  -73.5625,
...,  -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125,  -26.3594, -128.8750,
...,  -60.7500,  -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,   93.5625,   77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:31.770371
140590752046656 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.361267 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:40:31.783781
140590752046656 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1664_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:31.796890
140590752046656 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1664_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:31.807071
140590752046656 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1664_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:40:31.809995
140590752046656 simple_timer.cpp:55] [rocprofv3] output generation ::
0.033954 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:31.810979
140590752046656 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.035002 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 8/12][Approximate profiling time left: 32 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_2.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:33.089293
140627393916480 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.127760 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:33.089391
140627393916480 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:34.346517
140627393916480 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,   57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,   57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,   46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [  -5.0977,   99.9375,  -73.5625,
...,  -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125,  -26.3594, -128.8750,
...,  -60.7500,  -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,   93.5625,   77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:39.462408
140627393916480 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.373017 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:40:39.475809
140627393916480 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1685_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:39.488920
140627393916480 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1685_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:39.499079
140627393916480 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1685_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:40:39.502039
140627393916480 simple_timer.cpp:55] [rocprofv3] output generation ::
0.033956 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:39.503020
140627393916480 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.035001 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 9/12][Approximate profiling time left: 24 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_3.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:40.781419
139645562818112 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128095 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:40.781518
139645562818112 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:42.039044
139645562818112 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:47.156540
139645562818112 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.375022 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:40:47.169915
139645562818112 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1706_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:47.182992
139645562818112 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1706_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:47.192488
139645562818112 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1706_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:40:47.195448
139645562818112 simple_timer.cpp:55] [rocprofv3] output generation ::
0.033303 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:47.196420
139645562818112 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.034340 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 10/12][Approximate profiling time left: 15 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_4.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:48.473463
139661775289920 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.127847 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:48.473566
139661775289920 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:49.732658
139661775289920 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:54.785763
139661775289920 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.312198 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:40:54.799115
139661775289920 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1727_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:54.812163
139661775289920 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1727_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:40:54.821000
139661775289920 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1727_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:40:54.823966
139661775289920 simple_timer.cpp:55] [rocprofv3] output generation ::
0.032600 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:54.824923
139661775289920 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.033617 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 11/12][Approximate profiling time left: 7 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_5.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:40:56.098401
140351931285056 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128140 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:56.098501
140351931285056 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:40:57.354867
140351931285056 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,   57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,   57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,   46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,   99.9375,  -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125,  -26.3594, -128.8750,
..., -60.7500,  -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,   93.5625,   77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:41:02.414880
140351931285056 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.316379 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:41:02.428261
140351931285056 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1747_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:41:02.441254
140351931285056 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1747_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:41:02.450139
140351931285056 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1747_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:41:02.453089
140351931285056 simple_timer.cpp:55] [rocprofv3] output generation ::
0.032568 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:41:02.454039
140351931285056 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.033584 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 12/12][Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_6.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260316 20:41:03.726413
140638725008960 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.127988 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:41:03.726512
140638725008960 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:41:04.982084
140638725008960 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260316 20:41:10.044071
140638725008960 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.317559 sec

INFO      |-> [rocprofiler-sdk] E20260316 20:41:10.057580
140638725008960 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1768_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:41:10.070497
140638725008960 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1768_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260316 20:41:10.077690
140638725008960 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/b031b949a562/1768_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260316 20:41:10.080592
140638725008960 simple_timer.cpp:55] [rocprofv3] output generation ::
0.030819 sec

INFO      |-> [rocprofiler-sdk] W20260316 20:41:10.081559
140638725008960 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.031850 sec

INFO      |-> [rocprofiler-sdk]
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/fusion_matmulbias_32768x32768/MI300X_A1
INFO [roofline] No roofline data found. Generating...
Empirical Roofline Calculation
Copyright © 2025 Advanced Micro Devices, Inc. All rights reserved.
Total detected GPU devices: 1
GPU Device 0 (gfx942) with 304 CUs: Profiling...
99% [||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.1 ms, mean:4205.1 GB/sec, stdev=9.8 GB/sec
99% [||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:2611340115968 bytes, duration:411.6 ms, mean:6331.9 GB/sec, stdev=17.6
GB/sec
99% [||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]

```

L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:1632087572480 bytes, duration:63.6 ms, mean:25651.5 GB/sec, stdev=50.6
GB/sec
99% [|||||]
L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:127506841600 bytes, duration:3.8 ms, mean:33603.9 GB/sec, stdev=155.7
GB/sec
99% [|||||]
LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:79691776000 bytes, duration:1.3 ms, mean:63722.6 GB/sec, stdev=351.2
GB/sec
99% [|||||]
Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1081258016768, duration:13.0 ms, mean:82943.2 GFLOPS,
stdev=170.6 GFLOPS
99% [|||||]
Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39631.6 GFLOPS,
stdev=68.3 GFLOPS
99% [|||||]
Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:91.774 ms, mean:5769.4 GFLOPS,
stdev=8.9 GFLOPS
99% [|||||]
Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:265214230528, duration:2.012 ms, mean:131391.6 GFLOPS,
stdev=1423.7 GFLOPS
99% [|||||]
Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:122406567936, duration:1.1 ms, mean:106713.8 GFLOPS,
stdev=1435.3 GFLOPS
99% [|||||]
Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82624.9 GOPS,
stdev=171.5 GOPS
99% [|||||]
Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:265214230528, duration:3.7 ms, mean:72189.3 GOPS,
stdev=564.5 GOPS
99% [|||||]
Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:122406567936, duration:6.2 ms, mean:19758.1 GOPS,
stdev=54.7 GOPS
100% [|||||]
Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
100% [|||||]
Peak MFMA FLOPs (F6), GPU ID: 0, workgroupSize:256, workgroups:38912,

```

experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:10200547328000, duration:5.3 ms, mean:1926686.5 GFLOPS,
stdev=14507.5 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:5100273664000, duration:5.4 ms, mean:945742.2 GFLOPS,
stdev=3098.8 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:2550136832000, duration:4.9 ms, mean:527650.7 GFLOPS,
stdev=3870.6 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1275068416000, duration:9.1 ms, mean:141579.4 GFLOPS,
stdev=1078.9 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:637534208000, duration:4.3 ms, mean:145798.2 GFLOPS,
stdev=589.9 GFLOPS
 99% [||||| ]
Peak MFMA IOPs (I8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:5100273664000, duration:5.1 ms, mean:1005567.1 GOPS,
stdev=6221.7 GOPS

```

INFO [roofline] Empirical Roofline PDFs saved!

1.5.3 Analyze the Fusion MatMulBias Kernel Profiles

Using the Triton Proton

Small Input

```
[27]: !proton-viewer -m time/ms,flops16 profiles/fusion_matmulbias_1024x1024.hatchet
```

```

96.044 3627099881472.000 ROOT
 56.706 nan _ZN2at6native29vectorized_elementwise_kernel
ILi4ENS0_11FillFunctorIiEESt5arrayIPcLm1EEEEviT0_T1_
 39.338 3627099881472.000 matmulbias_kernel [M=1024,
N=1024, K=1024] [BM=128, BN=128 BK=64]

```

Legend (Metric: time/ms (inc) Min: 39.34 Max: 96.04)

```

90.37 - 96.04
79.03 - 90.37
67.69 - 79.03
56.35 - 67.69
45.01 - 56.35
39.34 - 45.01

```

```
name User code      Only in left graph
Only in right graph
```

Large Input

```
[28]: !proton-viewer -m time/ms,flops16 profiles/fusion_matmulbias_32768x32768.hatchet
```

```
4268.561 1196268651020288.000 ROOT
    0.557 nan _ZN2at6native29vectorized_elementwise_kernelI
Li4ENS0_11FillFunctorIiEEST5arrayIPcLm1EEEEviT0_T1_
4268.003 1196268651020288.000 matmulbias_kernel
[M=32768, N=32768, K=32768] [BM=128, BN=128 BK=64]
```

Legend (Metric: time/ms (inc) Min: 0.56 Max: 4268.56)

3841.76 - 4268.56
2988.16 - 3841.76
2134.56 - 2988.16
1280.96 - 2134.56
427.36 - 1280.96
0.56 - 427.36

```
name User code      Only in left graph
Only in right graph
```

Using the ROCm Systems Compute analyzer

Small Input

```
[29]: !rocprof-compute analyze -p workloads/fusion_matmulbias_1024x1024/MI300X_A1 -k 0
```

----- /-|----- -|-----
 |' _/ _ \ / _| ' \ | ' _/ _ \ | |----- / _/ _ \ | ' _ \ _ \ | ' \ | | | | _/ _
 \
 | | | () | (_ | |) | | | () | _|-----| (_ | () | | | | | | | | | | | | | | | |
 _/
 | _ \ _ _/ \ _ _ | . _ _/ | _ \ _ _/ | _ _ _ \ _ _/ | _ | | _ | | _ _/
 \ _ _ , _ | \ _ _ \ _ _ |
 | _ |
 | _ |

```
INFO Analysis mode = cli
INFO [analysis] deriving rocprofiler-compute metrics...
WARNING PC sampling: can not detect pc sampling method without /workspa
ce/user/workloads/fusion_matmulbias_1024x1024/MI300X_A1/ps_file_pc_sampling_host
_trap.csv
```

0. Top Stats

0.1 Top Kernels

Kernel_Name				Count	Sum(ns)
Mean(ns)	Median(ns)	Pct	S		
0	void at::native::vectorized_elementwise_ 43282.42 42403.09 54.55 * kernel<4, at::native::FillFunctor<int>, std::array<char*, 1ul> >(int, at::nat...			139.00	6016255.73
1	matmulbias_kernel 26847.17 24866.50 42.11			173.00	4644559.72
2	__amd_rocclr_copyBuffer 2999.23 3000.17 2.42			89.00	266931.59
3	void at::native::(anonymous namespace):: 6265.36 7293.16 0.17 distribution_elementwise_grid_stride_ker nel<float, 4, at::native::templates::...			3.00	18796.08
4	void at::native::(anonymous namespace):: 2336.76 2168.33 0.15 CatArrayBatchedCopy_contig<at::native::(anonymous namespace)::OpaqueType<2u>,...			7.00	16357.34
5	void at::native::vectorized_elementwise_ 3502.34 3514.51 0.10 kernel<4, at::native::compare_scalar_ker nel<double>(at::TensorIteratorBase&, ...			3.00	10507.01

6	void at::native::vectorized_elementwise_	2.00	6421.17
3210.59	3210.59 0.06		
	kernel<8, at::native::UnaryFunctor<c10:		
	:Half, c10::Half, bool, at::native::(...		
7	void at::native::vectorized_elementwise_	2.00	5195.09
2597.54	2597.54 0.05		
	kernel<8, at::native::AbsFunctor<c10::Ha		
	lf>, std::array<char*, 2ul> >(int, at...		
8	void at::native::vectorized_elementwise_	2.00	5121.82
2560.91	2560.91 0.05		
	kernel<8, at::native::BinaryFunctor<c10:		
	:Half, c10::Half, bool, at::native::(...		
9	void at::native::reduce_kernel<512, 1, a	1.00	4353.33
4353.33	4353.33 0.04		
	t::native::ReduceOp<double, at::native::		
	func_wrapper_t<double, at::native::Ma...		

0.2 Dispatch List

Dispatch_ID	Kernel_Name	GPU_ID
0	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double,	0

at::CUDAGeneratorImpl*)::...

```
1          1 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker      0
          nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
          , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
          eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...
```

```
2          2 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker      0
          nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
          , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
          eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...
```

```
3          3 matmulbias_kernel
0
```

```
4          4 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,      0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          l>)
```

```
5          5 matmulbias_kernel
0
```

```
6          6 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,      0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          l>)
```

```
7          7 matmulbias_kernel
```


0

```
8          8 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)
```

```
9          9 matmulbias_kernel
0
```

1. System Info

	Info
workload_name	fusion_matmulbias_1024x1024
command	python triton/matmulbias.py fusion
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Mon Mar 16 20:36:53 2026 (UTC)
version	3
hostname	b031b949a562
cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1
linux_distro	CentOS Stream 10 (Coughlan)
linux_kernel_version	5.15.0-170-generic
amd_gpu_kernel_version	nan
cpu_memory	230893520
gpu_memory	nan

rocm_version	7.0.3
vbios	113-M3000108-103
compute_partition	SPX
memory_partition	nan
gpu_series	MI300
gpu_model	MI300X_A1
gpu_arch	gfx942
gpu_chip_id	29877
gpu_l1	32
gpu_l2	4096
cu_per_gpu	304
simd_per_cu	4
se_per_gpu	32
wave_size	64
workgroup_max_size	1024
max_waves_per_cu	32
max_sclk	2100
max_mclk	1300
cur_sclk	2100
cur_mclk	1300
total_l2_chan	128
lds_banks_per_cu	32
sqc_per_gpu	160
pipes_per_gpu	4

num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID Peak	Metric Pct of Peak	Avg	Unit
2.1.0 81715.20	VALU FLOPs 0.00	0.00	Gflop/s
2.1.1 81715.20	VALU IOPs 0.83	681.31	Giop/s
2.1.2 2614886.40	MFMA FLOPs (F8) 0.00	0.00	Gflop/s
2.1.3 1307443.20	MFMA FLOPs (BF16) 0.00	0.00	Gflop/s
2.1.4 1307443.20	MFMA FLOPs (F16) 0.00	0.00	Gflop/s
2.1.5 163430.40	MFMA FLOPs (F32) 0.00	0.00	Gflop/s
2.1.6 163430.40	MFMA FLOPs (F64) 0.00	0.00	Gflop/s
2.1.7	MFMA FLOPs (F6F4)		Gflop/s
2.1.8	MFMA IOPs (Int8)	0.00	Giop/s

2614886.40	0.00		
2.1.9	Active CUs	242.00	Cus
304.00	79.61		
2.1.10	SALU Utilization	3.12	Pct
100.00	3.12		
2.1.11	VALU Utilization	2.29	Pct
100.00	2.29		
2.1.12	MFMA Utilization	0.00	Pct
100.00	0.00		
2.1.13	VMEM Utilization	0.83	Pct
100.00	0.83		
2.1.14	Branch Utilization	0.62	Pct
100.00	0.62		
2.1.15	VALU Active Threads	64.00	Threads
64.00	100.00		
2.1.16	IPC	0.10	Instr/cycle
5.00	1.99		
2.1.17	Wavefront Occupancy	2384.22	Wavefronts
9728.00	24.51		
2.1.18	Theoretical LDS Bandwidth	0.00	Gb/s
81715.20	0.00		
2.1.19	LDS Bank Conflicts/Access		Conflicts/access
32.00			
2.1.20	vL1D Cache Hit Rate	0.00	Pct

100.00	0.00		
2.1.21 81715.20	vL1D Cache BW 15.25	12458.31	Gb/s
2.1.22 100.00	L2 Cache Hit Rate 50.04	50.04	Pct
2.1.23 34406.40	L2 Cache BW 36.24	12470.24	Gb/s
2.1.24 5324.80	L2-Fabric Read BW 0.01	0.30	Gb/s
2.1.25 5324.80	L2-Fabric Write BW 116.98	6229.16	Gb/s
2.1.26	L2-Fabric Read Latency	341.68	Cycles
2.1.27	L2-Fabric Write Latency	407.72	Cycles
2.1.28 100.00	sL1D Cache Hit Rate 99.74	99.74	Pct
2.1.29 21504.00	sL1D Cache BW 0.45	97.33	Gb/s
2.1.30 100.00	L1I Hit Rate 99.96	99.96	Pct
2.1.31 21504.00	L1I BW 3.17	681.31	Gb/s
2.1.32	L1I Fetch Latency		Cycles

3.1 Memory Chart

Instr Buff	Instr Dispatch	Exec
LDS	L2 Cache	

SALU: 917504

```
Wave 0 Instr Buf 24
2: 304 <-----
```

>

1

RVGPRseq: 12.0

11

```

Wave N-1 Instr Buf      VALU:
720896      Vector L1 Cache
      Request      |
|
|-----|
>
|
v
      SGPRs :      32

      MFMA: 0
      Rd      :      0 Hit      :      0 %      Rd      :
0      Rd      :      4010

|-----|
>      <-----|      <-----|
-
Fabric

Wt      : 16777216Lat      :      N/A      Wr      : 4194304 Wr      :
4194304      Rd      :
205

      VMEM: 262144      LDS Alloc:      -
----->      -----
>      <----- Latency (cycles)      Rd:
213

      ----->
0      Atomic:      0 Coales:      25 %      Atomic:      0      Atomic:
0      Wr      : 4194304      <----- HBM
      <
----->      <-----
>      ----->      Rd      :      342      Wr:
4194304

      LDS: 0
      Stall :      77
cycles      Atomic:      0      Wr      :
408      ----->
      ----->      Scratch Alloc:

----->      Atomic:      0

0

Wave Occupancy      GWS: 0
      Scalar L1D
Cache      Latency (cycles)

```

```

10
--->                                Rd      :   1
60

Wave Life
      <-----
      ^
|
10640
BRANCH: 131072 Wavefronts:   Rd      : 65536 Hit    :  93.0
%   Wr      :    0   Rd      :   N/A
|
|----->                                0 <
-----
>                                |
|

Lat      :   N/A   Atomic:    0   Wr      :
N/A
v
^
      <-----
>

|                                Workgroups:

|

0   Instr L1 Cache
    GMI

|

|

|                                Fetch :
458752                                Hit    :    99 %   Req
: 3840

-----
-----<-----

Lat      : 2240.0 cycles

```

4. Roofline

4.1 Roofline

141012.3
141012GFLOP/s

36456.2

9425.1

2436.7

630.0 63529 GB/s

33556 GB/s

162.9

42.1 4211 GB/s

0.01 0.18 3.16 56.23 1000.00

Performance (GFLOP/sec) Arithmetic Intensity (FLOP)s/Byte)

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct
5.1.1	CPF Stall	0.73	0.06	1.38	Pct
5.1.2	CPF-L2 Utilization	8.53	3.03	11.12	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	0.73	0.06	1.38	Pct

5.2 Packet Processor

Metric_ID	Metric	Avg	Min	Max
Unit				
5.2.0	CPC SYNC FIFO Full Rate			
Pct				
5.2.1	CPC CANE Stall Rate			

Pct

5.2.2 CPC ADC Utilization
Pct

5.2.3 CPC Utilization 100.00 100.00 100.00
Pct

5.2.4 CPC Stall Rate 9.28 7.34 10.46
Pct

5.2.5 CPC Packet Decoding Utilization 27.86 24.72 30.27
Pct

5.2.6 CPC-Workgroup Manager Utilization 67.12 63.57 70.04
Pct

5.2.7 CPC-L2 Utilization 0.22 0.17 0.23
Pct

5.2.8 CPC-UTCL1 Stall 0.14 0.11 0.23
Pct

5.2.9 CPC-UTCL2 Utilization 0.14 0.10 0.21
Pct

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID	Metric	Avg	Min	Max	Unit
-----------	--------	-----	-----	-----	------

6.1.0	Schedule-Pipe Wave Occupancy				Wave
-------	------------------------------	--	--	--	------

6.1.1	Accelerator Utilization	99.90	94.82	113.59	Pct
6.1.2	Scheduler-Pipe Utilization	0.00	0.00	0.00	Pct
6.1.3	Scheduler-Pipe Wave Utilization				Pct
6.1.4	Workgroup Manager Utilization	79.17	77.44	83.27	Pct
6.1.5	Shader Engine Utilization	80.66	75.78	334.45	Pct
6.1.6	SIMD Utilization	77.12	68.93	81.16	Pct
6.1.7 Workgroups	Dispatched Workgroups	0.00	0.00	0.00	
6.1.8 Wavefronts	Dispatched Wavefronts	0.00	0.00	0.00	
6.1.9 Cycles/wave	VGPR Writes				
6.1.10 Cycles/wave	SGPR Writes				

6.2 Workgroup Manager - Resource Allocation

Metric_ID	Metric	Avg	Min	Max
Unit				

6.2.0 Pct	Not-scheduled Rate (Workgroup Manager)	0.00	0.00	0.00
6.2.1 Pct	Not-scheduled Rate (Scheduler-Pipe)	0.00	0.00	0.00
6.2.2 Pct	Scheduler-Pipe FIFO Full Rate			
6.2.3 Pct	Scheduler-Pipe Stall Rate	0.00	0.00	0.00
6.2.4 Pct	Scratch Stall Rate	0.00	0.00	0.00
6.2.5 Pct	Insufficient SIMD Waveslots	0.00	0.00	0.00
6.2.6 Pct	Insufficient SIMD VGPRs	0.00	0.00	0.00
6.2.7 Pct	Insufficient SIMD SGPRs	0.00	0.00	0.00
6.2.8 Pct	Insufficient CU LDS	0.00	0.00	0.00
6.2.9 Pct	Insufficient CU Barriers	0.00	0.00	0.00
6.2.10 Pct	Reached CU Workgroup Limit	0.00	0.00	0.00
6.2.11 Pct	Reached CU Wavefront Limit	0.00	0.00	0.00

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID Unit	Metric	Avg	Min	Max
7.1.0 Work items	Grid Size	4194304.00	4194304.00	4194304.00
7.1.1 Work items	Workgroup Size	256.00	256.00	256.00
7.1.2 Wavefronts	Total Wavefronts	0.00	0.00	0.00
7.1.3 Wavefronts	Saved Wavefronts	0.00	0.00	0.00
7.1.4 Wavefronts	Restored Wavefronts	0.00	0.00	0.00
7.1.5 Registers	VGPRs	12.00	12.00	12.00
7.1.6 Registers	AGPRs	4.00	4.00	4.00
7.1.7 Registers	SGPRs	32.00	32.00	32.00
7.1.8 Bytes	LDS Allocation	0.00	0.00	0.00

7.1.9	Scratch Allocation	0.00	0.00	0.00
Bytes/workitem				

7.2 Wavefront Runtime Stats

Metric_ID	Metric	Avg	Min
Max Unit			
7.2.0	Kernel Time (Nanosec)	43282.42	42262.42
57790.84 Ns			
7.2.1	Kernel Time (Cycles)	104048.43	100198.62
136720.62 Cycle			
7.2.2	Instructions per wavefront	37.00	37.00
37.00 Instr/wavefront			
7.2.3	Wave Cycles	697273871.42	682545348.00
948990444.00 Cycles per kernel			
7.2.4	Dependency Wait Cycles	157443488.40	148754352.00
647592844.00 Cycles per kernel			
7.2.5	Issue Wait Cycles	535589857.50	522815344.00
747615348.00 Cycles per kernel			
7.2.6	Active Cycles	8650752.00	8650752.00
8650752.00 Cycles per kernel			
7.2.7	Wavefront Occupancy	2384.22	173.67
7073.45 Wavefronts			

10. Compute Units - Instruction Mix
 10.1 Overall Instruction Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.1.0	VALU	720896.00	720896.00	720896.00	Instr per kernel
10.1.1	VMEM	262144.00	262144.00	262144.00	Instr per kernel
10.1.2	LDS	0.00	0.00	0.00	Instr per kernel
10.1.3	MFMA	0.00	0.00	0.00	Instr per kernel
10.1.4	SALU	917504.00	917504.00	917504.00	Instr per kernel
10.1.5	SMEM	65536.00	65536.00	65536.00	Instr per kernel
10.1.6	Branch	131072.00	131072.00	131072.00	Instr per kernel

10.2 VALU Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.2.0 kernel	INT32	393216.00	393216.00	393216.00	Instr per
10.2.1 kernel	INT64	65536.00	65536.00	65536.00	Instr per

10.2.2 kernel	F16-ADD	0.00	0.00	0.00	Instr per
10.2.3 kernel	F16-MUL	0.00	0.00	0.00	Instr per
10.2.4 kernel	F16-FMA	0.00	0.00	0.00	Instr per
10.2.5 kernel	F16-Trans	0.00	0.00	0.00	Instr per
10.2.6 kernel	F32-ADD	0.00	0.00	0.00	Instr per
10.2.7 kernel	F32-MUL	0.00	0.00	0.00	Instr per
10.2.8 kernel	F32-FMA	0.00	0.00	0.00	Instr per
10.2.9 kernel	F32-Trans	0.00	0.00	0.00	Instr per
10.2.10 kernel	F64-ADD	0.00	0.00	0.00	Instr per
10.2.11 kernel	F64-MUL	0.00	0.00	0.00	Instr per
10.2.12 kernel	F64-FMA	0.00	0.00	0.00	Instr per
10.2.13 kernel	F64-Trans	0.00	0.00	0.00	Instr per

10.2.14	Conversion	0.00	0.00	0.00	Instr per kernel
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10.3 VMEM Instr Mix

Metric_ID	Metric		Avg	Min
Max	Unit			

10.3.0	Global/Generic Instr		262144.00	262144.00
262144.00	Instr per kernel			

10.3.1	Global/Generic Read		0.00	0.00
0.00	Instr per kernel			

10.3.2	Global/Generic Write		262144.00	262144.00
262144.00	Instr per kernel			

10.3.3	Global/Generic Atomic		0.00	0.00
0.00	Instr per kernel			

10.3.4	Spill/Stack Instr		0.00	0.00
0.00	Instr per kernel			

10.3.5	Spill/Stack Coalesceable Instr			
	Instr per kernel			

10.3.6	Spill/Stack Read		0.00	0.00
0.00	Instr per kernel			

10.3.7	Spill/Stack Write		0.00	0.00
0.00	Instr per kernel			

10.3.8	Spill/Stack Atomic		0.00	0.00
0.00	Instr per kernel			

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.4.0	MFMA-I8	0.00	0.00	0.00	Instr per kernel
10.4.1	MFMA-F8	0.00	0.00	0.00	Instr per kernel
10.4.2	MFMA-F16	0.00	0.00	0.00	Instr per kernel
10.4.3	MFMA-BF16	0.00	0.00	0.00	Instr per kernel
10.4.4	MFMA-F32	0.00	0.00	0.00	Instr per kernel
10.4.5	MFMA-F64	0.00	0.00	0.00	Instr per kernel
10.4.6	MFMA-F6F4				Instr per kernel

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak	Pct of Peak
11.1.0	VALU FLOPs	0.00	Gflop	81715.20	0.00
11.1.1	VALU IOPs	681.31	Giop	81715.20	0.83
11.1.2	MFMA FLOPs (F8)	0.00	Gflop	2614886.40	0.00
11.1.3	MFMA FLOPs (BF16)	0.00	Gflop	1307443.20	0.00
11.1.4	MFMA FLOPs (F16)	0.00	Gflop	1307443.20	0.00

11.1.5	MFMA FLOPs (F32)	0.00	Gflop	163430.40	0.00
11.1.6	MFMA FLOPs (F64)	0.00	Gflop	163430.40	0.00
11.1.7	MFMA FLOPs (F6F4)		Gflop		
11.1.8	MFMA IOPs (INT8)	0.00	Giop	2614886.40	0.00

11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
11.2.0	IPC	0.10	0.07	0.10	Instr/cycle
11.2.1	IPC (Issued)	1.00	1.00	1.00	Instr/cycle
11.2.2	SALU Utilization	3.12	2.37	3.23	Pct
11.2.3	VALU Utilization	2.29	1.73	2.37	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	0.83	0.63	0.86	Pct

11.2.6	Branch Utilization	0.62	0.47	0.65	Pct
11.2.7	VALU Active Threads	64.00	64.00	64.00	Threads
11.2.8	MFMA Utilization	0.00	0.00	0.00	Pct
11.2.9	MFMA Instr Cycles				Cycles/instr
11.2.10	VMEM Latency				Cycles
11.2.11	SMEM Latency				Cycles

11.3 Arithmetic Operations

Metric_ID	Metric	Avg	Min	Max	Unit
11.3.0 per kernel	FLOPs (Total)	0.00	0.00	0.00	Ops
11.3.1 per kernel	IOPs (Total)	29360128.00	29360128.00	29360128.00	Ops
11.3.2 per kernel	F8 OPs	0.00	0.00	0.00	Ops
11.3.3 per kernel	F16 OPs	0.00	0.00	0.00	Ops
11.3.4	BF16 OPs	0.00	0.00	0.00	Ops

per kernel

11.3.5	F32 OPs	0.00	0.00	0.00	Ops
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per kernel

11.3.6	F64 OPs	0.00	0.00	0.00	Ops
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per kernel

11.3.7	F6F4 OPs				Ops
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per kernel

11.3.8	INT8 OPs	0.00	0.00	0.00	Ops
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per kernel

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	0.00	Pct of peak
12.1.1	Access Rate	0.00	Pct of peak
12.1.2	Theoretical Bandwidth (% of Peak)	0.00	Pct of peak
12.1.3	Bank Conflict Rate		Pct of peak

12.2 LDS Stats

Metric_ID	Metric	Avg	Min	Max	Unit
12.2.0	LDS Instrs	0.00	0.00	0.00	Instr per

kernel

12.2.1	LDS LOAD				Instr per
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kernel

12.2.2 kernel	LDS STORE					Instr per
12.2.3 kernel	LDS ATOMIC					Instr per
12.2.4	LDS LOAD Bandwidth					Gbps
12.2.5	LDS STORE Bandwidth					Gbps
12.2.6	LDS ATOMIC Bandwidth					Gbps
12.2.7 kernel	Theoretical Bandwidth	0.00	0.00	0.00		Bytes per
12.2.8	LDS Latency					Cycles
12.2.9	Bank Conflicts/Access Conflicts/access					
12.2.10 kernel	Index Accesses	0.00	0.00	0.00		Cycles per
12.2.11 kernel	Atomic Return Cycles	0.00	0.00	0.00		Cycles per
12.2.12 kernel	Bank Conflict	0.00	0.00	0.00		Cycles per
12.2.13 kernel	Addr Conflict	0.00	0.00	0.00		Cycles per

12.2.14	Unaligned Stall	0.00	0.00	0.00	Cycles per kernel
12.2.15	Mem Violations	0.00	0.00	0.00	Accesses per kernel
12.2.16	LDS Command FIFO Full Rate				Cycles per kernel
12.2.17	LDS Data FIFO Full Rate				Cycles per kernel

13. Instruction Cache

13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	3.17	Pct of peak
13.1.1	Cache Hit Rate	99.27	Pct of peak
13.1.2	L1I-L2 Bandwidth	0.01	Pct of peak

13.2 Instruction Cache Accesses

Metric_ID	Metric	Avg	Min	Max
Unit				
13.2.0	Req	458752.00	458752.00	458752.00
Req per kernel				
13.2.1	Hits	455364.29	454341.00	455632.00
Hits per kernel				
13.2.2	Misses - Non Duplicated	160.00	160.00	160.00

Misses per kernel

13.2.3	Misses - Duplicated	3181.53	2983.00	4113.00
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Misses per kernel

13.2.4	Cache Hit Rate	99.27	99.07	99.31
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Pct

13.2.5	Instruction Fetch Latency
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Cycles

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
13.3.0	L1I-L2 Bandwidth	245760.00	245760.00	245760.00	Bytes per kernel

14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.45	Pct of peak
14.1.1	Cache Hit Rate	92.76	Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
14.2.0	Req	65536.00	65536.00	65536.00	Req

per kernel

14.2.1	Hits	60792.42	60648.00	61233.00	Req
per kernel					

14.2.2	Misses - Non Duplicated	160.00	160.00	160.00	Req
per kernel					

14.2.3	Misses- Duplicated	4585.14	3271.00	4721.00	Req
per kernel					

14.2.4	Cache Hit Rate	92.76	92.56	94.65	Pct
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14.2.5	Read Req (Total)	65536.00	65536.00	65536.00	Req
per kernel					

14.2.6	Atomic Req	0.00	0.00	0.00	Req
per kernel					

14.2.7	Read Req (1 DWord)	0.00	0.00	0.00	Req
per kernel					

14.2.8	Read Req (2 DWord)	0.00	0.00	0.00	Req
per kernel					

14.2.9	Read Req (4 DWord)	65536.00	65536.00	65536.00	Req
per kernel					

14.2.10	Read Req (8 DWord)	0.00	0.00	0.00	Req
per kernel					

14.2.11	Read Req (16 DWord)	0.00	0.00	0.00	Req
per kernel					

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	10240.00	10240.00	10240.00	Bytes per kernel
14.3.1	Read Req	160.00	160.00	160.00	Req per kernel
14.3.2	Write Req	0.00	0.00	0.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel
14.3.4	Stall Cycles	0.00	0.00	0.00	Cycles per kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric_ID	Metric	Avg	Min
Max	Unit		
15.1.0	Address Processing Unit Busy	74.89	73.14
79.39	Pct		
15.1.1	Address Stall	0.00	0.00
0.00	Pct		
15.1.2	Data Stall	61.55	59.94
68.90	Pct		

15.1.3	Data-Processor → Address Stall	0.00	0.00
0.00	Pct		
15.1.4	Sequencer → TA Address Stall		
	Cycles per kernel		
15.1.5	Sequencer → TA Command Stall		
	Cycles per kernel		
15.1.6	Sequencer → TA Data Stall		
	Cycles per kernel		
15.1.7	Total Instructions	262144.00	262144.00
262144.00	Instructions per kernel		
15.1.8	Global/Generic Instructions	262144.00	262144.00
262144.00	Instructions per kernel		
15.1.9	Global/Generic Read Instructions	0.00	0.00
0.00	Instructions per kernel		
15.1.10	Global/Generic Read Instructions for LDS		
	Instructions per kernel		
15.1.11	Global/Generic Write Instructions	262144.00	262144.00
262144.00	Instructions per kernel		
15.1.12	Global/Generic Atomic Instructions	0.00	0.00
0.00	Instructions per kernel		
15.1.13	Spill/Stack Instructions	0.00	0.00
0.00	Instructions per kernel		
15.1.14	Spill/Stack Read Instructions	0.00	0.00
0.00	Instructions per kernel		

15.1.15	Spill/Stack Read Instructions for LDS		
	Instructions per kernel		
15.1.16	Spill/Stack Write Instructions	0.00	0.00
0.00	Instructions per kernel		
15.1.17	Spill/Stack Atomic Instructions	0.00	0.00
0.00	Instructions per kernel		
15.1.18	Spill/Stack Total Cycles	0.00	0.00
0.00	Cycles per kernel		
15.1.19	Spill/Stack Coalesced Read	0.00	0.00
0.00	Cycles per kernel		
15.1.20	Spill/Stack Coalesced Write	0.00	0.00
0.00	Cycles per kernel		

15.2 Data-Return Path

Metric_ID	Metric	Avg	Min
Max Unit			
15.2.0	Data-Return Busy	75.78	67.78
80.07 Pct			
15.2.1	Cache RAM → Data-Return Stall	75.07	73.39
79.76 Pct			
15.2.2	Workgroup manager → Data-Return Stall	0.00	0.00
0.00 Pct			
15.2.3	Coalescable Instructions	0.00	0.00
0.00 Instructions per kernel			

15.2.4	Read Instructions	0.00	0.00
0.00	Instructions per kernel		

15.2.5	Write Instructions	262144.00	262144.00
262144.00	Instructions per kernel		

15.2.6	Atomic Instructions	0.00	0.00
0.00	Instructions per kernel		

15.2.7	Write Ack Instructions
	Instructions per kernel

INFO Not showing table with empty column(s): 16.6 L1D Addr Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
16.1.0	Hit rate	0.00	Pct of peak
16.1.1	Bandwidth	15.25	Pct of peak
16.1.2	Utilization	94.52	Pct of peak
16.1.3	Coalescing	25.00	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3	Max
16.2.0	Stalled on L2 Data	92.06	92.79	93.12	93.40	95.24
16.2.1	Stalled on L2 Req	75.78	76.69	76.95	77.19	83.22

16.2.2	Stalled on Address				
16.2.3	Stalled on Data				
16.2.4	Stalled on Latency FIFO				
16.2.5	Stalled on Request FIFO				
16.2.6	Stalled on Read Return				
16.2.7 0.00	Tag RAM Stall (Read)	0.00	0.00	0.00	0.00
16.2.8 0.00	Tag RAM Stall (Write)	0.00	0.00	0.00	0.00
16.2.9 0.00	Tag RAM Stall (Atomic)	0.00	0.00	0.00	0.00
16.3 L1D Cache Accesses					
Metric_ID Unit	Metric	Avg	Min	Max	
16.3.0 Req per kernel	Total Req	16777216.00	16777216.00	16777216.00	
16.3.1 Req per kernel	Read Req	0.00	0.00	0.00	

16.3.2	Write Req	16777216.00	16777216.00	16777216.00
	Req per kernel			
16.3.3	Atomic Req	0.00	0.00	0.00
	Req per kernel			
16.3.4	Cache BW	536870912.00	536870912.00	536870912.00
	Bytes per kernel			
16.3.5	Cache Hit Rate	0.00	0.00	0.00
	Pct			
16.3.6	Cache Accesses	4194304.00	4194304.00	4194304.00
	Req per kernel			
16.3.7	Cache Hits	0.00	0.00	0.00
	Req per kernel			
16.3.8	Invalidations	304.00	304.00	304.00
	Req per kernel			
16.3.9	L1-L2 BW	268435456.00	268435456.00	268435456.00
	Bytes per kernel			
16.3.10	Tag RAM 0 Req			
	Req per kernel			
16.3.11	Tag RAM 1 Req			
	Req per kernel			
16.3.12	Tag RAM 2 Req			
	Req per kernel			
16.3.13	Tag RAM 3 Req			
	Req per kernel			

16.3.14	L1-L2 Read	0.00	0.00	0.00
Req per kernel				

16.3.15	L1-L2 Write	4194304.00	4194304.00	4194304.00
Req per kernel				

16.3.16	L1-L2 Atomic	0.00	0.00	0.00
Req per kernel				

16.3.17	L1 Access Latency			
Cycles				

16.3.18	L1-L2 Read Latency			
Cycles				

16.3.19	L1-L2 Write Latency			
Cycles				

16.4 L1D - L2 Transactions

Metric_ID	Metric	Xfer	Coherency	Avg	Min
Max	Unit				

16.4.0	NC - Read	Read	NC	0.00	0.00
0.00	Req per kernel				

16.4.1	UC - Read	Read	UC	0.00	0.00
0.00	Req per kernel				

16.4.2	CC - Read	Read	CC	0.00	0.00
0.00	Req per kernel				

16.4.3	RW - Read	Read	RW	0.00	0.00
0.00	Req per kernel				

16.4.4	RW - Write	Write	RW	4194304.00	4194304.00
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4194304.00 Req per kernel

16.4.5	NC - Write	Write	NC	0.00	0.00
0.00	Req per kernel				

16.4.6	UC - Write	Write	UC	0.00	0.00
0.00	Req per kernel				

16.4.7	CC - Write	Write	CC	0.00	0.00
0.00	Req per kernel				

16.4.8	NC - Atomic	Atomic	NC	0.00	0.00
0.00	Req per kernel				

16.4.9	UC - Atomic	Atomic	UC	0.00	0.00
0.00	Req per kernel				

16.4.10	CC - Atomic	Atomic	CC	0.00	0.00
0.00	Req per kernel				

16.4.11	RW - Atomic	Atomic	RW	0.00	0.00
0.00	Req per kernel				

16.5 L1D Addr Translation

Metric_ID	Metric	Avg	Min
Max	Units		

16.5.0	Req	4194304.00	4194304.00
4194304.00	Req per kernel		

16.5.1	Inflight Req
Req per kernel	

16.5.2	Hit Ratio	91.80	91.72
92.73	Pct		

16.5.3	Hits	3850476.03	3846979.00
3889510.00	Req per kernel		

16.5.4	Translation Misses	16376.53	16366.00
16383.00	Req per kernel		

16.5.5 Misses under Translation Miss
Req per kernel

16.5.6	Permission Misses	0.00	0.00
0.00	Req per kernel		

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache

17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
17.1.0	Utilization	86.52	Pct
17.1.1	Bandwidth	36.24	Pct
17.1.2	Hit Rate	50.04	Pct
17.1.3	L2-Fabric Read BW	0.30	Gb/s
17.1.4	L2-Fabric Write and Atomic BW	6229.16	Gb/s
17.1.5	HBM Bandwidth	5324.80	Gb/s

17.2 L2 - Fabric Transactions

Metric_ID	Metric	Avg	Min
Max	Unit		
17.2.0	Read BW	13144.40	12800.00
24832.00	Bytes per kernel		

17.2.1	HBM Read Traffic	104.07	92.59
108.50	Pct		
17.2.2	Remote Read Traffic	0.37	0.00
7.41	Pct		
17.2.3	Uncached Read Traffic	4.14	0.00
93.81	Pct		
17.2.4	Write and Atomic BW	268435456.00	268435456.00
268435456.00	Bytes per kernel		
17.2.5	HBM Write and Atomic Traffic	100.00	100.00
100.00	Pct		
17.2.6	Remote Write and Atomic Traffic	0.00	0.00
0.00	Pct		
17.2.7	Atomic Traffic	0.00	0.00
0.00	Pct		
17.2.8	Uncached Write and Atomic Traffic	0.00	0.00
0.00	Pct		
17.2.9	Read Latency	341.68	279.79
527.89	Cycles		
17.2.10	Write and Atomic Latency	407.72	392.23
563.45	Cycles		
17.2.11	Atomic Latency		
	Cycles		
17.2.12	Read Stall		
	Pct		

17.2.13 Write Stall
 Pct

17.3 L2 Cache Accesses

Metric_ID	Metric	Avg	Min
Max Unit			

17.3.0	Bandwidth	537384837.53	537382912.00
537417088.00	Bytes per kernel		

17.3.1 Read Bandwidth
Bytes per kernel

17.3.2 Write Bandwidth
Bytes per kernel

17.3.3 Atomic Bandwidth
Bytes per kernel

17.3.4	Req	4198319.04	4198304.00
4198571.00	Req per kernel		

17.3.5	Read Req	4009.69	4000.00
4194.00	Req per kernel		

17.3.6	Write Req	4194304.00	4194304.00
4194304.00	Req per kernel		

17.3.7	Atomic Req	0.00	0.00
0.00	Req per kernel		

17.3.8	Streaming Req	0.00	0.00
0.00	Req per kernel		

17.3.9	Bypassss Req		
	Req per kernel		
17.3.10	Probe Req	0.00	0.00
0.00	Req per kernel		
17.3.11	Input Buffer Req		
	Req per kernel		
17.3.12	Cache Hit	50.04	50.04
50.05	Pct		
17.3.13	Hits	2101048.00	2101048.00
2101048.00	Hits per kernel		
17.3.14	Misses	2097271.04	2097256.00
2097523.00	Misses per kernel		
17.3.15	Writeback	2097204.49	2097200.00
2097208.00	Cachelines per kernel		
17.3.16	Writeback (Internal)	1966080.08	1966080.00
1966088.00	Cachelines per kernel		
17.3.17	Writeback (vL1D Req)	131072.04	131071.00
131081.00	Cachelines per kernel		
17.3.18	Evict (Internal)	1835118.33	1835103.00
1835124.00	Cachelines per kernel		
17.3.19	Evict (vL1D Req)	0.00	0.00
0.00	Cachelines per kernel		
17.3.20	NC Req	0.00	0.00
0.00	Req per kernel		

17.3.21	UC Req	13.46	0.00
190.00	Req per kernel		

17.3.22	CC Req	0.00	0.00
0.00	Req per kernel		

17.3.23	RW Req	4198304.00	4198304.00
4198304.00	Req per kernel		

17.5 L2 - Fabric Interface Stalls

Metric_ID Transaction	Metric Avg	Min	Max	Unit	Type	
17.5.0	Read - PCIe Stall Pct				PCIe Stall	Read
17.5.1	Read - Infinity Fabric Stall Pct				Infinity Fabric Stall	Read
17.5.2	Read - HBM Stall Pct				HBM Stall	Read
17.5.3	Write - PCIe Stall Pct				PCIe Stall	Write
17.5.4	Write - Infinity Fabric Stall Pct				Infinity Fabric Stall	Write
17.5.5	Write - HBM Stall Pct				HBM Stall	Write
17.5.6	Write - Credit Starvation				Credit Starvation	Write
0.00	0.00	0.00	Pct			

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID	Metric	Avg	Min
Max Unit			
17.6.0	Read (32B)	0.00	0.00
0.00	Req per kernel		
17.6.1	Read (64B)	205.38	200.00
388.00	Req per kernel		
17.6.2	Read (128B)	0.00	0.00
0.00	Req per kernel		
17.6.3	Read (Uncached)	9.76	0.00
364.00	Req per kernel		
17.6.4	HBM Read	213.46	200.00
390.00	Req per kernel		
17.6.5	Remote Read	0.81	0.00
16.00	Req per kernel		
17.6.6	Read Bandwidth - PCIe		
Bytes per kernel			
17.6.7	Read Bandwidth - Infinity Fabric		
Bytes per kernel			
17.6.8	Read Bandwidth - HBM		
Bytes per kernel			
17.6.9	Write and Atomic (32B)	0.00	0.00
0.00	Req per kernel		
17.6.10	Write and Atomic (Uncached)	0.00	0.00

0.00 Req per kernel

17.6.11	Write and Atomic (64B)	4194304.00	4194304.00
4194304.00	Req per kernel		

17.6.12	HBM Write and Atomic	4194304.00	4194304.00
4194304.00	Req per kernel		

17.6.13	Remote Write and Atomic	0.00	0.00
0.00	Req per kernel		

17.6.14 Write Bandwidth - PCIe
Bytes per kernel

17.6.15 Write Bandwidth - Infinity Fabric
Bytes per kernel

17.6.16 Write Bandwidth - HBM
Bytes per kernel

17.6.17	Atomic	0.00	0.00
0.00	Req per kernel		

17.6.18 Atomic - HBM
Req per kernel

17.6.19 Atomic Bandwidth - PCIe
Bytes per kernel

17.6.20 Atomic Bandwidth - Infinity Fabric
Bytes per kernel

17.6.21 Atomic Bandwidth - HBM
Bytes per kernel

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	50.04	0.00	50.04	50.05	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.2.0	0.00	50.06	50.12	50.12	50.12	50.15
18.2.1	1.00	50.11	50.12	50.12	50.12	50.15
18.2.2	2.00	50.11	50.12	50.12	50.12	50.14
18.2.3	3.00	50.11	50.12	50.12	50.12	50.15
18.2.4	4.00	50.12	50.12	50.12	50.12	50.15
18.2.5	5.00	50.00	50.00	50.00	50.00	50.03
18.2.6	6.00	50.00	50.00	50.00	50.00	50.03
18.2.7	7.00	50.00	50.00	50.00	50.00	50.03
18.2.8	8.00	50.00	50.00	50.00	50.00	50.03
18.2.9	9.00	50.00	50.00	50.00	50.00	50.03
18.2.10	10.00	49.99	50.00	50.00	50.00	50.03
18.2.11	11.00	50.11	50.11	50.12	50.12	50.14
18.2.12	12.00	50.00	50.00	50.00	50.00	50.03
18.2.13	13.00	50.00	50.00	50.00	50.00	50.03
18.2.14	14.00	50.00	50.00	50.00	50.00	50.03
18.2.15	15.00	50.00	50.00	50.00	50.00	50.03

18.2.16	16.00	50.00	50.00	50.00	50.00	50.03
18.2.17	17.00	50.00	50.00	50.00	50.00	50.03
18.2.18	18.00	49.99	50.00	50.00	50.00	50.03
18.2.19	19.00	50.11	50.11	50.12	50.12	50.14
18.2.20	20.00	50.00	50.00	50.00	50.00	50.03
18.2.21	21.00	50.00	50.00	50.00	50.00	50.03
18.2.22	22.00	50.00	50.00	50.00	50.00	50.03
18.2.23	23.00	50.00	50.00	50.00	50.00	50.03
18.2.24	24.00	50.06	50.12	50.12	50.12	50.15
18.2.25	25.00	50.11	50.12	50.12	50.12	50.15
18.2.26	26.00	50.11	50.12	50.12	50.12	50.14
18.2.27	27.00	50.11	50.12	50.12	50.12	50.15
18.2.28	28.00	50.12	50.12	50.12	50.12	50.15
18.2.29	29.00	50.00	50.00	50.00	50.00	50.03
18.2.30	30.00	50.00	50.00	50.00	50.00	50.03
18.2.31	31.00	50.00	50.00	50.00	50.00	50.03
18.2.32	32.00	50.05	50.12	50.12	50.12	50.15
18.2.33	33.00	50.11	50.12	50.12	50.12	50.15
18.2.34	34.00	50.10	50.12	50.12	50.12	50.14
18.2.35	35.00	50.11	50.12	50.12	50.12	50.15
18.2.36	36.00	50.12	50.12	50.12	50.12	50.15
18.2.37	37.00	50.00	50.00	50.00	50.00	50.03
18.2.38	38.00	50.00	50.00	50.00	50.00	50.03
18.2.39	39.00	50.00	50.00	50.00	50.00	50.03

18.2.40	40.00	50.00	50.00	50.00	50.00	50.03
18.2.41	41.00	50.00	50.00	50.00	50.00	50.03
18.2.42	42.00	49.99	50.00	50.00	50.00	50.03
18.2.43	43.00	50.11	50.11	50.12	50.12	50.14
18.2.44	44.00	50.00	50.00	50.00	50.00	50.03
18.2.45	45.00	50.00	50.00	50.00	50.00	50.03
18.2.46	46.00	50.00	50.00	50.00	50.00	50.03
18.2.47	47.00	50.00	50.00	50.00	50.00	50.03
18.2.48	48.00	50.00	50.00	50.00	50.00	50.03
18.2.49	49.00	50.00	50.00	50.00	50.00	50.03
18.2.50	50.00	49.99	50.00	50.00	50.00	50.03
18.2.51	51.00	50.11	50.11	50.12	50.12	50.14
18.2.52	52.00	50.00	50.00	50.00	50.00	50.03
18.2.53	53.00	50.00	50.00	50.00	50.00	50.03
18.2.54	54.00	50.00	50.00	50.00	50.00	50.03
18.2.55	55.00	50.00	50.00	50.00	50.00	50.03
18.2.56	56.00	50.05	50.12	50.12	50.12	50.15
18.2.57	57.00	50.11	50.12	50.12	50.12	50.15
18.2.58	58.00	50.11	50.12	50.12	50.12	50.14
18.2.59	59.00	50.11	50.12	50.12	50.12	50.15
18.2.60	60.00	50.12	50.12	50.12	50.12	50.15
18.2.61	61.00	50.00	50.00	50.00	50.00	50.03
18.2.62	62.00	50.00	50.00	50.00	50.00	50.03
18.2.63	63.00	50.00	50.00	50.00	50.00	50.03

18.2.64	64.00	50.00	50.00	50.00	50.00	50.03
18.2.65	65.00	50.00	50.00	50.00	50.00	50.03
18.2.66	66.00	49.99	50.00	50.00	50.00	50.03
18.2.67	67.00	50.11	50.11	50.12	50.12	50.14
18.2.68	68.00	50.00	50.00	50.00	50.00	50.03
18.2.69	69.00	50.00	50.00	50.00	50.00	50.03
18.2.70	70.00	50.00	50.00	50.00	50.00	50.03
18.2.71	71.00	50.00	50.00	50.00	50.00	50.03
18.2.72	72.00	50.01	50.12	50.12	50.12	50.15
18.2.73	73.00	50.11	50.12	50.12	50.12	50.15
18.2.74	74.00	50.11	50.12	50.12	50.12	50.14
18.2.75	75.00	50.11	50.12	50.12	50.12	50.15
18.2.76	76.00	50.12	50.12	50.12	50.12	50.15
18.2.77	77.00	50.00	50.00	50.00	50.00	50.03
18.2.78	78.00	50.00	50.00	50.00	50.00	50.03
18.2.79	79.00	50.00	50.00	50.00	50.00	50.03
18.2.80	80.00	50.03	50.12	50.12	50.12	50.15
18.2.81	81.00	50.11	50.12	50.12	50.12	50.15
18.2.82	82.00	50.11	50.12	50.12	50.12	50.14
18.2.83	83.00	50.11	50.12	50.12	50.12	50.15
18.2.84	84.00	50.12	50.12	50.12	50.12	50.15
18.2.85	85.00	50.00	50.00	50.00	50.00	50.03
18.2.86	86.00	50.00	50.00	50.00	50.00	50.03
18.2.87	87.00	50.00	50.00	50.00	50.00	50.03

18.2.88	88.00	50.00	50.00	50.00	50.00	50.03
18.2.89	89.00	50.00	50.00	50.00	50.00	50.03
18.2.90	90.00	49.99	50.00	50.00	50.00	50.03
18.2.91	91.00	50.11	50.11	50.12	50.12	50.14
18.2.92	92.00	50.00	50.00	50.00	50.00	50.03
18.2.93	93.00	50.00	50.00	50.00	50.00	50.03
18.2.94	94.00	50.00	50.00	50.00	50.00	50.03
18.2.95	95.00	50.00	50.00	50.00	50.00	50.03
18.2.96	96.00	50.00	50.00	50.00	50.00	50.03
18.2.97	97.00	50.00	50.00	50.00	50.00	50.03
18.2.98	98.00	49.99	50.00	50.00	50.00	50.03
18.2.99	99.00	50.11	50.11	50.12	50.12	50.14
18.2.100	100.00	50.00	50.00	50.00	50.00	50.03
18.2.101	101.00	50.00	50.00	50.00	50.00	50.03
18.2.102	102.00	50.00	50.00	50.00	50.00	50.03
18.2.103	103.00	50.00	50.00	50.00	50.00	50.03
18.2.104	104.00	50.04	50.12	50.12	50.12	50.15
18.2.105	105.00	50.11	50.12	50.12	50.12	50.15
18.2.106	106.00	50.10	50.12	50.12	50.12	50.14
18.2.107	107.00	50.11	50.12	50.12	50.12	50.15
18.2.108	108.00	50.12	50.12	50.12	50.12	50.15
18.2.109	109.00	50.00	50.00	50.00	50.00	50.03
18.2.110	110.00	50.00	50.00	50.00	50.00	50.03
18.2.111	111.00	50.00	50.00	50.00	50.00	50.03

18.2.112	112.00	50.05	50.12	50.12	50.12	50.15
18.2.113	113.00	50.11	50.12	50.12	50.12	50.15
18.2.114	114.00	50.11	50.12	50.12	50.12	50.14
18.2.115	115.00	50.11	50.12	50.12	50.12	50.15
18.2.116	116.00	50.12	50.12	50.12	50.12	50.15
18.2.117	117.00	50.00	50.00	50.00	50.00	50.03
18.2.118	118.00	50.00	50.00	50.00	50.00	50.03
18.2.119	119.00	50.00	50.00	50.00	50.00	50.03
18.2.120	120.00	50.00	50.00	50.00	50.00	50.03
18.2.121	121.00	50.00	50.00	50.00	50.00	50.03
18.2.122	122.00	49.99	50.00	50.00	50.00	50.03
18.2.123	123.00	50.11	50.11	50.12	50.12	50.14
18.2.124	124.00	50.00	50.00	50.00	50.00	50.03
18.2.125	125.00	50.00	50.00	50.00	50.00	50.03
18.2.126	126.00	50.00	50.00	50.00	50.00	50.03
18.2.127	127.00	50.00	50.00	50.00	50.00	50.03

18.3 L2 Requests (per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.3.0	0.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.1	1.00	32768.00	32768.00	32768.00	32768.00	32789.00

18.3.2	2.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.3	3.00	32768.00	32768.00	32769.00	32769.00	32789.00
18.3.4	4.00	32768.00	32768.00	32768.00	32768.00	32820.00
18.3.5	5.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.6	6.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.7	7.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.8	8.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.9	9.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.10	10.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.11	11.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.12	12.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.13	13.00	32848.00	32848.00	32848.00	32848.00	32869.00

18.3.14	14.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.15	15.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.16	16.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.17	17.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.18	18.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.19	19.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.20	20.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.21	21.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.22	22.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.23	23.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.24	24.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.25	25.00	32768.00	32768.00	32768.00	32768.00	32789.00

18.3.26	26.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.27	27.00	32768.00	32768.00	32769.00	32769.00	32789.00
18.3.28	28.00	32768.00	32768.00	32768.00	32768.00	32817.00
18.3.29	29.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.30	30.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.31	31.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.32	32.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.33	33.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.34	34.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.35	35.00	32768.00	32768.00	32769.00	32769.00	32789.00
18.3.36	36.00	32768.00	32768.00	32768.00	32768.00	32824.00
18.3.37	37.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.3.38	38.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.39	39.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.40	40.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.41	41.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.42	42.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.43	43.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.44	44.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.45	45.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.46	46.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.47	47.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.48	48.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.49	49.00	32848.00	32848.00	32848.00	32848.00	32869.00

18.3.50	50.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.51	51.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.52	52.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.53	53.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.54	54.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.55	55.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.56	56.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.57	57.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.58	58.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.59	59.00	32768.00	32768.00	32769.00	32769.00	32789.00
18.3.60	60.00	32768.00	32768.00	32768.00	32768.00	32824.00
18.3.61	61.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.3.62	62.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.63	63.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.64	64.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.65	65.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.66	66.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.67	67.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.68	68.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.69	69.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.70	70.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.71	71.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.72	72.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.73	73.00	32768.00	32768.00	32768.00	32768.00	32789.00

18.3.74	74.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.75	75.00	32768.00	32768.00	32769.00	32769.00	32789.00
18.3.76	76.00	32768.00	32768.00	32768.00	32768.00	32805.00
18.3.77	77.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.78	78.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.79	79.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.80	80.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.81	81.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.82	82.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.83	83.00	32768.00	32768.00	32769.00	32769.00	32789.00
18.3.84	84.00	32768.00	32768.00	32768.00	32768.00	32806.00
18.3.85	85.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.3.86	86.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.87	87.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.88	88.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.89	89.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.90	90.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.91	91.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.92	92.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.93	93.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.94	94.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.95	95.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.96	96.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.97	97.00	32848.00	32848.00	32848.00	32848.00	32869.00

18.3.98	98.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.99	99.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.100	100.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.101	101.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.102	102.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.103	103.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.104	104.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.105	105.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.106	106.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.107	107.00	32768.00	32768.00	32769.00	32769.00	32789.00
18.3.108	108.00	32768.00	32768.00	32768.00	32768.00	32807.00
18.3.109	109.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.3.110	110.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.111	111.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.112	112.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.113	113.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.114	114.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.115	115.00	32768.00	32768.00	32769.00	32769.00	32789.00
18.3.116	116.00	32768.00	32768.00	32768.00	32768.00	32807.00
18.3.117	117.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.118	118.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.119	119.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.120	120.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.121	121.00	32848.00	32848.00	32848.00	32848.00	32869.00

18.3.122	122.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.123	123.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.124	124.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.125	125.00	32848.00	32848.00	32848.00	32848.00	32869.00
18.3.126	126.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.127	127.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	1.50	32768.00	0.00
18.4.1	1.00	1.68	32768.00	0.00
18.4.2	2.00	81.60	32768.00	0.00
18.4.3	3.00	1.93	32768.00	0.00
18.4.4	4.00	1.47	32768.00	0.00
18.4.5	5.00	1.19	32768.00	0.00
18.4.6	6.00	1.27	32768.00	0.00
18.4.7	7.00	1.58	32768.00	0.00
18.4.8	8.00	81.19	32768.00	0.00
18.4.9	9.00	81.19	32768.00	0.00

18.4.10	10.00	81.29	32768.00	0.00
18.4.11	11.00	81.17	32768.00	0.00
18.4.12	12.00	1.22	32768.00	0.00
18.4.13	13.00	81.20	32768.00	0.00
18.4.14	14.00	1.20	32768.00	0.00
18.4.15	15.00	1.17	32768.00	0.00
18.4.16	16.00	81.19	32768.00	0.00
18.4.17	17.00	81.19	32768.00	0.00
18.4.18	18.00	81.29	32768.00	0.00
18.4.19	19.00	81.17	32768.00	0.00
18.4.20	20.00	1.22	32768.00	0.00
18.4.21	21.00	81.20	32768.00	0.00
18.4.22	22.00	1.20	32768.00	0.00
18.4.23	23.00	1.17	32768.00	0.00
18.4.24	24.00	1.50	32768.00	0.00
18.4.25	25.00	1.66	32768.00	0.00
18.4.26	26.00	81.60	32768.00	0.00
18.4.27	27.00	1.93	32768.00	0.00
18.4.28	28.00	1.45	32768.00	0.00
18.4.29	29.00	1.19	32768.00	0.00
18.4.30	30.00	1.27	32768.00	0.00
18.4.31	31.00	1.58	32768.00	0.00
18.4.32	32.00	1.50	32768.00	0.00
18.4.33	33.00	1.68	32768.00	0.00

18.4.34	34.00	81.60	32768.00	0.00
18.4.35	35.00	1.93	32768.00	0.00
18.4.36	36.00	1.50	32768.00	0.00
18.4.37	37.00	1.19	32768.00	0.00
18.4.38	38.00	1.27	32768.00	0.00
18.4.39	39.00	1.59	32768.00	0.00
18.4.40	40.00	81.19	32768.00	0.00
18.4.41	41.00	81.19	32768.00	0.00
18.4.42	42.00	81.29	32768.00	0.00
18.4.43	43.00	81.17	32768.00	0.00
18.4.44	44.00	1.22	32768.00	0.00
18.4.45	45.00	81.20	32768.00	0.00
18.4.46	46.00	1.20	32768.00	0.00
18.4.47	47.00	1.17	32768.00	0.00
18.4.48	48.00	81.19	32768.00	0.00
18.4.49	49.00	81.19	32768.00	0.00
18.4.50	50.00	81.29	32768.00	0.00
18.4.51	51.00	81.17	32768.00	0.00
18.4.52	52.00	1.22	32768.00	0.00
18.4.53	53.00	81.20	32768.00	0.00
18.4.54	54.00	1.20	32768.00	0.00
18.4.55	55.00	1.17	32768.00	0.00
18.4.56	56.00	1.50	32768.00	0.00
18.4.57	57.00	1.68	32768.00	0.00

18.4.58	58.00	81.60	32768.00	0.00
18.4.59	59.00	1.93	32768.00	0.00
18.4.60	60.00	1.50	32768.00	0.00
18.4.61	61.00	1.19	32768.00	0.00
18.4.62	62.00	1.27	32768.00	0.00
18.4.63	63.00	1.58	32768.00	0.00
18.4.64	64.00	81.19	32768.00	0.00
18.4.65	65.00	81.19	32768.00	0.00
18.4.66	66.00	81.29	32768.00	0.00
18.4.67	67.00	81.17	32768.00	0.00
18.4.68	68.00	1.22	32768.00	0.00
18.4.69	69.00	81.20	32768.00	0.00
18.4.70	70.00	1.20	32768.00	0.00
18.4.71	71.00	1.17	32768.00	0.00
18.4.72	72.00	1.50	32768.00	0.00
18.4.73	73.00	1.68	32768.00	0.00
18.4.74	74.00	81.60	32768.00	0.00
18.4.75	75.00	1.93	32768.00	0.00
18.4.76	76.00	1.36	32768.00	0.00
18.4.77	77.00	1.19	32768.00	0.00
18.4.78	78.00	1.27	32768.00	0.00
18.4.79	79.00	1.58	32768.00	0.00
18.4.80	80.00	1.50	32768.00	0.00
18.4.81	81.00	1.68	32768.00	0.00

18.4.82	82.00	81.60	32768.00	0.00
18.4.83	83.00	1.93	32768.00	0.00
18.4.84	84.00	1.37	32768.00	0.00
18.4.85	85.00	1.19	32768.00	0.00
18.4.86	86.00	1.27	32768.00	0.00
18.4.87	87.00	1.57	32768.00	0.00
18.4.88	88.00	81.19	32768.00	0.00
18.4.89	89.00	81.19	32768.00	0.00
18.4.90	90.00	81.29	32768.00	0.00
18.4.91	91.00	81.17	32768.00	0.00
18.4.92	92.00	1.22	32768.00	0.00
18.4.93	93.00	81.20	32768.00	0.00
18.4.94	94.00	1.20	32768.00	0.00
18.4.95	95.00	1.17	32768.00	0.00
18.4.96	96.00	81.19	32768.00	0.00
18.4.97	97.00	81.19	32768.00	0.00
18.4.98	98.00	81.29	32768.00	0.00
18.4.99	99.00	81.17	32768.00	0.00
18.4.100	100.00	1.22	32768.00	0.00
18.4.101	101.00	81.20	32768.00	0.00
18.4.102	102.00	1.20	32768.00	0.00
18.4.103	103.00	1.17	32768.00	0.00
18.4.104	104.00	1.50	32768.00	0.00
18.4.105	105.00	1.66	32768.00	0.00

18.4.106	106.00	81.60	32768.00	0.00
18.4.107	107.00	1.93	32768.00	0.00
18.4.108	108.00	1.37	32768.00	0.00
18.4.109	109.00	1.19	32768.00	0.00
18.4.110	110.00	1.27	32768.00	0.00
18.4.111	111.00	1.58	32768.00	0.00
18.4.112	112.00	1.50	32768.00	0.00
18.4.113	113.00	1.68	32768.00	0.00
18.4.114	114.00	81.60	32768.00	0.00
18.4.115	115.00	1.93	32768.00	0.00
18.4.116	116.00	1.37	32768.00	0.00
18.4.117	117.00	1.19	32768.00	0.00
18.4.118	118.00	1.27	32768.00	0.00
18.4.119	119.00	1.58	32768.00	0.00
18.4.120	120.00	81.19	32768.00	0.00
18.4.121	121.00	81.19	32768.00	0.00
18.4.122	122.00	81.29	32768.00	0.00
18.4.123	123.00	81.17	32768.00	0.00
18.4.124	124.00	1.22	32768.00	0.00
18.4.125	125.00	81.20	32768.00	0.00
18.4.126	126.00	1.20	32768.00	0.00
18.4.127	127.00	1.17	32768.00	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
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L2-Fabric Atomic

18.5.0 0.00	0.00	4.43	32768.00
18.5.1 0.00	1.00	4.14	32768.00
18.5.2 0.00	2.00	4.41	32768.00
18.5.3 0.00	3.00	4.09	32768.00
18.5.4 0.00	4.00	4.14	32768.00
18.5.5 0.00	5.00	0.06	32768.00
18.5.6 0.00	6.00	0.28	32768.00
18.5.7 0.00	7.00	0.10	32768.00
18.5.8 0.00	8.00	0.12	32768.00
18.5.9 0.00	9.00	0.12	32768.00
18.5.10 0.00	10.00	0.26	32768.00
18.5.11	11.00	4.12	32768.00

0.00

18.5.12	12.00	0.22	32768.00
0.00			

18.5.13	13.00	0.10	32768.00
0.00			

18.5.14	14.00	0.44	32768.00
0.00			

18.5.15	15.00	0.19	32768.00
0.00			

18.5.16	16.00	0.12	32768.00
0.00			

18.5.17	17.00	0.12	32768.00
0.00			

18.5.18	18.00	0.26	32768.00
0.00			

18.5.19	19.00	4.12	32768.00
0.00			

18.5.20	20.00	0.21	32768.00
0.00			

18.5.21	21.00	0.11	32768.00
0.00			

18.5.22	22.00	0.45	32768.00
0.00			

18.5.23	23.00	0.20	32768.00
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0.00

18.5.24	24.00	4.43	32768.00
0.00			

18.5.25	25.00	4.14	32768.00
0.00			

18.5.26	26.00	4.42	32768.00
0.00			

18.5.27	27.00	4.09	32768.00
0.00			

18.5.28	28.00	4.14	32768.00
0.00			

18.5.29	29.00	0.06	32768.00
0.00			

18.5.30	30.00	0.28	32768.00
0.00			

18.5.31	31.00	0.10	32768.00
0.00			

18.5.32	32.00	4.43	32768.00
0.00			

18.5.33	33.00	4.14	32768.00
0.00			

18.5.34	34.00	4.42	32768.00
0.00			

18.5.35	35.00	4.09	32768.00
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0.00

18.5.36	36.00	4.14	32768.00
0.00			

18.5.37	37.00	0.06	32768.00
0.00			

18.5.38	38.00	0.28	32768.00
0.00			

18.5.39	39.00	0.10	32768.00
0.00			

18.5.40	40.00	0.12	32768.00
0.00			

18.5.41	41.00	0.12	32768.00
0.00			

18.5.42	42.00	0.26	32768.00
0.00			

18.5.43	43.00	4.12	32768.00
0.00			

18.5.44	44.00	0.20	32768.00
0.00			

18.5.45	45.00	0.11	32768.00
0.00			

18.5.46	46.00	0.47	32768.00
0.00			

18.5.47	47.00	0.19	32768.00
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0.00

18.5.48	48.00	0.12	32768.00
0.00			

18.5.49	49.00	0.12	32768.00
0.00			

18.5.50	50.00	0.26	32768.00
0.00			

18.5.51	51.00	4.12	32768.00
0.00			

18.5.52	52.00	0.19	32768.00
0.00			

18.5.53	53.00	0.11	32768.00
0.00			

18.5.54	54.00	0.45	32768.00
0.00			

18.5.55	55.00	0.19	32768.00
0.00			

18.5.56	56.00	4.43	32768.00
0.00			

18.5.57	57.00	4.14	32768.00
0.00			

18.5.58	58.00	4.42	32768.00
0.00			

18.5.59	59.00	4.09	32768.00
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0.00

18.5.60	60.00	4.14	32768.00
0.00			

18.5.61	61.00	0.06	32768.00
0.00			

18.5.62	62.00	0.28	32768.00
0.00			

18.5.63	63.00	0.10	32768.00
0.00			

18.5.64	64.00	0.12	32768.00
0.00			

18.5.65	65.00	0.12	32768.00
0.00			

18.5.66	66.00	0.26	32768.00
0.00			

18.5.67	67.00	4.12	32768.00
0.00			

18.5.68	68.00	0.20	32768.00
0.00			

18.5.69	69.00	0.11	32768.00
0.00			

18.5.70	70.00	0.49	32768.00
0.00			

18.5.71	71.00	0.18	32768.00
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0.00

18.5.72	72.00	4.43	32768.00
0.00			

18.5.73	73.00	4.14	32768.00
0.00			

18.5.74	74.00	4.42	32768.00
0.00			

18.5.75	75.00	4.09	32768.00
0.00			

18.5.76	76.00	4.14	32768.00
0.00			

18.5.77	77.00	0.06	32768.00
0.00			

18.5.78	78.00	0.28	32768.00
0.00			

18.5.79	79.00	0.10	32768.00
0.00			

18.5.80	80.00	4.43	32768.00
0.00			

18.5.81	81.00	4.15	32768.00
0.00			

18.5.82	82.00	4.42	32768.00
0.00			

18.5.83	83.00	4.09	32768.00
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0.00

18.5.84	84.00	4.14	32768.00
0.00			

18.5.85	85.00	0.06	32768.00
0.00			

18.5.86	86.00	0.28	32768.00
0.00			

18.5.87	87.00	0.10	32768.00
0.00			

18.5.88	88.00	0.12	32768.00
0.00			

18.5.89	89.00	0.12	32768.00
0.00			

18.5.90	90.00	0.26	32768.00
0.00			

18.5.91	91.00	4.12	32768.00
0.00			

18.5.92	92.00	0.19	32768.00
0.00			

18.5.93	93.00	0.11	32768.00
0.00			

18.5.94	94.00	0.50	32768.00
0.00			

18.5.95	95.00	0.19	32768.00
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0.00

18.5.96	96.00	0.12	32768.00
0.00			

18.5.97	97.00	0.12	32768.00
0.00			

18.5.98	98.00	0.26	32768.00
0.00			

18.5.99	99.00	4.12	32768.00
0.00			

18.5.100	100.00	0.19	32768.00
0.00			

18.5.101	101.00	0.11	32768.00
0.00			

18.5.102	102.00	0.47	32768.00
0.00			

18.5.103	103.00	0.19	32768.00
0.00			

18.5.104	104.00	4.43	32768.00
0.00			

18.5.105	105.00	4.14	32768.00
0.00			

18.5.106	106.00	4.42	32768.00
0.00			

18.5.107	107.00	4.09	32768.00
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0.00

18.5.108	108.00	4.14	32768.00
0.00			

18.5.109	109.00	0.06	32768.00
0.00			

18.5.110	110.00	0.28	32768.00
0.00			

18.5.111	111.00	0.10	32768.00
0.00			

18.5.112	112.00	4.43	32768.00
0.00			

18.5.113	113.00	4.14	32768.00
0.00			

18.5.114	114.00	4.42	32768.00
0.00			

18.5.115	115.00	4.09	32768.00
0.00			

18.5.116	116.00	4.14	32768.00
0.00			

18.5.117	117.00	0.06	32768.00
0.00			

18.5.118	118.00	0.28	32768.00
0.00			

18.5.119	119.00	0.10	32768.00
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0.00

18.5.120	120.00	0.12	32768.00
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0.00

18.5.121	121.00	0.12	32768.00
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0.00

18.5.122	122.00	0.26	32768.00
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0.00

18.5.123	123.00	4.12	32768.00
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0.00

18.5.124	124.00	0.21	32768.00
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0.00

18.5.125	125.00	0.11	32768.00
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0.00

18.5.126	126.00	0.49	32768.00
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0.00

18.5.127	127.00	0.19	32768.00
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0.00

18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	0.00	0.00	0.00	0.00	141.00
18.6.1	1.00	0.00	0.00	0.00	0.00	449.36
18.6.2	2.00	176.83	238.50	269.75	295.88	534.25
18.6.3	3.00	0.00	0.00	38.00	56.25	169.50
18.6.4	4.00	0.00	0.00	0.00	0.00	6396.50

18.6.5	5.00	0.00	0.00	0.00	0.00	342.00
18.6.6	6.00	0.00	0.00	0.00	0.00	468.00
18.6.7	7.00	0.00	0.00	0.00	0.00	0.00
18.6.8	8.00	589.00	1005.50	1006.00	1011.00	2779.00
18.6.9	9.00	516.00	1033.00	1038.50	1155.00	1694.00
18.6.10	10.00	539.00	1081.00	1086.50	1316.50	2112.00
18.6.11	11.00	181.67	272.75	274.50	275.62	539.50
18.6.12	12.00	0.00	0.00	0.00	0.00	592.00
18.6.13	13.00	1057.00	1060.00	1065.00	1068.50	1925.00
18.6.14	14.00	0.00	0.00	0.00	0.00	630.00
18.6.15	15.00	0.00	0.00	0.00	0.00	0.00
18.6.16	16.00	503.50	1017.00	1023.00	1026.00	2642.00
18.6.17	17.00	522.50	1030.75	1043.00	1090.00	1749.00
18.6.18	18.00	537.00	1088.50	1094.50	1441.25	2050.00
18.6.19	19.00	190.00	274.25	275.50	291.00	558.00
18.6.20	20.00	0.00	0.00	0.00	0.00	619.00
18.6.21	21.00	1064.00	1066.00	1071.00	1074.50	1859.00
18.6.22	22.00	0.00	0.00	0.00	0.00	674.00
18.6.23	23.00	0.00	0.00	0.00	0.00	0.00
18.6.24	24.00	0.00	0.00	0.00	0.00	147.50
18.6.25	25.00	0.00	0.00	0.00	0.00	254.67
18.6.26	26.00	178.50	229.60	271.50	299.20	508.50
18.6.27	27.00	0.00	0.00	45.40	78.88	221.00
18.6.28	28.00	0.00	0.00	0.00	0.00	5907.00

18.6.29	29.00	0.00	0.00	0.00	0.00	343.00
18.6.30	30.00	0.00	0.00	0.00	0.00	522.00
18.6.31	31.00	0.00	0.00	0.00	0.00	0.00
18.6.32	32.00	0.00	0.00	0.00	0.00	128.50
18.6.33	33.00	0.00	0.00	0.00	0.00	318.75
18.6.34	34.00	238.50	309.10	361.50	382.88	614.50
18.6.35	35.00	0.00	0.00	27.00	51.85	176.00
18.6.36	36.00	0.00	0.00	0.00	0.00	4234.75
18.6.37	37.00	0.00	0.00	0.00	0.00	340.00
18.6.38	38.00	0.00	0.00	0.00	0.00	497.00
18.6.39	39.00	0.00	0.00	0.00	0.00	0.00
18.6.40	40.00	928.50	1504.50	1635.00	1764.00	3198.00
18.6.41	41.00	749.00	1468.00	1517.00	1543.75	2010.00
18.6.42	42.00	837.50	1704.25	1807.00	1975.50	2679.00
18.6.43	43.00	302.00	403.88	430.50	454.32	750.25
18.6.44	44.00	0.00	0.00	0.00	0.00	550.00
18.6.45	45.00	1529.00	1627.50	1690.00	1828.50	2339.00
18.6.46	46.00	0.00	0.00	0.00	0.00	525.00
18.6.47	47.00	0.00	0.00	0.00	0.00	0.00
18.6.48	48.00	970.50	1532.00	1627.00	1761.50	3019.00
18.6.49	49.00	733.00	1447.50	1472.50	1506.25	1952.00
18.6.50	50.00	832.50	1634.25	1741.50	1958.25	2387.00
18.6.51	51.00	287.50	395.38	430.25	452.62	732.50
18.6.52	52.00	0.00	0.00	0.00	0.00	553.00

18.6.53	53.00	1583.00	1594.00	1719.00	1829.50	2543.00
18.6.54	54.00	0.00	0.00	0.00	0.00	508.00
18.6.55	55.00	0.00	0.00	0.00	0.00	0.00
18.6.56	56.00	0.00	0.00	0.00	0.00	127.50
18.6.57	57.00	0.00	0.00	0.00	0.00	256.58
18.6.58	58.00	238.33	299.90	359.50	378.75	606.50
18.6.59	59.00	0.00	0.00	24.40	46.50	181.00
18.6.60	60.00	0.00	0.00	0.00	0.00	4197.75
18.6.61	61.00	0.00	0.00	0.00	0.00	345.00
18.6.62	62.00	0.00	0.00	0.00	0.00	515.00
18.6.63	63.00	0.00	0.00	0.00	0.00	0.00
18.6.64	64.00	647.00	1116.00	1167.00	1211.50	2679.00
18.6.65	65.00	491.50	963.00	1018.00	1103.00	1591.00
18.6.66	66.00	607.50	1116.00	1254.00	1533.50	1785.00
18.6.67	67.00	185.33	269.75	296.00	324.75	558.00
18.6.68	68.00	0.00	0.00	0.00	0.00	643.00
18.6.69	69.00	1002.00	1188.50	1238.00	1333.50	1918.00
18.6.70	70.00	0.00	0.00	0.00	0.00	478.00
18.6.71	71.00	0.00	0.00	0.00	0.00	0.00
18.6.72	72.00	0.00	0.00	0.00	0.00	116.25
18.6.73	73.00	0.00	0.00	0.00	0.00	198.00
18.6.74	74.00	157.83	212.28	237.75	267.25	506.50
18.6.75	75.00	0.00	0.00	46.20	69.12	230.00
18.6.76	76.00	0.00	0.00	0.00	0.00	4966.50

18.6.77	77.00	0.00	0.00	0.00	0.00	476.00
18.6.78	78.00	0.00	0.00	0.00	0.00	453.00
18.6.79	79.00	0.00	0.00	0.00	0.00	0.00
18.6.80	80.00	0.00	0.00	0.00	0.00	113.75
18.6.81	81.00	0.00	0.00	0.00	0.00	179.54
18.6.82	82.00	154.17	204.40	231.50	269.25	503.50
18.6.83	83.00	0.00	0.00	46.20	69.12	224.00
18.6.84	84.00	0.00	0.00	0.00	0.00	5084.50
18.6.85	85.00	0.00	0.00	0.00	0.00	475.00
18.6.86	86.00	0.00	0.00	0.00	0.00	450.00
18.6.87	87.00	0.00	0.00	0.00	0.00	0.00
18.6.88	88.00	584.00	1044.00	1194.00	1256.50	2743.00
18.6.89	89.00	475.00	1004.00	1053.50	1176.50	1621.00
18.6.90	90.00	572.00	1199.75	1297.50	1529.50	2048.00
18.6.91	91.00	184.50	274.75	299.20	325.12	543.50
18.6.92	92.00	0.00	0.00	0.00	0.00	596.00
18.6.93	93.00	1045.00	1243.00	1295.00	1345.00	1906.00
18.6.94	94.00	0.00	0.00	0.00	0.00	478.00
18.6.95	95.00	0.00	0.00	0.00	0.00	0.00
18.6.96	96.00	243.00	478.50	583.00	647.00	2201.00
18.6.97	97.00	261.50	504.75	571.50	618.75	1329.00
18.6.98	98.00	291.00	586.25	665.50	948.00	1341.00
18.6.99	99.00	90.67	135.00	146.50	171.38	373.25
18.6.100	100.00	0.00	0.00	0.00	0.00	489.00

18.6.101	101.00	511.00	585.50	645.00	692.00	1352.00
18.6.102	102.00	0.00	0.00	0.00	0.00	577.00
18.6.103	103.00	0.00	0.00	0.00	0.00	0.00
18.6.104	104.00	0.00	0.00	0.00	0.00	135.00
18.6.105	105.00	0.00	0.00	0.00	0.00	199.00
18.6.106	106.00	91.33	116.90	131.50	166.00	448.00
18.6.107	107.00	0.00	0.00	41.25	103.12	287.50
18.6.108	108.00	0.00	0.00	0.00	0.00	6713.25
18.6.109	109.00	0.00	0.00	0.00	0.00	478.00
18.6.110	110.00	0.00	0.00	0.00	0.00	559.00
18.6.111	111.00	0.00	0.00	0.00	0.00	0.00
18.6.112	112.00	0.00	0.00	0.00	0.00	142.00
18.6.113	113.00	0.00	0.00	0.00	0.00	272.58
18.6.114	114.00	86.17	120.50	131.50	160.00	429.00
18.6.115	115.00	0.00	0.00	30.50	100.88	287.75
18.6.116	116.00	0.00	0.00	0.00	0.00	6747.25
18.6.117	117.00	0.00	0.00	0.00	0.00	483.00
18.6.118	118.00	0.00	0.00	0.00	0.00	557.00
18.6.119	119.00	0.00	0.00	0.00	0.00	0.00
18.6.120	120.00	338.00	477.00	502.00	593.50	1943.00
18.6.121	121.00	266.00	507.25	568.50	627.00	1160.00
18.6.122	122.00	264.50	581.50	684.00	897.00	1487.00
18.6.123	123.00	106.20	135.50	157.75	174.12	348.50
18.6.124	124.00	0.00	0.00	0.00	0.00	487.00

18.6.125	125.00	519.00	610.00	658.00	685.00	1282.00
18.6.126	126.00	0.00	0.00	0.00	0.00	576.00
18.6.127	127.00	0.00	0.00	0.00	0.00	0.00

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	328.59	338.98	345.23	350.53	462.33
18.7.1	1.00	320.53	329.02	334.22	338.88	450.74
18.7.2	2.00	338.00	348.29	353.29	357.59	479.93
18.7.3	3.00	335.15	346.07	351.46	355.39	472.87
18.7.4	4.00	358.57	370.30	375.68	383.65	517.13
18.7.5	5.00	348.19	359.76	364.24	371.69	497.98
18.7.6	6.00	371.91	383.33	387.49	393.86	522.70
18.7.7	7.00	358.90	367.19	372.36	377.16	501.53
18.7.8	8.00	411.20	442.78	456.46	462.93	635.63
18.7.9	9.00	459.29	463.66	465.25	467.17	649.08
18.7.10	10.00	404.95	444.15	455.70	465.87	652.65
18.7.11	11.00	404.62	439.46	453.26	462.97	642.99
18.7.12	12.00	384.38	402.56	413.04	425.16	584.55
18.7.13	13.00	379.78	398.87	409.17	418.76	588.53
18.7.14	14.00	387.20	415.72	429.21	443.31	619.84
18.7.15	15.00	400.52	431.38	447.64	464.44	660.99
18.7.16	16.00	410.89	444.12	456.91	464.59	628.59
18.7.17	17.00	455.73	463.77	465.55	467.55	649.76
18.7.18	18.00	397.11	444.80	456.40	466.23	653.97

18.7.19	19.00	403.41	439.75	451.09	464.11	637.27
18.7.20	20.00	383.20	404.71	413.08	425.70	590.44
18.7.21	21.00	383.31	400.38	408.61	420.17	586.87
18.7.22	22.00	385.89	416.45	428.62	442.82	627.15
18.7.23	23.00	398.97	432.81	447.26	462.84	660.31
18.7.24	24.00	321.66	338.51	345.09	351.41	458.17
18.7.25	25.00	317.60	329.05	334.17	338.50	445.21
18.7.26	26.00	333.21	347.58	353.13	358.61	478.04
18.7.27	27.00	329.89	344.90	350.28	355.77	475.41
18.7.28	28.00	354.49	369.64	376.25	382.56	514.19
18.7.29	29.00	345.14	359.20	365.28	370.93	501.74
18.7.30	30.00	368.62	382.72	388.07	392.94	525.67
18.7.31	31.00	354.26	367.14	372.26	376.96	505.73
18.7.32	32.00	426.16	457.48	465.55	470.44	659.95
18.7.33	33.00	460.50	465.19	467.78	469.41	656.71
18.7.34	34.00	407.68	458.19	466.08	471.29	658.25
18.7.35	35.00	411.00	449.95	461.37	471.08	659.68
18.7.36	36.00	388.71	414.43	424.66	438.86	624.09
18.7.37	37.00	384.97	410.49	417.95	428.30	611.45
18.7.38	38.00	394.21	418.61	429.21	440.18	634.64
18.7.39	39.00	402.58	440.58	456.56	467.20	661.64
18.7.40	40.00	332.70	345.68	351.51	358.20	487.08
18.7.41	41.00	324.67	333.70	338.69	344.11	469.71
18.7.42	42.00	339.17	351.15	357.03	362.77	494.16

18.7.43	43.00	334.42	348.58	354.60	361.12	488.97
18.7.44	44.00	359.03	377.74	385.14	393.87	539.47
18.7.45	45.00	346.44	366.76	372.39	379.91	520.39
18.7.46	46.00	376.88	392.40	398.33	404.39	550.28
18.7.47	47.00	353.17	367.41	372.03	377.34	517.46
18.7.48	48.00	323.50	342.46	347.59	353.05	504.87
18.7.49	49.00	315.94	330.42	334.50	340.04	475.69
18.7.50	50.00	331.28	347.70	353.09	359.13	504.94
18.7.51	51.00	329.40	346.70	351.00	356.98	497.73
18.7.52	52.00	354.77	373.79	381.49	390.70	540.88
18.7.53	53.00	346.00	363.38	369.80	377.19	524.55
18.7.54	54.00	371.55	389.14	394.64	401.53	554.13
18.7.55	55.00	349.42	363.75	369.28	374.49	513.39
18.7.56	56.00	418.31	454.38	460.71	466.37	654.80
18.7.57	57.00	448.86	460.82	463.47	465.56	650.85
18.7.58	58.00	395.75	454.14	461.58	466.93	652.50
18.7.59	59.00	396.50	445.59	457.92	467.36	657.14
18.7.60	60.00	383.60	408.65	421.74	434.47	624.04
18.7.61	61.00	380.76	404.60	413.61	426.71	595.66
18.7.62	62.00	387.84	414.93	425.36	436.42	628.17
18.7.63	63.00	399.30	437.95	454.30	464.46	656.94
18.7.64	64.00	331.22	343.02	348.04	354.97	483.89
18.7.65	65.00	322.58	333.33	337.56	343.14	467.29
18.7.66	66.00	337.48	349.31	355.02	359.87	491.30

18.7.67	67.00	335.96	348.75	353.49	359.24	487.47
18.7.68	68.00	357.60	375.74	381.03	388.20	530.07
18.7.69	69.00	346.91	366.53	370.65	377.26	514.02
18.7.70	70.00	374.38	390.23	395.43	401.96	547.78
18.7.71	71.00	353.33	368.60	372.64	377.91	514.84
18.7.72	72.00	414.13	457.12	468.38	472.88	659.81
18.7.73	73.00	464.25	469.86	471.30	472.88	660.86
18.7.74	74.00	417.93	450.29	461.22	471.33	641.91
18.7.75	75.00	405.98	447.25	457.53	468.08	664.25
18.7.76	76.00	396.90	412.00	424.21	434.32	628.95
18.7.77	77.00	392.55	408.84	418.94	427.91	625.77
18.7.78	78.00	391.27	416.25	427.42	439.45	622.52
18.7.79	79.00	393.06	431.98	446.82	459.17	669.45
18.7.80	80.00	403.90	451.88	462.62	468.18	656.17
18.7.81	81.00	458.46	465.15	466.71	468.37	654.97
18.7.82	82.00	407.78	442.61	455.81	466.89	638.12
18.7.83	83.00	401.79	439.77	450.94	463.09	655.14
18.7.84	84.00	388.19	407.97	419.06	430.87	611.11
18.7.85	85.00	379.83	404.50	415.00	424.75	616.62
18.7.86	86.00	388.52	412.23	423.17	434.11	614.48
18.7.87	87.00	384.61	428.18	442.87	455.93	665.42
18.7.88	88.00	327.07	339.66	344.85	352.13	477.26
18.7.89	89.00	318.21	327.61	334.34	339.66	458.94
18.7.90	90.00	333.18	344.62	351.07	357.14	483.19

18.7.91	91.00	331.85	343.20	349.00	355.58	478.61
18.7.92	92.00	349.88	370.11	377.62	383.50	524.45
18.7.93	93.00	343.35	359.62	367.07	373.23	507.46
18.7.94	94.00	372.24	384.62	391.36	397.31	538.53
18.7.95	95.00	350.76	362.06	368.24	373.65	506.89
18.7.96	96.00	416.16	462.68	470.16	473.43	666.77
18.7.97	97.00	461.26	468.11	469.73	471.68	661.21
18.7.98	98.00	406.20	449.27	462.14	470.90	655.86
18.7.99	99.00	403.47	439.32	456.24	466.44	656.11
18.7.100	100.00	378.84	411.29	423.65	435.95	617.26
18.7.101	101.00	373.56	406.30	415.72	425.81	607.92
18.7.102	102.00	387.85	421.51	435.21	448.34	610.76
18.7.103	103.00	404.03	442.17	456.13	471.14	651.23
18.7.104	104.00	319.51	339.77	345.47	350.19	492.95
18.7.105	105.00	313.90	328.38	334.01	338.28	475.26
18.7.106	106.00	331.26	348.46	353.63	358.58	500.99
18.7.107	107.00	327.65	344.95	350.55	355.15	494.57
18.7.108	108.00	349.47	371.87	377.77	386.38	547.45
18.7.109	109.00	337.10	360.36	366.43	373.22	523.23
18.7.110	110.00	369.72	388.53	394.63	400.25	552.33
18.7.111	111.00	345.71	364.84	370.57	375.60	515.54
18.7.112	112.00	316.97	337.45	342.95	348.18	493.15
18.7.113	113.00	311.99	327.23	331.86	336.70	471.93
18.7.114	114.00	328.96	347.31	351.24	356.51	504.35

18.7.115	115.00	326.36	344.31	348.27	354.56	500.28
18.7.116	116.00	345.58	370.15	377.78	385.66	543.30
18.7.117	117.00	335.72	359.66	365.95	371.99	520.50
18.7.118	118.00	364.16	387.49	392.59	398.82	554.92
18.7.119	119.00	346.06	364.69	368.95	374.24	515.08
18.7.120	120.00	403.39	457.70	466.62	470.46	662.49
18.7.121	121.00	458.26	465.43	467.10	469.51	659.61
18.7.122	122.00	390.76	446.28	458.13	468.03	655.46
18.7.123	123.00	396.49	437.55	451.95	464.03	648.62
18.7.124	124.00	378.55	408.32	421.05	432.26	602.62
18.7.125	125.00	377.08	404.52	413.14	424.56	582.46
18.7.126	126.00	385.68	420.44	432.59	447.12	624.98
18.7.127	127.00	397.47	440.22	454.17	469.42	647.11

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00
18.8.2	2.00	0.00	0.00	0.00	0.00	0.00
18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00
18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00

18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00
18.8.15	15.00	0.00	0.00	0.00	0.00	0.00
18.8.16	16.00	0.00	0.00	0.00	0.00	0.00
18.8.17	17.00	0.00	0.00	0.00	0.00	0.00
18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00
18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00
18.8.26	26.00	0.00	0.00	0.00	0.00	0.00
18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00
18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00

18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00
18.8.39	39.00	0.00	0.00	0.00	0.00	0.00
18.8.40	40.00	0.00	0.00	0.00	0.00	0.00
18.8.41	41.00	0.00	0.00	0.00	0.00	0.00
18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00
18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00
18.8.50	50.00	0.00	0.00	0.00	0.00	0.00
18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00
18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00

18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00
18.8.63	63.00	0.00	0.00	0.00	0.00	0.00
18.8.64	64.00	0.00	0.00	0.00	0.00	0.00
18.8.65	65.00	0.00	0.00	0.00	0.00	0.00
18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00
18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00
18.8.74	74.00	0.00	0.00	0.00	0.00	0.00
18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00
18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00

18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00
18.8.87	87.00	0.00	0.00	0.00	0.00	0.00
18.8.88	88.00	0.00	0.00	0.00	0.00	0.00
18.8.89	89.00	0.00	0.00	0.00	0.00	0.00
18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00
18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00
18.8.98	98.00	0.00	0.00	0.00	0.00	0.00
18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00
18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00

18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00
18.8.111	111.00	0.00	0.00	0.00	0.00	0.00
18.8.112	112.00	0.00	0.00	0.00	0.00	0.00
18.8.113	113.00	0.00	0.00	0.00	0.00	0.00
18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00
18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00
18.8.122	122.00	0.00	0.00	0.00	0.00	0.00
18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00
18.8.127	127.00	0.00	0.00	0.00	0.00	0.00

18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.12.0	0.00	0.00	0.00	0.00	0.00	0.00
18.12.1	1.00	0.00	0.00	0.00	0.00	0.00
18.12.2	2.00	0.00	0.00	0.00	0.00	0.00
18.12.3	3.00	0.00	0.00	0.00	0.00	0.00
18.12.4	4.00	0.00	0.00	0.00	0.00	0.00
18.12.5	5.00	0.00	0.00	0.00	0.00	0.00
18.12.6	6.00	0.00	0.00	0.00	0.00	0.00
18.12.7	7.00	0.00	0.00	0.00	0.00	0.00
18.12.8	8.00	0.00	0.00	0.00	0.00	0.00
18.12.9	9.00	0.00	0.00	0.00	0.00	0.00
18.12.10	10.00	0.00	0.00	0.00	0.00	0.00
18.12.11	11.00	0.00	0.00	0.00	0.00	0.00
18.12.12	12.00	0.00	0.00	0.00	0.00	0.00
18.12.13	13.00	0.00	0.00	0.00	0.00	0.00
18.12.14	14.00	0.00	0.00	0.00	0.00	0.00
18.12.15	15.00	0.00	0.00	0.00	0.00	0.00
18.12.16	16.00	0.00	0.00	0.00	0.00	0.00
18.12.17	17.00	0.00	0.00	0.00	0.00	0.00
18.12.18	18.00	0.00	0.00	0.00	0.00	0.00
18.12.19	19.00	0.00	0.00	0.00	0.00	0.00
18.12.20	20.00	0.00	0.00	0.00	0.00	0.00
18.12.21	21.00	0.00	0.00	0.00	0.00	0.00
18.12.22	22.00	0.00	0.00	0.00	0.00	0.00

18.12.23	23.00	0.00	0.00	0.00	0.00	0.00
18.12.24	24.00	0.00	0.00	0.00	0.00	0.00
18.12.25	25.00	0.00	0.00	0.00	0.00	0.00
18.12.26	26.00	0.00	0.00	0.00	0.00	0.00
18.12.27	27.00	0.00	0.00	0.00	0.00	0.00
18.12.28	28.00	0.00	0.00	0.00	0.00	0.00
18.12.29	29.00	0.00	0.00	0.00	0.00	0.00
18.12.30	30.00	0.00	0.00	0.00	0.00	0.00
18.12.31	31.00	0.00	0.00	0.00	0.00	0.00
18.12.32	32.00	0.00	0.00	0.00	0.00	0.00
18.12.33	33.00	0.00	0.00	0.00	0.00	0.00
18.12.34	34.00	0.00	0.00	0.00	0.00	0.00
18.12.35	35.00	0.00	0.00	0.00	0.00	0.00
18.12.36	36.00	0.00	0.00	0.00	0.00	0.00
18.12.37	37.00	0.00	0.00	0.00	0.00	0.00
18.12.38	38.00	0.00	0.00	0.00	0.00	0.00
18.12.39	39.00	0.00	0.00	0.00	0.00	0.00
18.12.40	40.00	0.00	0.00	0.00	0.00	0.00
18.12.41	41.00	0.00	0.00	0.00	0.00	0.00
18.12.42	42.00	0.00	0.00	0.00	0.00	0.00
18.12.43	43.00	0.00	0.00	0.00	0.00	0.00
18.12.44	44.00	0.00	0.00	0.00	0.00	0.00
18.12.45	45.00	0.00	0.00	0.00	0.00	0.00
18.12.46	46.00	0.00	0.00	0.00	0.00	0.00

18.12.47	47.00	0.00	0.00	0.00	0.00	0.00
18.12.48	48.00	0.00	0.00	0.00	0.00	0.00
18.12.49	49.00	0.00	0.00	0.00	0.00	0.00
18.12.50	50.00	0.00	0.00	0.00	0.00	0.00
18.12.51	51.00	0.00	0.00	0.00	0.00	0.00
18.12.52	52.00	0.00	0.00	0.00	0.00	0.00
18.12.53	53.00	0.00	0.00	0.00	0.00	0.00
18.12.54	54.00	0.00	0.00	0.00	0.00	0.00
18.12.55	55.00	0.00	0.00	0.00	0.00	0.00
18.12.56	56.00	0.00	0.00	0.00	0.00	0.00
18.12.57	57.00	0.00	0.00	0.00	0.00	0.00
18.12.58	58.00	0.00	0.00	0.00	0.00	0.00
18.12.59	59.00	0.00	0.00	0.00	0.00	0.00
18.12.60	60.00	0.00	0.00	0.00	0.00	0.00
18.12.61	61.00	0.00	0.00	0.00	0.00	0.00
18.12.62	62.00	0.00	0.00	0.00	0.00	0.00
18.12.63	63.00	0.00	0.00	0.00	0.00	0.00
18.12.64	64.00	0.00	0.00	0.00	0.00	0.00
18.12.65	65.00	0.00	0.00	0.00	0.00	0.00
18.12.66	66.00	0.00	0.00	0.00	0.00	0.00
18.12.67	67.00	0.00	0.00	0.00	0.00	0.00
18.12.68	68.00	0.00	0.00	0.00	0.00	0.00
18.12.69	69.00	0.00	0.00	0.00	0.00	0.00
18.12.70	70.00	0.00	0.00	0.00	0.00	0.00

18.12.71	71.00	0.00	0.00	0.00	0.00	0.00
18.12.72	72.00	0.00	0.00	0.00	0.00	0.00
18.12.73	73.00	0.00	0.00	0.00	0.00	0.00
18.12.74	74.00	0.00	0.00	0.00	0.00	0.00
18.12.75	75.00	0.00	0.00	0.00	0.00	0.00
18.12.76	76.00	0.00	0.00	0.00	0.00	0.00
18.12.77	77.00	0.00	0.00	0.00	0.00	0.00
18.12.78	78.00	0.00	0.00	0.00	0.00	0.00
18.12.79	79.00	0.00	0.00	0.00	0.00	0.00
18.12.80	80.00	0.00	0.00	0.00	0.00	0.00
18.12.81	81.00	0.00	0.00	0.00	0.00	0.00
18.12.82	82.00	0.00	0.00	0.00	0.00	0.00
18.12.83	83.00	0.00	0.00	0.00	0.00	0.00
18.12.84	84.00	0.00	0.00	0.00	0.00	0.00
18.12.85	85.00	0.00	0.00	0.00	0.00	0.00
18.12.86	86.00	0.00	0.00	0.00	0.00	0.00
18.12.87	87.00	0.00	0.00	0.00	0.00	0.00
18.12.88	88.00	0.00	0.00	0.00	0.00	0.00
18.12.89	89.00	0.00	0.00	0.00	0.00	0.00
18.12.90	90.00	0.00	0.00	0.00	0.00	0.00
18.12.91	91.00	0.00	0.00	0.00	0.00	0.00
18.12.92	92.00	0.00	0.00	0.00	0.00	0.00
18.12.93	93.00	0.00	0.00	0.00	0.00	0.00
18.12.94	94.00	0.00	0.00	0.00	0.00	0.00

18.12.95	95.00	0.00	0.00	0.00	0.00	0.00
18.12.96	96.00	0.00	0.00	0.00	0.00	0.00
18.12.97	97.00	0.00	0.00	0.00	0.00	0.00
18.12.98	98.00	0.00	0.00	0.00	0.00	0.00
18.12.99	99.00	0.00	0.00	0.00	0.00	0.00
18.12.100	100.00	0.00	0.00	0.00	0.00	0.00
18.12.101	101.00	0.00	0.00	0.00	0.00	0.00
18.12.102	102.00	0.00	0.00	0.00	0.00	0.00
18.12.103	103.00	0.00	0.00	0.00	0.00	0.00
18.12.104	104.00	0.00	0.00	0.00	0.00	0.00
18.12.105	105.00	0.00	0.00	0.00	0.00	0.00
18.12.106	106.00	0.00	0.00	0.00	0.00	0.00
18.12.107	107.00	0.00	0.00	0.00	0.00	0.00
18.12.108	108.00	0.00	0.00	0.00	0.00	0.00
18.12.109	109.00	0.00	0.00	0.00	0.00	0.00
18.12.110	110.00	0.00	0.00	0.00	0.00	0.00
18.12.111	111.00	0.00	0.00	0.00	0.00	0.00
18.12.112	112.00	0.00	0.00	0.00	0.00	0.00
18.12.113	113.00	0.00	0.00	0.00	0.00	0.00
18.12.114	114.00	0.00	0.00	0.00	0.00	0.00
18.12.115	115.00	0.00	0.00	0.00	0.00	0.00
18.12.116	116.00	0.00	0.00	0.00	0.00	0.00
18.12.117	117.00	0.00	0.00	0.00	0.00	0.00
18.12.118	118.00	0.00	0.00	0.00	0.00	0.00

Large Input

[illegible]

0. Top Stats

0.1 Top Kernels

Kernel_Name	Count	Sum(ns)
Mean(ns) Median(ns) Pct S		
0 matmulbias_kernel 244912943.12 241288813.50 99.90 *	17.00	4163520033.02
1 void at::native::(anonymous namespace):: distribution_elementwise_grid_stride_ker nel<float, 4, at::native::templates::...	3.00	3370760.00
2 void at::native::vectorized_elementwise_ kernel<4, at::native::FillFunctor<int>, std::array<char*, 1ul> >(int, at::nat...	12.00	561582.07
3 __amd_rocclr_copyBuffer 2957.69 2913.17 0.01	89.00	263233.97
4 void at::native::(anonymous namespace):: CatArrayBatchedCopy_contig<at::native::(anonymous namespace)::OpaqueType<2u>,...	7.00	19253.25
5 void at::native::vectorized_elementwise_ kernel<4, at::native::compare_scalar_ker nel<double>(at::TensorIteratorBase&, ...	3.00	10447.16
6 void at::native::vectorized_elementwise_ kernel<8, at::native::AUnaryFunctor<c10:	2.00	7406.91

```

:Half, c10::Half, bool, at::native::(...)

7 void at::native::vectorized_elementwise_      2.00      5766.33
2883.17      2883.17      0.00
kernel<8, at::native::BinaryFunctor<c10:

:Half, c10::Half, bool, at::native::(...)

8 void at::native::vectorized_elementwise_      2.00      5629.25
2814.62      2814.62      0.00
kernel<8, at::native::AbsFunctor<c10::Ha

lf>, std::array<char*, 2ul> >(int, at...

9 void rocprim::ROCPRIM_400001_NS::detail:      1.00      4550.50
4550.50      4550.50      0.00
:partition_kernel<(rocprim::ROCPRIM_4000

01_NS::detail::select_method)0, true,...
```

0.2 Dispatch List

Dispatch_ID	Kernel_Name	GPU_ID
0	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
1	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4,	0

```

at::native::templates::cuda::normal_and_transform<c10::Half, float
    , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
    eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...

```

```

2          2 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker          0
    nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
    , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
    eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...

```

```

3          3 matmulbias_kernel
0

```

```

4          4 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
    std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
    1>)

```

```

5          5 matmulbias_kernel
0

```

```

6          6 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
    std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
    1>)

```

```

7          7 matmulbias_kernel
0

```

```

8          8 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
    std::array<char*, 1ul> >(int,

```

```
at::native::FillFunctor<int>, std::array<char*, 1u
1>)
```

```
9          9  matmulbias_kernel
0
```

1. System Info

Info

workload_name	fusion_matmulbias_32768x32768
command fusion	python triton/matmulbias.py --M 32768 --N 32768
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Mon Mar 16 20:39:33 2026 (UTC)
version	3
hostname	b031b949a562
cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1

linux_distro	CentOS Stream 10 (Coughlan)
--------------	-----------------------------

linux_kernel_version	5.15.0-170-generic
----------------------	--------------------

amd_gpu_kernel_version	nan
------------------------	-----

cpu_memory	230893520
------------	-----------

gpu_memory	nan
------------	-----

rocm_version	7.0.3
--------------	-------

vbios	113-M3000108-103
-------	------------------

compute_partition	SPX
-------------------	-----

memory_partition	nan
------------------	-----

gpu_series	MI300
------------	-------

gpu_model	MI300X_A1
-----------	-----------

gpu_arch	gfx942
----------	--------

gpu_chip_id	29877
gpu_l1	32
gpu_l2	4096
cu_per_gpu	304
simd_per_cu	4
se_per_gpu	32
wave_size	64
workgroup_max_size	1024
max_waves_per_cu	32
max_sclk	2100
max_mclk	1300
cur_sclk	2100

cur_mclk	1300
total_l2_chan	128
lds_banks_per_cu	32
sqc_per_gpu	160
pipes_per_gpu	4
num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID Peak	Metric Pct of Peak	Avg	Unit
2.1.0 81715.20	VALU FLOPs 0.01	4.09	Gflop/s
2.1.1 81715.20	VALU IOPs 2.11	1720.39	Giop/s

2.1.2	MFMA FLOPs (F8)	0.00	Gflop/s
2614886.40	0.00		
2.1.3	MFMA FLOPs (BF16)	0.00	Gflop/s
1307443.20	0.00		
2.1.4	MFMA FLOPs (F16)	287737.81	Gflop/s
1307443.20	22.01		
2.1.5	MFMA FLOPs (F32)	0.00	Gflop/s
163430.40	0.00		
2.1.6	MFMA FLOPs (F64)	0.00	Gflop/s
163430.40	0.00		
2.1.7	MFMA FLOPs (F6F4)		Gflop/s
2.1.8	MFMA IOPs (Int8)	0.00	Giop/s
2614886.40	0.00		
2.1.9	Active CUs	304.00	Cus
304.00	100.00		
2.1.10	SALU Utilization	5.99	Pct
100.00	5.99		
2.1.11	VALU Utilization	20.24	Pct
100.00	20.24		
2.1.12	MFMA Utilization	39.25	Pct
100.00	39.25		
2.1.13	VMEM Utilization	1.23	Pct
100.00	1.23		

2.1.14 100.00	Branch Utilization 2.16	2.16	Pct
2.1.15 64.00	VALU Active Threads 100.00	64.00	Threads
2.1.16 5.00	IPC 7.63	0.38	Instr/cycle
2.1.17 9728.00	Wavefront Occupancy 18.02	1753.37	Wavefronts
2.1.18 81715.20	Theoretical LDS Bandwidth 18.99	15520.37	Gb/s
2.1.19 32.00	LDS Bank Conflicts/Access 0.00	0.00	Conflicts/access
2.1.20 100.00	vL1D Cache Hit Rate 50.10	50.10	Pct
2.1.21 81715.20	vL1D Cache BW 11.20	9149.87	Gb/s
2.1.22 100.00	L2 Cache Hit Rate 48.67	48.67	Pct
2.1.23 34406.40	L2 Cache BW 13.27	4566.24	Gb/s
2.1.24 5324.80	L2-Fabric Read BW 43.92	2338.58	Gb/s
2.1.25 5324.80	L2-Fabric Write BW 0.16	8.78	Gb/s

2.1.26	L2-Fabric Read Latency	582.82	Cycles
2.1.27	L2-Fabric Write Latency	227.86	Cycles
2.1.28 100.00	sL1D Cache Hit Rate		Pct
2.1.29 21504.00	sL1D Cache BW 0.00	0.00	Gb/s
2.1.30 100.00	L1I Hit Rate 100.00	100.00	Pct
2.1.31 21504.00	L1I BW 7.39	1588.12	Gb/s
2.1.32	L1I Fetch Latency		Cycles

3. Memory Chart

3.1 Memory Chart

(Normalization: per_kernel)

Instr Buff Instr Dispatch Exec

LDS

L2 Cache

SALU: 5145717097

-----> Active
CUs Req : 52759101Util : 35 %
 xGMI/PCIE

Wave 0 Instr Buf 30
4: 304 <-----

SMEM: 0
Lat :
N/A Hit : 49
%

>
~
|
RVGPRseq: 31.0

 | |
Wave N-1 Instr Buf VALU:
17537186876 Vector L1 Cache
Request |

|

>
|
v
SGPRs : 48

MFMA:
4294967296 Rd : 68853694Hit : 50 % Rd
: 8590066151 Rd : 8590099443

> <----- <-----
-
Fabric

Wt : 26843545Lat : N/A Wr : 134217728 Wr :

```

134217728    Rd      :
4475905515

VMEM: 1080033280 LDS Alloc:  -
----->
> <----- Latency (cycles)      Rd:
4455164565
----->
0  Atomic:    0 Coales:    25 %    Atomic:    0    Atomic:
0    Wr      : 33555204    <----- HBM
                                <
-----> <-----
> -----> Rd      :    583    Wr:
33555204

LDS:
5275910144    Stall :    0
cycles        Atomic:    0    Wr      :
228 ----->
-----> Scratch Alloc:

-----> Atomic:    0
                                0

Wave Occupancy      GWS: 0
Scalar L1D
Cache      Latency (cycles)

6 -----
---> Rd      :
0

Wave Life
<-----
^
|
1329583
BRANCH: 1469825988 Wavefronts: Rd      :
0 Hit      :    N/A    Wr      :    0    Rd      :
N/A      |
|
-----> 0 <
-----
> |
|

Lat      :    N/A    Atomic:    0    Wr      :
N/A      |
v

```


141579GFLOP/s

131392 GFLOP/s
65959.2

15141.7

3476.0

798.0

63723 GB/s

33604 GB/s

183.2 25652 GB/s

42.1 4205 GB/s

0.01 0.18 3.16 56.23 1000.00

Performance (GFLOP/sec) Arithmetic Intensity (FLOP)s/Byte)

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct
5.1.1	CPF Stall	0.00	0.00	0.00	Pct
5.1.2	CPF-L2 Utilization	0.00	0.00	0.02	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	0.00	0.00	0.00	Pct

5.2 Packet Processor

Metric_ID	Metric	Avg	Min	Max
Unit				
5.2.0	CPC SYNC FIFO Full Rate			
Pct				
5.2.1	CPC CANE Stall Rate			
Pct				
5.2.2	CPC ADC Utilization			
Pct				
5.2.3	CPC Utilization	100.00	100.00	100.00
Pct				
5.2.4	CPC Stall Rate	0.34	0.09	0.52
Pct				
5.2.5	CPC Packet Decoding Utilization	1.93	0.43	3.31
Pct				
5.2.6	CPC-Workgroup Manager Utilization	97.97	96.64	100.04
Pct				

5.2.7 Pct	CPC-L2 Utilization	0.00	0.00	0.00
5.2.8 Pct	CPC-UTCL1 Stall	0.00	0.00	0.00
5.2.9 Pct	CPC-UTCL2 Utilization	0.00	0.00	0.00

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID Unit	Metric	Avg	Min	Max	
6.1.0 Wave	Schedule-Pipe Wave Occupancy				
6.1.1	Accelerator Utilization	100.00	100.00	100.00	Pct
6.1.2	Scheduler-Pipe Utilization	0.00	0.00	0.00	Pct
6.1.3	Scheduler-Pipe Wave Utilization				Pct
6.1.4	Workgroup Manager Utilization	98.55	97.64	100.61	Pct
6.1.5	Shader Engine Utilization	98.55	97.64	100.61	Pct

6.1.6	SIMD Utilization	98.32	97.15	100.31	Pct
6.1.7 Workgroups	Dispatched Workgroups	0.00	0.00	0.00	
6.1.8 Wavefronts	Dispatched Wavefronts	0.00	0.00	0.00	
6.1.9 Cycles/wave	VGPR Writes				
6.1.10 Cycles/wave	SGPR Writes				

6.2 Workgroup Manager - Resource Allocation

Metric_ID Unit	Metric	Avg	Min	Max
6.2.0 Pct	Not-scheduled Rate (Workgroup Manager)	0.00	0.00	0.00
6.2.1 Pct	Not-scheduled Rate (Scheduler-Pipe)	0.00	0.00	0.00
6.2.2 Pct	Scheduler-Pipe FIFO Full Rate			
6.2.3 Pct	Scheduler-Pipe Stall Rate	0.00	0.00	0.00
6.2.4 Pct	Scratch Stall Rate	0.00	0.00	0.00

6.2.5 Pct	Insufficient SIMD Waveslots	0.00	0.00	0.00
6.2.6 Pct	Insufficient SIMD VGPRs	0.00	0.00	0.00
6.2.7 Pct	Insufficient SIMD SGPRs	0.00	0.00	0.00
6.2.8 Pct	Insufficient CU LDS	0.00	0.00	0.00
6.2.9 Pct	Insufficient CU Barriers	0.00	0.00	0.00
6.2.10 Pct	Reached CU Workgroup Limit	0.00	0.00	0.00
6.2.11 Pct	Reached CU Wavefront Limit	0.00	0.00	0.00

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID Unit	Metric	Avg	Min	Max
7.1.0 Work items	Grid Size	24672376.47	16777216.00	33554432.00
7.1.1 Work items	Workgroup Size	376.47	256.00	512.00
7.1.2	Total Wavefronts	0.00	0.00	0.00

Wavefronts

7.1.3	Saved Wavefronts	0.00	0.00	0.00
-------	------------------	------	------	------

Wavefronts

7.1.4	Restored Wavefronts	0.00	0.00	0.00
-------	---------------------	------	------	------

Wavefronts

7.1.5	VGPRs	30.59	20.00	40.00
-------	-------	-------	-------	-------

Registers

7.1.6	AGPRs	163.76	132.00	192.00
-------	-------	--------	--------	--------

Registers

7.1.7	SGPRs	48.00	48.00	48.00
-------	-------	-------	-------	-------

Registers

7.1.8	LDS Allocation	0.00	0.00	0.00
-------	----------------	------	------	------

Bytes

7.1.9	Scratch Allocation	0.00	0.00	0.00
-------	--------------------	------	------	------

Bytes/workitem

7.2 Wavefront Runtime Stats

Metric_ID	Metric	Avg	Min
Max	Unit		
7.2.0	Kernel Time (Nanosec)	244912943.12	232290345.58
256045669.33	Ns		
7.2.1	Kernel Time (Cycles)	291528168.81	261281335.62
392682976.12	Cycle		
7.2.2	Instructions per wavefront	90147.35	70593.00
107529.00	Instr/wavefront		

7.2.3	Wave Cycles	494215384968.71	370267647656.00
620316572208.00	Cycles per kernel		
7.2.4	Dependency Wait Cycles	137573494592.71	92129868364.00
195624491452.00	Cycles per kernel		
7.2.5	Issue Wait Cycles	215432231297.88	152170826152.00
285448263592.00	Cycles per kernel		
7.2.6	Active Cycles	141477704292.00	123815931876.00
161311878880.00	Cycles per kernel		
7.2.7	Wavefront Occupancy	1753.37	1185.68
2389.05	Wavefronts		

10. Compute Units - Instruction Mix
10.1 Overall Instruction Mix

Metric_ID	Metric	Avg	Min	Max
Unit				
10.1.0	VALU	13242219580.24	11501830144.00	15200157696.00
Instr per kernel				
10.1.1	VMEM	1080033280.00	1080033280.00	1080033280.00
Instr per kernel				
10.1.2	LDS	5275910144.00	3761766400.00	6979321856.00
Instr per kernel				
10.1.3	MFMA	4294967296.00	4294967296.00	4294967296.00
Instr per kernel				

10.1.4	SALU	5145717097.41	4188798976.00	6222249984.00
Instr per kernel				

10.1.5	SMEM	0.00	0.00	0.00
Instr per kernel				

10.1.6	Branch	1469825987.76	1343750144.00	1611661312.00
Instr per kernel				

10.2 VALU Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max
Unit				

10.2.0	INT32	5563590053.65	3795320832.00	7552892928.00
Instr per kernel				

10.2.1	INT64	1079863657.41	1079246848.00	1080557568.00
Instr per kernel				

10.2.2	F16-ADD	12582912.00	12582912.00	12582912.00
Instr per kernel				

10.2.3	F16-MUL	0.00	0.00	0.00
Instr per kernel				

10.2.4	F16-FMA	0.00	0.00	0.00
Instr per kernel				

10.2.5	F16-Trans	0.00	0.00	0.00
Instr per kernel				

10.2.6	F32-ADD	0.00	0.00	0.00
Instr per kernel				

10.2.7	F32-MUL	1542023.53	1048576.00	2097152.00
Instr per kernel				

10.2.8	F32-FMA	0.00	0.00	0.00
Instr per kernel				

10.2.9	F32-Trans	1542023.53	1048576.00	2097152.00
Instr per kernel				

10.2.10	F64-ADD	0.00	0.00	0.00
Instr per kernel				

10.2.11	F64-MUL	0.00	0.00	0.00
Instr per kernel				

10.2.12	F64-FMA	0.00	0.00	0.00
Instr per kernel				

10.2.13	F64-Trans	0.00	0.00	0.00
Instr per kernel				

10.2.14	Conversion	19861263.06	18874368.00	20971520.00
Instr per kernel				

10.3 VMEM Instr Mix

Metric_ID	Metric		Avg	Min
Max	Unit			

10.3.0	Global/Generic Instr		1077936128.00	1077936128.00
1077936128.00	Instr per kernel			

10.3.1	Global/Generic Read		1073741824.00	1073741824.00
1073741824.00	Instr per kernel			

10.3.2	Global/Generic Write		4194304.00	4194304.00
--------	----------------------	--	------------	------------

4194304.00 Instr per kernel

10.3.3	Global/Generic Atomic	0.00	0.00
0.00	Instr per kernel		

10.3.4	Spill/Stack Instr	2097152.00	2097152.00
2097152.00	Instr per kernel		

10.3.5 Spill/Stack Coalesceable Instr
Instr per kernel

10.3.6	Spill/Stack Read	2097152.00	2097152.00
2097152.00	Instr per kernel		

10.3.7	Spill/Stack Write	0.00	0.00
0.00	Instr per kernel		

10.3.8	Spill/Stack Atomic	0.00	0.00
0.00	Instr per kernel		

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.4.0	MFMA-I8	0.00	0.00	0.00	
Instr per kernel					
10.4.1	MFMA-F8	0.00	0.00	0.00	
Instr per kernel					
10.4.2	MFMA-F16	4294967296.00	4294967296.00	4294967296.00	
Instr per kernel					
10.4.3	MFMA-BF16	0.00	0.00	0.00	
Instr per kernel					

10.4.4	MFMA-F32	0.00	0.00	0.00
Instr per kernel				

10.4.5	MFMA-F64	0.00	0.00	0.00
Instr per kernel				

10.4.6	MFMA-F6F4			
Instr per kernel				

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID Peak	Metric	Avg	Unit	Peak	Pct of
11.1.0 0.01	VALU FLOPs	4.09	Gflop	81715.20	
11.1.1 2.11	VALU IOPs	1720.39	Giop	81715.20	
11.1.2 0.00	MFMA FLOPs (F8)	0.00	Gflop	2614886.40	
11.1.3 0.00	MFMA FLOPs (BF16)	0.00	Gflop	1307443.20	
11.1.4 22.01	MFMA FLOPs (F16)	287737.81	Gflop	1307443.20	
11.1.5 0.00	MFMA FLOPs (F32)	0.00	Gflop	163430.40	

11.1.6	MFMA FLOPs (F64)	0.00	Gflop	163430.40
--------	------------------	------	-------	-----------

0.00

11.1.7	MFMA FLOPs (F6F4)		Gflop	
--------	-------------------	--	-------	--

11.1.8	MFMA IOPs (INT8)	0.00	Giop	2614886.40
--------	------------------	------	------	------------

0.00

11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
11.2.0	IPC	0.38	0.24	0.48	Instr/cycle
11.2.1	IPC (Issued)	0.86	0.84	0.88	Instr/cycle
11.2.2	SALU Utilization	5.99	3.51	7.83	Pct
11.2.3	VALU Utilization	20.24	13.24	24.55	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	1.23	0.90	1.36	Pct
11.2.6	Branch Utilization	2.16	1.35	2.70	Pct

11.2.7	VALU Active Threads	64.00	64.00	64.00	Threads
11.2.8	MFMA Utilization	39.25	28.78	43.26	Pct
11.2.9	MFMA Instr Cycles	32.00	32.00	32.00	Cycles/instr
11.2.10	VMEM Latency				Cycles
11.2.11	SMEM Latency				Cycles
11.3 Arithmetic Operations					
Metric_ID	Metric	Avg		Min	
Max	Unit				
11.3.0	FLOPs (Total)	70369746863043.77	70369683701760.00		
70369817919488.00	Ops per kernel				
11.3.1	IOPs (Total)	425181037507.76	311972331520.00		
552540831744.00	Ops per kernel				
11.3.2	F8 OPs	0.00	0.00		
0.00	Ops per kernel				
11.3.3	F16 OPs	70369549484032.00	70369549484032.00		
70369549484032.00	Ops per kernel				
11.3.4	BF16 OPs	0.00	0.00		
0.00	Ops per kernel				
11.3.5	F32 OPs	197379011.76	134217728.00		

268435456.00 Ops per kernel

11.3.6 F64 OPs 0.00 0.00
0.00 Ops per kernel

11.3.7 F6F4 OPs
Ops per kernel

11.3.8 INT8 OPs 0.00 0.00
0.00 Ops per kernel

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	34.53	Pct of peak
12.1.1	Access Rate	22.57	Pct of peak
12.1.2	Theoretical Bandwidth (% of Peak)	18.99	Pct of peak
12.1.3	Bank Conflict Rate	0.00	Pct of peak

12.2 LDS Stats

Metric_ID	Metric	Avg	Min
	Max Unit		
12.2.0	LDS Instrs	5275910144.00	3761766400.00
	6979321856.00 Instr per kernel		
12.2.1	LDS LOAD		
	Instr per kernel		
12.2.2	LDS STORE		
	Instr per kernel		

12.2.3	LDS ATOMIC Instr per kernel		
12.2.4	LDS LOAD Bandwidth Gbps		
12.2.5	LDS STORE Bandwidth Gbps		
12.2.6	LDS ATOMIC Bandwidth Gbps		
12.2.7	Theoretical Bandwidth 4398046511104.00 Bytes per kernel	3815952119928.47	3298534883328.00
12.2.8	LDS Latency Cycles		
12.2.9	Bank Conflicts/Access 0.00 Conflicts/access	0.00	0.00
12.2.10	Index Accesses 34359738368.00 Cycles per kernel	29812125936.94	25769803776.00
12.2.11	Atomic Return Cycles 0.00 Cycles per kernel	0.00	0.00
12.2.12	Bank Conflict 0.00 Cycles per kernel	0.00	0.00
12.2.13	Addr Conflict 0.00 Cycles per kernel	0.00	0.00
12.2.14	Unaligned Stall 0.00 Cycles per kernel	0.00	0.00

12.2.15	Mem Violations	0.00	0.00
	0.00 Accesses per kernel		

12.2.16	LDS Command FIFO Full Rate
	Cycles per kernel

12.2.17	LDS Data FIFO Full Rate
	Cycles per kernel

13. Instruction Cache

13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	7.39	Pct of peak
13.1.1	Cache Hit Rate	100.00	Pct of peak
13.1.2	L1I-L2 Bandwidth	0.00	Pct of peak

13.2 Instruction Cache Accesses

Metric_ID	Metric	Avg	Min
Max Unit			
13.2.0	Req	6100507226.35	5287182336.00
7015497728.00	Req per kernel		
13.2.1	Hits	6100502007.41	5287146107.00
7015494690.00	Hits per kernel		
13.2.2	Misses - Non Duplicated	855.59	320.00
6865.00	Misses per kernel		
13.2.3	Misses - Duplicated	4310.06	2605.00

28774.00 Misses per kernel

13.2.4	Cache Hit Rate	100.00	100.00
100.00	Pct		

13.2.5 Instruction Fetch Latency
Cycles

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
13.3.0	L1I-L2 Bandwidth	2105223.53	921600.00	13260800.00	Bytes
	per kernel				

14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.00	Pct of peak
14.1.1	Cache Hit Rate		Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
14.2.0	Req	0.00	0.00	0.00	Req per kernel
14.2.1	Hits	0.00	0.00	0.00	Req per kernel

14.2.2	Misses - Non Duplicated	0.00	0.00	0.00	Req per kernel
14.2.3	Misses- Duplicated	0.00	0.00	0.00	Req per kernel
14.2.4	Cache Hit Rate				Pct
14.2.5	Read Req (Total)	0.00	0.00	0.00	Req per kernel
14.2.6	Atomic Req	0.00	0.00	0.00	Req per kernel
14.2.7	Read Req (1 DWord)	0.00	0.00	0.00	Req per kernel
14.2.8	Read Req (2 DWord)	0.00	0.00	0.00	Req per kernel
14.2.9	Read Req (4 DWord)	0.00	0.00	0.00	Req per kernel
14.2.10	Read Req (8 DWord)	0.00	0.00	0.00	Req per kernel
14.2.11	Read Req (16 DWord)	0.00	0.00	0.00	Req per kernel

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	0.00	0.00	0.00	Bytes per kernel

14.3.1	Read Req	0.00	0.00	0.00	Req per kernel
14.3.2	Write Req	0.00	0.00	0.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel
14.3.4	Stall Cycles	28.18	0.00	479.00	Cycles per kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric_ID Min	Metric Max Unit	Avg
15.1.0 35.74	Address Processing Unit Busy 41.66 Pct	38.50
15.1.1 9.58	Address Stall 14.68 Pct	12.03
15.1.2 0.60	Data Stall 0.77 Pct	0.74
15.1.3 0.00	Data-Processor → Address Stall 0.00 Pct	0.00
15.1.4	Sequencer → TA Address Stall Cycles per kernel	
15.1.5	Sequencer → TA Command Stall Cycles per kernel	
15.1.6	Sequencer → TA Data Stall Cycles per kernel	

15.1.7	Total Instructions	1080033280.00
1080033280.00	1080033280.00 Instructions per kernel	
15.1.8	Global/Generic Instructions	1077936128.00
1077936128.00	1077936128.00 Instructions per kernel	
15.1.9	Global/Generic Read Instructions	1073741824.00
1073741824.00	1073741824.00 Instructions per kernel	
15.1.10	Global/Generic Read Instructions for LDS Instructions per kernel	
15.1.11	Global/Generic Write Instructions	4194304.00
4194304.00	4194304.00 Instructions per kernel	
15.1.12	Global/Generic Atomic Instructions	0.00
0.00	0.00 Instructions per kernel	
15.1.13	Spill/Stack Instructions	2097152.00
2097152.00	2097152.00 Instructions per kernel	
15.1.14	Spill/Stack Read Instructions	2097152.00
2097152.00	2097152.00 Instructions per kernel	
15.1.15	Spill/Stack Read Instructions for LDS Instructions per kernel	
15.1.16	Spill/Stack Write Instructions	0.00
0.00	0.00 Instructions per kernel	
15.1.17	Spill/Stack Atomic Instructions	0.00
0.00	0.00 Instructions per kernel	
15.1.18	Spill/Stack Total Cycles	33554432.00
33554432.00	33554432.00 Cycles per kernel	

15.1.19	Spill/Stack Coalesced Read	0.00
0.00	0.00 Cycles per kernel	

15.1.20	Spill/Stack Coalesced Write	0.00
0.00	0.00 Cycles per kernel	

15.2 Data-Return Path

Metric_ID	Metric	Avg
Min	Max Unit	

15.2.0	Data-Return Busy	71.77
69.30	78.42 Pct	

15.2.1	Cache RAM → Data-Return Stall	50.53
47.83	62.55 Pct	

15.2.2	Workgroup manager → Data-Return Stall	0.00
0.00	0.00 Pct	

15.2.3	Coalescable Instructions	0.00
0.00	0.00 Instructions per kernel	

15.2.4	Read Instructions	1075838976.00
1075838976.00	1075838976.00 Instructions per kernel	

15.2.5	Write Instructions	4194304.00
4194304.00	4194304.00 Instructions per kernel	

15.2.6	Atomic Instructions	0.00
0.00	0.00 Instructions per kernel	

15.2.7	Write Ack Instructions	
	Instructions per kernel	

INFO Not showing table with empty column(s): 16.6 L1D Addr
Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
16.1.0	Hit rate	50.10	Pct of peak
16.1.1	Bandwidth	11.20	Pct of peak
16.1.2	Utilization	79.09	Pct of peak
16.1.3	Coalescing	25.00	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3
Max					
16.2.0	Stalled on L2 Data	40.94	41.28	50.34	51.10
61.93					
16.2.1	Stalled on L2 Req	0.00	0.00	0.03	0.03
0.12					
16.2.2	Stalled on Address				
16.2.3	Stalled on Data				
16.2.4	Stalled on Latency FIFO				
16.2.5	Stalled on Request FIFO				

16.2.6	Stalled on Read Return				
16.2.7 0.00	Tag RAM Stall (Read)	0.00	0.00	0.00	0.00
16.2.8 0.26	Tag RAM Stall (Write)	0.17	0.22	0.22	0.26
16.2.9 0.00	Tag RAM Stall (Atomic)	0.00	0.00	0.00	0.00
16.3 L1D Cache Accesses					
Metric_ID Max	Metric Unit	Avg		Min	
16.3.0 69122129920.00	Total Req Req per kernel	69122129920.00		69122129920.00	
16.3.1 68853694464.00	Read Req Req per kernel	68853694464.00		68853694464.00	
16.3.2 268435456.00	Write Req Req per kernel	268435456.00		268435456.00	
16.3.3 0.00	Atomic Req Req per kernel	0.00		0.00	
16.3.4 2237677961216.00	Cache BW Bytes per kernel	2237677961216.00		2237677961216.00	
16.3.5 50.10	Cache Hit Rate Pct	50.10		50.10	
16.3.6	Cache Accesses	17481859072.00		17481859072.00	

17481859072.00 Req per kernel

16.3.7	Cache Hits	8757575193.41	8757572430.00
8757575680.00	Req per kernel		

16.3.8	Invalidations	482.82	304.00
3344.00	Req per kernel		

16.3.9	L1-L2 BW	1108118401867.29	1108118339584.00
1108118755584.00	Bytes per kernel		

16.3.10	Tag RAM 0 Req
Req per kernel	

16.3.11	Tag RAM 1 Req
Req per kernel	

16.3.12	Tag RAM 2 Req
Req per kernel	

16.3.13	Tag RAM 3 Req
Req per kernel	

16.3.14	L1-L2 Read	8590066150.59	8590065664.00
8590068914.00	Req per kernel		

16.3.15	L1-L2 Write	134217728.00	134217728.00
134217728.00	Req per kernel		

16.3.16	L1-L2 Atomic	0.00	0.00
0.00	Req per kernel		

16.3.17	L1 Access Latency
Cycles	

16.3.18	L1-L2 Read Latency
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Cycles

16.3.19 L1-L2 Write Latency
Cycles

16.4 L1D - L2 Transactions

Metric_ID Min	Metric Max	Unit	Xfer	Coherency	Avg
16.4.0 0.00	NC - Read 0.00	Read Req per kernel	NC		0.00
16.4.1 0.00	UC - Read 0.00	Read Req per kernel	UC		0.00
16.4.2 0.00	CC - Read 0.00	Read Req per kernel	CC		0.00
16.4.3 8590065664.00	RW - Read 8590068994.00	Read Req per kernel	RW		8590066165.71
16.4.4 134217728.00	RW - Write 134218215.00	Write Req per kernel	RW		134217756.65
16.4.5 0.00	NC - Write 0.00	Write Req per kernel	NC		0.00
16.4.6 0.00	UC - Write 0.00	Write Req per kernel	UC		0.00
16.4.7 0.00	CC - Write 0.00	Write Req per kernel	CC		0.00
16.4.8 0.00	NC - Atomic 0.00	Atomic Req per kernel	NC		0.00

16.4.9	UC - Atomic	Atomic	UC	0.00
0.00	0.00	Req per kernel		
16.4.10	CC - Atomic	Atomic	CC	0.00
0.00	0.00	Req per kernel		
16.4.11	RW - Atomic	Atomic	RW	0.00
0.00	0.00	Req per kernel		

16.5 L1D Addr Translation

Metric_ID	Metric		Avg	Min
	Max	Units		
16.5.0	Req		17481859072.00	17481859072.00
17481859072.00	Req per kernel			
16.5.1	Inflight Req			
	Req per kernel			
16.5.2	Hit Ratio		99.59	99.59
99.59	Pct			
16.5.3	Hits		17410385920.00	17410385920.00
17410385920.00	Req per kernel			
16.5.4	Translation Misses		67475456.47	67475456.00
67475464.00	Req per kernel			
16.5.5	Misses under Translation Miss			
	Req per kernel			
16.5.6	Permission Misses		0.00	0.00
0.00	Req per kernel			

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache

17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
17.1.0	Utilization	81.84	Pct
17.1.1	Bandwidth	13.27	Pct
17.1.2	Hit Rate	48.67	Pct
17.1.3	L2-Fabric Read BW	2338.58	Gb/s
17.1.4	L2-Fabric Write and Atomic BW	8.78	Gb/s
17.1.5	HBM Bandwidth	5324.80	Gb/s

17.2 L2 - Fabric Transactions

Metric_ID	Metric	Avg
Min	Max Unit	
17.2.0	Read BW	572146424237.18
566285976256.00	588607723776.00 Bytes per kernel	
17.2.1	HBM Read Traffic	99.56
95.56	100.39 Pct	
17.2.2	Remote Read Traffic	0.50
0.00	4.44 Pct	
17.2.3	Uncached Read Traffic	0.00
0.00	0.00 Pct	
17.2.4	Write and Atomic BW	2147508570.35
2147483648.00	2147617408.00 Bytes per kernel	

17.2.5	HBM Write and Atomic Traffic	100.00
100.00	100.00 Pct	
17.2.6	Remote Write and Atomic Traffic	0.00
0.00	0.00 Pct	
17.2.7	Atomic Traffic	0.00
0.00	0.00 Pct	
17.2.8	Uncached Write and Atomic Traffic	0.00
0.00	0.00 Pct	
17.2.9	Read Latency	582.82
469.42	1095.63 Cycles	
17.2.10	Write and Atomic Latency	227.86
161.85	515.57 Cycles	
17.2.11	Atomic Latency Cycles	
17.2.12	Read Stall Pct	
17.2.13	Write Stall Pct	

17.3 L2 Cache Accesses

Metric_ID	Metric	Avg	Min
Max Unit			
17.3.0	Bandwidth	1116712591819.29	1116710164608.00
1116735976448.00	Bytes per kernel		

17.3.1	Read Bandwidth		
Bytes per kernel			
17.3.2	Write Bandwidth		
Bytes per kernel			
17.3.3	Atomic Bandwidth		
Bytes per kernel			
17.3.4	Req	8724317123.59	8724298161.00
8724499816.00	Req per kernel		
17.3.5	Read Req	8590099442.82	8590080434.00
8590282434.00	Req per kernel		
17.3.6	Write Req	134217728.00	134217728.00
134217728.00	Req per kernel		
17.3.7	Atomic Req	0.00	0.00
0.00	Req per kernel		
17.3.8	Streaming Req	0.00	0.00
0.00	Req per kernel		
17.3.9	Bypassss Req		
Req per kernel			
17.3.10	Probe Req	0.00	0.00
0.00	Req per kernel		
17.3.11	Input Buffer Req		
Req per kernel			
17.3.12	Cache Hit	48.67	47.27
49.10	Pct		

17.3.13	Hits	4258770131.76	4210419191.00
4283501554.00	Hits per kernel		
17.3.14	Misses	4492682293.88	4440875252.00
4712982248.00	Misses per kernel		
17.3.15	Writeback	16778666.29	16777328.00
16784302.00	Cachelines per kernel		
17.3.16	Writeback (Internal)	16731853.29	16711465.00
16740982.00	Cachelines per kernel		
17.3.17	Writeback (vL1D Req)	46370.24	43259.00
71640.00	Cachelines per kernel		
17.3.18	Evict (Internal)	4492317087.94	4438983253.00
4712720103.00	Cachelines per kernel		
17.3.19	Evict (vL1D Req)	0.00	0.00
0.00	Cachelines per kernel		
17.3.20	NC Req	0.00	0.00
0.00	Req per kernel		
17.3.21	UC Req	376.29	0.00
5716.00	Req per kernel		
17.3.22	CC Req	0.00	0.00
0.00	Req per kernel		
17.3.23	RW Req	8724316730.35	8724298248.00
8724493512.00	Req per kernel		

17.5 L2 - Fabric Interface Stalls

Metric_ID	Metric	Type
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Transaction	Avg	Min	Max	Unit		
17.5.0	Read - PCIe Stall Pct				PCIe Stall	Read
17.5.1	Read - Infinity Fabric Stall Pct				Infinity Fabric Stall	Read
17.5.2	Read - HBM Stall Pct				HBM Stall	Read
17.5.3	Write - PCIe Stall Pct				PCIe Stall	Write
17.5.4	Write - Infinity Fabric Stall Pct				Infinity Fabric Stall	Write
17.5.5	Write - HBM Stall Pct				HBM Stall	Write
17.5.6	Write - Credit Starvation 0.00 0.00 0.00 Pct				Credit Starvation	Write

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID	Metric	Avg
Min	Max Unit	
17.6.0	Read (32B)	0.00
0.00	0.00 Req per kernel	
17.6.1	Read (64B)	12023151.76
-174197488.00	195415084.00 Req per kernel	
17.6.2	Read (128B)	4463882363.47
4424093875.00	4651491137.00 Req per kernel	

17.6.3	Read (Uncached)	709.29
0.00	10698.00 Req per kernel	
17.6.4	HBM Read	4455164565.35
4424090263.00	4500813425.00 Req per kernel	
17.6.5	Remote Read	23500716.88
0.00	208509816.00 Req per kernel	
17.6.6	Read Bandwidth - PCIe Bytes per kernel	
17.6.7	Read Bandwidth - Infinity Fabric Bytes per kernel	
17.6.8	Read Bandwidth - HBM Bytes per kernel	
17.6.9	Write and Atomic (32B)	765.41
0.00	4180.00 Req per kernel	
17.6.10	Write and Atomic (Uncached)	0.00
0.00	0.00 Req per kernel	
17.6.11	Write and Atomic (64B)	33554438.71
33554432.00	33554504.00 Req per kernel	
17.6.12	HBM Write and Atomic	33555204.12
33554432.00	33558612.00 Req per kernel	
17.6.13	Remote Write and Atomic	0.00
0.00	0.00 Req per kernel	
17.6.14	Write Bandwidth - PCIe Bytes per kernel	

17.6.15	Write Bandwidth - Infinity Fabric Bytes per kernel		
17.6.16	Write Bandwidth - HBM Bytes per kernel		
17.6.17	Atomic	0.00	0.00
0.00	0.00 Req per kernel		
17.6.18	Atomic - HBM Req per kernel		
17.6.19	Atomic Bandwidth - PCIe Bytes per kernel		
17.6.20	Atomic Bandwidth - Infinity Fabric Bytes per kernel		
17.6.21	Atomic Bandwidth - HBM Bytes per kernel		

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	48.64	0.66	47.41	49.10	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
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18.2.0	0.00	47.41	48.49	49.10	49.10	49.10
18.2.1	1.00	47.41	48.49	49.10	49.10	49.10
18.2.2	2.00	47.41	48.49	49.10	49.10	49.10
18.2.3	3.00	47.41	48.49	49.10	49.10	49.10
18.2.4	4.00	47.41	48.49	49.10	49.10	49.10
18.2.5	5.00	47.41	48.49	49.10	49.10	49.10
18.2.6	6.00	47.41	48.49	49.10	49.10	49.10
18.2.7	7.00	47.41	48.49	49.10	49.10	49.10
18.2.8	8.00	47.41	48.49	49.10	49.10	49.10
18.2.9	9.00	47.41	48.49	49.10	49.10	49.10
18.2.10	10.00	47.41	48.49	49.10	49.10	49.10
18.2.11	11.00	47.41	48.49	49.10	49.10	49.10
18.2.12	12.00	47.41	48.49	49.10	49.10	49.10
18.2.13	13.00	47.41	48.49	49.10	49.10	49.10
18.2.14	14.00	47.41	48.49	49.10	49.10	49.10
18.2.15	15.00	47.41	48.49	49.10	49.10	49.10
18.2.16	16.00	46.75	47.83	49.09	49.09	49.10
18.2.17	17.00	46.75	47.83	49.09	49.09	49.10
18.2.18	18.00	46.75	47.83	49.09	49.09	49.10
18.2.19	19.00	46.75	47.83	49.09	49.09	49.10
18.2.20	20.00	46.75	47.83	49.09	49.09	49.10
18.2.21	21.00	46.75	47.83	49.09	49.09	49.10
18.2.22	22.00	46.75	47.83	49.09	49.09	49.10
18.2.23	23.00	46.75	47.83	49.09	49.09	49.10

18.2.24	24.00	46.75	47.83	49.09	49.09	49.10
18.2.25	25.00	46.75	47.83	49.09	49.09	49.10
18.2.26	26.00	46.75	47.83	49.09	49.09	49.10
18.2.27	27.00	46.75	47.83	49.09	49.09	49.10
18.2.28	28.00	46.75	47.83	49.09	49.09	49.10
18.2.29	29.00	46.75	47.83	49.09	49.09	49.10
18.2.30	30.00	46.75	47.83	49.09	49.09	49.10
18.2.31	31.00	46.75	47.83	49.09	49.09	49.10
18.2.32	32.00	46.84	48.54	49.10	49.10	49.10
18.2.33	33.00	46.84	48.54	49.10	49.10	49.10
18.2.34	34.00	46.84	48.54	49.10	49.10	49.10
18.2.35	35.00	46.84	48.54	49.10	49.10	49.10
18.2.36	36.00	46.84	48.54	49.10	49.10	49.10
18.2.37	37.00	46.84	48.54	49.10	49.10	49.10
18.2.38	38.00	46.84	48.54	49.10	49.10	49.10
18.2.39	39.00	46.84	48.54	49.10	49.10	49.10
18.2.40	40.00	46.84	48.54	49.10	49.10	49.10
18.2.41	41.00	46.84	48.54	49.10	49.10	49.10
18.2.42	42.00	46.84	48.54	49.10	49.10	49.10
18.2.43	43.00	46.84	48.54	49.10	49.10	49.10
18.2.44	44.00	46.84	48.54	49.10	49.10	49.10
18.2.45	45.00	46.84	48.54	49.10	49.10	49.10
18.2.46	46.00	46.84	48.54	49.10	49.10	49.10
18.2.47	47.00	46.84	48.54	49.10	49.10	49.10

18.2.48	48.00	47.23	48.68	49.09	49.09	49.10
18.2.49	49.00	47.23	48.68	49.09	49.09	49.10
18.2.50	50.00	47.23	48.68	49.09	49.09	49.10
18.2.51	51.00	47.23	48.68	49.09	49.09	49.10
18.2.52	52.00	47.23	48.68	49.09	49.09	49.10
18.2.53	53.00	47.23	48.68	49.09	49.09	49.10
18.2.54	54.00	47.23	48.68	49.09	49.09	49.10
18.2.55	55.00	47.23	48.68	49.09	49.09	49.10
18.2.56	56.00	47.23	48.68	49.09	49.09	49.10
18.2.57	57.00	47.23	48.68	49.09	49.09	49.10
18.2.58	58.00	47.23	48.68	49.09	49.09	49.10
18.2.59	59.00	47.23	48.68	49.09	49.09	49.10
18.2.60	60.00	47.23	48.68	49.09	49.09	49.10
18.2.61	61.00	47.23	48.68	49.09	49.09	49.10
18.2.62	62.00	47.23	48.68	49.09	49.09	49.10
18.2.63	63.00	47.23	48.68	49.09	49.09	49.10
18.2.64	64.00	44.11	46.07	49.09	49.10	49.10
18.2.65	65.00	44.11	46.07	49.10	49.10	49.10
18.2.66	66.00	44.11	46.07	49.10	49.10	49.10
18.2.67	67.00	44.11	46.07	49.10	49.10	49.10
18.2.68	68.00	44.11	46.07	49.10	49.10	49.10
18.2.69	69.00	44.11	46.07	49.10	49.10	49.10
18.2.70	70.00	44.11	46.07	49.10	49.10	49.10
18.2.71	71.00	44.11	46.07	49.10	49.10	49.10

18.2.72	72.00	44.11	46.07	49.10	49.10	49.10
18.2.73	73.00	44.11	46.07	49.10	49.10	49.10
18.2.74	74.00	44.11	46.07	49.10	49.10	49.10
18.2.75	75.00	44.11	46.07	49.09	49.10	49.10
18.2.76	76.00	44.11	46.07	49.10	49.10	49.10
18.2.77	77.00	44.11	46.07	49.10	49.10	49.10
18.2.78	78.00	44.11	46.07	49.10	49.10	49.10
18.2.79	79.00	44.11	46.07	49.10	49.10	49.10
18.2.80	80.00	47.00	48.37	49.09	49.10	49.10
18.2.81	81.00	47.00	48.37	49.09	49.10	49.10
18.2.82	82.00	47.00	48.37	49.09	49.10	49.10
18.2.83	83.00	47.00	48.37	49.09	49.10	49.10
18.2.84	84.00	47.00	48.37	49.09	49.10	49.10
18.2.85	85.00	47.00	48.37	49.09	49.10	49.10
18.2.86	86.00	47.00	48.37	49.09	49.10	49.10
18.2.87	87.00	47.00	48.37	49.09	49.10	49.10
18.2.88	88.00	47.00	48.37	49.09	49.10	49.10
18.2.89	89.00	47.00	48.37	49.09	49.10	49.10
18.2.90	90.00	47.00	48.37	49.09	49.10	49.10
18.2.91	91.00	47.00	48.37	49.09	49.10	49.10
18.2.92	92.00	47.00	48.37	49.09	49.10	49.10
18.2.93	93.00	47.00	48.37	49.09	49.10	49.10
18.2.94	94.00	47.00	48.37	49.09	49.10	49.10
18.2.95	95.00	47.00	48.37	49.09	49.10	49.10

18.2.96	96.00	46.10	47.46	49.10	49.10	49.10
18.2.97	97.00	46.10	47.46	49.10	49.10	49.10
18.2.98	98.00	46.10	47.46	49.10	49.10	49.10
18.2.99	99.00	46.10	47.46	49.10	49.10	49.10
18.2.100	100.00	46.10	47.46	49.10	49.10	49.10
18.2.101	101.00	46.10	47.46	49.10	49.10	49.10
18.2.102	102.00	46.10	47.46	49.10	49.10	49.10
18.2.103	103.00	46.10	47.46	49.10	49.10	49.10
18.2.104	104.00	46.10	47.46	49.10	49.10	49.10
18.2.105	105.00	46.10	47.46	49.10	49.10	49.10
18.2.106	106.00	46.10	47.46	49.10	49.10	49.10
18.2.107	107.00	46.10	47.46	49.10	49.10	49.10
18.2.108	108.00	46.10	47.46	49.10	49.10	49.10
18.2.109	109.00	46.10	47.46	49.10	49.10	49.10
18.2.110	110.00	46.10	47.46	49.10	49.10	49.10
18.2.111	111.00	46.10	47.46	49.10	49.10	49.10
18.2.112	112.00	47.33	48.09	49.08	49.10	49.10
18.2.113	113.00	47.33	48.09	49.08	49.10	49.10
18.2.114	114.00	47.33	48.09	49.08	49.10	49.10
18.2.115	115.00	47.33	48.09	49.08	49.10	49.10
18.2.116	116.00	47.33	48.09	49.08	49.10	49.10
18.2.117	117.00	47.33	48.09	49.08	49.10	49.10
18.2.118	118.00	47.33	48.09	49.08	49.10	49.10
18.2.119	119.00	47.33	48.09	49.08	49.10	49.10

18.2.120	120.00	47.33	48.09	49.08	49.10	49.10
18.2.121	121.00	47.33	48.09	49.08	49.10	49.10
18.2.122	122.00	47.33	48.09	49.08	49.10	49.10
18.2.123	123.00	47.33	48.09	49.08	49.10	49.10
18.2.124	124.00	47.33	48.09	49.08	49.10	49.10
18.2.125	125.00	47.33	48.09	49.08	49.10	49.10
18.2.126	126.00	47.33	48.09	49.08	49.10	49.10
18.2.127	127.00	47.33	48.09	49.08	49.10	49.10

18.3 L2 Requests (per normUnit)

Metric_ID Q3	Channel Max	Min	Q1	Median
18.3.0 68158704.00	0.00 68160176.00	68158544.00	68158548.00	68158704.00
18.3.1 68158644.00	1.00 68159848.00	68158544.00	68158546.00	68158644.00
18.3.2 68158704.00	2.00 68160186.00	68158505.00	68158507.00	68158704.00
18.3.3 68158704.00	3.00 68160171.00	68158544.00	68158548.00	68158704.00
18.3.4 68158705.00	4.00 68160165.00	68158545.00	68158548.00	68158704.00
18.3.5 68158704.00	5.00 68160150.00	68158544.00	68158546.00	68158704.00
18.3.6	6.00	68158544.00	68158545.00	68158704.00

68158704.00 68160225.00

18.3.7 7.00 68158544.00 68158549.00 68158704.00
68158704.00 68160179.00

18.3.8 8.00 68158626.00 68158631.00 68158704.00
68158704.00 68160189.00

18.3.9 9.00 68158626.00 68158629.00 68158704.00
68158704.00 68160185.00

18.3.10 10.00 68158625.00 68158628.00 68158704.00
68158704.00 68160190.00

18.3.11 11.00 68158625.00 68158627.00 68158704.00
68158704.00 68160195.00

18.3.12 12.00 68158626.00 68158627.00 68158704.00
68158704.00 68160188.00

18.3.13 13.00 68158625.00 68158628.00 68158704.00
68158704.00 68160180.00

18.3.14 14.00 68158545.00 68158547.00 68158624.00
68158624.00 68159634.00

18.3.15 15.00 68158626.00 68158629.00 68158704.00
68158704.00 68160259.00

18.3.16 16.00 68158625.00 68158628.00 68158704.00
68158704.00 68160188.00

18.3.17 17.00 68158624.00 68158627.00 68158704.00
68158704.00 68160214.00

18.3.18 18.00 68158625.00 68158626.00 68158704.00

68158704.00 68160201.00

18.3.19 19.00 68158626.00 68158628.00 68158704.00
68158704.00 68160191.00

18.3.20 20.00 68158624.00 68158626.00 68158704.00
68158704.00 68160191.00

18.3.21 21.00 68158625.00 68158628.00 68158704.00
68158704.00 68160220.00

18.3.22 22.00 68158544.00 68158546.00 68158624.00
68158624.00 68159645.00

18.3.23 23.00 68158625.00 68158628.00 68158704.00
68158704.00 68160263.00

18.3.24 24.00 68158545.00 68158549.00 68158704.00
68158704.00 68160183.00

18.3.25 25.00 68158546.00 68158547.00 68158644.00
68158644.00 68159893.00

18.3.26 26.00 68158504.00 68158508.00 68158704.00
68158704.00 68160192.00

18.3.27 27.00 68158544.00 68158550.00 68158704.00
68158704.00 68160180.00

18.3.28 28.00 68158546.00 68158547.00 68158704.00
68158704.00 68160165.00

18.3.29 29.00 68158544.00 68158546.00 68158704.00
68158704.00 68160185.00

18.3.30 30.00 68158545.00 68158546.00 68158704.00

68158704.00 68160246.00

18.3.31 31.00 68158545.00 68158549.00 68158704.00
68158704.00 68160178.00

18.3.32 32.00 68158549.00 68158560.00 68158704.00
68158704.00 68160187.00

18.3.33 33.00 68158549.00 68158554.00 68158644.00
68158644.00 68159894.00

18.3.34 34.00 68158505.00 68158510.00 68158704.00
68158704.00 68160210.00

18.3.35 35.00 68158548.00 68158554.00 68158704.00
68158704.00 68160194.00

18.3.36 36.00 68158546.00 68158550.00 68158704.00
68158704.00 68160184.00

18.3.37 37.00 68158547.00 68158551.00 68158704.00
68158704.00 68160193.00

18.3.38 38.00 68158544.00 68158548.00 68158704.00
68158704.00 68160256.00

18.3.39 39.00 68158546.00 68158554.00 68158704.00
68158704.00 68160191.00

18.3.40 40.00 68158629.00 68158637.00 68158704.00
68158704.00 68160208.00

18.3.41 41.00 68158629.00 68158636.00 68158704.00
68158704.00 68160224.00

18.3.42 42.00 68158627.00 68158634.00 68158704.00

68158704.00	68160212.00			
18.3.43	43.00	68158629.00	68158635.00	68158704.00
68158704.00	68160200.00			
18.3.44	44.00	68158628.00	68158634.00	68158704.00
68158704.00	68160188.00			
18.3.45	45.00	68158627.00	68158632.00	68158704.00
68158704.00	68160228.00			
18.3.46	46.00	68158547.00	68158551.00	68158624.00
68158624.00	68159652.00			
18.3.47	47.00	68158630.00	68158635.00	68158704.00
68158704.00	68160278.00			
18.3.48	48.00	68158624.00	68158627.00	68158704.00
68158704.00	68160225.00			
18.3.49	49.00	68158624.00	68158628.00	68158704.00
68158704.00	68160240.00			
18.3.50	50.00	68158624.00	68158629.00	68158704.00
68158704.00	68160186.00			
18.3.51	51.00	68158625.00	68158627.00	68158704.00
68158704.00	68160191.00			
18.3.52	52.00	68158625.00	68158627.00	68158704.00
68158704.00	68160228.00			
18.3.53	53.00	68158624.00	68158627.00	68158704.00
68158704.00	68160242.00			
18.3.54	54.00	68158545.00	68158547.00	68158624.00

68158624.00 68159630.00

18.3.55 55.00 68158627.00 68158631.00 68158704.00
68158704.00 68160274.00

18.3.56 56.00 68158545.00 68158551.00 68158704.00
68158704.00 68160220.00

18.3.57 57.00 68158545.00 68158548.00 68158644.00
68158644.00 68159913.00

18.3.58 58.00 68158504.00 68158507.00 68158704.00
68158704.00 68160183.00

18.3.59 59.00 68158545.00 68158550.00 68158704.00
68158704.00 68160179.00

18.3.60 60.00 68158545.00 68158549.00 68158704.00
68158704.00 68160205.00

18.3.61 61.00 68158545.00 68158548.00 68158704.00
68158704.00 68160215.00

18.3.62 62.00 68158544.00 68158546.00 68158704.00
68158704.00 68160225.00

18.3.63 63.00 68158544.00 68158549.00 68158704.00
68158704.00 68160179.00

18.3.64 64.00 68158624.00 68158625.00 68158704.00
68158704.00 68160238.00

18.3.65 65.00 68158624.00 68158624.00 68158704.00
68158704.00 68160201.00

18.3.66 66.00 68158624.00 68158625.00 68158704.00

68158704.00 68160226.00

18.3.67 67.00 68158624.00 68158626.00 68158704.00
68158704.00 68160197.00

18.3.68 68.00 68158624.00 68158624.00 68158704.00
68158704.00 68160237.00

18.3.69 69.00 68158624.00 68158624.00 68158704.00
68158704.00 68160197.00

18.3.70 70.00 68158544.00 68158544.00 68158624.00
68158624.00 68159668.00

18.3.71 71.00 68158624.00 68158625.00 68158704.00
68158704.00 68160251.00

18.3.72 72.00 68158544.00 68158546.00 68158704.00
68158704.00 68160225.00

18.3.73 73.00 68158544.00 68158545.00 68158644.00
68158644.00 68159878.00

18.3.74 74.00 68158504.00 68158505.00 68158704.00
68158704.00 68160221.00

18.3.75 75.00 68158544.00 68158545.00 68158704.00
68158704.00 68160177.00

18.3.76 76.00 68158544.00 68158545.00 68158704.00
68158704.00 68160208.00

18.3.77 77.00 68158544.00 68158544.00 68158704.00
68158704.00 68160166.00

18.3.78 78.00 68158544.00 68158544.00 68158704.00

68158704.00 68160278.00

18.3.79 79.00 68158544.00 68158545.00 68158704.00
68158704.00 68160172.00

18.3.80 80.00 68158544.00 68158548.00 68158704.00
68158704.00 68160189.00

18.3.81 81.00 68158544.00 68158548.00 68158644.00
68158644.00 68159874.00

18.3.82 82.00 68158504.00 68158507.00 68158704.00
68158704.00 68160194.00

18.3.83 83.00 68158544.00 68158549.00 68158704.00
68158704.00 68160182.00

18.3.84 84.00 68158545.00 68158548.00 68158704.00
68158704.00 68160178.00

18.3.85 85.00 68158544.00 68158550.00 68158704.00
68158704.00 68160171.00

18.3.86 86.00 68158544.00 68158546.00 68158704.00
68158704.00 68160261.00

18.3.87 87.00 68158544.00 68158553.00 68158704.00
68158704.00 68160182.00

18.3.88 88.00 68158626.00 68158631.00 68158704.00
68158704.00 68160199.00

18.3.89 89.00 68158625.00 68158628.00 68158704.00
68158704.00 68160202.00

18.3.90 90.00 68158626.00 68158631.00 68158704.00

68158704.00 68160200.00

18.3.91 91.00 68158624.00 68158628.00 68158704.00
68158704.00 68160199.00

18.3.92 92.00 68158626.00 68158629.00 68158704.00
68158704.00 68160195.00

18.3.93 93.00 68158626.00 68158629.00 68158704.00
68158704.00 68160205.00

18.3.94 94.00 68158544.00 68158547.00 68158624.00
68158624.00 68159641.00

18.3.95 95.00 68158626.00 68158630.00 68158704.00
68158704.00 68160251.00

18.3.96 96.00 68158625.00 68158627.00 68158704.00
68158704.00 68160205.00

18.3.97 97.00 68158624.00 68158626.00 68158704.00
68158704.00 68160234.00

18.3.98 98.00 68158624.00 68158629.00 68158704.00
68158704.00 68160206.00

18.3.99 99.00 68158624.00 68158629.00 68158704.00
68158704.00 68160210.00

18.3.100 100.00 68158624.00 68158625.00 68158704.00
68158704.00 68160191.00

18.3.101 101.00 68158624.00 68158626.00 68158704.00
68158704.00 68160228.00

18.3.102 102.00 68158544.00 68158547.00 68158624.00

68158624.00 68159631.00

18.3.103 103.00 68158624.00 68158631.00 68158704.00
68158704.00 68160264.00

18.3.104 104.00 68158544.00 68158550.00 68158704.00
68158704.00 68160182.00

18.3.105 105.00 68158544.00 68158546.00 68158644.00
68158644.00 68159887.00

18.3.106 106.00 68158504.00 68158507.00 68158704.00
68158704.00 68160184.00

18.3.107 107.00 68158544.00 68158547.00 68158704.00
68158704.00 68160200.00

18.3.108 108.00 68158544.00 68158547.00 68158704.00
68158704.00 68160170.00

18.3.109 109.00 68158544.00 68158548.00 68158704.00
68158704.00 68160196.00

18.3.110 110.00 68158544.00 68158545.00 68158704.00
68158704.00 68160244.00

18.3.111 111.00 68158544.00 68158549.00 68158704.00
68158704.00 68160196.00

18.3.112 112.00 68158544.00 68158547.00 68158704.00
68158704.00 68160182.00

18.3.113 113.00 68158544.00 68158546.00 68158644.00
68158644.00 68159907.00

18.3.114 114.00 68158506.00 68158509.00 68158704.00

68158704.00 68160185.00

18.3.115 115.00 68158544.00 68158549.00 68158704.00
68158704.00 68160179.00

18.3.116 116.00 68158544.00 68158547.00 68158704.00
68158704.00 68160171.00

18.3.117 117.00 68158544.00 68158547.00 68158704.00
68158704.00 68160208.00

18.3.118 118.00 68158544.00 68158548.00 68158704.00
68158704.00 68160236.00

18.3.119 119.00 68158544.00 68158549.00 68158704.00
68158704.00 68160174.00

18.3.120 120.00 68158626.00 68158629.00 68158704.00
68158704.00 68160201.00

18.3.121 121.00 68158626.00 68158628.00 68158704.00
68158704.00 68160244.00

18.3.122 122.00 68158625.00 68158628.00 68158704.00
68158704.00 68160200.00

18.3.123 123.00 68158624.00 68158628.00 68158704.00
68158704.00 68160195.00

18.3.124 124.00 68158624.00 68158628.00 68158704.00
68158704.00 68160190.00

18.3.125 125.00 68158624.00 68158628.00 68158704.00
68158704.00 68160243.00

18.3.126 126.00 68158545.00 68158546.00 68158624.00

68158624.00 68159630.00

18.3.127 127.00 68158626.00 68158630.00 68158704.00
68158704.00 68160253.00

18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	67110251.06	1048576.00	0.00
18.4.1	1.00	67110144.82	1048576.00	0.00
18.4.2	2.00	67110198.41	1048576.00	0.00
18.4.3	3.00	67110254.82	1048576.00	0.00
18.4.4	4.00	67110252.18	1048576.00	0.00
18.4.5	5.00	67110253.53	1048576.00	0.00
18.4.6	6.00	67110250.88	1048576.00	0.00
18.4.7	7.00	67110257.12	1048576.00	0.00
18.4.8	8.00	67110073.59	1048576.00	0.00
18.4.9	9.00	67110073.94	1048576.00	0.00
18.4.10	10.00	67110072.94	1048576.00	0.00
18.4.11	11.00	67110074.94	1048576.00	0.00
18.4.12	12.00	67110071.59	1048576.00	0.00
18.4.13	13.00	67110059.47	1048576.00	0.00
18.4.14	14.00	67110075.00	1048576.00	0.00
18.4.15	15.00	67110079.71	1048576.00	0.00
18.4.16	16.00	67110071.82	1048576.88	0.00
18.4.17	17.00	67110071.47	1048576.88	0.00
18.4.18	18.00	67110071.94	1048576.76	0.00

18.4.19	19.00	67110071.53	1048576.88	0.00
18.4.20	20.00	67110071.47	1048576.94	0.00
18.4.21	21.00	67110057.47	1048577.00	0.00
18.4.22	22.00	67110072.12	1048576.94	0.00
18.4.23	23.00	67110075.65	1048576.94	0.00
18.4.24	24.00	67110250.82	1048577.06	0.00
18.4.25	25.00	67110142.00	1048576.94	0.00
18.4.26	26.00	67110198.47	1048577.00	0.00
18.4.27	27.00	67110253.59	1048576.94	0.00
18.4.28	28.00	67110250.71	1048577.00	0.00
18.4.29	29.00	67110249.65	1048576.88	0.00
18.4.30	30.00	67110250.59	1048577.00	0.00
18.4.31	31.00	67110254.06	1048576.82	0.00
18.4.32	32.00	67110253.06	1048576.00	0.00
18.4.33	33.00	67110143.12	1048576.00	0.00
18.4.34	34.00	67110202.24	1048576.00	0.00
18.4.35	35.00	67110254.82	1048576.00	0.00
18.4.36	36.00	67110252.00	1048576.00	0.00
18.4.37	37.00	67110252.00	1048576.00	0.00
18.4.38	38.00	67110253.65	1048576.00	0.00
18.4.39	39.00	67110254.65	1048576.00	0.00
18.4.40	40.00	67110073.35	1048576.00	0.00
18.4.41	41.00	67110072.94	1048576.00	0.00
18.4.42	42.00	67110075.88	1048576.00	0.00

18.4.43	43.00	67110072.47	1048576.00	0.00
18.4.44	44.00	67110073.59	1048576.00	0.00
18.4.45	45.00	67110059.24	1048576.00	0.00
18.4.46	46.00	67110077.41	1048576.00	0.00
18.4.47	47.00	67110076.94	1048576.00	0.00
18.4.48	48.00	67110073.12	1048576.00	0.00
18.4.49	49.00	67110073.24	1048576.00	0.00
18.4.50	50.00	67110072.00	1048576.00	0.00
18.4.51	51.00	67110073.65	1048576.00	0.00
18.4.52	52.00	67110072.00	1048576.00	0.00
18.4.53	53.00	67110060.35	1048576.00	0.00
18.4.54	54.00	67110073.06	1048576.00	0.00
18.4.55	55.00	67110078.12	1048576.00	0.00
18.4.56	56.00	67110253.35	1048576.00	0.00
18.4.57	57.00	67110144.71	1048576.00	0.00
18.4.58	58.00	67110198.88	1048576.00	0.00
18.4.59	59.00	67110255.65	1048576.00	0.00
18.4.60	60.00	67110252.29	1048576.00	0.00
18.4.61	61.00	67110252.76	1048576.00	0.00
18.4.62	62.00	67110250.41	1048576.00	0.00
18.4.63	63.00	67110256.88	1048576.00	0.00
18.4.64	64.00	67110069.82	1048576.00	0.00
18.4.65	65.00	67110070.35	1048576.00	0.00
18.4.66	66.00	67110070.41	1048576.00	0.00

18.4.67	67.00	67110069.59	1048576.00	0.00
18.4.68	68.00	67110070.18	1048576.00	0.00
18.4.69	69.00	67110057.76	1048576.00	0.00
18.4.70	70.00	67110072.82	1048576.00	0.00
18.4.71	71.00	67110073.35	1048576.00	0.00
18.4.72	72.00	67110249.82	1048576.00	0.00
18.4.73	73.00	67110142.41	1048576.00	0.00
18.4.74	74.00	67110196.76	1048576.00	0.00
18.4.75	75.00	67110251.94	1048576.00	0.00
18.4.76	76.00	67110249.82	1048576.00	0.00
18.4.77	77.00	67110250.12	1048576.00	0.00
18.4.78	78.00	67110250.06	1048576.00	0.00
18.4.79	79.00	67110252.29	1048576.00	0.00
18.4.80	80.00	67110252.06	1048576.00	0.00
18.4.81	81.00	67110145.47	1048576.00	0.00
18.4.82	82.00	67110200.06	1048576.00	0.00
18.4.83	83.00	67110253.00	1048576.00	0.00
18.4.84	84.00	67110250.82	1048576.00	0.00
18.4.85	85.00	67110253.41	1048576.00	0.00
18.4.86	86.00	67110251.12	1048576.00	0.00
18.4.87	87.00	67110253.53	1048576.00	0.00
18.4.88	88.00	67110072.59	1048576.00	0.00
18.4.89	89.00	67110074.71	1048576.00	0.00
18.4.90	90.00	67110072.88	1048576.00	0.00

18.4.91	91.00	67110071.35	1048576.00	0.00
18.4.92	92.00	67110071.88	1048576.00	0.00
18.4.93	93.00	67110061.06	1048576.00	0.00
18.4.94	94.00	67110073.47	1048576.00	0.00
18.4.95	95.00	67110075.76	1048576.00	0.00
18.4.96	96.00	67110073.18	1048576.00	0.00
18.4.97	97.00	67110070.94	1048576.00	0.00
18.4.98	98.00	67110074.41	1048576.00	0.00
18.4.99	99.00	67110072.41	1048576.00	0.00
18.4.100	100.00	67110071.65	1048576.00	0.00
18.4.101	101.00	67110058.35	1048576.00	0.00
18.4.102	102.00	67110076.24	1048576.00	0.00
18.4.103	103.00	67110075.71	1048576.00	0.00
18.4.104	104.00	67110252.41	1048576.00	0.00
18.4.105	105.00	67110143.00	1048576.00	0.00
18.4.106	106.00	67110201.12	1048576.00	0.00
18.4.107	107.00	67110254.41	1048576.00	0.00
18.4.108	108.00	67110252.29	1048576.00	0.00
18.4.109	109.00	67110250.59	1048576.00	0.00
18.4.110	110.00	67110254.18	1048576.00	0.00
18.4.111	111.00	67110255.06	1048576.00	0.00
18.4.112	112.00	67110252.18	1048576.00	0.00
18.4.113	113.00	67110143.18	1048576.00	0.00
18.4.114	114.00	67110199.65	1048576.00	0.00

18.4.115	115.00	67110254.47	1048576.00	0.00
18.4.116	116.00	67110251.65	1048576.00	0.00
18.4.117	117.00	67110250.71	1048576.00	0.00
18.4.118	118.00	67110251.47	1048576.00	0.00
18.4.119	119.00	67110255.24	1048576.00	0.00
18.4.120	120.00	67110072.82	1048576.00	0.00
18.4.121	121.00	67110072.29	1048576.00	0.00
18.4.122	122.00	67110072.76	1048576.00	0.00
18.4.123	123.00	67110072.88	1048576.00	0.00
18.4.124	124.00	67110071.53	1048576.00	0.00
18.4.125	125.00	67110058.65	1048576.00	0.00
18.4.126	126.00	67110074.35	1048576.00	0.00
18.4.127	127.00	67110077.18	1048576.00	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID L2-Fabric Atomic	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
18.5.0 0.00	0.00	34806736.94	262156.24
18.5.1 0.00	1.00	34806735.94	262153.18
18.5.2 0.00	2.00	34806682.06	262155.65
18.5.3 0.00	3.00	34806721.53	262152.35

18.5.4 0.00	4.00	34806734.41	262156.47
18.5.5 0.00	5.00	34806727.59	262153.29
18.5.6 0.00	6.00	34806682.53	262155.65
18.5.7 0.00	7.00	34806719.41	262152.35
18.5.8 0.00	8.00	34806612.65	262156.35
18.5.9 0.00	9.00	34806631.88	262152.59
18.5.10 0.00	10.00	34806575.12	262156.00
18.5.11 0.00	11.00	34806606.82	262151.88
18.5.12 0.00	12.00	34806621.47	262156.35
18.5.13 0.00	13.00	34806631.12	262152.71
18.5.14 0.00	14.00	34806572.00	262155.76
18.5.15 0.00	15.00	34806606.76	262151.88

18.5.16 0.00	16.00	34857997.59	262148.35
18.5.17 0.00	17.00	34858015.18	262147.29
18.5.18 0.00	18.00	34858035.47	262148.00
18.5.19 0.00	19.00	34857999.71	262146.59
18.5.20 0.00	20.00	34858000.65	262147.88
18.5.21 0.00	21.00	34858011.24	262146.82
18.5.22 0.00	22.00	34858031.65	262148.00
18.5.23 0.00	23.00	34857998.65	262146.47
18.5.24 0.00	24.00	34857969.29	262147.53
18.5.25 0.00	25.00	34857981.94	262146.59
18.5.26 0.00	26.00	34858012.24	262148.00
18.5.27 0.00	27.00	34857954.12	262146.35

18.5.28 0.00	28.00	34857973.18	262147.53
18.5.29 0.00	29.00	34857976.71	262146.82
18.5.30 0.00	30.00	34858015.88	262148.00
18.5.31 0.00	31.00	34857963.18	262146.47
18.5.32 0.00	32.00	34659325.71	262144.47
18.5.33 0.00	33.00	34659299.71	262145.41
18.5.34 0.00	34.00	34659287.12	262144.47
18.5.35 0.00	35.00	34659346.35	262145.41
18.5.36 0.00	36.00	34659327.29	262144.47
18.5.37 0.00	37.00	34659297.24	262145.65
18.5.38 0.00	38.00	34659290.76	262144.47
18.5.39 0.00	39.00	34659346.82	262145.41

18.5.40 0.00	40.00	34659472.59	262144.47
18.5.41 0.00	41.00	34659445.12	262145.41
18.5.42 0.00	42.00	34659425.94	262144.47
18.5.43 0.00	43.00	34659489.35	262145.53
18.5.44 0.00	44.00	34659475.24	262144.47
18.5.45 0.00	45.00	34659441.00	262145.41
18.5.46 0.00	46.00	34659420.18	262144.47
18.5.47 0.00	47.00	34659493.59	262145.53
18.5.48 0.00	48.00	34700382.29	262157.18
18.5.49 0.00	49.00	34700380.71	262152.59
18.5.50 0.00	50.00	34700382.00	262158.00
18.5.51 0.00	51.00	34700375.53	262152.71

18.5.52 0.00	52.00	34700381.18	262156.88
18.5.53 0.00	53.00	34700381.12	262152.59
18.5.54 0.00	54.00	34700375.76	262158.12
18.5.55 0.00	55.00	34700377.71	262152.71
18.5.56 0.00	56.00	34700533.00	262156.35
18.5.57 0.00	57.00	34700532.29	262152.59
18.5.58 0.00	58.00	34700544.24	262157.65
18.5.59 0.00	59.00	34700523.18	262152.59
18.5.60 0.00	60.00	34700538.47	262156.47
18.5.61 0.00	61.00	34700529.24	262152.59
18.5.62 0.00	62.00	34700550.53	262157.24
18.5.63 0.00	63.00	34700525.29	262152.59

18.5.64 0.00	64.00	35409887.12	262195.35
18.5.65 0.00	65.00	35409839.06	262193.76
18.5.66 0.00	66.00	35409715.12	262194.94
18.5.67 0.00	67.00	35409859.76	262191.88
18.5.68 0.00	68.00	35409890.65	262195.35
18.5.69 0.00	69.00	35409839.71	262193.76
18.5.70 0.00	70.00	35409706.00	262195.18
18.5.71 0.00	71.00	35409864.82	262191.65
18.5.72 0.00	72.00	35410166.53	262194.94
18.5.73 0.00	73.00	35410159.65	262194.24
18.5.74 0.00	74.00	35410020.06	262194.59
18.5.75 0.00	75.00	35410150.47	262192.59

18.5.76 0.00	76.00	35410168.59	262195.06
18.5.77 0.00	77.00	35410147.88	262194.24
18.5.78 0.00	78.00	35410027.76	262195.06
18.5.79 0.00	79.00	35410146.71	262192.24
18.5.80 0.00	80.00	34687408.24	262156.59
18.5.81 0.00	81.00	34687410.41	262152.47
18.5.82 0.00	82.00	34687375.71	262157.65
18.5.83 0.00	83.00	34687378.00	262152.59
18.5.84 0.00	84.00	34687411.53	262156.47
18.5.85 0.00	85.00	34687410.12	262152.59
18.5.86 0.00	86.00	34687382.06	262157.41
18.5.87 0.00	87.00	34687384.24	262152.47

18.5.88 0.00	88.00	34687580.76	262156.94
18.5.89 0.00	89.00	34687591.59	262152.71
18.5.90 0.00	90.00	34687554.88	262157.29
18.5.91 0.00	91.00	34687551.18	262152.47
18.5.92 0.00	92.00	34687585.06	262156.94
18.5.93 0.00	93.00	34687589.71	262152.71
18.5.94 0.00	94.00	34687549.76	262157.29
18.5.95 0.00	95.00	34687553.65	262152.35
18.5.96 0.00	96.00	35110703.41	262155.65
18.5.97 0.00	97.00	35110697.00	262152.24
18.5.98 0.00	98.00	35110768.18	262155.76
18.5.99 0.00	99.00	35110642.65	262153.41

18.5.100 0.00	100.00	35110700.06	262155.88
18.5.101 0.00	101.00	35110688.06	262152.12
18.5.102 0.00	102.00	35110759.82	262155.76
18.5.103 0.00	103.00	35110640.65	262153.18
18.5.104 0.00	104.00	35110724.65	262155.88
18.5.105 0.00	105.00	35110719.12	262152.24
18.5.106 0.00	106.00	35110794.82	262156.00
18.5.107 0.00	107.00	35110677.06	262153.41
18.5.108 0.00	108.00	35110721.24	262155.88
18.5.109 0.00	109.00	35110722.24	262152.35
18.5.110 0.00	110.00	35110796.12	262156.00
18.5.111 0.00	111.00	35110683.06	262153.29

18.5.112 0.00	112.00	34668617.35	262152.35
18.5.113 0.00	113.00	34668648.71	262151.88
18.5.114 0.00	114.00	34668645.12	262152.71
18.5.115 0.00	115.00	34668669.94	262152.35
18.5.116 0.00	116.00	34668618.41	262152.35
18.5.117 0.00	117.00	34668647.47	262151.88
18.5.118 0.00	118.00	34668649.76	262152.71
18.5.119 0.00	119.00	34668675.47	262153.06
18.5.120 0.00	120.00	34668610.82	262152.24
18.5.121 0.00	121.00	34668628.12	262152.41
18.5.122 0.00	122.00	34668622.06	262152.47
18.5.123 0.00	123.00	34668656.82	262152.65

18.5.124 0.00	124.00	34668609.06	262152.35
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18.5.125 0.00	125.00	34668621.82	262152.29
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18.5.126 0.00	126.00	34668615.29	262152.35
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18.5.127 0.00	127.00	34668654.59	262152.53
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18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	391.57	456.12	600.46	626.98	1059.92
18.6.1	1.00	400.73	464.06	613.57	640.48	1085.23
18.6.2	2.00	405.31	469.42	619.26	646.82	1100.71
18.6.3	3.00	405.59	469.28	620.13	647.74	1102.09
18.6.4	4.00	411.61	478.18	619.66	645.33	1090.00
18.6.5	5.00	405.07	468.66	618.32	644.13	1093.90
18.6.6	6.00	432.87	498.33	644.43	670.77	1140.63
18.6.7	7.00	417.83	482.66	636.90	662.79	1141.76
18.6.8	8.00	390.14	456.23	597.24	623.04	1048.51
18.6.9	9.00	404.94	468.71	617.32	643.53	1091.91
18.6.10	10.00	399.44	462.79	610.17	638.04	1071.04
18.6.11	11.00	405.47	468.91	621.34	649.20	1101.84
18.6.12	12.00	411.65	478.04	618.87	644.75	1084.95

18.6.13	13.00	404.81	467.99	618.68	644.66	1092.11
18.6.14	14.00	425.93	490.47	630.57	657.30	1103.95
18.6.15	15.00	410.64	474.50	625.80	652.30	1114.49
18.6.16	16.00	483.16	509.02	532.11	578.15	1044.16
18.6.17	17.00	495.48	526.87	553.59	587.96	1086.62
18.6.18	18.00	489.23	520.81	545.80	582.05	1067.12
18.6.19	19.00	494.88	531.72	555.35	587.97	1097.10
18.6.20	20.00	501.17	529.19	555.12	597.79	1080.01
18.6.21	21.00	494.27	527.21	554.53	587.07	1087.26
18.6.22	22.00	515.96	539.70	565.47	609.98	1100.39
18.6.23	23.00	500.73	533.35	560.72	593.46	1110.12
18.6.24	24.00	482.64	510.35	535.06	572.46	1055.30
18.6.25	25.00	491.27	523.18	549.68	583.25	1081.04
18.6.26	26.00	494.83	529.22	555.32	587.92	1096.94
18.6.27	27.00	495.05	529.62	555.48	588.00	1098.58
18.6.28	28.00	504.08	529.63	555.72	596.88	1085.89
18.6.29	29.00	495.40	526.84	553.73	587.78	1089.71
18.6.30	30.00	522.30	552.92	581.77	617.27	1136.57
18.6.31	31.00	507.54	545.44	567.72	601.34	1137.14
18.6.32	32.00	354.66	535.92	636.29	672.79	1193.07
18.6.33	33.00	368.36	549.95	655.30	695.88	1229.96
18.6.34	34.00	361.74	543.83	649.20	686.78	1215.65
18.6.35	35.00	367.91	549.86	658.31	699.61	1237.24
18.6.36	36.00	373.97	555.61	656.98	696.14	1224.12

18.6.37	37.00	367.51	549.14	656.69	696.76	1230.50
18.6.38	38.00	390.52	572.42	674.28	715.84	1260.50
18.6.39	39.00	373.24	555.20	659.16	697.68	1236.57
18.6.40	40.00	350.47	532.91	636.62	675.89	1201.62
18.6.41	41.00	362.48	543.69	650.82	692.14	1226.33
18.6.42	42.00	365.85	547.06	656.34	697.52	1239.70
18.6.43	43.00	367.02	548.35	657.58	698.74	1240.24
18.6.44	44.00	374.20	555.32	656.91	697.93	1233.06
18.6.45	45.00	368.37	549.97	656.82	697.92	1238.27
18.6.46	46.00	396.53	578.90	681.64	724.65	1278.91
18.6.47	47.00	381.41	563.22	673.55	718.15	1276.44
18.6.48	48.00	404.30	501.81	515.28	532.58	989.21
18.6.49	49.00	414.62	511.85	527.75	546.38	1013.61
18.6.50	50.00	416.86	515.23	531.95	550.58	1024.96
18.6.51	51.00	418.23	516.03	532.86	551.53	1026.16
18.6.52	52.00	426.03	520.49	534.41	553.36	1021.68
18.6.53	53.00	420.39	517.43	533.57	551.97	1026.93
18.6.54	54.00	447.00	542.99	557.08	577.75	1066.86
18.6.55	55.00	432.24	531.89	549.12	570.59	1065.95
18.6.56	56.00	403.31	500.33	513.33	529.62	981.01
18.6.57	57.00	419.33	517.85	532.34	550.97	1020.94
18.6.58	58.00	413.02	511.53	525.09	541.59	1001.74
18.6.59	59.00	419.28	517.74	534.92	553.60	1027.06
18.6.60	60.00	425.29	519.69	533.76	552.09	1015.13

18.6.61	61.00	418.93	516.75	532.76	551.13	1021.28
18.6.62	62.00	441.00	534.50	549.28	569.36	1050.75
18.6.63	63.00	423.80	519.92	533.94	551.65	1025.46
18.6.64	64.00	548.48	574.13	659.60	700.47	1095.79
18.6.65	65.00	558.02	585.11	673.98	715.29	1121.61
18.6.66	66.00	558.42	583.40	679.14	720.80	1133.33
18.6.67	67.00	559.92	584.85	680.46	722.40	1134.47
18.6.68	68.00	562.83	588.97	676.87	718.05	1121.46
18.6.69	69.00	557.72	583.20	676.58	718.04	1126.55
18.6.70	70.00	584.73	612.32	701.56	743.55	1167.95
18.6.71	71.00	570.03	596.33	693.11	735.54	1167.41
18.6.72	72.00	542.81	566.38	655.46	694.26	1079.41
18.6.73	73.00	557.75	582.93	675.40	715.64	1121.82
18.6.74	74.00	552.24	576.22	668.79	708.29	1100.74
18.6.75	75.00	557.71	582.36	678.11	719.38	1127.83
18.6.76	76.00	563.03	588.14	676.45	716.23	1113.12
18.6.77	77.00	556.96	581.66	676.42	716.62	1119.96
18.6.78	78.00	578.53	605.12	694.27	734.96	1150.37
18.6.79	79.00	562.65	587.68	678.51	718.89	1123.03
18.6.80	80.00	448.06	502.83	545.22	565.24	1021.96
18.6.81	81.00	461.26	520.14	558.75	581.27	1062.75
18.6.82	82.00	454.82	513.48	552.74	575.26	1042.60
18.6.83	83.00	460.61	522.32	558.44	584.21	1068.71
18.6.84	84.00	466.85	521.59	564.17	584.88	1055.06

18.6.85	85.00	460.47	520.06	557.87	581.32	1061.84
18.6.86	86.00	482.68	536.88	580.30	601.18	1091.11
18.6.87	87.00	466.33	521.77	563.80	584.24	1065.30
18.6.88	88.00	447.07	505.73	546.61	568.45	1037.13
18.6.89	89.00	459.78	518.66	558.52	580.53	1062.87
18.6.90	90.00	461.26	523.14	558.92	585.18	1073.81
18.6.91	91.00	462.95	523.86	560.53	585.58	1075.17
18.6.92	92.00	466.95	522.52	564.01	584.76	1062.46
18.6.93	93.00	461.13	520.73	558.57	581.69	1067.61
18.6.94	94.00	488.26	544.75	586.46	607.61	1107.44
18.6.95	95.00	473.13	535.88	571.08	597.48	1106.76
18.6.96	96.00	429.55	499.49	609.15	624.93	1081.84
18.6.97	97.00	442.90	512.05	628.41	647.07	1121.76
18.6.98	98.00	436.84	504.47	622.91	640.09	1103.86
18.6.99	99.00	442.71	510.95	634.01	653.22	1131.89
18.6.100	100.00	449.00	519.56	629.72	647.35	1112.82
18.6.101	101.00	442.35	511.07	629.40	647.91	1120.18
18.6.102	102.00	463.71	532.29	641.95	659.07	1133.41
18.6.103	103.00	448.01	516.42	636.83	656.10	1141.13
18.6.104	104.00	427.43	498.58	612.88	629.64	1093.72
18.6.105	105.00	439.52	509.34	626.75	644.90	1118.44
18.6.106	106.00	444.04	512.09	632.56	651.25	1133.88
18.6.107	107.00	444.93	513.44	633.85	652.67	1135.81
18.6.108	108.00	449.22	520.33	630.17	647.98	1118.31

18.6.109	109.00	442.53	511.73	628.92	646.91	1122.32
18.6.110	110.00	470.22	539.91	654.68	674.70	1166.30
18.6.111	111.00	454.66	524.29	646.99	667.99	1165.86
18.6.112	112.00	481.84	514.98	520.85	547.29	1008.08
18.6.113	113.00	491.88	525.93	533.25	561.49	1031.31
18.6.114	114.00	494.36	529.58	537.51	567.56	1046.46
18.6.115	115.00	495.57	530.20	538.72	568.66	1047.79
18.6.116	116.00	500.77	530.76	539.73	565.69	1032.38
18.6.117	117.00	494.15	527.36	534.92	563.79	1035.06
18.6.118	118.00	521.65	554.36	561.10	591.91	1079.08
18.6.119	119.00	506.62	543.68	549.92	585.67	1077.99
18.6.120	120.00	478.86	509.44	519.27	540.88	995.29
18.6.121	121.00	494.06	528.28	535.19	564.26	1034.66
18.6.122	122.00	487.58	520.87	528.25	555.72	1016.39
18.6.123	123.00	493.64	530.23	536.63	569.90	1044.13
18.6.124	124.00	499.53	529.93	539.83	565.34	1027.08
18.6.125	125.00	492.96	527.11	535.52	565.29	1033.16
18.6.126	126.00	514.91	540.46	553.17	575.85	1046.00
18.6.127	127.00	499.65	532.82	541.08	573.30	1053.37

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	158.18	161.76	239.54	255.40	497.38
18.7.1	1.00	163.38	167.63	244.95	260.80	504.78
18.7.2	2.00	164.26	168.54	247.27	263.62	509.54

18.7.3	3.00	166.56	171.18	249.18	263.70	509.51
18.7.4	4.00	167.67	171.71	247.33	262.30	509.34
18.7.5	5.00	165.87	170.43	247.97	263.73	515.66
18.7.6	6.00	178.28	182.52	261.82	278.29	533.36
18.7.7	7.00	171.68	176.59	260.80	276.82	542.11
18.7.8	8.00	158.56	162.44	240.90	255.41	508.33
18.7.9	9.00	165.47	170.23	249.54	264.52	527.01
18.7.10	10.00	160.95	165.35	248.57	263.73	515.08
18.7.11	11.00	165.48	170.33	254.53	270.22	531.48
18.7.12	12.00	166.24	170.56	247.02	262.45	512.64
18.7.13	13.00	164.12	169.14	248.25	262.20	515.74
18.7.14	14.00	173.90	177.95	251.11	265.93	511.26
18.7.15	15.00	166.69	171.31	248.03	262.87	516.39
18.7.16	16.00	153.73	156.80	223.96	236.50	506.18
18.7.17	17.00	159.95	163.83	232.54	245.46	521.28
18.7.18	18.00	156.01	159.31	231.42	243.97	512.03
18.7.19	19.00	161.13	164.64	237.13	251.91	525.10
18.7.20	20.00	162.20	165.73	230.61	244.40	508.05
18.7.21	21.00	159.97	163.80	231.23	246.24	512.01
18.7.22	22.00	169.45	172.96	234.20	245.16	510.17
18.7.23	23.00	162.10	165.34	231.65	245.48	511.64
18.7.24	24.00	153.30	156.38	224.23	237.79	491.84
18.7.25	25.00	158.11	161.59	230.31	244.00	500.90
18.7.26	26.00	159.22	162.42	231.54	243.53	508.84

18.7.27	27.00	160.35	163.62	232.50	245.45	504.13
18.7.28	28.00	161.69	164.95	231.88	244.37	508.17
18.7.29	29.00	159.68	163.28	232.27	244.97	509.20
18.7.30	30.00	172.04	175.41	245.43	256.81	530.10
18.7.31	31.00	165.19	168.64	244.00	257.03	538.08
18.7.32	32.00	160.91	165.65	240.96	260.29	500.05
18.7.33	33.00	168.09	172.86	250.76	267.86	515.70
18.7.34	34.00	163.82	168.06	246.11	261.79	503.29
18.7.35	35.00	169.15	173.40	250.24	266.76	516.61
18.7.36	36.00	170.05	174.39	249.66	267.55	508.83
18.7.37	37.00	168.12	172.49	250.77	268.03	514.11
18.7.38	38.00	177.71	182.26	259.00	277.39	526.73
18.7.39	39.00	169.32	173.82	254.20	269.81	518.76
18.7.40	40.00	159.02	164.25	240.81	258.75	490.23
18.7.41	41.00	164.50	169.52	247.22	262.68	501.46
18.7.42	42.00	165.62	170.17	253.77	269.81	515.15
18.7.43	43.00	166.81	171.66	254.13	270.72	514.27
18.7.44	44.00	169.08	173.94	248.44	265.50	502.88
18.7.45	45.00	167.67	172.22	249.05	265.77	507.79
18.7.46	46.00	179.83	184.37	259.62	276.51	522.93
18.7.47	47.00	173.13	178.05	256.35	272.63	525.19
18.7.48	48.00	150.28	155.96	221.12	237.74	483.08
18.7.49	49.00	155.70	161.91	227.14	242.49	493.00
18.7.50	50.00	155.86	162.46	232.17	246.80	503.53

18.7.51	51.00	159.01	164.86	232.57	249.19	503.77
18.7.52	52.00	161.15	166.98	228.35	243.48	491.09
18.7.53	53.00	159.46	165.38	229.00	245.08	496.81
18.7.54	54.00	170.67	177.63	237.46	252.81	509.06
18.7.55	55.00	164.68	171.00	236.44	251.05	507.69
18.7.56	56.00	152.09	157.86	222.08	237.83	489.07
18.7.57	57.00	158.87	164.74	230.25	246.94	506.17
18.7.58	58.00	154.14	161.00	224.10	240.60	490.39
18.7.59	59.00	158.77	164.78	230.90	247.37	503.40
18.7.60	60.00	159.81	165.89	229.16	246.22	500.97
18.7.61	61.00	157.68	163.64	229.95	246.75	505.09
18.7.62	62.00	167.06	173.53	238.35	254.41	517.21
18.7.63	63.00	159.64	165.42	230.84	249.02	506.16
18.7.64	64.00	154.85	157.76	244.16	261.79	504.03
18.7.65	65.00	159.95	163.22	251.25	267.25	517.57
18.7.66	66.00	160.66	163.97	257.91	274.57	525.43
18.7.67	67.00	163.38	167.04	258.38	274.64	528.97
18.7.68	68.00	164.71	167.84	250.70	267.12	513.01
18.7.69	69.00	162.68	166.13	252.23	268.66	517.20
18.7.70	70.00	174.62	178.17	262.53	278.38	529.42
18.7.71	71.00	167.81	171.55	259.23	275.22	533.12
18.7.72	72.00	155.60	158.33	244.83	260.80	513.71
18.7.73	73.00	162.09	165.17	253.02	268.36	533.01
18.7.74	74.00	157.69	160.91	247.70	262.84	510.19

18.7.75	75.00	161.78	165.42	252.40	269.41	526.87
18.7.76	76.00	163.37	166.38	251.49	268.73	522.40
18.7.77	77.00	160.93	164.26	253.25	270.21	528.01
18.7.78	78.00	170.86	173.73	261.97	277.59	537.93
18.7.79	79.00	162.86	166.06	255.59	272.81	528.52
18.7.80	80.00	153.54	157.09	218.79	234.54	503.32
18.7.81	81.00	160.75	164.10	226.91	244.43	522.33
18.7.82	82.00	156.09	159.77	221.03	237.71	497.88
18.7.83	83.00	161.80	165.18	226.80	243.02	516.39
18.7.84	84.00	162.63	166.13	226.56	243.27	513.33
18.7.85	85.00	160.87	164.30	227.06	243.95	517.24
18.7.86	86.00	169.92	173.54	234.79	251.13	526.25
18.7.87	87.00	162.42	165.69	228.56	246.36	516.54
18.7.88	88.00	152.78	156.80	220.14	236.55	495.32
18.7.89	89.00	158.67	162.39	225.46	242.45	507.88
18.7.90	90.00	158.93	162.63	230.94	246.67	516.81
18.7.91	91.00	161.04	164.65	231.01	248.07	517.44
18.7.92	92.00	161.86	165.33	225.12	241.73	501.05
18.7.93	93.00	160.43	163.83	224.95	243.07	507.49
18.7.94	94.00	171.99	175.66	234.31	250.94	518.66
18.7.95	95.00	165.82	169.36	231.00	248.30	521.87
18.7.96	96.00	155.27	162.39	239.42	256.19	505.75
18.7.97	97.00	162.12	168.72	248.00	264.57	523.58
18.7.98	98.00	158.02	164.68	247.34	264.62	513.26

18.7.99	99.00	163.12	170.18	253.20	270.85	526.77
18.7.100	100.00	164.73	171.48	246.44	261.93	505.17
18.7.101	101.00	162.49	168.45	246.83	262.27	511.22
18.7.102	102.00	171.44	178.52	249.35	265.13	506.91
18.7.103	103.00	163.79	171.18	246.59	261.74	512.26
18.7.104	104.00	155.04	161.75	240.82	256.06	493.34
18.7.105	105.00	160.10	166.08	245.45	261.55	504.25
18.7.106	106.00	161.11	167.67	247.20	264.16	507.43
18.7.107	107.00	162.30	168.61	247.93	262.80	509.81
18.7.108	108.00	164.13	170.88	246.07	262.69	507.75
18.7.109	109.00	161.75	167.98	247.41	263.35	512.30
18.7.110	110.00	174.11	181.13	259.83	278.53	534.32
18.7.111	111.00	166.88	173.93	258.02	275.52	539.71
18.7.112	112.00	152.42	157.01	224.69	240.25	484.21
18.7.113	113.00	157.53	162.73	230.39	244.61	490.57
18.7.114	114.00	158.41	163.43	232.96	245.42	498.14
18.7.115	115.00	161.12	166.03	233.03	247.02	498.44
18.7.116	116.00	162.48	167.21	233.43	245.34	494.11
18.7.117	117.00	160.25	165.33	232.59	244.55	496.85
18.7.118	118.00	172.38	177.31	245.76	258.99	519.54
18.7.119	119.00	165.62	171.16	243.74	257.15	522.19
18.7.120	120.00	152.98	157.68	225.92	237.97	491.55
18.7.121	121.00	159.63	164.64	234.21	246.04	506.59
18.7.122	122.00	155.57	160.54	232.14	245.77	500.48

18.7.123	123.00	159.92	164.88	237.18	252.31	513.00
18.7.124	124.00	161.16	165.32	232.04	245.56	495.12
18.7.125	125.00	158.85	163.52	231.92	245.19	498.53
18.7.126	126.00	167.71	172.90	234.82	247.63	496.63
18.7.127	127.00	160.65	165.84	231.71	245.88	500.39

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00
18.8.2	2.00	0.00	0.00	0.00	0.00	0.00
18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00
18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00
18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00
18.8.15	15.00	0.00	0.00	0.00	0.00	0.00
18.8.16	16.00	0.00	0.00	0.00	0.00	0.00

18.8.17	17.00	0.00	0.00	0.00	0.00	0.00
18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00
18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00
18.8.26	26.00	0.00	0.00	0.00	0.00	0.00
18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00
18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00
18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00
18.8.39	39.00	0.00	0.00	0.00	0.00	0.00
18.8.40	40.00	0.00	0.00	0.00	0.00	0.00

18.8.41	41.00	0.00	0.00	0.00	0.00	0.00
18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00
18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00
18.8.50	50.00	0.00	0.00	0.00	0.00	0.00
18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00
18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00
18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00
18.8.63	63.00	0.00	0.00	0.00	0.00	0.00
18.8.64	64.00	0.00	0.00	0.00	0.00	0.00

18.8.65	65.00	0.00	0.00	0.00	0.00	0.00
18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00
18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00
18.8.74	74.00	0.00	0.00	0.00	0.00	0.00
18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00
18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00
18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00
18.8.87	87.00	0.00	0.00	0.00	0.00	0.00
18.8.88	88.00	0.00	0.00	0.00	0.00	0.00

18.8.89	89.00	0.00	0.00	0.00	0.00	0.00
18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00
18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00
18.8.98	98.00	0.00	0.00	0.00	0.00	0.00
18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00
18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00
18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00
18.8.111	111.00	0.00	0.00	0.00	0.00	0.00
18.8.112	112.00	0.00	0.00	0.00	0.00	0.00

18.8.113	113.00	0.00	0.00	0.00	0.00	0.00
18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00
18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00
18.8.122	122.00	0.00	0.00	0.00	0.00	0.00
18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00
18.8.127	127.00	0.00	0.00	0.00	0.00	0.00

18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID	Channel	Min	Q1	Median
Q3	Max			
18.12.0	0.00	34563072.00	34563072.00	34563400.00
34780449.00	35453779.00			
18.12.1	1.00	34563072.00	34563072.00	34563384.00
34780412.00	35453704.00			
18.12.2	2.00	34563072.00	34563072.00	34563400.00
34780391.00	35453735.00			

18.12.3	3.00	34563072.00	34563072.00	34563384.00
34780440.00	35453613.00			
18.12.4	4.00	34563072.00	34563072.00	34563400.00
34780475.00	35453806.00			
18.12.5	5.00	34563072.00	34563075.00	34563384.00
34780403.00	35453749.00			
18.12.6	6.00	34563072.00	34563072.00	34563400.00
34780384.00	35453727.00			
18.12.7	7.00	34563072.00	34563072.00	34563384.00
34780451.00	35453619.00			
18.12.8	8.00	34563072.00	34563072.00	34563400.00
34780785.00	35453097.00			
18.12.9	9.00	34563072.00	34563075.00	34563384.00
34780772.00	35453015.00			
18.12.10	10.00	34563072.00	34563072.00	34563400.00
34780744.00	35452974.00			
18.12.11	11.00	34563072.00	34563072.00	34563384.00
34780766.00	35452926.00			
18.12.12	12.00	34563072.00	34563072.00	34563400.00
34780735.00	35453056.00			
18.12.13	13.00	34563072.00	34563072.00	34563384.00
34780772.00	35452989.00			
18.12.14	14.00	34563072.00	34563072.00	34563400.00
34780748.00	35452994.00			

18.12.15	15.00	34563072.00	34563072.00	34563384.00
34780747.00	35452925.00			
18.12.16	16.00	34563452.00	34565394.00	34566312.00
34984011.00	35695633.00			
18.12.17	17.00	34563447.00	34565398.00	34566315.00
34984172.00	35695141.00			
18.12.18	18.00	34563445.00	34565399.00	34566305.00
34983759.00	35695669.00			
18.12.19	19.00	34563452.00	34565400.00	34566325.00
34984411.00	35695247.00			
18.12.20	20.00	34563438.00	34565409.00	34566312.00
34983989.00	35695660.00			
18.12.21	21.00	34563416.00	34565409.00	34566331.00
34984181.00	35695130.00			
18.12.22	22.00	34563445.00	34565387.00	34566302.00
34983752.00	35695617.00			
18.12.23	23.00	34563420.00	34565397.00	34566313.00
34984395.00	35695212.00			
18.12.24	24.00	34563437.00	34565408.00	34566314.00
34984896.00	35695945.00			
18.12.25	25.00	34563416.00	34565408.00	34566329.00
34985100.00	35695505.00			
18.12.26	26.00	34563444.00	34565385.00	34566304.00
34984594.00	35695994.00			

18.12.27	27.00	34563420.00	34565399.00	34566315.00
34985213.00	35695506.00			
18.12.28	28.00	34563454.00	34565394.00	34566314.00
34984946.00	35695933.00			
18.12.29	29.00	34563446.00	34565399.00	34566314.00
34985132.00	35695520.00			
18.12.30	30.00	34563445.00	34565396.00	34566305.00
34984629.00	35696063.00			
18.12.31	31.00	34563454.00	34565397.00	34566324.00
34985209.00	35695509.00			
18.12.32	32.00	34563072.00	34563072.00	34563205.00
34802604.00	35308882.00			
18.12.33	33.00	34563072.00	34563072.00	34563216.00
34802570.00	35308763.00			
18.12.34	34.00	34563072.00	34563072.00	34563205.00
34802660.00	35308813.00			
18.12.35	35.00	34563072.00	34563072.00	34563213.00
34802459.00	35309010.00			
18.12.36	36.00	34563072.00	34563072.00	34563205.00
34802599.00	35308906.00			
18.12.37	37.00	34563072.00	34563072.00	34563216.00
34802572.00	35308793.00			
18.12.38	38.00	34563072.00	34563072.00	34563205.00
34802665.00	35308831.00			

18.12.39	39.00	34563072.00	34563072.00	34563213.00
34802477.00	35309010.00			
18.12.40	40.00	34563072.00	34563072.00	34563205.00
34802430.00	35309302.00			
18.12.41	41.00	34563072.00	34563072.00	34563216.00
34802387.00	35309259.00			
18.12.42	42.00	34563072.00	34563072.00	34563206.00
34802470.00	35309293.00			
18.12.43	43.00	34563072.00	34563072.00	34563213.00
34802281.00	35309508.00			
18.12.44	44.00	34563072.00	34563072.00	34563205.00
34802430.00	35309282.00			
18.12.45	45.00	34563072.00	34563072.00	34563216.00
34802383.00	35309207.00			
18.12.46	46.00	34563072.00	34563072.00	34563206.00
34802470.00	35309279.00			
18.12.47	47.00	34563072.00	34563072.00	34563213.00
34802259.00	35309492.00			
18.12.48	48.00	34563087.00	34564767.00	34568687.00
34661801.00	35277751.00			
18.12.49	49.00	34563089.00	34564770.00	34568671.00
34661695.00	35277225.00			
18.12.50	50.00	34563083.00	34564760.00	34568688.00
34661674.00	35277537.00			

18.12.51	51.00	34563092.00	34564773.00	34568673.00
34661600.00	35277122.00			
18.12.52	52.00	34563076.00	34564756.00	34568648.00
34661824.00	35277768.00			
18.12.53	53.00	34563096.00	34564753.00	34568638.00
34661709.00	35277204.00			
18.12.54	54.00	34563082.00	34564752.00	34568654.00
34661666.00	35277516.00			
18.12.55	55.00	34563096.00	34564761.00	34568628.00
34661588.00	35277138.00			
18.12.56	56.00	34563075.00	34564755.00	34568649.00
34662401.00	35278192.00			
18.12.57	57.00	34563096.00	34564756.00	34568638.00
34662277.00	35277635.00			
18.12.58	58.00	34563082.00	34564751.00	34568651.00
34662280.00	35278009.00			
18.12.59	59.00	34563096.00	34564761.00	34568631.00
34662202.00	35277601.00			
18.12.60	60.00	34563087.00	34564768.00	34568687.00
34662374.00	35278199.00			
18.12.61	61.00	34563089.00	34564770.00	34568670.00
34662287.00	35277640.00			
18.12.62	62.00	34563083.00	34564761.00	34568688.00
34662303.00	35278011.00			

18.12.63	63.00	34563092.00	34564772.00	34568669.00
34662194.00	35277593.00			
18.12.64	64.00	34563072.00	34563072.00	34566493.00
36465283.00	37798446.00			
18.12.65	65.00	34563072.00	34563072.00	34566565.00
36465163.00	37797578.00			
18.12.66	66.00	34563072.00	34563072.00	34566504.00
36465677.00	37798336.00			
18.12.67	67.00	34563072.00	34563072.00	34566569.00
36464973.00	37797535.00			
18.12.68	68.00	34563072.00	34563072.00	34566480.00
36465260.00	37798433.00			
18.12.69	69.00	34563072.00	34563072.00	34566542.00
36465186.00	37797585.00			
18.12.70	70.00	34563072.00	34563072.00	34566487.00
36465721.00	37798269.00			
18.12.71	71.00	34563072.00	34563072.00	34566547.00
36464996.00	37797487.00			
18.12.72	72.00	34563072.00	34563072.00	34566474.00
36467883.00	37797372.00			
18.12.73	73.00	34563072.00	34563072.00	34566546.00
36467824.00	37796678.00			
18.12.74	74.00	34563072.00	34563072.00	34566479.00
36468359.00	37797225.00			

18.12.75	75.00	34563072.00	34563072.00	34566541.00
36467666.00	37796488.00			
18.12.76	76.00	34563072.00	34563072.00	34566492.00
36467931.00	37797404.00			
18.12.77	77.00	34563072.00	34563072.00	34566553.00
36467822.00	37796664.00			
18.12.78	78.00	34563072.00	34563072.00	34566496.00
36468301.00	37797270.00			
18.12.79	79.00	34563072.00	34563072.00	34566551.00
36467624.00	37796489.00			
18.12.80	80.00	34563086.00	34564169.00	34567646.00
34755110.00	35905899.00			
18.12.81	81.00	34563088.00	34564163.00	34567616.00
34755716.00	35905871.00			
18.12.82	82.00	34563084.00	34564174.00	34567633.00
34755564.00	35905624.00			
18.12.83	83.00	34563088.00	34564158.00	34567625.00
34755771.00	35905684.00			
18.12.84	84.00	34563087.00	34564195.00	34567625.00
34755150.00	35905931.00			
18.12.85	85.00	34563088.00	34564161.00	34567627.00
34755768.00	35905912.00			
18.12.86	86.00	34563087.00	34564182.00	34567634.00
34755556.00	35905641.00			

18.12.87	87.00	34563088.00	34564169.00	34567621.00
34755756.00	35905701.00			
18.12.88	88.00	34563086.00	34564195.00	34567624.00
34755349.00	35906085.00			
18.12.89	89.00	34563088.00	34564162.00	34567629.00
34755958.00	35906006.00			
18.12.90	90.00	34563087.00	34564179.00	34567635.00
34755808.00	35905757.00			
18.12.91	91.00	34563088.00	34564169.00	34567622.00
34756072.00	35905805.00			
18.12.92	92.00	34563086.00	34564170.00	34567648.00
34755333.00	35906077.00			
18.12.93	93.00	34563088.00	34564165.00	34567617.00
34755943.00	35905956.00			
18.12.94	94.00	34563084.00	34564174.00	34567637.00
34755804.00	35905767.00			
18.12.95	95.00	34563088.00	34564157.00	34567624.00
34756060.00	35905806.00			
18.12.96	96.00	34563072.00	34563271.00	34563864.00
35205667.00	36060884.00			
18.12.97	97.00	34563072.00	34563262.00	34563873.00
35205383.00	36061042.00			
18.12.98	98.00	34563072.00	34563269.00	34563862.00
35205764.00	36060815.00			

18.12.99	99.00	34563072.00	34563263.00	34563877.00
35205627.00	36060755.00			
18.12.100	100.00	34563072.00	34563270.00	34563870.00
35205669.00	36060834.00			
18.12.101	101.00	34563072.00	34563259.00	34563870.00
35205386.00	36061061.00			
18.12.102	102.00	34563072.00	34563269.00	34563866.00
35205779.00	36060790.00			
18.12.103	103.00	34563072.00	34563262.00	34563869.00
35205642.00	36060720.00			
18.12.104	104.00	34563072.00	34563270.00	34563869.00
35205700.00	36061637.00			
18.12.105	105.00	34563072.00	34563259.00	34563869.00
35205195.00	36061807.00			
18.12.106	106.00	34563072.00	34563268.00	34563866.00
35205630.00	36061503.00			
18.12.107	107.00	34563072.00	34563262.00	34563868.00
35205611.00	36061470.00			
18.12.108	108.00	34563072.00	34563271.00	34563864.00
35205663.00	36061643.00			
18.12.109	109.00	34563072.00	34563263.00	34563872.00
35205222.00	36061746.00			
18.12.110	110.00	34563072.00	34563269.00	34563863.00
35205607.00	36061591.00			

18.12.111	111.00	34563072.00	34563264.00	34563877.00
35205629.00	36061472.00			
18.12.112	112.00	34563102.00	34563633.00	34570838.00
34787492.00	35034538.00			
18.12.113	113.00	34563088.00	34563655.00	34570824.00
34787652.00	35035068.00			
18.12.114	114.00	34563099.00	34563630.00	34570837.00
34787595.00	35034630.00			
18.12.115	115.00	34563088.00	34563658.00	34570819.00
34787761.00	35035197.00			
18.12.116	116.00	34563107.00	34563641.00	34570861.00
34787459.00	35034519.00			
18.12.117	117.00	34563091.00	34563662.00	34570840.00
34787616.00	35035059.00			
18.12.118	118.00	34563102.00	34563653.00	34570849.00
34787575.00	35034612.00			
18.12.119	119.00	34563088.00	34563658.00	34570837.00
34787748.00	35035206.00			
18.12.120	120.00	34563107.00	34563642.00	34570860.00
34787871.00	35033137.00			
18.12.121	121.00	34563089.00	34563662.00	34570842.00
34788012.00	35033700.00			
18.12.122	122.00	34563102.00	34563655.00	34570849.00
34788007.00	35033348.00			

18.12.123	123.00	34563088.00	34563659.00	34570839.00
34788153.00	35033799.00			

18.12.124	124.00	34563102.00	34563634.00	34570838.00
34787887.00	35033151.00			

18.12.125	125.00	34563088.00	34563657.00	34570825.00
34788038.00	35033699.00			

18.12.126	126.00	34563099.00	34563629.00	34570839.00
34788021.00	35033359.00			

18.12.127	127.00	34563088.00	34563659.00	34570819.00
34788146.00	35033802.00			

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